

Planning and Preparation

As a VI administrator, size your environment and create a specification of the software, tools, and external services that are required to build a VCF private cloud.

VCF Planning and Preparation Workbook

The workbook is a Microsoft Excel file that helps you gather the inputs that are required for deploying a VCF fleet including workload domains, VCF Identity Broker, vSphere Supervisor, and VCF Automation. You use the [VCF Planning and Preparation Workbook](#) for understanding the prerequisites for deploying the VCF fleet and VCF Instances, sizing your environment, as a dynamic VCF environment specification, for updating requirements according to your design, and more.

VCF Planning Tools

You can use the [VMware Configurations Maximums](#) and [VMware Ports and Protocols](#) tools to plan the VCF setup that can run your end user workloads according to the requirements of your organization.

Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#). For information about how to get onboarded to VCF, see [Overview of VCF Deploy, Converge, and Upgrade](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Deployment, Convergence, and Upgrade

Get to VMware Cloud Foundation or VMware vSphere Foundation 9. Use **VCF Installer** and **VCF Operations** for automated deployment workflows and manually deploy the final components to complete your platform.

You can deploy a new VCF or vSphere Foundation platform on pre-installed ESX hosts by using VCF Installer.

You can also use your existing virtual infrastructure, in multiple scenarios, as building blocks and converge it to a VCF or vSphere Foundation platform. For converging you update your existing infrastructure to version 9 prior to deploying more components by using VCF Installer.

You can upgrade your existing VCF 5.x environment, with a starting state that includes or not management components, to VCF 9.

In This Chapter

- [Overview of Deploy, Converge, and Upgrade](#)
- [Components Nodes in VCF and vSphere Foundation](#)
- [What is the VMware Cloud Foundation Installer?](#)
- [Deploying a New VMware Cloud Foundation or VMware vSphere Foundation Platform](#)
- [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#)
- [Upgrading Your VCF Management Domain to VMware Cloud Foundation 9.0](#)

Getting to Know VCF

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SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Overview of Deploy, Converge, and Upgrade

There are multiple paths to get to vSphere Foundation and VCF 9.0.x. Depending on your starting state, you follow one of the paths.

Paths to building an environment with version 9.0.x

Depending on the starting state of your environment, you can deploy, converge, or upgrade the environment to VMware Cloud Foundation 9.0.x. The scope of the deploy, converge, and upgrade guidance is management domain only as this is the minimum required logical block to form a VCF Instance.

Deploying a new VCF Instance with version 9.0.x

To install a new VCF fleet or VCF Instance without converging or upgrading any existing components, see [Deploying a New VMware Cloud Foundation or VMware vSphere Foundation Platform](#).

For visual guides to the sequence of installation operations, see the following journey maps.

- [Build Journey - Install a New vSphere Foundation Deployment](#)
- [Build Journey - Install a New VMware Cloud Foundation Deployment](#)

Starting with VCF 9.0 and later, the deployment process changes and you now install most of the components by using VCF Installer and VCF Operations workflows. Deployment of standalone components by deploying an OVA is typically not supported, apart from specific scenarios where you are explicitly instructed to do so.

- For information about the scope of components for installation by VCF Installer, see [What is the VMware Cloud Foundation Installer?](#).
- For information about the scope of components for installation after the VCF Installer workflow, see [Completing the Deployment of Your Platform](#).

By following the deployment guidance, you can directly deploy VCF 9.0.2 without first deploying 9.0.0 or 9.0.1 and then upgrading to 9.0.2.

Paths to building workload domains with version 9.0.x

After you create your management domain, you can proceed with the analogous, workload domain-specific guidance.

- To create a workload domain, see [Create a New Workload Domain](#).
- To import existing infrastructure as a workload domain in a VCF Instance, see [Import an Existing vCenter to Create a Workload Domain](#).
- To upgrade your workload domain components, see [Upgrading VMware Cloud Foundation 5.x Workload Domains to 9.0](#).

Converging your existing components to vSphere Foundation or VCF with version 9.0.x

Before converging

Before deciding which convergence path to follow, you can independently upgrade your existing pre-version 9.0 standalone components, that are not yet part of a VCF Instance, to version 9.0.x. For example, you can upgrade your vCenter, ESX, and Aria Operations to version 9.0. This component configuration is the minimum requirement to evaluate the capabilities of version 9.0.x, for up to 90 days, before you license the products from VCF Operations.

- [Upgrading vCenter to version 9.0](#)
- [Upgrading ESX to version 9.0](#)
- [Upgrading Aria Operations to version 9.0](#)

Note:

Starting with VCF and vSphere Foundation 9.0, you manage your licenses through VCF Operations across your entire fleet and can manage licenses for multiple VCF Operations instances from the VCF Business Services console ([vcf.broadcom.com](#)), a part of the Broadcom Support Portal. For more information, see [Licensing Overview](#).

Convergence prerequisites

After you chose a convergence path, based on your existing pre-version 9.0.x or independently upgraded version 9.0.x standalone components, you must proceed with creating your management domain by converging the components to a supported vSphere Foundation or VCF deployment model. For either path, first verify that you meet the minimum version requirements and supported configurations. See [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).

Converging to vSphere Foundation 9.0.x

If you decided to converge your standalone components to vSphere Foundation 9.0.x, you must follow a specific convergence order. See [Converge a vCenter Instance and ESX Hosts to vSphere Foundation Platform](#).

For a visual guide to the sequence of convergence operations, see the [Build Journey - Converge Existing vSphere to vSphere Foundation 9 with a New VCF Operations Instance](#) journey map.

Converging to VCF 9.0.x

If you decided to converge your virtual infrastructure to a VCF fleet or VCF Instance, again, you must follow a specific convergence order for VCF product components. Depending on your existing components, proceed with one of the converging scenarios. See [Supported Scenarios to Converge to VCF](#).

For a visual guide to the sequence of convergence operations, see the [Build Journey - Converge Your Infrastructure to VMware Cloud Foundation from Existing vSphere and VMware Aria Operations](#) journey map.

The table below lists all VCF components and each table row includes a link to a dedicated upgrade procedure. You must upgrade the management planes first, followed by core SDDC components in a distinct upgrade order.

#	VCF Component	Converging existing standalone components to VCF 9.0.x
1	Aria Suite Lifecycle	Yes*
2	Aria Operations / VCF Operations	Yes, automated or manual
3	Aria Automation / VCF Automation	Yes
4	VMware Identity Manager / VCF Identity Broker	No**
5	Aria Automation Orchestrator / VCF Operations orchestrator	Yes
6	Aria Operations for Networks / VCF Operations for networks	Yes
7	Aria Operations for Logs / VCF Operations for logs	No***
8	SDDC Manager	N/A
9	HCX / VCF Operations HCX	Yes
10	NSX	- VCF 9.0.0 - No - VCF 9.0.1 - Yes****
11	vCenter	Yes
12	ESX	Yes
13	vSAN	Yes

* VMware Aria Suite Lifecycle has no upgrade path to version 9. A VCF Operations fleet management appliance 9 is deployed and the existing connection with VMware Aria Operations 8.x is imported to the new appliance.

After you upgrade VMware Aria Automation 8.x to VCF Automation 9.0, the VCF Operations fleet management appliance starts managing VCF Automation 9.0, but VMware Aria Suite Lifecycle 8.x continues to manage VMware Identity Manager. Do not delete VMware Identity Manager as it is still an authentication source for VCF Automation 9.0. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x.

** No upgrade path for VMware Identity Manager. You must manually deploy VCF Identity Broker.

*** No upgrade path for VMware Aria Operations for Logs. Redeploy of VCF Operations for logs 9.0 is required.

**** Starting with VCF 9.0.1, NSX 4.1.0.2 or later deployments that are registered with vCenter appliances that do not use Enhanced Linked Mode, are upgraded to version 9.0.1 as a part of the VCF Installer workflows.

Upgrading a VCF Instance with version 5.x to version 9.0.x

If you have an existing VCF 5.x environment, see [Upgrading Your VCF Management Domain to VMware Cloud Foundation 9.0](#).

For a visual guide to the sequence of upgrade operations, see the [Holistic Journey - Fully Adopt VMware Cloud Foundation from Existing vSphere and VMware Aria Operations](#) journey map.

The table below lists all VCF components and each table row includes a link to a dedicated upgrade procedure. You must upgrade the management planes first, followed by core SDDC components in a distinct upgrade order.

#	VCF Component	Upgrading to version 9.0.x
1	Aria Suite Lifecycle	Yes*
2	Aria Operations / VCF Operations	Yes
3	Aria Automation / VCF Automation	Yes
4	VMware Identity Manager / VCF Identity Broker	No**
5	Aria Automation Orchestrator / VCF Operations orchestrator	Yes
6	Aria Operations for Networks / VCF Operations for networks	Yes
7	Aria Operations for Logs / VCF Operations for logs	No***
8	SDDC Manager	Yes
9	HCX / VCF Operations HCX	Yes
10	NSX	Yes****
11	vCenter	Yes
12	ESX	Yes
13	vSAN	Yes

* VMware Aria Suite Lifecycle has no upgrade path to version 9. A VCF Operations fleet management appliance 9 is deployed and the existing connection with VMware Aria Operations 8.x is imported to the new appliance.

After you upgrade VMware Aria Automation 8.x to VCF Automation 9.0, the VCF Operations fleet management appliance starts managing VCF Automation 9.0, but VMware Aria Suite Lifecycle 8.x continues to manage VMware Identity Manager. Do not delete VMware Identity Manager as it is still an authentication source for VCF Automation 9.0. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x.

** No upgrade path for VMware Identity Manager. You must manually deploy VCF Identity Broker.

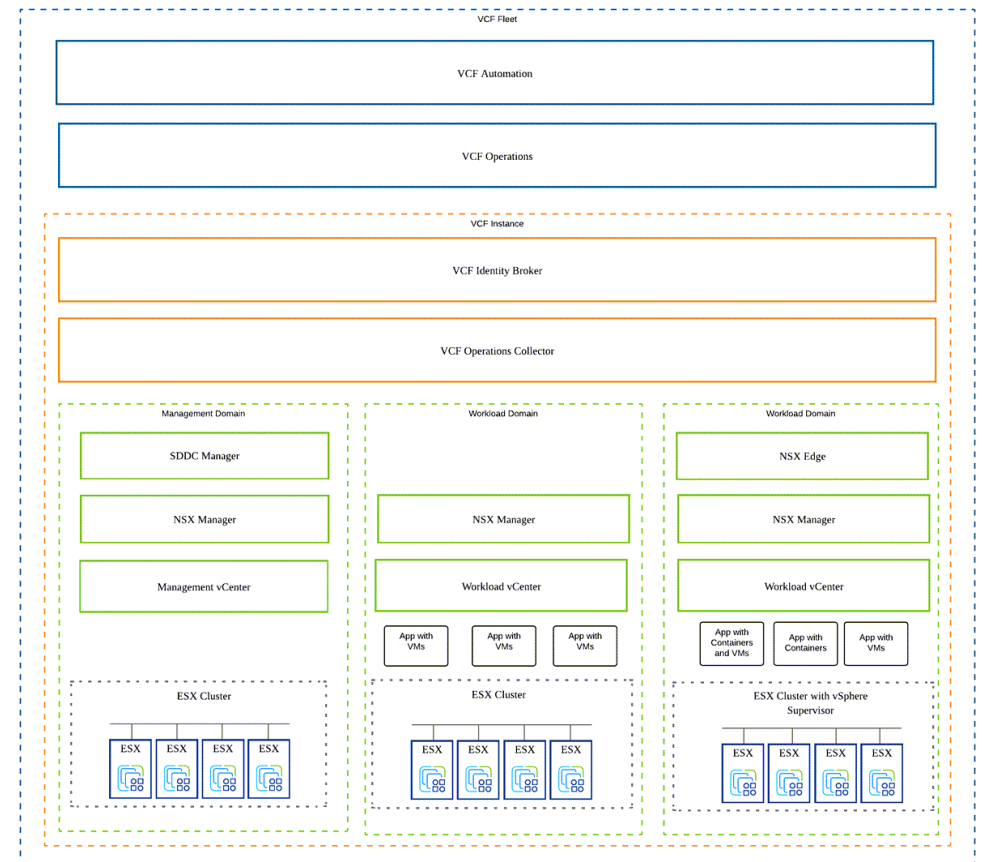
*** No upgrade path for VMware Aria Operations for Logs. Redeploy of VCF Operations for logs 9.0.x is required.

**** NSX 4.x deployments that are not part of VCF instances can be upgraded to version 9.0.x after they are first imported to existing VCF instances as workload domains. See [Import an Existing vCenter to Create a Workload Domain](#) and [Upgrading VMware Cloud Foundation 5.x Workload Domains to 9.0](#). Starting with VCF 9.0.1, NSX 4.1.0.2 or later deployments that are not part of VCF instances and are registered with vCenter appliances that do not use Enhanced Linked Mode, can be upgraded to version 9.0.1 as a part of a management domain by using the VCF Installer convergence workflow.

Components Nodes in VCF and vSphere Foundation

The functionality of VCF and vSphere Foundation is implemented by a set of components nodes. For VCF, these nodes are deployed in the management domain of the VCF Instances. For vSphere Foundation, you can run the nodes on a management vCenter.

Figure 123: Example VCF Nodes Configuration in a VCF Fleet



Component Node	Platform Type	Description
VCF Operations	- VCF - vSphere Foundation	See VCF Operations Overview .
VCF Operations collector (also known as VCF Operations cloud proxy)	- VCF - vSphere Foundation	A part of VCF Operations. Collects and monitors data locally in the VCF Instance.
VCF Operations for logs	VCF	A part of VCF Operations.

Component Node	Platform Type	Description
	vSphere Foundation	Adds intelligent logging and log analysis to VCF Operations.
VCF Operations for networks	VCF	Adds intelligent network analytics to VCF Operations.
VCF Operations for networks collector	VCF	A part of VCF Operations. Collects and monitors network data locally in the VCF Instance.
VCF Operations fleet management	VCF	A part of VCF Operations. Provides support for deployment, lifecycle management, and configuration for the VCF Operations, VCF Automation, and VCF Identity Broker.
VCF Identity Broker	VCF	Connects an identity provider (IdP) to the components of the VCF fleet for single sign-on.
VCF Operations HCX	VCF	A part of VCF Operations. Supports workload migration between vCenter instances or between clusters.
VCF Automation	VCF	See VCF Automation Overview .
SDDC Manager	VCF	A part of VCF Operations. Provides support for the following functionality in VCF Operations: - Lifecycle management for ESX, vCenter, and NSX - Deployment of workload domains - Import of vCenter instances as workload domains - Configuration of vSAN stretched clusters - Automated certificate replacement and password management
NSX Manager	VCF	See NSX Overview .
NSX Edge	VCF	Provides north-south traffic connectivity between the physical data center networks and the NSX SDN networks, and east-west traffic flow between virtualized workloads. It provides stateful services such as load balancing and DHCP.
vCenter	- VCF - vSphere Foundation	See vSphere Overview .

For information about the architecture of these nodes, see [VCF Design](#).

Component Comparison Between VCF 5.x and VCF 9.0

For users familiar with VCF 5.x, the following table maps components of VCF 5.x to their equivalents in 9.0.

VCF 5.x Component	VCF 9.0 Component
VMware Cloud Builder	VCF Installer

VCF 5.x Component	VCF 9.0 Component
SDDC Manager	SDDC Manager
VMware Aria Operations	VCF Operations
VMware Aria Operations for logs	VCF Operations for logs
VMware Aria Operations for networks	VCF Operations for networks
VMware Aria Lifecycle	VCF Operations fleet management
Workspace ONE Access	-
-	VCF Identity Broker
VMware HCX	VCF Operations HCX
VMware Aria Automation	VCF Automation
VMware Aria Automation Orchestrator	VCF Operations orchestrator
VMware Cloud Director	VCF Automation
VMware vSphere with Tanzu/VMware IaaS Control Plane	vSphere Supervisor

What is the VMware Cloud Foundation Installer?

VMware Cloud Foundation Installer is a dedicated virtual machine, that helps you plan, configure, and deploy all the required VMware Cloud Foundation and VMware vSphere Foundation components. VMware Cloud Foundation Installer is delivered together with the SDDC Manager appliance and both are located in the `VCF-SDDC-Manager-Appliance-9.x.x.ova` file.

Use the VCF Installer to:

- Download and manage the binaries required to deploy VCF or VMware vSphere Foundation.
- Deploy and configure a new VCF or VMware vSphere Foundation platform. The installation process automates the deployment and configuration of the components in each solution. You can use existing VMware Cloud Foundation Operations and VMware Cloud Foundation Automation instances or deploy new ones.
- Use existing vSphere infrastructure to create a VCF or VMware vSphere Foundation platform.

VCF Installer works with major, minor, and maintenance releases, but does not support express patch releases. After the VCF Installer workflows complete, you can apply express patch releases manually.

Depending on where you deploy it, the same VCF Installer appliance is able to deploy multiple VCF platforms. If you deploy the appliance inside the management infrastructure (on one of the hosts that will form the management domain), then the appliance switches into SDDC Manager mode and can no longer be used in installer mode.

Table 812: VCF Installer Deployment Scope per Platform

Component	Component Appliances	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
VMware vCenter	vCenter	Yes	Yes	Yes
VCF Operations	operations (primary node)	Yes	Yes	Yes
	operations (replica node)	No	Yes	No
	operations (data node)	No	Yes	No
	operations-collector (cloud proxy)	Yes	Yes	No
	fleet management	Yes	Yes	No

Component	Component Appliances	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
VMware NSX	NSX (node 1)	Yes	Yes	No
	NSX (node 2)	No	Yes	No
	NSX (node 3)	No	Yes	No
VCF Automation*	automation	Yes	Yes	No
	automation	No	Yes	No
	automation	No	Yes	No
SDDC Manager	SDDC Manager	Yes	Yes	No

*You can skip the VCF Automation deployment by VCF Installer and deploy it later as a Day-N operation.

The VCF Installer can either deploy the required components or use existing components to create your VCF or VMware vSphere Foundation platform. For scenarios with existing components, see [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).

See [What Is VMware Cloud Foundation?](#) for more information about VCF.

For information about deploying optional components for VCF and vSphere Foundation, see [Completing the Deployment of Your Platform](#).

Deploying a New VMware Cloud Foundation or VMware vSphere Foundation Platform

The process of deploying VMware Cloud Foundation or VMware vSphere Foundation involves preparation, deployment, and post-deployment tasks.

Preparation:

- [Preparing ESX Hosts for VMware Cloud Foundation or vSphere Foundation](#)
- [Deploy the VMware Cloud Foundation Installer Appliance](#)
- [Downloading Binaries to VCF Installer](#)

Deployment:

- [Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard](#)
- [Start a New vSphere Foundation Deployment by Using the VCF Installer Deployment Wizard](#)

Post-deployment:

- [Completing the Deployment of Your Platform](#)

Preparing to Deploy a New VMware Cloud Foundation or vSphere Foundation Platform

Before you can use the VCF Installer to deploy a new VMware Cloud Foundation or VMware vSphere Foundation platform, you must prepare your environment.

Before you start deploying a new platform, you must deploy and configure the necessary infrastructure. You can use the *Planning and Preparation Workbook* to gather all of the information you need to deploy VCF or vSphere Foundation. You can also refer to the [VMware Cloud Foundation Design](#) for specific requirements and recommendations for the design of each component. Before you deploy a new platform, you must:

- Review the Prerequisite Checklist in the Planning and Preparation Workbook. See [Planning and Preparation](#).

- Image your ESX hosts. See [Preparing ESX Hosts for VMware Cloud Foundation or vSphere Foundation](#).
- Download the VCF Installer appliance from the Broadcom Support portal.
- Download the VMware Cloud Foundation Download Tool from the Broadcom Support portal if the VMware Cloud Foundation Installer appliance does not have direct access to the online depot (dl.broadcom.com).

Preparing ESX Hosts for VMware Cloud Foundation or vSphere Foundation

Before you can begin the process of deploying a new VMware Cloud Foundation or VMware vSphere Foundation platform, you must prepare the ESX hosts.

Note:

The number of required hosts depends on the type of storage and the deployment model that you plan to use. You can use the *Planning and Preparation Workbook* to determine the minimum number of ESX hosts required for your environment. See [Planning and Preparation](#).

Preparing the ESX hosts involves installing the correct version of ESX and performing some basic configuration tasks.

For the supported ESX version, see the *VMware Cloud Foundation Release Notes*.

If your environment requires a custom ISO file for ESX, you can create one using VMware PowerCLI or vSphere Lifecycle. See [Creating a Custom ESX ISO Image Using VMware PowerCLI](#) and [Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager](#).

You might need to create a custom ISO image for ESX in the following situations:

- The ESX version specified in the VMware Cloud Foundation Release Notes does not have an associated ISO file on the Broadcom Support Portal. This can be the case for ESX patch releases.
- You need an async patch version of ESX.
- You need a vendor-specific (OEM) ISO file.

Note:

VMware Cloud Foundation does not support stateless ESX hosts.

With VMware Cloud Foundation 9.0.1 and later, you can use non-certified ESA disks during vSAN ESA cluster configuration for proof-of-concept purposes only. For more information, see [KB 408300](#).

Install ESX on Hosts Using an ISO Image

Install ESX on all hosts you plan to use in your VMware Cloud Foundation or VMware vSphere Foundation platform.

Download the ESX ISO from the Broadcom Support Portal. For the supported ESX versions, see the *VMware Cloud Foundation Release Notes*. If the required version of ESX does not have an ISO available on the Broadcom Support Portal, you can create one. See [Creating a Custom ESX ISO Image Using VMware PowerCLI](#) or [Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager](#).

1. Mount the ESX ISO on the host and restart the machine.
2. Set the BIOS or UEFI to boot from the mounted ISO.
3. On the welcome screen, press **Enter** to continue.
4. Accept the End User License Agreement by pressing **Enter**.
5. On the **Select a Disk to Install or Upgrade** screen, select the drive on which to install ESX on and press **Enter**.
6. Select the keyboard type for the host.
You can change the keyboard type after installation in the direct console.

7. Enter the root password for the host.
8. In the **Confirm Install** screen, press **F11** to confirm the start of the installation.
9. On the **Installation Complete** screen, press **Enter** to reboot the host.
10. Set the first boot device to be the drive on which you installed ESX.
11. Repeat this procedure for all remaining hosts.

[Configure the Network on VMware Cloud Foundation or vSphere Foundation Hosts.](#)

Configure the Network on VMware Cloud Foundation or vSphere Foundation Hosts

Use the ESX Direct Console User Interface (DCUI) for host network configuration and administrative access.

Perform the following tasks to configure the host network settings:

- Configure the network adapter (vmk0) and VLAN ID for the Management Network.
- Configure the IP address, subnet mask, gateway, DNS server, and FQDN for the ESX host.

Repeat this procedure for all hosts that you are adding to your VCF or VMware vSphere Foundation platform. If you completed the *Planning and Preparation Workbook*, refer to it for the required values.

1. Open the DCUI of the ESX host.
 - a) Open a console window to the host.
 - b) Press F2 to enter the DCUI.
 - c) Log in using the `esx_root_user_password`.
2. Configure the network.
 - a) Select **Configure Management Network** and press Enter.
 - b) Select **VLAN (Optional)** and press Enter.
 - c) Enter the VLAN ID for the ESX Management Network and press Enter.
 - d) Select **IPv4 Configuration** and press Enter.
 - e) Select **Set static IPv4 address and network configuration** and press the Space bar.
 - f) Enter the IPv4 Address, Subnet Mask and Default Gateway and press Enter.
 - g) Select **DNS Configuration** and press Enter.
 - h) Select **Use the following DNS Server address and hostname** and press the Space bar.
 - i) Enter the Primary DNS Server, Alternate DNS Server and Hostname (FQDN) and press Enter.
 - j) Select **Custom DNS Suffixes** and press Enter.
 - k) Ensure that there are no suffixes listed and press Enter.
3. Press Escape to exit and press Y to confirm the changes.
4. Repeat this procedure for all remaining hosts.

[Configure the Virtual Machine Network Port Group on VMware Cloud Foundation or vSphere Foundation Hosts.](#)

Configure the Virtual Machine Network Port Group on VMware Cloud Foundation or vSphere Foundation Hosts

Configure the Virtual Machine Network port group for each ESX host by using the VMware Host Client.

You configure the VLAN ID of the VM Network port group on the vSphere Standard Switch. This configuration provides connectivity to the Management network to allow communication during the automated deployment.

Repeat this procedure for all hosts that you are adding to your VCF or VMware vSphere Foundation environment. If you completed the *Planning and Preparation Workbook*, refer to it for the required values.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. Click **OK** to join the Customer Experience Improvement Program.
3. Configure a VLAN for the VM Network port group.
 - a) In the navigation pane, click **Networking**.
 - b) Click the **Port groups** tab, select the **VM network** port group, and click **Edit Settings**.
 - c) On the **Edit port group - VM network** page, enter the VM Management Network VLAN ID, and click **Save**.
4. Repeat this procedure for all remaining hosts.

[Configure NTP on VMware Cloud Foundation or vSphere Foundation Hosts.](#)

Configure NTP on VMware Cloud Foundation or vSphere Foundation Hosts

Complete the initial configuration of all ESX hosts by configuring the NTP service to avoid time synchronization issues.

Repeat this procedure for all ESX hosts you are adding. If you completed the *Planning and Preparation Workbook*, refer to it for the required values.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. Configure and start the NTP service.
 - a) In the navigation pane, click **Manage**, and click the **System** tab.
 - b) Click **Time & date** and click **Edit NTP Settings**.
 - c) On the **Edit NTP Settings** page, select the **Use Network Time Protocol (enable NTP client)** radio button, and change the NTP service startup policy to **Start and stop with host**.
 - d) In the **NTP servers** text box, enter the NTP Server FQDN or IP Address, and click **Save**.
 - e) Click the **Services** tab, select **ntpd**, and click **Start**.
3. Repeat this procedure for all remaining hosts.

[Regenerate the Self-Signed Certificate on ESX Hosts.](#)

Regenerate the Self-Signed Certificate on ESX Hosts

Once you have configured the ESX hosts' identity by providing a hostname, you must regenerate the self-signed certificate to ensure the correct common name is defined in the certificate.

During the installation of ESX, the installer generates a self-signed certificate for each ESX host, but the process is performed prior to the ESX hostname being configured. This means all ESX hosts have a common name in their self-signed certificate of `localhost.localdomain`. All communication between the VCF Installer and the ESX hosts is performed securely over HTTPS and as a result it validates the identity when making a connection by comparing the common name of the certificate against the FQDN provided within the VCF Installer deployment wizard.

To ensure that the connection attempts and validation do not fail, you must manually regenerate the self-signed certificate after the hostname has been configured.

Note:

If corporate policy requires that you use external CA-signed certificates instead of VMCA-signed certificates for ESX hosts, you can add external certificates to the hosts using the [SDDC Manager API](#).

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. In the **Actions** menu, click **Services > Enable Secure Shell (SSH)**.
3. Log in to the ESX host using an SSH client such as Putty.
4. Regenerate the self-signed certificate by executing the following command:

```
/sbin/generate-certificates
```
5. Reboot the ESX host.
6. Log back in to the VMware Host Client and click **Services > Disable Secure Shell (SSH)** from the **Actions** menu.
7. Repeat this procedure for all remaining hosts.

Configure ESX Hosts with Signed Certificates

If corporate policy requires that you use external CA-signed certificates instead of VMCA-signed certificates for ESX hosts, you can manually add external certificates to the hosts.

External CA-signed certificate and key are available.

When you install ESX software on a server to create an ESX host, the host initially has an autogenerated certificate. By default, when the ESX host is added to a vSphere cluster during deployment of the management domain or other operations involving ESX hosts (for example, host commissioning, workload domain creation, and so on), the autogenerated certificate is replaced with a certificate that is signed by the VMware Certificate Authority (VMCA).

When you use external certificates during deployment, they are not replaced by VMCA-signed certificates. Once you perform deployment with external certificates for ESX hosts, all future ESX hosts added to VCF must also use external certificates.

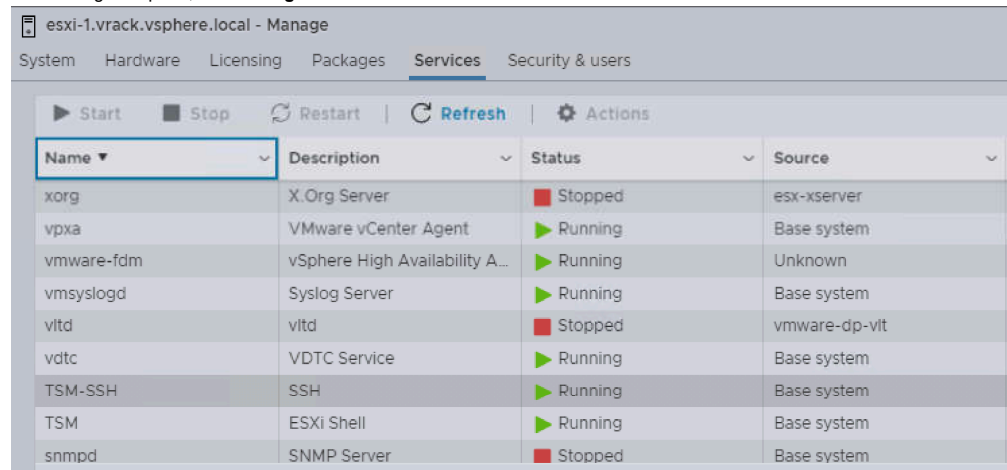
To deploy the management domain using external certificates, you must use a JSON specification file with the VCF Installer. You cannot use the VCF Installer deployment wizard. In the JSON specification file, use the `SecuritySpec` section in the `SddcSpec` to specify a value of `Custom` for `esxiCertsMode`. For example:

```
"securitySpec" : {
  "esxiCertsMode" : "Custom",
  "rootCaCerts" : [ {
    "alias" : "Rainpole-CA",
    "certChain" : [ "-----BEGIN CERTIFICATE-----
MIDCczCCAlugAwIBAgIQI9xwbTkI9J5GhMffcP5CHDANBgkqhkiG9w0BAQsFADBM
MRIwEAYKCZImiZPyLQGGBGRYCaW8xGDAWBgoJkiaJk/IsZAEZFgghyYw1ucG9sZTEc
MBoGA1UEAxMTcmFpbnBvbG9tZGMwMXJwbC1DQTAEFw0yMDAzMzAxNDQ2MTNwFw0y
NTAzMzAxNDU2MTNwFwEjAQBg9oJkiaJk/IsZAEZFgJpbzEYMBYGCgmSjomT8ixk
ARkKWHJhaW5wb2x1MRwwGgYDVQQDEXNyYw1ucG9sZS1kYzAxcmBsLUNBMTIIBIjAN
BgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEazpwkz7aPlQcfevcCelHc9DPswHkd
kjY96Vh3GvYlesavEcy/q/BoVvh3KgLMly8r7cy2cNPO3FANKOfqVdVx3ghfEUyL
g6lW9BskAlwryzJrmjhOJJVqvB8CWjy+eCp7MejHGdEud6WdEvK8CaBcPngEgOKM
eLrNLGe8OCw8yY4GTTrjU+H7PYQZtyD0kxy5f47ueaDXat4ENRGcAuHEfCoMGfAr
bDue1oO4diHd900bcYm5ggBNX0jhRudNULXPTayZl2ksImV0+QkaVeptQImXfCgb
kgnHQJ5CxK26up7fB5eAsmGLAsJLbnHuM7P9xvV09EvWjFCgLGX/oBBDYtQIDAQAB
o1EwTzALBgNVHQ8EBAMCAYYwDwyDVR0TAQH/BAUwAwEB/zAdBgNVHQ4EFgQU7oOq
QBK8ygm8mHnAfb+u6/GO0ZucwEAYJKwYBBAGCNxUBBAMCAQAwDQYJKoZIhvcNAQEL
BQADggEBALYxZGj4vWjFDNlatOUsBx2jrmxbExgMayRpNlSc2aj+7vzxHxUW5VbX
x9nc/BfkTiCK6c7Y9VYb+mgjb8z0kNv58sT4ar1yI1ln63VOCoyyLcaFB8HyEJpD
wUhz4RNPoSijZMpm+M5EuSLfWlHeJo7N8sLqHgvvk1dFpbK8fIHbPS5KJwJbbPe
w9UuNRdcxN9hFWKBC0SvfgX+1CJxVdvgfi65rSHpWinJzrXXdh999DfpDESrzwH
0pqE3GtMct1Nqalp2QJFdahbT+kxj7QWHTjUy1SENDHjdln7a8WH8RGxvEy/97YZ
+crXmxvQ/bAgHk9vcrERBRjfyIs7v88=
```

```
-----END CERTIFICATE-----" ] } } }
```

See the [VCF Installer API Reference Guide](#) for more information.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. In the navigation pane, click **Manage** and click the **Services** tab.



esxi-1.vrack.vsphere.local - Manage			
System Hardware Licensing Packages Services Security & users			
▶ Start ■ Stop ↺ Restart ↻ Refresh ⚙ Actions			
Name ▼	Description ▼	Status ▼	Source ▼
xorg	X.Org Server	Stopped	esx-xserver
vpwa	VMware vCenter Agent	Running	Base system
vmware-fdm	vSphere High Availability A...	Running	Unknown
vmsyslogd	Syslog Server	Running	Base system
vitd	vitd	Stopped	vmware-dp-vit
vdte	VDTC Service	Running	Base system
TSM-SSH	SSH	Running	Base system
TSM	ESXi Shell	Running	Base system
snmpd	SNMP Server	Stopped	Base system

3. Select the **TSM-SSH** service and click **Start** if not started.
4. Log in to the ESX Shell for the first host, either directly from the DCUI or from an SSH client, as a user with administrator privileges.
5. In the directory `/etc/vmware/ssl`, rename the existing certificates using the following commands:

```
mv rui.crt orig.rui.crt
mv rui.key orig.rui.key
```
6. Copy the external certificate and key that you want to use to `/etc/vmware/ssl`.
7. Rename the external certificate and key to `rui.crt` and `rui.key`.
8. Restart the host management agents by running the following commands:

```
/etc/init.d/hostd restart
/etc/init.d/vpwa restart
```
9. In the VMware Host Client, select the **TSM-SSH** service for the ESX host and click **Stop**.
10. Repeat for all the ESX hosts that you are adding to VMware Cloud Foundation.

Creating a Custom ESX ISO Image Using VMware PowerCLI

You can use VMware PowerCLI to create a custom ISO for installing ESX.

VMware PowerCLI 13.0 or later.

Download the zip files for the following:

- ESX patch for the ESX version specified in the VMware Cloud Foundation Release Notes or in the list of supported async patches in [KB 88287](#). You can download patches from the Broadcom Support Portal.
- OEM add-on for ESX from the Broadcom Support Portal. If the ESX version specified in the BOM is not available, contact your vendor to determine which OEM add-on version to use.

1. Gather the required information for the software spec that is used to create the custom ISO.

- a) In VMware PowerCLI, use the [Get-DepotBaseImages](#) cmdlet to get the base image version from the zip file for the ESX patch that you downloaded from the patches portal.

For example:

```
Get-DepotBaseImages "c:\temp\VMware-ESX-9.0.0-xxxxxxx-depot.zip"
```

- b) Use the [Get-DepotAddons](#) cmdlet to get the add-on name and version from the zip file for the OEM add-on for ESX that you downloaded from the Broadcom Support Portal. (if applicable)

For example:

```
Get-DepotAddons "c:\temp\HPE-9.0.0.10.x-xxx-Synergy-Addon-depot.zip"
```

2. Create the software spec using the information you gathered in step 1.

The software spec is a JSON file that contains information about the ESX version and vendor add-on (if applicable).

For example:

```
{
  "add_on": {
    "name": "HPE-Custom-Syn-AddOn",
    "version": "9.0.0.x.x-xx"
  },
  "base_image": {
    "version": "9.0.0-0.xx.xxxxxxxx"
  },
  "components": null,
  "hardware_support": null,
  "solutions": null
}
```

3. In VMware PowerCLI, use the [New-IsoImage](#) cmdlet to generate a custom ISO.

For example:

```
New-IsoImage -SoftwareSpec "c:\temp\HPE-70Uld-custom.JSON" -Depots "c:\temp\VMware-ESX-9.0-17551050-depot.zip" , "c:\temp\HPE-9.0.0.xx.x.x-xxxxxxx-Synergy-Addon-depot.zip" -Destination "c:\temp\HPE-9.0-custom.iso"
```

Provide the path to the software spec you created in step 2.

The depot(s) include the path to the zip files for the supported ESX version and vendor add-on.

The destination include the path and file name for the custom ISO file.

For more information about the [New-IsoImage](#) cmdlet, see <https://developer.broadcom.com/powercli/latest/vmware.imagebuilder/commands/new-isoimage>.

Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager

If you have access to a vSphere environment, you can use vSphere Lifecycle Manager to create and export a custom ISO.

Import the ESX patch and vendor add-on (if applicable) zip files to the vSphere Lifecycle Manager depot. See [Import Updates to the vSphere Lifecycle Manager Depot](#).

- Log in to vCenter using the vSphere Client.
- Create a new temporary cluster, selecting the **Manage all hosts in the cluster with a single image** check box.
- Select the ESX version and vendor add-on (optional) and click **OK**.
- Export the vSphere Lifecycle Manager image as an ISO.
See [Export an Image](#).
- Delete the temporary cluster.

Deploy the VMware Cloud Foundation Installer Appliance

Deploy the VCF Installer appliance on a network that can access the ESX hosts and VM management network that you plan to use for the deployment. The appliance must be able to access all required external services, such as DNS and NTP.

Before you deploy the VCF Installer appliance, verify that your environment meets the requirements for this process.

Resource Requirements	4 vCPUs 16 GB RAM 914 GB storage
Network Requirements	- Verify that the static IP address and FQDN for the VCF Installer appliance are available. - Verify that connectivity is in place from the VCF Installer appliance and the management VLAN used in the deployment.
Deployment Platform	You can deploy the VCF Installer appliance on an ESX host or on a laptop running VMware Workstation or VMware Fusion.

Where you deploy the VCF Installer appliance impacts how it can be used. You can deploy the appliance:

- Outside of the management infrastructure on any machine with connectivity to the management infrastructure. In this case, you can use the appliance to deploy multiple VCF or VMware vSphere Foundation platforms.
- On one of the ESX hosts that will form the management domain. In this case, you can only use the appliance to deploy a single VCF platform.

This procedure describes how to deploy the VCF Installer appliance directly to an ESX host.

- In a web browser, log in to the ESX host using the VMware Host Client.
- In the navigation pane, select **Host**, and click **Create/Register VM**.
- On the Select creation type dialog box, select **Deploy a virtual machine from an OVF or OVA file** and click **Next**.
- On the Select OVF and VMDK files page, enter a name for the virtual machine, browse to the VCF-SDDC-Manager-Appliance-9.0.1.0.<build_number>.ova file that you downloaded from the Broadcom Support Portal, and click **Next**.
- On the Select Storage page, select a datastore and click **Next**.

I'm deploying the VCF Installer appliance...	Storage considerations
Outside of the management infrastructure	Choose any valid datastore.
On one of the ESX hosts that will form the management domain	Choose a local disk. After the management domain storage is configured using the VCF Installer, the VCF Installer appliance is migrated to that storage with Storage vMotion and switched to SDDC Manager mode. The local disk remains empty and is not added to the management domain storage. In this case, the local disk should not be considered when planning for capacity.

6. On the License agreements dialog box, click **I agree** and then click **Next**.

7. On the Deployment options dialog box, enter the following values and click **Next**.

Setting	Value
Network mappings	<i>your_portgroup</i>
Disk provisioning	Thin
Power on automatically	Selected

8. On the Additional settings dialog box, expand **Application**, enter the following values, and click **Next**.

Setting	Details
Root user password/Root user password confirm	Note: The root user password must be a minimum of 15 characters and include at least one uppercase, one lowercase, one digit, and one special character. Supported special characters: @ ! # \$ % ? ^ - Passwords must not contain a dictionary word or part of a dictionary word. - Passwords must not contain the user name or parts of the user name.
Local user password/Local user password confirm	You use the local user account (admin@local) to log in to the VCF Installer. Note: The local user password must be a minimum of 15 characters and include at least one uppercase, one lowercase, one digit, and one special character. Supported special characters: @ ! # \$ % ? ^ - Passwords must not contain a dictionary word or part of a dictionary word. - Passwords must not contain the user name or parts of the user name.
Host Name	Enter the fully qualified domain name (FQDN) for the VCF Installer appliance. Note: If you are deploying the VCF Installer on one of the ESX hosts that will form the management domain, the FQDN is used for SDDC Manager after deployment.
NTP Servers	Enter the NTP server(s). Use a comma if entering multiple NTP servers. NTP servers can be entered using FQDNs or IP addresses.

9. On the Additional settings dialog box, expand **Networking Configuration**, enter the following values, and click **Next**.

Setting	Details
Network 1 IP Address	Enter the IP address for the VCF Installer appliance.
Network 1 Subnet Mask	Enter the subnet mask for the VCF Installer appliance.
Network Default Gateway	Enter the default gateway for the VCF Installer appliance.
DNS Domain Name	Enter the DNS domain name. For example, <i>rainpole.io</i> .

Setting	Details
Domain Search Path	Enter the domain search path(s). Use a comma if entering multiple search paths. For example <i>rainpole.io, sfo.rainpole.io</i> .
Domain Name Servers	Enter the IP address of the primary and secondary DNS servers (comma separated). Do not specify more than two servers.

10. On the Ready to complete page, review the virtual machine configuration.

Note: Make sure your passwords meet the requirements specified above before clicking **Finish** or your deployment will not succeed.

11. After the VCF Installer appliance is deployed, SSH in to the VM as the `vcf` user and the local user password provided in step 8.

12. Ensure that you can ping the ESX hosts.

13. Verify that the VCF Installer appliance has access to the required external services, such as DNS and NTP by performing forward and reverse DNS lookups for each host and the specified NTP servers.

14. In a web browser, log in to the VCF Installer appliance administration interface: `https://installer_appliance_FQDN`. Enter the `admin@local` user and the password you provided when you deployed the appliance and then click **Log In**.

Downloading Binaries to VCF Installer

Before you can deploy VMware vSphere Foundation or VMware Cloud Foundation, you must download the required binaries to the VCF Installer appliance.

If the VCF Installer appliance can connect to the internet (either directly or through a proxy server), you can connect it to an online depot and download the binaries using a download token generated from the Broadcom Support Portal.

If the VCF Installer appliance cannot connect to the internet, you can connect it to an offline depot or use the VMware Cloud Foundation Download Tool.

VMware Cloud Foundation required binaries:

- VMware Cloud Foundation Automation
- VMware NSX
- VMware Cloud Foundation Operations
- VMware Cloud Foundation Operations fleet management
- VMware Cloud Foundation Operations collector
- VMware vCenter
- SDDC Manager

VMware vSphere Foundation required binaries:

- VMware Cloud Foundation Operations
- VMware vCenter

Note:

You do not need to download all of the binaries if you are reusing existing components. Even if you skip deployment of VCF Operations and VCF Automation in order to deploy them later with custom networking, you should still download the binaries, since they are required for the custom networking deployment.

Connect to an Online Depot to Download Binaries

If the VCF Installer can connect to the internet (either directly or through a proxy server), you can connect to an online depot and download the binaries using your Broadcom Support Portal account.

- See [KB 327186](#) for information about the URLs that must be accessible to download bundles.
- Obtain a download token from the Broadcom Support Portal.

You can only connect the VCF Installer appliance to one type of depot. If the appliance is connected to an online depot and you configure a connection to an offline depot, the online depot connection is disabled and deleted.

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Depot Settings and Binary Management**.
3. Click **Configure** to connect to the online depot.
4. Enter the download token you obtained from the Broadcom Support Portal.
5. To configure a proxy server, toggle the **Enable proxy server** setting to the on position.
 - a) Select **HTTP** or **HTTPS**.
 - b) Enter the proxy server FQDN/IP address and port number.

Note:

If SSL interception is enabled in the proxy, you must manually add the Root CA certificates to the VCF Installer Trust Store. For more information, see [Add a Trusted Certificate to the SDDC Manager Trust Store](#).

 - c) If your proxy server requires authentication, toggle the **Authenticated** setting to the on position and enter the user name and password.

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.
6. Click **Configure**.

You are now ready to download binaries.

Connect to an Offline Depot to Download Binaries

If the VCF Installer appliance cannot connect to the internet, you can connect to an offline depot.

You must set up an offline depot before you can connect to it. See [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).

You can only connect the VCF Installer appliance to one type of depot. If the appliance is connected to an online depot and you configure a connection to an offline depot, the online depot connection is disabled and deleted.

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.

2. Click **Depot Settings and Binary Management**.
3. Click **Configure** to connect to the offline depot.
4. Enter the following information for the offline depot:
 - FQDN or IP address
 - Port
 - User name
 - Password
5. Click **Configure**.

You are now ready to download binaries.

Set Up an Offline Depot Web Server for VMware Cloud Foundation

Before you can connect a VCF Instance or VCF fleet to an offline depot, you must configure an internal web server to host binaries and ESX component data.

The offline depot web server must meet the following requirements:

- Operating System: A Linux or Windows virtual machine. For Linux, any distribution is supported (for example, CentOS, RHEL, Ubuntu). For example, you can use Photon OS. See <https://packages.broadcom.com/photon/>.
- Storage: A dedicated disk for storing offline depot content. Minimum recommended size: 1 TB.
- Web Server and SSL: A web server to serve the offline depot content over HTTP/HTTPS. For HTTPS, the server must support TLSv1.2 and TLSv1.3. The choice of web server is flexible. For example: Apache HTTP Server, NGINX, and so on.
- Networking and Access: The virtual machine must have a static IP address. DNS records for the offline depot server are recommended but not required.
- User Access: A dedicated non-root user on the virtual machine. The user must have read/write access to the offline depot directory and appropriate permissions to run VCF Download Tool commands.

The following procedure describes how to set up an Apache HTTP Server on a Photon OS 5 virtual machine. Depending on your environment, you might need to modify some of the commands. For example, if corporate policies require you to use a different operating system or web server.

The procedure uses `/var/www/html` as the document root directory for the web server. The directory must exist or you must manually create it by running the `mkdir -p /var/www/html` command. Alternatively, to use a different document root directory, in the procedure, replace `/var/www/html` with your preferred root directory.

The following procedure uses `depot` as an example name of the depot VM and `depot.rainpole.io` as offline depot URL.

For a guided step-by-step video demonstration, see [VCF 9.0 Demo Offline Depot](#).

1. Deploy the Photon OS OVA file on a VM.
Do not power on the VM.

2. Add a hard drive with a capacity of 1 TB or more.
3. Power on and log in to the Photon OS 5 VM.
4. Specify networking details.

a) Navigate to `/etc/systemd/network/` and create a `.network` file.

For example, `10-static-en.network`.

b) In a text editor, open the `.network` file and add your network configuration.

For example:

```
[Match]
Name=eth0

[Network]
Address=172.16.10.14/24
Gateway=172.16.10.1
DNS=172.16.10.4 172.16.10.5
```

c) Set the correct permissions for the `.network` file.

```
chmod 644 10-static-en.network
```
5. Update the hostname by using the `hostnamectl set-hostname offline depotURL` command, where is `offline depotURL` is your offline depot URL.

For example:

```
hostnamectl set-hostname depot.rainpole.io
```
6. Restart the network and DNS lookup services.

```
systemctl restart systemd-networkd
systemctl restart systemd-resolved
```
7. Install Apache HTTP Server, update the operating system, and reboot the VM.

```
tdnf install httpd tar jq --assumeyes
tdnf update --assumeyes
reboot
```
8. Format the disk you added to the VM.

```
mkdir -p /var/www/html
mkfs.ext4 /dev/sdb
echo "/dev/sdb /var/www/html ext4 defaults 1 1" >> /etc/fstab
mount -a
```

9. Prepare the SSL Certificate.

By following the commands, you create a directory and the private key.

```
mkdir /root/http-certificates
openssl genkey -out /root/http-certificates/server.key -algorithm RSA -pkeyopt rsa_keygen_bits:2048
```

a) Provide SAN information, create a configuration file with the required attributes, for example `conf.cfg`, and add the file to the `openssl req` command.

Replace the following example values in the template with your own values: `conf.cfg`, `SE`, `Stockholm`, `Stockholm`, `Rainpole`, `IT`, `depot.rainpole.io`, and `172.16.10.11`.

Note:

OpenSSL does not prompt you for Subject Alternative Names (SANs) during the certificate request (CSR) creation process. SANs are treated as extensions, rather than standard subject information fields.

```
cat << 'EOF' > /root/http-certificates/conf.cfg
[req]
distinguished_name = req_distinguished_name
req_extensions = req_ext
prompt = no

[req_distinguished_name]
C = SE
ST = Stockholm
L = Stockholm
O = Rainpole
OU = IT
CN = depot.rainpole.io

[req_ext]
subjectAltName = @alt_names

[alt_names]
IP.1 = 172.16.10.11
DNS.1 = depot.rainpole.io
EOF
```
- b) Create a
- `.csr`
- file by using the
- `.key`
- and
- `.cfg`
- files.
- ```
openssl req -new -key /root/http-certificates/server.key -out /root/http-certificates/request.csr -config conf.cfg
```

10. Sign the certificate.

| Signing Method                 | Steps                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| External Certificate Authority | <div>1. Sign <code>request.csr</code> using your organization's external certificate authority.</div> <div>2. Combine the machine, intermediate, and root certificates into one <code>server.crt</code> file (PEM format), in this order:<pre>-----BEGIN CERTIFICATE----- [Machine Cert] [Intermediate Cert] [Root Cert] -----END CERTIFICATE-----</pre></div> |

| Signing Method                      | Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VMware Certificate Authority (VMCA) | <ol style="list-style-type: none"> <li>3. Copy <code>server.crt</code> to <code>/root/http-certificates/</code>.</li> <li>1. Transfer <code>request.csr</code> to the vCenter (for example, to <code>/root/</code>).</li> <li>2. SSH into vCenter as root and run the following commands: <pre>openssl x509 -req -days 365 -in /root/request.csr \ \ -CA /var/lib/vmware/vmca/root.cer \ -CAkey /var/lib/vmware/vmca/privatekey.pem \ -CAcreateserial -out /root/server.crt -sha256 \ -extensions req_ext -extfile conf.cfg  cat /var/lib/vmware/vmca/root.cer &gt;&gt; /root/server.crt</pre> </li> <li>3. Copy <code>server.crt</code> to <code>/root/http-certificates/</code>.</li> <li>4. Clean the <code>.crt</code> certificate files. <pre>rm -f /root/server.crt /root/request.csr</pre> </li> </ol> |

#### 11. Move the signed `.key` and `.crt` certificate files to the Apache configuration directory.

```
mv server.key server.crt /etc/httpd/conf/
chmod 0400 /etc/httpd/conf/server.key /etc/httpd/conf/server.crt
chown root:root /etc/httpd/conf/server.key /etc/httpd/conf/server.crt
```

#### 12. To configure the Apache server, edit the `/etc/httpd/conf/httpd.conf` and `/etc/httpd/conf/extra/httpd-ssl.conf` files.

##### a) Run the following commands to enable SSL modules and replace the operations@rainpole.io and depot.rainpole.io:443 example values with your own.

```
sed -i 's|#LoadModule ssl_module|LoadModule ssl_module|' /etc/httpd/conf/httpd.conf
sed -i 's|#LoadModule socache_shmcb_module|LoadModule socache_shmcb_module|' /etc/httpd/conf/httpd.conf
sed -i 's|#Include conf/extra/httpd-ssl.conf|Include conf/extra/httpd-ssl.conf|' /etc/httpd/conf/httpd.conf
sed -i 's|DocumentRoot "/etc/httpd/html"|DocumentRoot "/var/www/html"|' /etc/httpd/conf/httpd.conf
sed -i 's|<Directory "/etc/httpd/html"|<Directory "/var/www/html"|' /etc/httpd/conf/httpd.conf
sed -i 's|DocumentRoot "/etc/httpd/html"|DocumentRoot "/var/www/html"|' /etc/httpd/conf/extra/httpd-ssl.conf
sed -i 's|ServerAdmin you@example.com|ServerAdmin operations@rainpole.io|' /etc/httpd/conf/extra/httpd-ssl.conf
sed -i 's|ServerName www.example.com:443|ServerName depot.rainpole.io:443|' /etc/httpd/conf/extra/httpd-ssl.conf
```

##### b) In the `/etc/httpd/conf/httpd.conf` file, locate the `DocumentRoot "/var/www/html"` section and change `Required all denied` to `Required all granted`.

##### c) In the `/etc/httpd/conf/extra/httpd-ssl.conf` file, locate the `</VirtualHost>` tag and above the tag, paste the following configuration code.

```
<Directory /var/www/html/PROD/COMP>
 AuthType Basic
 AuthName "Basic Authentication"
 AuthUserFile /etc/httpd/conf/.htpasswd
 require valid-user
</Directory>
```

```
<Directory /var/www/html/PROD/metadata>
 AuthType Basic
 AuthName "Basic Authentication"
 AuthUserFile /etc/httpd/conf/.htpasswd
 require valid-user
</Directory>
<Directory "/var/www/html/PROD/COMP/Compatibility/VxrailCompatibilityData.json">
 # VxRail VVS Cookie Validation (VCF 5.0)
 <If "%{HTTP:Cookie}" == 'ngssoession=ngsso-token' ">
 Require all granted
 </If>
</Directory>
<Directory /var/www/html/PROD/vsan/hcl>
 Require all granted
</Directory>
Those Alias statements are needed only for VCF 5.1.0.0.
Alias /products/vl/bundles/lastupdatedtime /var/www/html/PROD/vsan/hcl/lastupdatedtime.json
Alias /products/vl/bundles/all /var/www/html/PROD/vsan/hcl/all.json
Needed only if UMDS downloads are presented
<Directory /var/www/html/umds-patch-store>
 Require all granted
</Directory>
```

#### 13. Configure the authentication.

Replace *Username* with the user name that VCF Operations and SDDC Manager use to authenticate against the depot.

```
htpasswd -c /etc/httpd/conf/.htpasswd Username
chown apache /etc/httpd/conf/.htpasswd
chmod 0400 /etc/httpd/conf/.htpasswd
```

#### 14. Test the configuration, enable and start the Apache HTTP server.

```
httpd -t
systemctl start httpd
systemctl status httpd
systemctl enable httpd
```

#### 15. To open the HTTPS port in the firewall, edit `/etc/systemd/scripts/ip4save` file.

In the file, above `COMMIT`, add the following text.

```
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
```

For example:

```
init
*filter
:INPUT DROP [0:0]
:FORWARD DROP [0:0]
:OUTPUT DROP [0:0]
Allow local-only connections
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
```

```
#keep commented till upgrade issues are sorted
#-A INPUT -j LOG --log-prefix "FIREWALL:INPUT "
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A OUTPUT -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
COMMIT
```

#### 16. Restart the service.

```
systemctl restart iptables
```

#### 17. Open a web browser and verify access to <https://<offline-depot-fqdn>/>.

#### 18. Clean up the temporary certificate generation files.

```
rm -rf /root/http-certificates
```

#### 19. Remove the default Apache index file.

```
rm -f /var/www/html/index.html
```

#### 20. Set ownership and permissions for document root directory.

```
chown apache -R /var/www/html/
find /var/www/html -type d -exec chmod 0500 {} \;
find /var/www/html -type f -exec chmod 0400 {} \;
chmod 755 /var/www/
chmod 755 /var/www/html/
```

### Download Install Binaries to an Offline Depot

If you are connecting the VCF Installer to an offline depot, you must download the install binaries to the offline depot before you can use them to deploy VMware vSphere Foundation or VMware Cloud Foundation.

- [Set Up an Offline Depot Web Server for VMware Cloud Foundation.](#)
- Generate a download token on the Broadcom Support Portal.
- The offline depot must have:
  - The latest version of the VCF Download Tool. You can download it from the Broadcom Support portal.
  - Internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
  - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
- Connect the VCF Installer to the offline depot. See [Connect to an Offline Depot to Download Binaries.](#)

You can use the VCF Download Tool to download install bundles to the offline depot.

If the computer hosting the offline depot can only access the internet using a proxy server, use the following options when downloading:

| Option         | Description                                                                                                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --proxy-server | Provide the proxy server FQDN and port.<br>For example: --proxy-server proxy.example.com:3128.                                                                                                                                            |
| --proxy-https  | Add this option if the proxy server uses HTTPS.<br>To use this option, navigate to the <i>toolroot</i> folder of the VCF Download Tool and add the proxy certificate to the Bundle Transfer Utility JRE default trust store. For example: |

| Option                     | Description                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <i>toolroot/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore toolroot/jre/lin64/lib/security/cacerts</i>                                                                                                                         |
| --proxy-user               | For a proxy server that requires authentication, enter the user name.                                                                                                                                                                           |
| --proxy-user-password-file | For a proxy server that requires authentication, enter the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password.<br>For example, --proxy-user-password-file ../../password.txt. |

**Note:** VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

On the computer hosting offline depot, download the install binaries and metadata to the offline depot root directory.

| I want to deploy...       | Example Command                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VMware Cloud Foundation   | <i>./vcf-download-tool binaries download --vcf-version 9.0.0 -d /var/www/offline-bundles --depot-download-token-file /path/to/downloadtoken.txt --automated-install --type INSTALL</i><br><br>The download command downloads both binaries and metadata. Make sure that the offline depot root directory ( <i>/var/www/offline-bundles</i> in the example) has enough storage. |
| VMware vSphere Foundation | <i>./vcf-download-tool binaries download --vcf-version 9.0.0 --sku VVF -d /var/www/offline-bundles --depot-download-token-file /path/to/downloadtoken.txt</i><br><br>The download command downloads both binaries and metadata. Make sure that the offline depot root directory ( <i>/var/www/offline-bundles</i> in the example) has enough storage.                          |

For information about additional binaries commands and options, see [VCF Download Tool Binaries Commands and Options](#).

### Download Install Binaries from an Online or Offline Depot

After you connect the VCF Installer appliance to an online or offline depot, download the required binaries to the VCF Installer appliance.

Configure a connection to an online or offline depot.

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: [https://installer\\_appliance\\_FQDN](https://installer_appliance_FQDN).  
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Depot Settings**.
3. In the Binary Management section, select a product and version from the drop-down menus.  
The UI displays all the required binaries for your product and version, including information about the file size and download status.
4. Select the binaries to download and click **Download**.  
The VCF Installer downloads and validates the binaries. Progress is displayed in the Download Status column.

After the binaries are successfully downloaded, you are ready to deploy.

### Use the VMware Cloud Foundation Download Tool to Download Install Binaries

If you cannot connect the VCF Installer appliance to an online or offline depot, use the VMware Cloud Foundation Download Tool to download binaries to a computer with access to the internet and then upload them to the VCF Installer appliance.

- A Windows or Linux computer with internet connectivity (either directly or through a proxy) for downloading the binaries and other required files.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VMware Cloud Foundation Download Tool for long-running operations.
- Obtain a download token from the Broadcom Support Portal.
- Create a text file to store the download token and copy it to the Windows or Linux computer you are using to download bundles and to the VCF Installer appliance.
- Create a text file on the VCF Installer appliance that contains the admin@local password.

#### Note:

The VMware Cloud Foundation Download Tool is the only supported method for downloading binaries for VCF 9.0. Do not use third-party tools or other methods to download binaries.

Using the VMware Cloud Foundation Download Tool to download binaries involves the following steps:

- Download the latest version of the VMware Cloud Foundation Download Tool from the Broadcom Support Portal.
- On a computer with access to the internet, use the VMware Cloud Foundation Download Tool to download the binaries and metadata for VMware vSphere Foundation or VCF.
- Copy the binaries and metadata to the VCF Installer appliance.
- On the VCF Installer appliance, use the VMware Cloud Foundation Download Tool to upload the binaries and metadata to the internal repository.

If the computer with internet access can only access the internet using a proxy server, use the following options when downloading:

| Option                     | Description                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --proxy-server             | Provide the proxy server FQDN and port.<br>For example: --proxy-server proxy.example.com:3128.                                                                                                                                                                                                                                                                                                     |
| --proxy-https              | Add this option if the proxy server uses HTTPS.<br>To use this option, navigate to the <code>toolroot</code> folder of the VMware Cloud Foundation Download Tool and add the proxy certificate to the Bundle Transfer Utility JRE default trust store. For example:<br><pre>toolroot/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore toolroot/jre/lin64/lib/security/cacerts</pre> |
| --proxy-user               | For a proxy server that requires authentication, enter the user name.                                                                                                                                                                                                                                                                                                                              |
| --proxy-user-password-file | For a proxy server that requires authentication, enter the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password.<br>For example, --proxy-user-password-file ../../password.txt.                                                                                                                                                    |

**Note:** VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

Example that combines the options:

```
./vcf-download-tool binaries download --depot-download-token-file /path/to/downloadtoken.txt --vcf-version 9.0.0 -d /root/offline-bundles --proxy-server proxy.example.com:3128 --proxy-user vmwuser --proxy-user-password-file ../../password.txt --proxy-https --automated-install --type INSTALL
```

1. Copy the most recent version of the VMware Cloud Foundation Download Tool to a computer with internet access.
2. Extract `vcf-download-tool-9.0.0.0.<build_number>.tar.gz`.  
Confirm that you have execute permission on all folders.
3. Download the binaries and metadata to the computer with internet access

| I want to deploy...       | Example Command                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VMware Cloud Foundation   | <pre>./vcf-download-tool binaries download --vcf-version 9.0.0 -d /root/offline-bundles --depot-download-token-file /path/to/downloadtoken.txt --automated-install --type INSTALL</pre><br>The download command downloads both binaries and metadata. Make sure that the download destination (/root/offline-bundles in the example) has enough storage. |
| VMware vSphere Foundation | <pre>./vcf-download-tool binaries download --vcf-version 9.0.0 --sku VVF -d /root/offline-bundles --depot-download-token-file /path/to/downloadtoken.txt</pre><br>The download command downloads both binaries and metadata. Make sure that the download destination (/root/offline-bundles in the example) has enough storage.                          |

4. Copy the entire download directory (for example, `/root/offline-bundles`) to the VCF Installer appliance.

You can select any location on the VCF Installer appliance that has enough free space available.

5. Copy the VMware Cloud Foundation Download Tool to the VCF Installer appliance.

- a) SSH in to the VCF Installer appliance using the `vcf` user account.
- b) Enter `su` to switch to the root user.
- c) Create the `vcf-download` directory.

```
mkdir /opt/vmware/vcf/lcm/vcf-download
```

**Note:** If the `/opt/vmware/vcf/lcm/vcf-download` directory already exists with an older version of the VCF Download Tool, you need to delete contents of the existing directory before proceeding.

- d) Copy the VMware Cloud Foundation Download Tool file (`vcf-download-tool-9.0.0.0.<build_number>.tar.gz`) to the `/opt/vmware/vcf/lcm/vcf-download` directory.
- e) Extract the contents of `vcf-download-tool-9.0.0.0.<build_number>.tar.gz`.

```
tar -xvf vcf-download-tool-9.0.0.0.<build_number>.tar.gz
```

- f) Set the permissions for the `vcf-download` directory.

```
cd /opt/vmware/vcf/lcm/
chown vcf_lcm:vcf -R vcf-download
chmod 750 -R vcf-download
```



6. From the VCF Installer appliance, use the VMware Cloud Foundation Download Tool to upload the binaries and metadata.

```
./vcf-download-tool binaries upload --depot-store /download/directory/ --sddc-manager-fqdn FQDN --sddc-manager-user admin@local --sddc-manager-user-password-file /path/to/pw.txt
```

For `/download/directory/`, enter the full path to the location you copied the download directory in step 4.

For `FQDN`, enter the fully-qualified domain name (FQDN) of the VCF Installer appliance.

For `username`, enter `admin@local`.

After the binaries are successfully uploaded, you are ready to deploy.

## Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard

You can use the VCF Installer deployment wizard to deploy a new VCF fleet or deploy a new VCF Instance within an existing fleet.

- Download the required binaries for VCF. See [Downloading Binaries to VCF Installer](#).
- To deploy a new VCF instance in existing VCF fleet, you select your VCF Operations instance as existing for the new deployment.

You can download and complete the Planning and Preparation Workbook to help plan and gather all the information required for deployment. See [Planning and Preparation](#). If you plan to use an existing VCF Operations instance in your deployment, enter information about that instance in the workbook.

A VCF fleet refers to an environment consisting of one or more VCF instances and standalone vCenter instances that are managed by the same VCF Operations instance. When you deploy the first VCF instance in a VCF fleet, you can deploy a new VCF Operations instance or connect to an existing one. When you expand a VCF fleet with an additional VCF instance, you must connect to an existing VCF Operations instance.

Use the VCF Installer deployment wizard to specify deployment information specific for your environment such as networks, hosts, and other information. The VMware Cloud Foundation platform is automatically deployed and configured using the information provided.

Specific deployment configurations can be achieved through the SDDC Manager API only. To deploy a VCF fleet or a VCF instance that uses Link Aggregation Group uplinks you must use a JSON specification file. See [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#) for more information.

VCF Installer allows you to skip VCF Automation during the initial deployment and deploy it as a Day-N operation. See [Deploying VCF Automation as a Day-N Operation](#).

- In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface at `https://installer_appliance_FQDN`.  
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
- Click **Deployment Wizard > VMware Cloud Foundation**.
- Select what you want to deploy.
  - A new VCF fleet
  - A VCF Instance in an existing VCF fleet

Some of the options in the following steps will be different depending on which option you select.

4. Select the existing components.

| Component      | New VCF Fleet                                                                                                                                                                                                                                                                                                                             | VCF Instance in an existing VCF fleet                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF Operations | Select this option to connect the VCF fleet to an existing VCF Operations instance. Otherwise a new VCF Operations instance is deployed.<br>If you select a VCF Operations instance that is connected to a VCF Operations fleet management appliance or a VCF Automation instance, these components will also be added to this VCF fleet. | n/a<br>You connect to your existing VCF Operations later in the deployment wizard.                                                                  |
| VMware vCenter | Do not select. To use an existing vCenter, see <a href="#">Deploy Components by Using VCF Installer to Complete the Converging to VCF Process</a> .                                                                                                                                                                                       | Do not select. To use an existing vCenter, see <a href="#">Deploy Components by Using VCF Installer to Complete the Converging to VCF Process</a> . |

The VCF Installer validates the selected components to ensure that they are compatible.

5. Enter the general configuration details for your deployment.

Additional information:

| Option                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF Instance Name                                                     | The VCF Instance Name is used to provide a unique name when creating the VMware Cloud Foundation Adapter within VCF Operations. A descriptive name provides for easy identification when you have multiple VCF Instances.                                                                                                                                                                                                                                                                                                                                                                  |
| Management domain name                                                | This name is used as prefix for the default names of objects created or deployed by the VCF Installer. You can change these default names as you progress through the deployment wizard.                                                                                                                                                                                                                                                                                                                                                                                                   |
| Custom networking configuration for VCF Operations and VCF Automation | By default, the VCF Installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVPG) as vCenter.<br>To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF Installer deploys the other components. See <a href="#">Deploying VCF Management Components with Custom Networking</a> .<br><b>Note:</b><br>This option is not available when deploying a VCF Instance in an existing VCF fleet or when using an existing VCF Operations instance. |
| Deployment model                                                      | You can choose the deployment model for any new appliances that get deployed. <b>High Availability (Three-node)</b> is recommended for production environments.<br><b>Note:</b> The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance.<br>See the VMware Cloud Foundation <a href="#">Design</a> documentation for more information.                                                                                                                                                                               |

| Option                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <b>Note:</b> New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Password creation                              | <p>If you don't want to manually enter passwords for newly installed appliances, you can auto-generate passwords. For auto-generated passwords, the complexity is the following.</p> <ul style="list-style-type: none"> <li>Length = 20 characters</li> <li>Minimum special characters = 1</li> <li>Minimum numbers = 1</li> <li>Minimum upper case letters = 1</li> <li>Minimum lower case letters = 0</li> <li>Allowed special characters = '!', '@', '#', '\$', '^', _</li> <li>Maximum consecutive characters = 2</li> </ul> <p>After a successful deployment, you can view the generated passwords.</p> |
| Customer Experience Improvement Program (CEIP) | <p>Select an option to activate or deactivate CEIP across all deployed components.</p> <p>VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See <a href="#">Customer Experience Improvement Program</a> for more information.</p>                                                                                                                                                                                                                                                                               |

6. Enter the VCF Operations details.

| I'm deploying...                                            | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A new VCF fleet with a new VCF Operations instance          | Enter all the required information.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| A new VCF fleet with an existing VCF Operations instance    | <ul style="list-style-type: none"> <li>Connect to your existing VCF Operations instance.</li> <li>If a VCF Operations fleet management appliance is connected to the VCF Operations instance, enter the administrator password.</li> <li>If no VCF Operations fleet management appliance is connected to the VCF Operations instance, enter the required information to deploy a new one.</li> <li>Enter details for the new VCF Operations collector appliance.</li> </ul> |
| A new VCF instance with an existing VCF Operations instance | <p>Connect to the VCF Operations instance that manages the VCF fleet in which you want to deploy this VCF Instance. After connecting, enter details for the new VCF Operations collector appliance.</p> <p><b>Note:</b></p> <p>The existing VCF Operations instance must be connected to VCF Automation and VCF Operations fleet management.</p>                                                                                                                            |

**Note:**

This page does not appear if you selected the custom networking option for VCF Operations and VCF Automation.

7. Enter the VCF Automation details.

Select the **I want to connect a VCF Automation instance later** checkbox if you want to use an existing Aria Automation 8.x instance that needs to be upgraded to VCF Automation 9.0. When deployment is complete, upgrade

Aria Automation to VCF Automation and then use VCF Operations to import the VCF Automation instance. See [Import an Existing VMware Aria Automation Instance Into VCF Operations and Upgrade to VCF Automation 9.0](#).

You can also select this check box to deploy VCF Automation as Day-N operation, instead of using the VCF Installer deployment workflows. See [Deploying VCF Automation as a Day-N Operation](#).

If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and you must enter the root password.

**Note:**

This screen does not appear if you selected the custom networking option for VCF Operations and VCF Automation.

When creating a new VCF Automation instance in a High Availability deployment, you must specify 4 node IP addresses. Three of the IP addresses are used for active nodes and the fourth IP address is used when recreating a node for upgrade purposes. The node IPs do not require DNS records.

8. Enter the vCenter details.

If you are manually entering root or administrator passwords, do not use the % special character.

9. Enter the NSX Manager details.

10. Choose a storage option and enter the storage details.

Your selection determines the initial shared storage type for the default cluster in the management domain. This initial shared storage type is known as principal storage. Once created, the principal storage type for a cluster cannot be changed. However, you can add additional clusters with different principal storage types to the management domain later.

| Option                            | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>vSAN</b>                       | <ul style="list-style-type: none"> <li>Select vSAN ESA or vSAN OSA.</li> <li>Enter a name for the vSAN datastore or use the pre-populated name.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>VMFS on Fibre Channel (FC)</b> | <p>Enter a name for the VMFS on FC datastore or use the pre-populated name.</p> <p>The datastore must already be created and mounted on all ESX hosts.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>NFS v3</b>                     | <ul style="list-style-type: none"> <li>Enter a name for the NFS v3 datastore or use the pre-populated name.</li> <li>Enter the path to the NFS share.</li> <li>Enter the IPv4 address of the NFS server.</li> <li>(optional) Choose to bind the NFS datastore to the VMkernel adapter specified when the datastore gets created.</li> </ul> <p>Indicates whether to bind the created NFS datastore to the VMkernel NIC created based on NFS Network Spec. This is to prevent unintentional flow of NFS traffic through any other VMkernel NIC, if such connectivity exists.</p> <p>The NFS Server must already be created and meet the NFS Storage Guidelines and Requirements found in the <a href="#">vSphere Storage Guide</a>.</p> |

## 11. Enter the ESX host details.

ESX hosts require passwords with predefined requirements. For more information, see [ESX Passwords and Account Lockout](#).

**Note:**

The VCF Installer requires that all ESX hosts share a common password. If your ESX hosts have different passwords, you must deploy using a JSON spec. See [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#).

## 12. Enter details about the networks.

To use the ESX Management Network information (gateway, VLAN, MTU) to create the VM Management Network distributed port group, select the checkbox. Otherwise, enter the information you want the VCF Installer to use to create a distributed port group for the VM Management Network.

Specify IP inclusion ranges for the vSAN and vMotion networks of the management domain. IP addresses from the specified range are automatically assigned to ESX hosts. Ensure that the IP ranges include sufficient IP addresses for the deployment. The number of IP addresses must be at least equal to the number of ESX hosts deployed.

As an example, if you specify the range start value as 192.168.1.1 and end as 192.168.1.20, a total of 20 IP addresses would be available. Do not use special IP addresses, such as the network or broadcast address.

IPs for the vMotion Network must be part of the VLAN configured with the vMotion portgroup. IPs for the vSAN Network must be part of the VLAN configured for the vSAN portgroup. All IPs within the range must be available for use or IP conflicts will occur. It is a good practice to validate this prior to starting a deployment.

## 13. Enter the vSphere Distributed Switch details.

| Option                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default                                    | This profile is recommended. It provides a unified fabric for all traffic types using a single vSphere Distributed Switch.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Storage Traffic Separation                 | This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.                                                                                                                                                                                                                                                                                                                                                                                                    |
| NSX Traffic Separation                     | This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic, while the other is used for all other traffic.                                                                                                                                                                                                                                                                                                                                                                                       |
| Storage Traffic and NSX Traffic Separation | This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic.                                                                                                                                                                                                                                                                                                                                 |
| Custom Switch Configuration                | Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations.<br>Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. Management, vMotion, vSAN and NSX-Overlay network types can be configured only once for a cluster.<br>NSX-VLAN and Public network traffic types can be configured in multiple vSphere Distributed Switches. |

## 14. Enter the SDDC Manager details.

**Note:**

If VCF Installer is deployed on one of the management domain ESX hosts, details populate automatically.

## 15. Review the deployment information.

To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

## 16. Review the validation information.

VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warnings | Review any warnings and either address them or click <b>Acknowledge</b> to proceed with deployment.                                                                                                                                                                                                                                                                                                                   |
| Errors   | Review all failed validations and resolve the issue(s) using the remediation information provided in the error message.<br>You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage. |

17. After addressing all validation errors and warnings, click **Deploy**.

## 18. Monitor the deployment progress.

If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

## 19. When the deployment completes successfully, you are ready for the next steps.

- Download the JSON specification file for the deployment (optional).  
**Note:** The JSON specification file does not include passwords.
- Click **Review Passwords** to review the passwords.  
**Note:**  
You can also retrieve the passwords using the [SDDC Manager API](#) and [VCF Operations API](#).
- Open the VCF Operations UI to apply licensing. See [Registering VCF Operations with the VCF Business Services console](#).

## Start a New vSphere Foundation Deployment by Using the VCF Installer Deployment Wizard

You can use the VMware Cloud Foundation Installer deployment wizard to deploy a new VMware vSphere Foundation platform.

Download the required binaries for VMware vSphere Foundation. See [Downloading Binaries to VCF Installer](#).

You can download and complete the Planning and Preparation Workbook to help plan and gather all the information required for a VMware vSphere Foundation deployment. See [Planning and Preparation](#).

Use the VCF Installer deployment wizard to specify deployment information specific to your environment such as networks, hosts, and other information. The VMware vSphere Foundation software components are automatically deployed and configured using the information provided.

The following procedure describes how to deploy using the deployment wizard. You can also deploy using a custom JSON specification file. See [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#) for more information.

To create a VMware vSphere Foundation environment using existing vSphere infrastructure, see [Converge a vCenter Instance and ESX Hosts to vSphere Foundation Platform](#).

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.  
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard > VMware vSphere Foundation**.
3. Do not select any existing components.  
To use an existing vCenter to create a VMware vSphere Foundation environment, see [Converge a vCenter Instance and ESX Hosts to vSphere Foundation Platform](#).
4. Enter the general configuration details for your deployment.  
Additional information:

| Option                                         | Description                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Password creation                              | You can auto-generate passwords for newly installed appliances if you don't want to manually enter passwords. At the end of the deployment wizard, you can view the generated passwords.                                                                                                                              |
| Customer Experience Improvement Program (CEIP) | Select an option to activate or deactivate CEIP across all deployed components.<br>VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. For more information, see <a href="#">Customer Experience Improvement Program</a> . |

5. Enter the VCF Operations details.
6. Enter the vCenter details.
7. Choose a storage option and enter the storage details.  
Your selection determines the initial shared storage type for the primary cluster in the management domain.
8. Enter the ESX host details.

**Note:**

The VCF Installer requires that all ESX hosts share a common password. If your ESX hosts have different passwords, you must deploy using a JSON specification file. See [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#).

9. Enter details about the networks.

To use the ESX Management Network information (gateway, VLAN, MTU) to create the VM Management Network distributed port group, select the checkbox. Otherwise, enter the information you want the VCF Installer to use to create a distributed port group for the VM Management Network.

Specify IP inclusion ranges for the vSAN and vMotion networks of the management domain. IP addresses from the specified range are automatically assigned to hosts. Ensure that the IP ranges include sufficient IP addresses for the deployment. The number of IP addresses must be at least equal to the number of hosts deployed.

As an example, if you specify the range start value as 192.168.1.1 and end as 192.168.1.20, a total of 20 IP addresses would be available. Do not use special IP addresses, such as the network or broadcast address.

IPs for the vMotion Network must be part of the VLAN configured with the vMotion portgroup. IPs for the vSAN Network must be part of the VLAN configured for the vSAN portgroup. All IPs within the range must be available for use or IP conflicts will occur. It is a good practice to validate this prior to starting a deployment.

10. Enter the vSphere Distributed Switch details.

| Option                                     | Description                                                                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default                                    | This profile is recommended and the default configuration. It provides a unified fabric for all traffic types using a single vSphere Distributed Switch.                                                                                              |
| Storage Traffic Separation                 | This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.                                                                        |
| NSX Traffic Separation                     | This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, while the other is used for all other traffic.                                                       |
| Storage Traffic and NSX Traffic Separation | This profile creates three vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, the second switch is dedicated for storage traffic, while the third is used for all other traffic. |
| Custom Switch Configuration                | Create a custom vSphere Distributed Switch configuration.                                                                                                                                                                                             |

11. Review the deployment information.

To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

12. Review the validation information.

The VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warnings | Review any warnings and either address them or click <b>Acknowledge</b> to proceed with deployment.                                                                                                                                                                                                                                                                                                                   |
| Errors   | Review all failed validations and resolve the issue(s) using the remediation information provided in the error message.<br>You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage. |

13. After addressing all validation errors and warnings, click **Deploy**.

14. Monitor the deployment progress.

If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

15. When the deployment completes successfully, you are ready for the next steps.

- Download the JSON specification file for the deployment (optional).  
**Note:** The JSON specification file does not include passwords.
- Click **Review Passwords** to review the passwords.  
**Note:**  
You can also retrieve the passwords using the [SDDC Manager API](#) and [VCF Operations API](#).
- Open the VCF Operations UI to apply licensing. See [Registering VCF Operations with the VCF Business Services console](#).
- Register your vCenter instance in VCF Operations. See [Configuring a vCenter Account in VCF Operations](#).

## Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation

VCF Installer can deploy your platform by using a JSON file instead of the deployment wizard.

Download the required binaries for VMware Cloud Foundation or VMware vSphere Foundation. See [Downloading Binaries to VCF Installer](#).

You can download and complete the Planning and Preparation Workbook to help plan and gather all the information required for a VCF or VMware vSphere Foundation deployment. See [Planning and Preparation](#). If you plan to use any existing components in your deployment, enter information about those components in the workbook.

A JSON specification file can be used to:

- Deploy a new VCF fleet
- Deploy a new VCF Instance
- Converge existing vSphere infrastructure to a new VCF fleet
- Converge existing vSphere infrastructure to a new VCF Instance
- Deploy a new vSphere Foundation platform
- Converge existing vSphere infrastructure to a new vSphere Foundation platform

A JSON specification file must be used to:

- Deploy a new VCF fleet where the Distributed switch design uses LACP networking.
- Deploy a new VCF Instance where the Distributed switch design uses LACP networking.

Using a JSON specification file allows for additional levels of customization that are not available in the deployment wizard. For example, the deployment wizard assumes that your ESX hosts share the same root password. If your ESX hosts have different root passwords, use the JSON specification file to deploy your environment.

In addition, if you are deploying multiple platforms, you may find it easier to start with an existing JSON specification file and modify it for additional deployments.

See the [VMware Cloud Foundation Installer API Reference Guide](#) for information about the properties and parameters you can use to create a JSON specification file. For reference, see the following [example JSON file](#) with predefined values.

The following table provides some guidance on certain properties you should specify, depending on the type of environment you want to create and what existing components you are using:

| I want to...                                                                  | workflowType property | VCF Operations vcfOperationsSpec useExistingDeployment property | VMware vCenter vcenterSpec useExistingDeployment property | VCF Operations fleet management vcfOperationsFleetManagementSpec useExistingDeployment property | VCF Automation vcfAutomationSpec useExistingDeployment property | NSX Manager nsxtSpec useExistingDeployment property |
|-------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------|
| Deploy a new VCF fleet                                                        | VCF                   | optional (true or false)                                        | false                                                     | optional (true or false)                                                                        | optional (true or false)                                        | n/a                                                 |
| Deploy a new VCF instance                                                     | VCF_EXTEND            | true                                                            | false                                                     | true                                                                                            | true                                                            | n/a                                                 |
| Converge existing vSphere infrastructure to a new VCF fleet                   | VCF                   | optional (true or false)                                        | true                                                      | optional (true or false)                                                                        | optional (true or false)                                        | false                                               |
| Converge existing vSphere infrastructure to a new VCF instance                | VCF_EXTEND            | true                                                            | true                                                      | true                                                                                            | true                                                            | false                                               |
| Deploy a new vSphere Foundation platform                                      | VVF                   | false                                                           | false                                                     | n/a                                                                                             | n/a                                                             | n/a                                                 |
| Converge existing vSphere infrastructure to a new vSphere Foundation platform | VVF                   | false                                                           | true                                                      | n/a                                                                                             | n/a                                                             | n/a                                                 |

To deploy a new VCF fleet or instance where the Distributed switch design uses LACP networking, you must use a JSON specification file. You can configure LAG/LACP networking on one or more Distributed switch by setting the *lagSpecs* attribute in the *dvsSpecs* section for each. Note that a single Distributed switch cannot support a mix of LAG and non-LAG uplinks.

Use this sample JSON snippet to configure a management domain with LACP networking. You must specify LAG usages or references explicitly (in DVS uplink associations and PortGroup and Network uplink definitions).

```
"dvsSpecs": [
 {
 "dvsName": "VCF1-cl01-vds02",
 "vmnics": [
 "vmnic0",
 "vmnic1"
],
 "networks": [
 "VM_MANAGEMENT",
 "MANAGEMENT",
 "VMOTION",
 "VSAN"
```

```
],
 "niocSpecs": null,
 "mtu": 9000,
 "isUsedByNxt": true,
 "nsxtSwitchConfig": {
 "transportZones": [
 {
 "name": "VCF1-cl01-Overlay-Zone01",
 "transportType": "OVERLAY"
 }
]
 },
 "vmnicsToUplinks": [
 {
 "id": "vmnic0",
 "uplink": "lag-0"
 },
 {
 "id": "vmnic1",
 "uplink": "lag-1"
 }
],
 "nsxTeamings": [
 {
 "policy": "FAILOVER_ORDER",
 "activeUplinks": [
 "lag"
]
 }
],
 "lagSpecs": [
 {
 "name": "lag",
 "uplinksCount": 2,
 "lacpMode": "ACTIVE",
 "loadBalancingMode": "VLAN",
 "lacpTimeoutMode": "SLOW"
 }
]
 }
},
...
...
...
"networkSpecs": [
 {
 "portGroupKey": "VCF-cl01-vds02-pg-esxi-mgmt",
 "networkType": "MANAGEMENT",
 "subnet": "10.0.0.0/22",
 "gateway": "10.0.0.250",
 "subnetMask": null,
 "includeIpAddress": null,
 "includeIpAddressRanges": null,
```

```
 "vlanId": "0",
 "mtu": 1500,
 "teamingPolicy": "failover_explicit",
 "activeUplinks": [
 "lag"
],
 "standbyUplinks": null
 }
 ...
 ...
 ...
}
```

To later deploy a workload domain with LACP networking, see [Create a Workload Domain with LACP Networking](#).

Refer to the *VMware Cloud Foundation Installer API Reference Guide* to create a JSON specification file using the information from your Planning and Preparation Workbook.

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.  
Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deploy Using JSON Spec** and upload the file.
3. Review the Summary and JSON Preview and click **Next**.
4. Review the validation information.

The VCF Installer validates the information you provided in the JSON specification and reports any errors or warnings.

|          |                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warnings | Review any warnings and either address them or click <b>Acknowledge</b> to proceed with deployment.                                                                                 |
| Errors   | Review all failed validations and resolve the issue(s) using the remediation information provided in the error message.<br>You can modify and reupload the JSON specification file. |

5. After addressing all validation errors and warnings, click **Deploy**.
6. Monitor the deployment progress.  
If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.
7. When the deployment completes successfully, you are ready for the next steps.
  - Download the JSON specification file for the deployment (optional).  
**Note:** The JSON specification file does not include passwords.
  - Click **Review Passwords** to review the passwords.  
**Note:**  
You can also retrieve the passwords using the [SDDC Manager API](#) and [VCF Operations API](#).
  - Open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

## Troubleshooting VMware Cloud Foundation or vSphere Foundation Deployments

If you encounter errors during deployment, you can use log files and the Supportability and Serviceability (SoS) Utility to help with troubleshooting.

### Log Files and Locations

If you experience validation or deployment failures, check the log files for the domain manager service on the VCF Installer appliance.

```
/var/log/vmware/vcf/domainmanager/domainmanager.log
```

If you experience issues with VCF Operations or VCF Automation, check the log files on the VCF Operations fleet management appliance.

```
/var/log/vrlcm/vmware_vrlcm.log
```

### Local Account Options

If your local account is locked, you can log in the VM of the in VCF Installer appliance and restart the `commonsvcs` service. After the restart, allow approximately 5 minutes for the service to boot.

```
systemctl restart commonsvcs
```

### Supportability and Serviceability (SoS) Utility

You can run the SoS Utility on the VCF Installer appliance to perform health checks and collect logs, which you can use to help debug a deployment.

#### Note:

After a successful deployment, you should only run the SoS Utility on the SDDC Manager appliance.

The SoS Utility is not a debug tool, but it does provide health check operations that can facilitate debugging a failed deployment. To run the SoS Utility, SSH in to the VCF Installer appliance using the admin account and then enter `su` to switch to the root user. Navigate to the `/opt/vmware/sddc-support` directory and type `./sos` followed by the options required for your desired operation.

```
./sos --option-1 --option-2...--option-n
```

### SoS Utility Help Options

Use the SoS Utility help to options to see information about the SoS tool itself.

| Option                                    | Description                                          |
|-------------------------------------------|------------------------------------------------------|
| <code>--help</code><br><code>-h</code>    | Provides a summary of the available SoS tool options |
| <code>--version</code><br><code>-v</code> | Provides the SoS tool's version number.              |

### SoS Utility Generic Options

These are generic options for the SoS Utility.

| Option                      | Description                                                        |
|-----------------------------|--------------------------------------------------------------------|
| <code>--debug-mode</code>   | Runs the SoS tool in debug mode.                                   |
| <code>--enable-stats</code> | Enables collection of SoS execution stats.                         |
| <code>--force</code>        | Allows SoS operations while workflows are running.<br><b>Note:</b> |

| Option                               | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | In most cases, you should not use this option, although it is required to run health check options on the VCF Installer appliance.                                                                                                                                                                                                                                                                             |
| <code>--history</code>               | Displays the last twenty SoS operations performed.                                                                                                                                                                                                                                                                                                                                                             |
| <code>--log-dir LOGDIR</code>        | Specifies the directory to store the logs.                                                                                                                                                                                                                                                                                                                                                                     |
| <code>--log-folder LOGFOLDER</code>  | Specifies the name of the log directory.                                                                                                                                                                                                                                                                                                                                                                       |
| <code>--setup-json SETUP_JSON</code> | Custom setup-json file for log collection.<br>SoS prepares the inventory automatically based on the environment where it is running. If you want to collect logs for a pre-defined set of components, you can create a <code>setup.json</code> file and pass the file as input to SoS. A sample JSON file is available on the VCF Installer appliance in the <code>/opt/vmware/sddc-support/</code> directory. |
| <code>--short</code>                 | Displays health results only for failures and warnings.                                                                                                                                                                                                                                                                                                                                                        |
| <code>--skip-known-host-check</code> | Skips the specified check for SSL thumbprints for hosts in the known hosts file.                                                                                                                                                                                                                                                                                                                               |
| <code>--zip</code>                   | Creates a zipped tar file for the output.                                                                                                                                                                                                                                                                                                                                                                      |

### SoS Utility Log File Options

| Option                               | Description                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--api-logs</code>              | Collects output from APIs.                                                                                                                                                                                                                                                                      |
| <code>--vcf-installer-logs</code>    | Collects VCF Installer appliance support logs.                                                                                                                                                                                                                                                  |
| <code>--esx-logs</code>              | Collects logs from the ESX hosts only.<br>Logs are collected from each ESX host available in the deployment.                                                                                                                                                                                    |
| <code>--no-clean-old-logs</code>     | Use this option to prevent the tool from removing any output from a previous collection run.<br>By default, before writing the output to the directory, the tool deletes the prior run's output files that might be present. If you want to retain the older output files, specify this option. |
| <code>--no-health-check</code>       | Skips the health check executed as part of log collection.                                                                                                                                                                                                                                      |
| <code>--nsx-logs</code>              | Collects logs from the NSX Manager instances only.                                                                                                                                                                                                                                              |
| <code>--sddc-manager-logs</code>     | Collects logs from the SDDC Manager only.                                                                                                                                                                                                                                                       |
| <code>--vc-logs</code>               | Collects logs from the vCenter instances only.<br>Logs are collected from each vCenter server available in the deployment.                                                                                                                                                                      |
| <code>--vm-screenshots</code>        | Collects screen shots from all VMs.                                                                                                                                                                                                                                                             |
| <code>--automation-logs</code>       | Collects VCF Automation support logs only.                                                                                                                                                                                                                                                      |
| <code>--operations-logs</code>       | Collects VCF Operations support logs only.                                                                                                                                                                                                                                                      |
| <code>--operations-fleet-logs</code> | Collects VCF Operations fleet management support logs only.                                                                                                                                                                                                                                     |
| <code>--collect-all-logs</code>      | Collects logs for all components.                                                                                                                                                                                                                                                               |

### SoS Utility Health Check Options

The SoS Utility can be used to perform health checks on various components or services, including connectivity, compute, and storage.

#### Note:



The health check options are primarily designed to run on the SDDC Manager appliance. Running them on the VCF Installer appliance requires the `--force` parameter, which instructs the SoS Utility to identify the SDDC Manager appliance deployed by the VCF Installer, and then execute the health check remotely. For example:

```
./sos --health-check --force
```

| Option                             | Description                                                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--certificate-health</code>  | Verifies that the component certificates are valid (within the expiry date).                                                                                                            |
| <code>--connectivity-health</code> | Performs a connectivity health check to inspect whether the different components of the system such as the ESX hosts, vCenter servers, NSX Managers, and SDDC Manager VM can be pinged. |
| <code>--compute-health</code>      | Performs a compute health check.                                                                                                                                                        |
| <code>--general-health</code>      | Verifies ESX entries across all sources, checks the Postgres DB operational status for hosts, checks ESX for error dumps, and gets NSX Manager and cluster status.                      |
| <code>--get-host-ips</code>        | Returns server information.                                                                                                                                                             |
| <code>--health-check</code>        | Performs all available health checks.                                                                                                                                                   |
| <code>--ntp-health</code>          | Verifies whether the time on the components is synchronized with the NTP server in the VCF Installer appliance.                                                                         |
| <code>--services-health</code>     | Performs a services health check to confirm whether services are running                                                                                                                |

## Completing the Deployment of Your Platform

After the automated by VCF Installer deployment of components for your VMware Cloud Foundation or vSphere Foundation platform, you must deploy the remaining components to complete the platform. You deploy the orchestrator appliance manually for both VCF and vSphere Foundation. For VCF you deploy the remaining appliances by using automation provided by VCF Operations. For vSphere Foundation you deploy the operations-logs appliances manually.

Deployment of a three-node VCF Identity Broker is optional per the requirements of your organization. If you do not need the three-node appliance deployment, you can use the same functionality that is embedded in vCenter.

**Table 813: Post-VCF Installer Deployment Scope per Platform**

| Component           | Component Appliances                   | VMware Cloud Foundation |                   | vSphere Foundation |
|---------------------|----------------------------------------|-------------------------|-------------------|--------------------|
|                     |                                        | Simple                  | High Availability |                    |
| VCF Identity Broker | identity broker (node 1)               | No                      | Yes               | No                 |
|                     | identity broker (node 2)               | No                      | Yes               | No                 |
|                     | identity broker (node 3)               | No                      | Yes               | No                 |
| VCF Operations      | operations-networks                    | Yes                     | Yes               | No                 |
|                     | operations-networks (collector node 1) | Yes                     | Yes               | No                 |
|                     | operations-networks (collector node 2) | No                      | Yes               | No                 |
|                     | operations-logs (primary node)         | Yes                     | Yes               | Yes                |
|                     | operations-logs (worker node 1)        | No                      | Yes               | Yes                |
|                     |                                        |                         |                   |                    |

| Component | Component Appliances                      | VMware Cloud Foundation |                   | vSphere Foundation |
|-----------|-------------------------------------------|-------------------------|-------------------|--------------------|
|           |                                           | Simple                  | High Availability |                    |
|           | operations-logs (worker node 2)           | No                      | Yes               | Yes                |
|           | operations-orchestrator (primary node)    | Yes                     | Yes               | Yes                |
|           | operations-orchestrator (secondary node1) | No                      | Yes               | Yes                |
|           | operations-orchestrator (secondary node2) | No                      | Yes               | Yes                |

## Deploying VCF Identity Broker

VCF Identity Broker provides Single Sign-On for VMware Cloud Foundation. Deploying VCF Identity Broker in appliance mode in fleet management allows users to access VCF management components with a single set of credentials.

### Note:

Deploying VCF Identity Broker in appliance mode is suitable for environments that require high availability utilizing a multi-node cluster. VCF Identity Broker can also be deployed as an embedded service within the management domain vCenter instance. For more information about VCF Identity Broker models, see [VCF Identity Broker Detailed Design](#).

VCF Identity Broker is not available in VMware vSphere Foundation. For VMware vSphere Foundation, identity and access management is embedded in the vCenter and uses the VMware Workspace ONE Access container service or "vcws1a-broker" for SSO. See Broadcom knowledge base article [436518](#).

Before starting the appliance deployment, download the VCF Identity Broker installation binary to the fleet management appliance. See [Downloading VCF Management Components Binaries](#). Then select **Fleet Management > Lifecycle** and click **Add on identity broker**.

To deploy VCF Identity Broker your account must have a role assigned in vSphere with the following permissions:

| Privilege   | Value                        |                             |
|-------------|------------------------------|-----------------------------|
| CNS         | Searchable                   |                             |
| Datastore   | All privileges               |                             |
| Global      | Manage Custom Attribute      |                             |
|             | Set Custom Attribute         |                             |
|             | Settings                     |                             |
| Host        | Local Operations             | Add Host to vCenter         |
|             |                              | Create Virtual Machine      |
|             |                              | Delete Virtual Machine      |
|             |                              | Reconfigure Virtual Machine |
|             | Inventory                    | Modify cluster              |
|             | Configuration                | Change settings             |
| Network     | Assign network               |                             |
| Permissions | Modify permission            |                             |
|             | Modify role                  |                             |
| Resource    | Assign vApp to Resource Pool |                             |

| Privilege                  | Value                                                     |
|----------------------------|-----------------------------------------------------------|
|                            | Assign Virtual Machine to Resource Pool                   |
| VM Storage Policies        | View VM storage policies                                  |
| vApp                       | All privileges                                            |
| Virtual Machines           | All privileges                                            |
| Content Library            | All privileges                                            |
| Service Account Management | Administer (Create / Delete accounts and Reset passwords) |
|                            | Create / Delete service accounts                          |
|                            | Reset password / Get status of password (valid / invalid) |

The deployment moves through the following stages.

1. Deployment
  - Installation type

|                    |                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>New Install</b> | Choose this option to perform a fresh installation.                                                                        |
| <b>Import</b>      | Imports an existing <b>identity broker</b> deployment, for example if VCF Identity Broker was removed for troubleshooting. |

- Version: 9.0.0.0
  - Deployment type: Small. For a small deployment, the cluster requires: 3 VMs, 8 vCPUs per VM, 16 GB RAM per VM, 100 GB disk per VM.
2. Certificate: Select the certificate for deployment. If a certificate is not available, click **Add Certificate** to create a new one. See [Managing Certificates in VMware Cloud Foundation](#).
  3. Infrastructure
    - Select vCenter.
      - vCenter servers with any of the following conditions are not listed:
        - VCF Identity Broker deployed in vCenter using the VCF Operations API.
        - vCenter version greater than 9.0
        - vCenter associated with SDDC Manager
        - vCenter used by another VCF Identity Broker instance. To use the vCenter, delete the associated VCF Identity Broker instance.
      - Select Cluster: Select the cluster in the management domain vCenter for deployment.
      - Select Folder
      - Select Resource Pool
      - Select Network
      - Select Datastore
    - 4. Network
      - Domain Name
      - Domain Search Path
      - DNS Servers: If the required DNS server is not listed, add it.
      - Time Sync Mode: Point to an NTP Server or Host Time
      - IPv4 Details:
        - Default IPv4 Gateway
        - IPv4 Netmask

5. Components
  - Install identity broker:
    - Add Password: If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.
    - Add Certificate: Select the certificate for deployment. To create a new certificate, click **Add Certificate**.
  - Component Properties
    - FQDN: Component FQDN
    - Certificate: Accept the automatic selection.
    - Component Password: Needed for the `vmware-system-user` and the `operator` account. Password must be a minimum of 15 characters.
  - Components
    - VM Node Prefix: A unique identifier used to name component nodes. To avoid conflicts and maintain proper node identification within the cluster, ensure that the node prefix is unique.
    - Internal Cluster CIDR: IP Address range used for internal cluster communication within the cluster. Choose a range that does not conflict with existing networks in infrastructure.
    - Primary VIP: Used for accessing component services. Should resolve to the Component Properties FQDN
    - Additional VIPs: (optional) Add up to 2 VIPs.
    - Cluster Node IP Pool: The Node IP Pool is a range of IP addresses allocated to cluster nodes. These IP's ensure that each node has a unique address for internal communication within the cluster. Additional IP's from the pool will be used during rolling cluster upgrades.
      - A minimum of 4 IP addresses needed, maximum number of IP's is 7.
      - Three of these IP addresses will be used for deploying cluster nodes hosting VCF Identity Broker component. Spare IP addresses will be used during rolling upgrades and patches.

To execute the Precheck, click **Next**.

1. Precheck: If the precheck completes successfully, click **Next**.
2. Summary: To trigger deployment after reviewing the summary, click **Submit**.

## Deploying VCF Operations for Networks

VCF Operations for networks provides a comprehensive overview of your VMware Cloud Foundation platform network health, issues, and operations all in one place. Deployed as an appliance using VCF Operations fleet management, it consists of Platform and Collector nodes.

Before starting the installation, download the `operations-networks` binary to the fleet management appliance. See [Downloading VCF Management Components Binaries](#). Then select **Fleet Management > Lifecycle** and click **Add** on **operations-networks**.

The deployment moves through the following stages.

1. Deployment
  - Installation type

|                    |                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>New Install</b> | Choose this option to perform a fresh installation.                                                                                                                                                                                                                                                                                                             |
| <b>Import</b>      | Imports an existing <b>operations-networks 9.0</b> deployment. Not applicable for a new installation.<br>If you have an existing VCF Operations for Networks deployment that is not managed by VMware Aria Suite Lifecycle, you can use this option to import your VCF Operations for networks into VCF Operations then upgrade to VCF Operations for networks. |

|                                            |                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Import from legacy fleet management</b> | Imports VMware Aria Operations for Networks 6.x for upgrade to operations-networks 9.0. |
|--------------------------------------------|-----------------------------------------------------------------------------------------|

- Version: 9.0.0.0
  - Deployment type: Standard or Cluster
2. Certificate: Select the certificate for deployment. If no certificate is available, you can generate or import one.
  3. Infrastructure
    - Select vCenter: Select the vCenter for the Management Domain.
    - Select Cluster
    - Select Folder (optional)
    - Select Resource Pool (optional)
    - Select Network
    - Select Datastore
    - Select Disk Mode
    - Use Content Library
  4. Network
    - Domain Name
    - Domain Search Path
    - DNS Servers
    - Time Sync Mode: Use NTP Server, Use Host Time
    - IPv4 Details:
      - Default IPv4 Gateway
      - IPv4 Netmask
  5. Component Properties
    - FIPS Compliance Mode: Default is on. If not needed, turn it off.
    - Certificate: Pre-selected from the previous step. To create a new certificate, click **Add Certificate**.
    - Anti-Affinity/Affinity Rule: Selection is on by default. If selected, choose the DRS Anti-Affinity rule type:
      - Keep virtual machines in separate hosts: For anti-affinity, select this rule type.
      - Keep virtual machines in same host: For affinity, select this rule type.
 If the Anti-Affinity/Affinity Rule selection is deselected, DRS Anti-Affinity rule type does not appear.
    - Component Password: Select a password to be used by admin, support, and console users. If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.
    - NTP Servers: Pre-selected from previous step. Change as needed.
  6. Components
    - Platform VM: Enter the Name, IP Address, and Node Size for the Platform VM
    - Collector VM: Enter the Name, IP Address, and Node Size for the Collector VM

**Note:**

FQDNs are not required for the VCF Operations for Network deployment, but you should configure DNS records for IP addresses consumed by Platform and Collector VMs on their respective DNS Servers.
  7. Precheck: If the precheck completes successfully, click **Next**.
  8. Summary: To trigger deployment after reviewing the summary, click **Submit**.

## Deploying VCF Operations for logs

VCF Operations for logs is designed to help you collect, analyze, and visualize logs from across your VMware Cloud Foundation platform.

You deploy the VCF Operations for logs appliance by using VCF Operations fleet management. This provides:

- Centralized log aggregation: Collect logs from all VCF components, such as vCenter, ESX, NSX, vSAN, VCF Automation, VCF Operations, VCF Identity Broker, VCF Operations for networks, and VCF Operations HCX
- Real-time analytics: Search and filter logs instantly for troubleshooting or auditing.
- Dashboards and alerts: Content packs for VCF components offer pre-built dashboards, alerts, and queries.

Before starting the installation, download the `operations-logs` binary to the fleet management appliance. See [Downloading VCF management components](#). Then select **Fleet Management > Lifecycle** and click **Add** on **operations-logs**.

The deployment moves through the following stages.

1. Deployment
  - Installation type

|                    |                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------|
| <b>New Install</b> | Choose this option to perform a fresh installation.                                                        |
| <b>Import</b>      | Imports an existing <b>operations-logs 9.0</b> or later deployment. Not applicable for a new installation. |

- Version: 9.0.0.0
  - Deployment type: Standard or Cluster
2. Certificate: Select the certificate for deployment. If no suitable certificate is available, you can generate or import one.
  3. Infrastructure
    - Select vCenter: Lists management domains for deployment.
    - Select Cluster
    - Select Folder (optional)
    - Select Resource Pool (optional)
    - Select Network
    - Select Datastore
    - Select Disk Mode
    - Use Content Library
  4. Network
    - Domain Name
    - Domain Search Path
    - DNS Servers: If none exist, then add one and select it.
    - Time Sync Mode: Use NTP Server, Use Host Time
    - IPv4 Details:
      - Default IPv4 Gateway
      - IPv4 Netmask

## 5. Components

## – Enter Component Properties

- Node Size: Select Small, Medium or Large. Medium is the minimum size recommended for production, testing, or development-type deployments.
- FIPS Compliance Mode: Default is on. If not needed, turn it off.
- Certificate: Pre-selected from the previous step. To create a new certificate, click **Add Certificate**.
- Set Affinity & Anti-Affinity rule if needed. If selected choose the DRS Anti-Affinity rule type:
  - Keep virtual machines in separate hosts: For anti-affinity, select this rule type.
  - Keep virtual machines in same host: For affinity, select this rule type.
- Configure Cluster VIP: If set to YES, the Cluster VIP section appears. Enter:
  - FQDN of the of the Cluster VIP
  - IP address that points to the FQDN of the Cluster VIP
- Upgrade VM Compatibility
- Always Use English
- Admin Email
- Component Password: Select a password to be used by root and admin local accounts. If you do not have a password, click **Add Password** to create a new one. Password must be a minimum of 15 characters.
- Time Sync Mode: Preselected.

## – Enter Component/Node properties

- VM Name
- FQDN
- IP Address

6. Precheck: If the precheck completes successfully, click **Next**.

7. Summary: To trigger deployment after reviewing the summary, click **Submit**.

## Deploying VCF Operations Orchestrator

Before you can access the VCF Operations orchestrator content in your VCF environment, deploy the VCF Operations orchestrator appliance.

- Verify that you have a running vCenter instance. The vCenter version must be 9 or later.
- Verify that the host on which you are deploying the VCF Operations orchestrator appliance meets the minimum hardware requirements.
  - 4 CPUs
  - 12 GB of memory
  - 200 GB hard disk

**Note:**

Do not reduce the default memory size, because the VCF Operations orchestrator server requires at least 8 GB of free memory.

- If your system is isolated and without Internet access, you must download the .ova file for the appliance from the Broadcom support portal.

1. Log in to the vSphere Client as an **administrator**.

2. Select an inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host.

3. Select **Actions > Deploy OVF Template**.

4. Enter the file path or the URL to the .ova file and click **Next**.

5. Enter a name and location for the VCF Operations orchestrator appliance, and click **Next**.

6. Select a host, cluster, resource pool, or vApp as a destination on which you want the appliance to run, and click **Next**.

7. Review the deployment details, and click **Next**.

8. Accept the terms in the license agreement and click **Next**.

9. Select the storage format you want to use for the VCF Operations orchestrator appliance.

| Format                                | Description                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Thick Provisioned Lazy Zeroed</b>  | Creates a virtual disk in a default thick format. The space required for the virtual disk is allocated when the virtual disk is created. If any data remains on the physical device, it is not erased during creation, but is zeroed out on demand later on first write from the virtual machine.                                           |
| <b>Thick Provisioned Eager Zeroed</b> | Supports clustering features such as Fault Tolerance. The space required for the virtual disk is allocated when the virtual disk is created. If any data remains on the physical device, it is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create disks in other formats. |
| <b>Thin Provisioned Format</b>        | Saves hard disk space. For the thin disk, you provision as much datastore space as the disk requires based on the value that you select for the disk size. The thin disk starts small and, at first, uses only as much datastore space as the disk needs for its initial operations.                                                        |

10. Click **Next**.

11. Configure the network settings and enter the **root** password.

When configuring the network settings of the VCF Operations orchestrator appliance, you must use the IPv4 protocol. For both DHCP and Static network configurations, you must add a fully qualified domain name (FQDN) for your VCF Operations orchestrator appliance.

If the host name displayed in the shell of the deployed VCF Operations orchestrator appliance is *photon-machine*, the preceding network configuration requirements are not met.

12. Optional: Configure additional network settings for the VCF Operations orchestrator appliance, such as enabling SSH access.

**Note:** When configuring a Kubernetes network, the values of the internal cluster CIDR and internal service CIDR must allow for at least 1024 hosts. Because of this requirement, the network mask value must be 22 or less. Network mask values higher than 22 are invalid. The Kubernetes network properties have the following default values:

| Kubernetes network property      | Default value | Property description                                          |
|----------------------------------|---------------|---------------------------------------------------------------|
| Kubernetes internal cluster CIDR | 10.244.0.0/22 | The CIDR used for pods running inside the Kubernetes cluster. |

|                                  |               |                                                                      |
|----------------------------------|---------------|----------------------------------------------------------------------|
| Kubernetes internal service CIDR | 10.244.4.0/22 | The CIDR used for Kubernetes services inside the Kubernetes cluster. |
|----------------------------------|---------------|----------------------------------------------------------------------|

**Note:**

You can also change the Kubernetes CIDR network properties after deployment.

13. Click **Next**.

14. Review the **Ready to complete** page and click **Finish**.

The VCF Operations orchestrator appliance is successfully deployed.

Log in to the VCF Operations orchestrator appliance command line as **root** and confirm that you can perform a forward or reverse DNS lookup.

- To perform a forward DNS lookup, run the `nslookup your_orchestrator_FQDN` command. The command must return the VCF Operations orchestrator appliance IP address.
- To perform a reverse DNS lookup, run the `nslookup your_orchestrator_IP` command. The command must return the VCF Operations orchestrator appliance FQDN.

For information on how to use VCF Operations orchestrator and how to integrate it with different VCF components, go to [Getting Started with VCF Operations orchestrator](#).

**Note:**

If you have not enabled SSH during deployment, you can also perform DNS lookups from the virtual machine console in vSphere Client.

If you encounter problems with your VCF Operations orchestrator deployment, go to [KB 93142](#).

**Configuring the VCF Operations Orchestrator Authentication Provider**

To use your deployed VCF Operations orchestrator, you must authenticate the deployment through the command line interface (CLI). To use these commands, you must log in to the VCF Operations orchestrator as a **root** user.

**Retrieving the current authentication provider**

You can retrieve the current authentication provider by running the following command:

```
vracli vro authentication
```

**Configure the authentication provider by using a guided wizard**

To configure the authentication provider by using a guided configuration wizard, run the following command:

```
vracli vro authentication wizard
```

After running the authentication wizard command, you are prompted to provide the necessary authentication provider information such as the type of authentication provider, hostname, and password.

To finish the configuring the authentication provider, restart the the VCF Operations orchestrator server:

```
kubectl -n prelude delete pods -lapp=vco-app
```

**Configure the authentication provider by using predefined parameters**

To configure the authentication provider by using predefined configuration parameters, run the `vracli vro authentication set` command. The command can have the following parameters:

| Parameter                                            | Importance                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-p</code> or <code>--provider</code>           | Required                                                                                   | This parameter defines the authentication provider type. The parameter value can be either <code>tm</code> , <code>vsphere</code> , or <code>vidb</code> depending on the authentication provider - VCF Automation, vSphere, or VCF SSO.                                                                                                                                                                             |
| <code>-hn</code> or <code>--hostname</code>          | Required                                                                                   | The hostname or URL of the authentication provider you want to configure. Both options are applicable.                                                                                                                                                                                                                                                                                                               |
| <code>-u</code> or <code>--username</code>           | Required                                                                                   | The username of the administrator associated with the authentication provider.                                                                                                                                                                                                                                                                                                                                       |
| <code>--password-file</code>                         | Optional                                                                                   | The path to a file containing the password of the administrator account for the authentication provider. If left empty, you receive a prompt for adding the password data. The password file must be stored inside the <code>/data/vco/usr/lib/vco</code> directory of the VCF Operations orchestrator appliance. When adding the parameter in the command, exclude the <code>/data/vco</code> part of the filepath. |
| <code>--admin-group</code>                           | Required for vSphere authentication providers. Ignored for other authentication providers. | Parameter for adding the VCF Operations orchestrator administrators group of the specified vSphere deployment.                                                                                                                                                                                                                                                                                                       |
| <code>--admin-group-domain</code>                    | Required for vSphere authentication providers. Ignored for other authentication providers. | This parameter defines the administrator group domain.                                                                                                                                                                                                                                                                                                                                                               |
| <code>-k</code> or <code>--ignore-certificate</code> | Optional                                                                                   | Using this parameter, the authentication process is configured to automatically trust the certificate of the authentication provider.                                                                                                                                                                                                                                                                                |
| <code>-f</code> or <code>--force</code>              | Optional                                                                                   | Using this parameter, you are not prompted for confirmation if the specified authentication provider is already configured.                                                                                                                                                                                                                                                                                          |
| <code>--fqdn</code>                                  | Optional                                                                                   | This parameter defines the external address of the VCF Operations orchestrator server.<br><b>Note:</b> You can retrieve the FQDN address for your environment by running the <code>nslookup &lt;your_orchestrator_IP&gt;</code> command.                                                                                                                                                                             |

**Example authentication configurations**

```
vracli vro authentication set -f -k -p tm -u <username> -hn <vcfa_hostname> --fqdn <deployment_fqdn> --tenant <tenant_ID>
Enter credentials for authentication (could be API token, Bearer Token, Password): *****
```

The authentication provider has been successfully registered.

```
vracli vro authentication set -f -k -p vsphere -u <username> -hn <vsphere_hostname> --fqdn <deployment_fqdn>
--tenant <tenant_ID> --admin-group=<admin_group_ID> --admin-group-domain=<admin_group_domain>
```

To finish the configuring the authentication provider, restart the the VCF Operations orchestrator server:

```
kubectl -n prelude scale deployment vco-app --replicas=0
kubectl -n prelude scale deployment vco-app --replicas=1
```

### Unregister an authentication provider

You can unregister the current authentication provider by running the `vracli vro authentication unregister` command. This command can have the following parameters:

| Parameter                                  | Importance | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-u</code> or <code>--username</code> | Required   | The username of the administrator associated with the authentication provider.                                                                                                                                                                                                                                                                                                              |
| <code>--password-file</code>               | Optional   | The path to a file containing the password of the administrator account for the authentication provider. If left empty, you receive a prompt for adding the password data. The password file must be stored inside the <code>/data/vco/usr/lib/vco</code> directory of the appliance. When including the parameter in the command, exclude the <code>/data/vco</code> part of the filepath. |

### CLI command logs

VCF Operations orchestrator CLI commands print their logs in the `/services-logs/prelude/vco-app/file-logs/vco-server-app_cfg-cli.log` file. When a command returns a result different than zero and the standard output does not show a specific error, the exception is visible in this file.

### Configure a VCF Operations Orchestrator Cluster

You can configure your new VCF Operations orchestrator deployment to run in high availability by deploying three nodes and connecting them as a cluster.

- Download and deploy three standalone VCF Operations orchestrator instances.

#### Note:

A clustered VCF Operations orchestrator environment can consist of three nodes.

- Verify that SSH access is enabled for all VCF Operations orchestrator nodes. To activate SSH access, run the `/usr/bin/toggle-ssh enable` command.
- Configure a load balancer server, such as Avi Load Balancer. For more information configuring the Avi Load Balancer, see the [VMware Avi Load Balancer documentation](#).

A VCF Operations orchestrator cluster consists of three VCF Operations orchestrator instances that share a common PostgreSQL database. The database of the configured VCF Operations orchestrator cluster can only run in asynchronous mode.

To create a VCF Operations orchestrator cluster, you must select one VCF Operations orchestrator instance to be the primary node of the cluster. After configuring the primary node, you join the secondary nodes to it.

The VCF Operations orchestrator cluster you created is pre-configured with automatic failover.

**Note:** Failure of the automatic failover can lead to loss of database data.

- Configure the primary node.
  - Log in to the VCF Operations orchestrator command line of the primary node over SSH as **root**.
  - To configure the cluster load balancer server, run the `vracli load-balancer set load_balancer_FQDN` command.
  - Configure the authentication provider by running the `vracli vro authentication set` command.
 

Optionally, you can add the FQDN address of the load balancer server by adding the `--fqdn` parameter to the `vracli authentication set` command. For more information on how to configure the authentication provider, go to [Configuring the Authentication Provider](#).
- Join secondary nodes to primary node.
  - Log in to the VCF Operations orchestrator command line of the secondary node over SSH as **root**.
  - To join the secondary node to the primary node, run the `vracli cluster join primary_node_hostname_or_IP` command.
  - Enter the root password of the primary node.
  - Repeat the procedure for other secondary node.
- Optional: If your primary node uses a custom certificate, you must either set the certificate in the appliance or generate a new certificate. For more information, go to [Generate a custom TLS certificate for VCF Operations orchestrator](#).
 

**Note:** The file containing the certificate chain must be PEM-encoded.
- Finish the cluster deployment.
  - Log in to the VCF Operations orchestrator command line of the primary node over SSH as **root**.
  - To confirm that all nodes are in a ready state, run the `kubectl -n prelude get nodes` command.
  - Run the `/opt/scripts/deploy.sh` script and wait for the deployment to finish.

You have created a VCF Operations orchestrator cluster. After creating the cluster, you can access your VCF Operations orchestrator environment only from the FQDN address of your load balancer server.

#### Note:

You cannot update the configuration of a cluster node if it has a different root password from the password used by the load balancer. To edit the configuration of this node, remove it from the load balancer, edit the configuration and add the node back to the load balancer.

To monitor the state of the VCF Operations orchestrator cluster, log in to the VCF Operations orchestrator client and navigate to the **System** tab of the dashboard.

#### Note:

If you are unable to access the VCF Operations orchestrator client, you can also monitor the states of your cluster nodes by running the `kubectl get pods -n prelude` command from the VCF Operations orchestrator appliance command line.

## Removing VCF Operations Orchestrator Cluster Nodes

You can reduce cluster capacity by removing two of the nodes in the deployment. Your VCF Operations orchestrator deployment can have either one or three nodes.

Create an VCF Operations orchestrator cluster. Go to [Configure a VCF Operations Orchestrator Cluster](#).

After removing a node from your VCF Operations orchestrator cluster, that node will no longer be functional. If you want to use this node again, you must delete its VCF Operations orchestrator appliance from your vCenter and deploy it again. See [Deploying VCF Operations Orchestrator](#).

1. Log in to the VCF Operations orchestrator appliance command line of the node you want to remove as **root**.
2. To remove the nodes from your cluster, run the `vracli cluster leave` command.
3. Log in to the VCF Operations orchestrator command line of the remaining node as **root**.
4. Run the `kubectl -n prelude get nodes` command and confirm that the removed node is no longer part of the cluster.
5. To finishing reducing the capacity of your cluster, run the `deploy.sh` command.

## Recovering a VCF Operations Orchestrator Cluster Node

You can recover a problematic cluster node from the VCF Operations orchestrator appliance command line.

To recover a problematic node in your VCF Operations orchestrator cluster, you must locate the node, remove it from the cluster, and then add it to the cluster again.

1. Identify the primary node of your VCF Operations orchestrator cluster.
  - a) Log in to the VCF Operations orchestrator appliance command line of one of your nodes over SSH as **root**.
  - b) Find the node with the **primary** role by running the `kubectl -n prelude exec postgres-0` command.

```
kubectl -n prelude exec postgres-0 - chpst -u postgres repmgr cluster show --
terse --compact
```

- c) Retrieve the name of the pod in which the primary node is located.  
In most cases, the name of the pod is `postgres-0.postgres.prelude.svc.cluster.local`.
  - d) Find the FQDN address of the primary node by running the `kubectl -n prelude get pods` command.  

```
kubectl -n prelude get pods -o wide
```
  - e) Find the database pod with the name you retrieved and get the FQDN address for the corresponding node.
2. Locate the problematic node by running the `kubectl -n prelude get node` command.  
The problematic node has a **NotReady** status.

3. Log in to the VCF Operations orchestrator command line of the primary node over SSH as **root**.
4. Remove the problematic node from the cluster by running the `vracli cluster remove <NODE-FQDN>` command.
5. Log in to the VCF Operations orchestrator appliance command line of the problematic node over SSH as **root**.
6. Add the node to the cluster again by running the `vracli cluster join <MASTER-DB-NODE-FQDN>` command.

## Deploying the VCF Operations for logs Appliance for VMware vSphere Foundation

Download the virtual appliance to deploy VCF Operations for logs for VMware vSphere Foundation (VVF) . VMware distributes the VCF Operations for logs virtual appliance as an `.ova` file. Deploy the VCF Operations for logs virtual appliance for VMware vSphere Foundation (VVF) by using the vSphere Client.

- Verify that you have a VMware vSphere Foundation (VVF) license.
- Verify that you have a copy of the VCF Operations for logs virtual appliance `.ova` file.
- Verify that you have permissions to deploy OVF templates to the inventory.
- Verify that you can log in to the VCF Operations UI.

1. In the vSphere Client, select **File > Deploy OVF Template**.
2. Follow the prompts in the **Deploy OVF Template** wizard.
3. On the **Select Configuration** page, select the size of the VCF Operations for logs virtual appliance based on the size of the environment for which you intend to collect logs.

**Small** is the minimum requirement for production environments.

VCF Operations for logs provides preset VM (virtual machine) sizes that you can select from to meet the ingestion requirements of your environment. These presets are certified size combinations of compute and disk resources, though you can add extra resources afterward. A small configuration is suitable only for demos.

**Note:** If you select **Large**, you must upgrade the virtual hardware on the VCF Operations for logs virtual machine after the deployment.

4. On the Select Storage page, select a disk format.
  - **Thick Provision Lazy Zeroed** creates a virtual disk in a default thick format. Space required for the virtual disk is allocated when the virtual disk is created. The data remaining on the physical device is not erased during creation, but is zeroed out on demand later, on first write from the virtual appliance.
  - **Thick Provision Eager Zeroed** creates a type of thick virtual disk that supports clustering features such as Fault Tolerance. Space required for the virtual disk is allocated at creation time. In contrast to the flat format, the data remaining on the physical device is zeroed out when the virtual disk is created. It might take much longer to create disks in this format than to create other types of disks.

**Important:** Deploy the VCF Operations for logs virtual appliance with thick provisioned eager zeroed disks whenever possible for better performance and operation of the virtual appliance.

- **Thin Provision** creates a disk in thin format. The disk expands as the amount of data saved on it increases. If your storage device does not support thick provisioning disks or you want to conserve unused disk space on the VCF Operations for logs virtual appliance, deploy the virtual appliance with thin provisioned disks.

**Note:** Shrinking disks on the VCF Operations for logs virtual appliance is not supported and might result in data corruption or data loss.



5. Optional: On the **Select Networks** page, set the networking parameters for the VCF Operations for logs virtual appliance. You can select the IPv4 or IPv6 protocol.

If you do not provide network settings, such as an IP address, DNS servers, and gateway information, VCF Operations for logs uses DHCP to set those settings.

**CAUTION:** Do not specify more than two domain name servers. If you specify more than two domain name servers, all configured domain name servers are ignored in the VCF Operations for logs virtual appliance.

Use a comma-separated list to specify domain name servers.

6. Optional: On the **Customize template** page, set network properties if you are not using DHCP.

Under Application, select the **Prefer IPv6 addresses** check box if you want to run the virtual machine in a dual stack network.

**CAUTION:** Do not select the **Prefer IPv6 addresses** check box if you want to use pure IPv4 even with IPv6 supported in your network. Select the check box only if your network has a dual stack or pure stack support for IPv6.

7. Optional: On the **Customize template** page, select **Other Properties** and set the root password for the VCF Operations for logs virtual appliance.

The root password is required for SSH. You can also set this password through the VMware Remote Console.

8. Follow the prompts to complete the deployment.

After you power on the virtual appliance, an initialization process begins. The initialization process takes several minutes to complete. At the end of the process, the virtual appliance restarts.

9. Navigate to the **Console** tab and verify the IP address of the VCF Operations for logs virtual appliance.

| IP Address Prefix | Description                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| https://          | The DHCP configuration on the virtual appliance is correct.                                                                                                                                                                                               |
| http://           | The DHCP configuration on the virtual appliance failed.<br>1. Power off the VCF Operations for logs virtual appliance.<br>2. Right-click the virtual appliance and select <b>Edit Settings</b> .<br>3. Set a static IP address for the virtual appliance. |

- Log in to the VCF Operations UI to configure vCenter instances to pull tasks, events, and alerts. For more information, see [Configuring Log Sources](#).

- In the VCF Operations UI, navigate to **Administration > Control Panel** and click the **Operations-Logs Appliance Integration** tile. Enter the address, username and password of the VCF Operations for logs deployment. This completes the VCF Operations integration with VCF Operations for logs.

The VCF Operations for logs Web interface is available at `https://operations-for-logs-host/` where `operations-for-logs-host` is the IP address or host name of the VCF Operations for logs virtual appliance.

- If you want to join an existing deployment, see [Join an Existing Deployment](#).
- Download the VCF Operations for logs 9.0 documentation from [here](#).

## Start a New VCF Operations for Logs Deployment

When you access the VCF Operations for logs web interface for the first time after the virtual appliance deployment or after removing a worker node from a cluster, you must finish the initial configuration steps.

- In the vSphere Client, note the IP address of the VCF Operations for logs virtual appliance.
- Verify that you are using a supported browser.

More recent browser versions also work, but have not been validated. Cookies must be activated in your browser. Supported browsers are:

- Mozilla Firefox 45.0 and above

- Google Chrome 51.0 and above
- Safari 9.1 and above
- Internet Explorer 11.0 and above
  - Internet Explorer Document mode must be set to **Standards Mode**. Other modes are not supported.
  - Browser Mode:** Compatibility View is not supported.
  - To use Internet Explorer with the VCF Operations for logs web interface, Windows local storage integrity level must be configured as low.

All settings that you modify during the initial configuration are also available in the Configuration web user interface.

1. Use a supported browser to navigate to the web user interface of VCF Operations for logs.

The URL format is `https://operations_for_logs-host/`, where `operations_for_logs-host` is the IP address or host name of the VCF Operations for logs virtual appliance.

The initial configuration wizard opens.

2. Click **Start New Deployment**.

3. Set the password for the administrator (user name admin) and click **Save and Continue**. Optionally, you can provide an email address for the administrator.

The administrator is a user linked to the Super Admin role.

4. On the General Configuration page, enter the email address to receive system notifications from VCF Operations for logs.

5. If you are using webhooks to send notifications to VCF Operations or a third-party application, enter a space-separated list of URLs in the **Send HTTP Post System Notifications To** text box.

6. Optional: To leave the CEIP, deselect the **Join the VMware Customer Experience Program** option. Click **Save and Continue**.

### Note:

If you join the CEIP, VCF Operations for logs uses a third-party tool called Pendo to collect analytics cookies. Pendo collects data based on your interaction with the user interface by tracking where you click, to help VMware understand how VCF Operations for logs is used. This data is used to improve the VMware services and design them better.

7. On the Time Configuration page, set how time is synchronized on the VCF Operations for logs virtual appliance and click **Test**.

| Option   | Description                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESX host | By default, VCF Operations for logs is configured to synchronize time with the <b>ESX host</b> where you deployed the VCF Operations for logs virtual appliance. |

8. Click **Save and Continue**.

9. Optional: To enable outgoing alert and system notification emails, specify the properties of an SMTP server.

To verify that the SMTP configuration is correct, enter a valid email address and click **Test**. VCF Operations for logs sends a test email to the address that you provided.

10. Optional: To provide a custom SSL certificate, upload a certificate file to the cluster in a PEM format. You can also view the details of the existing certificate.

The system adds the certificate to the truststores of all the nodes of the cluster and saves it for later use.

11. Click **Save and Continue**.

After the VCF Operations for logs process restarts, you are redirected to the **Dashboards** page of VCF Operations for logs.

- Log in to the VCF Operations UI. Navigate to **Administration > Control Panel** and click the **Operations-Logs Appliance Integration** tile. Enter the address, username and password of the VCF Operations for logs deployment. This completes the VCF Operations integration with VCF Operations for logs.
- In the VCF Operations UI, navigate to **Infrastructure Operations > Configurations > Log Collections**. Configure the vCenter instance. This configuration is to pull tasks, events, and alerts, and to configure ESX hosts to send syslog feeds to VCF Operations for logs.
- Install the VCF Operations for logs Linux Agent for monitoring journal service.
- Set up integration with a vCenter instance version 9.0 or later that is licensed with a VMware vSphere Foundation 9.0 license or a VMware Cloud Foundation 9.0 license. See Broadcom knowledge base article [404339](#).

## Join an Existing Deployment

After you deploy and set up a standalone VCF Operations for logs node, you can deploy a new VCF Operations for logs instance and add it to the existing node to form a VCF Operations for logs cluster.

- In the vSphere Client, note the IP address of the worker VCF Operations for logs virtual appliance.
- Verify that you have the IP address or host name of the primary VCF Operations for logs virtual appliance.
- Verify that you have a user account on the primary VCF Operations for logs virtual appliance with the Super Admin role, a role that has the relevant permissions.
- Verify that the versions of the VCF Operations for logs primary and worker nodes are in sync. Do not add an older version VCF Operations for logs worker to a newer version VCF Operations for logs primary node.
- Login to the web user interface of the VCF Operations for logs primary node and generate a secure token on the **Management > Cluster** page.

VCF Operations for logs can scale out by using multiple virtual appliance instances in clusters. Clusters enable linear scaling of ingestion throughput, increase query performance, and allow high-availability ingestion. In cluster mode, VCF Operations for logs provides primary and worker nodes. Both primary and worker nodes are responsible for a subset of data. Primary nodes can query all subsets of data and aggregate the results. You might require more nodes to support site needs. You can use from three to 18 nodes in a cluster. This means that a fully functional cluster must have a minimum of three healthy nodes. Most nodes in a larger cluster must be healthy. For example, if three nodes of a six-node cluster fail, none of the nodes functions fully until the failing nodes are removed.

1. Use a supported browser to navigate to the web user interface of the VCF Operations for logs worker.  
The URL format is `https://operations_for_logs-host/`, where `operations_for_logs` is the IP address or host name of the VCF Operations for logs worker virtual appliance.  
The initial configuration wizard opens.
2. Click **Join Existing Deployment**.
3. Enter the following details and click **Go**.
  1. IP address or host name of the VCF Operations for logs primary node.
  2. The secure token generated on the **Management > Cluster** page.

If the primary node provides an untrusted SSL certificate, a dialog box appears with the details of the certificate. Click **Accept** to send a request to the VCF Operations for logs primary node to join the existing deployment.

If you click **Cancel**, the join request is not sent to the primary node. You must accept the certificate to ensure that the worker node joins the existing deployment.

The worker node joins the existing deployment and VCF Operations for logs begins to operate in a cluster.

- Add more worker nodes as needed. The cluster must have a minimum of three nodes.

## Manual Deployment and Configuration of Components for Advanced Architectures for VMware Cloud Foundation and VMware vSphere Foundation

Certain VCF components support architectures that are not automated by VCF Installer. You manually deploy and configure components to achieve the required states.

### Configure VCF Operations Cluster for Continuous Availability

VCF Installer supports the deployment of VCF Operations in either Simple mode or High Availability mode during the initial VCF deployment. After the initial deployment, the VCF Operations cluster can be converted to a continuous availability architecture

Identify three separate fault domains for the installation of the VCF Operations primary node and data nodes, the VCF Operations primary replica node and data nodes, and the witness node, respectively. The fault domains can be located in either a different workload domain, a separate availability zone, or a different region. In a continuous availability architecture, there must be at least one primary node and one data node within the primary fault domain. Additionally, the replica fault domain should have the same number of nodes as the primary fault domain, and there should be one witness node in the witness fault domain. The VCF Operations nodes in the primary fault domain, the VCF Operations nodes in the replica fault domain, and the witness node in the witness fault domain must be able to communicate over the network, despite being in separate fault domains. Consult the VCF Operations sizing guidelines for the sizing and network latency requirements between the three distinct fault domains.

1. Deploy VCF Operations using VCF Installer in either Simple mode or High Availability mode within the fault domain designated for the primary node.
2. Deploy the witness node in the fault domain designated for it, by following these steps:
  - a) Download the VCF Operations ova file from the Broadcom support portal.
  - b) Upload the VCF Operations ova file to the vCenter in the fault domain designated for the witness node.
  - c) Deploy the VCF Operations OVF template and select **Witness** as the configuration for deployment.
  - d) Note down the IP address and the hostname or FQDN of the witness node.
  - e) Log in to the VCF Operations administrator console in the primary fault domain.
  - f) Add the newly deployed witness node to the VCF Operations cluster by using the IP address and hostname or FQDN of the witness node, as recorded in the previous steps.
3. If VCF Operations was deployed in a High Availability architecture during the initial setup, follow the steps below:
  - a) Deactivate High Availability in the VCF Operations administrator console.
  - b) The primary replica cluster node is converted into a standard data node within the cluster, which can then be removed if an extra data node is not needed. The remaining nodes in the cluster maintain their current roles.
  - c) Ensure that the VCF Operations cluster has at-least one primary node and one data node in the primary fault domain. Consult the VCF Operations sizing guidelines for the maximum number of data nodes that can be added

in the primary fault domain. Note that the number of VCF Operations nodes in the primary fault domain and the replica fault domain must be equal.

4. Deploy the same number of VCF Operations nodes in the designated replica fault domain as in the primary fault domain, following the steps below:
  - a) If the designated replica fault domain is managed by the VCF Operations in the primary fault domain, then use the lifecycle management capability within VCF Operations to add the same number of VCF Operations nodes in the replica fault domain as in the primary fault domain.
  - b) If the designated replica fault domain is NOT managed by the VCF Operations in the primary fault domain, then continue with the the steps.
    - a) Download the VCF Operations ova file from the Broadcom support portal.
    - b) Upload the VCF Operations ova file to the vCenter in the fault domain designated for the replica fault domain.
    - c) Use the VCF Operations OVF template to deploy an equal number of VCF Operations nodes as in the primary fault domain, ensuring that the selected size matches that of the nodes deployed in the primary fault domain.
    - d) Note down the IP address and the hostname of the VCF Operations nodes deployed in the replica fault domain.
    - e) Log in to the VCF Operations administrator console in the primary fault domain.
    - f) Add the newly deployed VCF Operations nodes in the replica fault domain using their IP address and hostnames to the VCF Operations cluster in the primary fault domain.
5. Click **Activate CA** in the VCF Operations Admin console.
6. The Continuous Availability wizard opens. The Witness node exists outside the fault domains. The primary node is already assigned to Fault Domain. Perform the following steps:
  - a) To pair with the VCF Operations nodes in the primary fault domain, move the VCF Operations data nodes deployed in the replica fault domain to Fault Domain.
  - b) Ensure an even distribution of VCF Operations nodes between the primary and replica fault domains to enable Continuous Availability. The number of nodes in both fault domains must be equal.
7. After successfully activating CA, navigate to **Lifecycle** under **Fleet Management** within the VCF Operations console and click **trigger inventory sync** on the Operations component.

The Lifecycle page for the VCF Operations component now displays Continuous Availability as the configuration for the VCF Operations cluster.

#### VCF Operations Analytics Node Data Storage and Fault Domain Behavior

VCF Operations uses a robust architecture to ensure data availability and fault tolerance across analytics nodes. Data is stored in pairs of analytics nodes distributed across two fault domains, ensuring continuous availability and synchronization. Each object is stored in two nodes (primary and replica) across fault domains, providing redundancy and resilience in case of failures.

**Table 814: Key Concepts of Data Storage and Fault Domain Behavior**

| Key Concept                      | Details                                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Storage and Synchronization | <p>When an object is discovered, VCF Operations determines the primary node to store the data and automatically replicates it to its paired node in the other fault domain.</p> <p>Synchronization between the primary and replica nodes ensures data consistency across fault domains</p> |
| Fault Domain Resilience          | <p>Each fault domain contains half of the analytics nodes, and each node has a paired replica in the other fault domain.</p> <p>The cluster continues to function without data loss as long as one node from each pair is available.</p>                                                   |

| Key Concept                      | Details                                                                                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous Availability          | The cluster can tolerate the loss of one fault domain or individual nodes without data loss, but the system will operate in a degraded state until the issue is resolved. |
| Latency and Network Requirements | Latency between fault domains must be <10 ms, with peaks up to 20 ms during 20-second intervals. High latency can impact synchronization but does not stop the process.   |
| Witness Node Role                | The witness node ensures cluster health and determines which fault domain remains online during communication failures.                                                   |

**Table 815: Key Scenario and Behavior**

| Scenario                                   | Behavior                                                                                                                                                                     |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Storage                               | Data is stored in a primary node and replicated to its paired node in the other fault domain.                                                                                |
| Loss of One Fault Domain                   | The cluster operates in a degraded state but continues without data loss. The fault domain must be repaired or replaced to restore full functionality.                       |
| Loss of Two Nodes in One Fault Domain      | The cluster continues to operate as long as one node from each pair is available.                                                                                            |
| Loss of Connectivity Between Fault Domains | One fault domain will go offline automatically. Once connectivity is restored, the admin must manually bring the fault domain back online to synchronize data.               |
| Witness Node Unavailability                | If both fault domains are healthy, the witness node's unavailability has no impact. If communication issues occur, the witness determines which fault domain remains online. |
| Node Replacement                           | Use the "Replace nodes of cluster" functionality in the admin UI. Sync time depends on object count, historical data, network bandwidth, and cluster load.                   |
| Latency Exceeds 20 ms                      | Synchronization continues but may be delayed. Latency must return to <10 ms for optimal performance.                                                                         |
| Primary Node Failover                      | If the primary node becomes unavailable, the primary replica node is promoted. The original primary node becomes the replica when it returns online.                         |
| Entire Fault Domain Replacement            | If a fault domain is unrecoverable, it can be replaced with newly deployed nodes. The admin UI allows replacing the primary replica node.                                    |
| Load Balancer Support                      | Load balancers are supported for continuous availability.                                                                                                                    |
| Node Reboot or Offline                     | Nodes automatically resume operation after reboot or being brought offline/online. No additional steps are required.                                                         |

#### Configure VCF Operations Cluster for High Availability

VCF Installer supports the manual deployment of VCF Operations in Simple mode for VMware vSphere Foundation. After the initial deployment, the VCF Operations cluster can be converted to a high availability architecture. To activate high availability (HA) for a VCF Operations cluster, specify one of the data nodes to become a replica of the primary node.

- Create and configure the primary node.

- Create and configure a data node with a static IP address.
- Note the fully qualified domain name (FQDN) or IP address of the primary node.

**Note:** If the cluster is running, activating HA restarts the cluster.

You can add HA to the VCF Operations cluster at installation time or after VCF Operations is up and running. Adding HA at installation is less intrusive because the cluster has not yet started.

1. In a Web browser, navigate to the primary node administration interface.  
`https://primary-node-name-or-ip-address/admin`
2. Enter the VCF Operations administrator user name of `admin`.
3. Enter the VCF Operations administrator password and click **Log In**.
4. To add a new data node, click **Add new Nodes**.
  - a) In the **Add new nodes to cluster** wizard, click the **Add Nodes** icon.
  - b) For **Node Name**, enter a generic name to identify the node.
  - c) For **Node Address**, enter the IP address of the node you want to add.
  - d) For **Current Cluster Role**, select **Data**, to add a data node to be used as the primary replica node.
  - e) Click **Save**, review and close the thumbprint information, and then click **Next**.
  - f) For **Primary Node Password**, enter the administrator password used during log in, and then click **Finish**.

If the cluster was online, the administration interface displays that the cluster initialization is in progress as the new data node gets added to the VCF Operations cluster. You must wait for the cluster to come **Online** again before activating HA.

5. Under High Availability, click **Activate**.
6. In the **Activate High Availability** wizard, select a data node to serve as the replica for the primary node, and click **OK**.
7. To activate HA, you must restart your cluster. Review the **Confirm Cluster Restart** message and click **Yes**.  
If the cluster was online, the administration interface displays progress as VCF Operations configures, synchronizes, and rebalances the cluster for HA.
8. If the primary node and replica node go offline, and the primary remains offline for any reason while the replica goes online, the replica node does not take over the primary role, take the entire cluster offline, including data nodes and log in to the replica node command-line console as a root.
9. Open `$ALIVE_BASE/persistence/persistence.properties` in a text editor.

10. Locate and set the following properties:

```
db.role=PRIMARY
db.driver=/data/vcops/xdm/vcops.bootstrap
```

11. Save and close `persistence.properties`.

12. In the administration interface, bring the replica node online, and verify that it becomes the primary node and bring the remaining cluster nodes online.

High Availability is activated on your VCF Operations cluster.

After creating a primary replica node, you have the following options.

- New, unstarted clusters:
  - Create and add data nodes.
  - Click **Start** VCF Operations to start the cluster, and log in to finish configuring the product.  
The cluster might take from 10 to 30 minutes to start, depending on the size of your cluster and nodes. Do not make changes or perform any actions on cluster nodes while the cluster is starting.

- Established, running clusters:
  - Create and add data nodes.
- To deactivate high availability, click **Deactivate HA** and follow the steps in the admin UI. The deactivation timeline depends on the cluster size and the data retention period. Please wait for the deactivation to complete.

## Deploying VCF Management Components with Custom Networking

If you skipped deployment of the VCF management components during initial deployment of a VCF Instance, you can use the SDDC Manager API to deploy them later.

Using the SDDC Manager API, you can deploy the VCF management components to specific vSphere distributed port groups or NSX segment. There are four main options. You can deploy the VCF management components to a:

- Shared management network
- Dedicated management network
- NSX overlay segment
- NSX VLAN segment

You can use custom networking to separate user-facing networks from management networks to meet regulatory and security requirements.

For more information about each of these options, see [Fleet Level Components Networking Models](#).

## Deploy VCF Management Components on the Shared Management Network

Use the SDDC Manager API to deploy the VCF management components using the same distributed port group as the VCF core components.

- Review Design Guide. See [Fleet Level Components on Shared Management Network Model](#).
- Download binaries for the VCF management components to SDDC Manager.

Use this procedure to configure all VCF management components (VCF Operations, VCF Operations fleet management, VCF Operations collector, and VCF Automation) to use the same distributed port group as the core components (for example, vCenter).

See the [SDDC Manager API Reference Guide](#) for more information.

1. Create a JSON file with information about the VCF management components to use for deployment.

**Note:** Do not include the `vcfManagementComponentsInfrastructureSpec` section in the JSON file. This ensures that the VCF management components use the same distributed port group as the VCF core components.

For example:

```
{
 "vcfOperationsFleetManagementSpec": {
 "hostname": "flt-fm01.rainpole.io",
 "rootUserPassword": "<password>",
 "adminUserPassword": "<password>",
 "useExistingDeployment": false
 },
 "vcfOperationsSpec": {
 "nodes": [
 {
 "hostname": "flt-ops01a.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "master"
 }
]
 }
}
```

```

 },
 {
 "hostname": "flt-ops01b.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "replica"
 },
 {
 "hostname": "flt-ops01c.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "data"
 }
],
 "useExistingDeployment": false,
 "applianceSize": "medium",
 "adminUserPassword": "<password>"
},
"vcfOperationsCollectorSpec": {
 "hostname": "sfo-opsc01.sfo.rainpole.io",
 "rootUserPassword": "<password>",
 "version": "9.0.0.0",
 "applianceSize": "small"
},
"vcfAutomationSpec": {
 "ipPool": [
 "10.11.10.51",
 "10.11.10.52",
 "10.11.10.53",
 "10.11.10.54"
],
 "nodePrefix": "flt-auto",
 "hostname": "flt-auto01.rainpole.io",
 "internalClusterCidr": "198.18.0.0/15",
 "adminUserPassword": "<password>"
}
}

```

2. Log in to the SDDC Manager UI and navigate to **Developer Center**.

3. Validate your JSON file.

```
POST /v1/vcf-management-components/validations
```

4. Deploy the VCF management components.

```
POST /v1/vcf-management-components
```

5. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

### Deploy VCF Management Components on a Dedicated Management Network

Use the SDDC Manager API to deploy the VCF management components using a dedicated distributed port group.

- Review Design Guide. See [Fleet Level Components on Dedicated Management Network Model](#).

- Download binaries for the VCF management components to SDDC Manager.
- Create a distributed port group with a dedicated VLAN on the management vCenter vSphere Distributed Switch.

Use this procedure to configure all VCF management components (VCF Operations, VCF Operations fleet management, VCF Automation) to use a different distributed port group than the VCF core components (for example, vCenter). The VCF Operations collector appliances are deployed to the VM management network.

See the [SDDC Manager API Reference Guide](#) for more information.

1. Create a JSON file with information about the VCF management components to use for deployment.

For example:

```

{
 "vcfOperationsFleetManagementSpec": {
 "hostname": "flt-fm01.rainpole.io",
 "rootUserPassword": "<password>",
 "adminUserPassword": "<password>",
 "useExistingDeployment": false
 },
 "vcfOperationsSpec": {
 "nodes": [
 {
 "hostname": "flt-ops01a.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "master"
 },
 {
 "hostname": "flt-ops01b.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "replica"
 },
 {
 "hostname": "flt-ops01c.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "data"
 }
],
 "useExistingDeployment": false,
 "applianceSize": "medium",
 "adminUserPassword": "<password>"
 },
 "vcfOperationsCollectorSpec": {
 "hostname": "sfo-opsc01.sfo.rainpole.io",
 "rootUserPassword": "<password>",
 "version": "9.0.0.0",
 "applianceSize": "small"
 },
 "vcfAutomationSpec": {
 "ipPool": [
 "10.11.99.51",
 "10.11.99.52",
 "10.11.99.53",
 "10.11.99.54"
],
 "nodePrefix": "flt-auto",

```

```

 "hostname": "flt-auto01.rainpole.io",
 "internalClusterCidr": "198.18.0.0/15",
 "adminUserPassword": "<password>"
 },
 "vcfManagementComponentsInfrastructureSpec": {
 "localRegionNetwork": {
 "networkName": "<VM-Management-port-group-name>",
 "subnetMask": "255.255.255.0",
 "gateway": "10.11.10.1"
 },
 "xRegionNetwork": {
 "networkName": "<Dedicated-Management-port-group-name>",
 "subnetMask": "255.255.255.0",
 "gateway": "10.11.99.1"
 }
 }
 }
}

```

Replace *<port-group-name>* with the name of the dedicated distributed virtual port group.

- Log in to the SDDC Manager UI and navigate to **Developer Center**.

- Validate your JSON file.

```
POST /v1/vcf-management-components/validations
```

- Deploy the VCF management components.

```
POST /v1/vcf-management-components
```

- When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

## Deploy VCF Management Components on an NSX Overlay Segment

Use the SDDC Manager API to deploy an NSX edge cluster and the VCF management components on an NSX overlay segment.

- Review Design Guide. See [Fleet Level Components on NSX Overlay Segment Model](#).
- Download binaries for the VCF management components to SDDC Manager.
- Create a json for the deployment of the NSX edge cluster using SDDC Manager API.

### Important:

It is not supported to create a Centralized Transit Gateway and import later to NSX Global manager. As an alternative use the SDDC manager API to create the NSX edge cluster.

Use this procedure to deploy an edge cluster using SDDC manager API and then deploy all VCF management components (VCF Operations, VCF Operations fleet management, and VCF Automation) to use an NSX overlay segment. The VCF Operations collector appliances are deployed to the VM management network.

See the [SDDC Manager API Reference Guide](#) for more information.

In this section you will deploy an NSX edge cluster using BGP routing that provides connectivity for the NSX segment later deployed for the VCF Management Components.

- Create a JSON file with for the deployment of the edge cluster using SDDC manager API, this will require a Tier-0 and Tier-1 gateway.

For example

```
{
```

```

 "edgeClusterName": "sfo-m01-ec01",
 "edgeClusterProfileType": "CUSTOM",
 "edgeClusterProfileSpec": {
 "bfdAllowedHop": 255,
 "bfdDeclareDeadMultiple": 3,
 "bfdProbeInterval": 1000,
 "edgeClusterProfileName": "sfo-m01-ecp01",
 "standbyRelocationThreshold": 30
 },
 "edgeClusterType": "NSX-T",
 "edgeRootPassword": "<password>",
 "edgeAdminPassword": "<password>",
 "edgeAuditPassword": "<password>",
 "edgeFormFactor": "MEDIUM",
 "tier0ServicesHighAvailability": "ACTIVE_ACTIVE",
 "mtu": 8900,
 "asn": 65101,
 "tier0RoutingType": "EBGP",
 "tier0Name": "flt-m01-ec01-t0-gw01",
 "tier1Name": "flt-m01-ec01-t1-gw01",
 "edgeNodeSpecs": [
 {
 "edgeNodeName": "sfo-m01-r01-en01.sfo.rainpole.io",
 "managementIP": "10.11.10.75/24",
 "managementGateway": "10.11.10.1",
 "edgeTepGateway": "10.11.19.1",
 "edgeTep1IP": "10.11.19.2/24",
 "edgeTep2IP": "10.11.19.3/24",
 "edgeTepVlan": 1119,
 "clusterId": "<ESX cluster_ID>",
 "interRackCluster": "false",
 "uplinkNetwork": [
 {
 "uplinkVlan": 1117,
 "uplinkInterfaceIP": "10.11.17.2/24",
 "peerIP": "10.11.17.10/24",
 "asnPeer": 65111,
 "bgpPeerPassword": "<password>"
 },
 {
 "uplinkVlan": 1118,
 "uplinkInterfaceIP": "10.11.18.2/24",
 "peerIP": "10.11.18.10/24",
 "asnPeer": 65111,
 "bgpPeerPassword": "<password>"
 }
],
 "vmManagementPortgroupName": "sfo-m01-cl01-vds01-pg-vm-mgmt",
 "vmManagementPortgroupVlan": 1110
 },
 {
 "edgeNodeName": "sfo-m01-r01-en02.sfo.rainpole.io",
 "managementIP": "10.11.10.76/24",

```

```

"managementGateway": "10.11.10.1",
"edgeTepGateway": "10.11.19.1",
"edgeTep1IP": "10.11.19.4/24",
"edgeTep2IP": "10.11.19.5/24",
"edgeTepVlan": 1119,
"clusterId": "<ESX cluster_ID>",
"interRackCluster": "false",
"uplinkNetwork": [
 {
 "uplinkVlan": 1117,
 "uplinkInterfaceIP": "10.11.17.3/24",
 "peerIP": "10.11.17.10/24",
 "asnPeer": 65111,
 "bgpPeerPassword": "<password>"
 },
 {
 "uplinkVlan": 1118,
 "uplinkInterfaceIP": "10.11.18.3/24",
 "peerIP": "10.11.18.10/24",
 "asnPeer": 65111,
 "bgpPeerPassword": "<password>"
 }
],
"vmManagementPortgroupName": "sfo-m01-cl01-vds01-pg-vm-mgmt",
"vmManagementPortgroupVlan": 1110
}
]
}

```

2. Log in to the SDDC Manager UI and navigate to **Developer Center**.

3. Validate your JSON file.

```
POST /v1/edge-clusters/validations
```

4. Deploy the edge cluster using the validated json from step 4.

```
POST /v1/edge-clusters
```

In this section you will create the NSX segment for the VCF Management components and attach it to the Tier-1 gateway.

5. Log in to the NSX Manager using admin privileges.

6. Select **Networking>Segments**.

7. Click **Add Segment** provide the following details.

- Name for the segment.
- For the **Connected Gateway** select the Tier-1 created during the edge cluster creation.
- For the **Transport Zone** select the Overlay transport zone.
- Under **Subnets** enter the IPv4 gateway for the VCF Management network segment.

8. Click **Save**.

In this section you will deploy the Management VMs onto the NSX segment using the SDDC Manager API.

9. Create a JSON file with information about the VCF management components to use for deployment.

For example:

```

{
 "vcfOperationsFleetManagementSpec": {
 "hostname": "flt-fm01.rainpole.io",
 "rootUserPassword": "<password>",
 "adminUserPassword": "<password>",
 "useExistingDeployment": false
 },
 "vcfOperationsSpec": {
 "nodes": [
 {
 "hostname": "flt-ops01a.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "master"
 },
 {
 "hostname": "flt-ops01b.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "replica"
 },
 {
 "hostname": "flt-ops01c.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "data"
 }
],
 "useExistingDeployment": false,
 "applianceSize": "medium",
 "adminUserPassword": "<password>"
 },
 "vcfOperationsCollectorSpec": {
 "hostname": "sfo-opsc01.sfo.rainpole.io",
 "rootUserPassword": "<password>",
 "version": "9.0.0.0",
 "applianceSize": "small"
 },
 "vcfAutomationSpec": {
 "ipPool": [
 "192.168.11.51",
 "192.168.11.52",
 "192.168.11.53",
 "192.168.11.54"
],
 "nodePrefix": "flt-auto",
 "hostname": "flt-auto01.rainpole.io",
 "internalClusterCidr": "198.18.0.0/15",
 "adminUserPassword": "<password>"
 },
 "vcfManagementComponentsInfrastructureSpec": {
 "localRegionNetwork": {
 "networkName": "<VM-Management-port-group-name>",

```



```

 "subnetMask": "255.255.255.0",
 "gateway": "10.11.10.1"
 },
 "xRegionNetwork": {
 "networkName": "<overlay-segment-name>",
 "subnetMask": "255.255.255.0",
 "gateway": "192.168.11.1"
 }
}

```

Replace `<overlay-segment-name>` with the name of the overlay segment.

**Note:** You can use the same segment for both `localRegionNetwork` and `xRegionNetwork`.

10. Log in to the SDDC Manager UI and navigate to **Developer Center**.

11. Validate your JSON file.

```
POST /v1/vcf-management-components/validations
```

12. Deploy the VCF management components.

```
POST /v1/vcf-management-components
```

13. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

## Deploy VCF Management Components on an NSX VLAN Segment

Use the SDDC Manager API to deploy the VCF management components on an NSX VLAN segment. The VCF Operations collector appliances are deployed to the VM management network.

- Review Design Guide. See [Fleet Level Components on NSX VLAN Segment Model](#).
- Download binaries for the VCF management components to SDDC Manager.
- Create a VLAN transport zone in the management domain NSX Manager. See [Create Transport Zones](#).
- In NSX Manager, create a segment on the newly created VLAN transport zone. See [Create an NSX Segment](#).
  - For the **Transport Zone**, select the VLAN transport zone.
- In NSX Manager, attach the VLAN transport zone to transport node profile for the management domain primary cluster.

Use this procedure to configure all VCF management components (VCF Operations, VCF Operations fleet management, and VCF Automation) to use an NSX VLAN segment. The VCF Operations collector appliances are deployed to the VM management network.

See the [SDDC Manager API Reference Guide](#) for more information.

1. Create a JSON file with information about the VCF management components to use for deployment.

For example:

```

{
 "vcfOperationsFleetManagementSpec": {
 "hostname": "flt-fm01.rainpole.io",
 "rootUserPassword": "<password>",
 "adminUserPassword": "<password>",
 "useExistingDeployment": false
 },
 "vcfOperationsSpec": {
 "nodes": [
 {
 "hostname": "flt-ops01a.rainpole.io",

```

```

 "rootUserPassword": "<password>",
 "type": "master"
 },
 {
 "hostname": "flt-ops01b.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "replica"
 },
 {
 "hostname": "flt-ops01c.rainpole.io",
 "rootUserPassword": "<password>",
 "type": "data"
 }
},
"useExistingDeployment": false,
"applianceSize": "medium",
"adminUserPassword": "<password>"
},
"vcfOperationsCollectorSpec": {
 "hostname": "sfo-opsc01.sfo.rainpole.io",
 "rootUserPassword": "<password>",
 "version": "9.0.0.0",
 "applianceSize": "small"
},
"vcfAutomationSpec": {
 "ipPool": [
 "10.11.99.51",
 "10.11.99.52",
 "10.11.99.53",
 "10.11.99.54"
],
 "nodePrefix": "flt-auto",
 "hostname": "flt-auto01.rainpole.io",
 "internalClusterCidr": "198.18.0.0/15",
 "adminUserPassword": "<password>"
},
"vcfMangementComponentsInfrastructureSpec": {
 "localRegionNetwork": {
 "networkName": "<VM-Management-port-group-name>",
 "subnetMask": "255.255.255.0",
 "gateway": "10.11.10.1"
 },
 "xRegionNetwork": {
 "networkName": "<vlan-segment-name>",
 "subnetMask": "255.255.255.0",
 "gateway": "10.11.99.1"
 }
}
}

```

Replace `<vlan-segment-name>` with the name of the VLAN segment.

**Note:** You can use the same segment for both `localRegionNetwork` and `xRegionNetwork`.

2. Log in to the SDDC Manager UI and navigate to **Developer Center**.

3. Validate your JSON file.

```
POST /v1/vcf-management-components/validations
```

4. Deploy the VCF management components.

```
POST /v1/vcf-management-components
```

5. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

## Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform

You can use your existing virtual infrastructure as building blocks for your VCF or vSphere Foundation platforms. The process for converging includes complying with the general requirements and scenario-specific prerequisites, manual upgrade procedures of your existing components, and deployments and configurations performed by VCF Installer. In the supported scenarios, the manual high-level steps upgrade your existing components to version 9.0.x and prepare your infrastructure for the VCF Installer workflows. By using the VCF Installer workflows, you can converge the components to a new VCF Fleet or a new VCF Instance in an existing VCF fleet.

### Note:

For a VCF platform, the following configurations, requirements, and converge scenarios are applicable for your management domain only. You import and upgrade workload domains in VCF Operations, after you deployed the management domain.

### Minimum Component Versions

The minimum component versions of your existing infrastructure are listed in the specific supported scenarios. See [Supported Scenarios to Converge to VCF](#).

### Supported Configurations

The following table lists supported and not supported configurations for converging your existing infrastructure to a management domain. Before you proceed with a supported scenario, verify that your existing configuration is supported.

### Note:

Starting with VCF 9.0.1, convergence of existing NSX instances without Enhanced Linked Mode (ELM) to a management domain is supported.

For VCF 9.0.0, convergence of existing NSX instances is not supported. Instead, you deploy a new NSX 9.0 instance during the convergence operation by using VCF Installer. The version of the deployed NSX is the latest compatible NSX version for the related vCenter instance. To change the NSX version, see [KB 429205](#).

- You can choose between an HA deployment model with a cluster of 3x NSX Managers or a Simple-node deployment model with 1 NSX Manager.
- You can choose to configure overlay traffic on a management VMkernel or on a VLAN-backed segment.

If you are converging an NSX Federation configuration, only the Local Manager is converted. To upgrade such environment, see [Upgrading NSX in a Federated Environment](#).

| Category | Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Not Supported |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Storage  | <ul style="list-style-type: none"> <li>Enough free space for the full deployment of VMware Cloud Foundation, according to the <i>Planning and Preparation Workbook</i></li> <li>Datastore shared across all hosts in the cluster, accessible from all hosts in the cluster, and writable by all hosts in the cluster</li> <li>Minimum number of ESX hosts for new deployments of VCF Operations, VCF Automation, and NSX Manager <ul style="list-style-type: none"> <li>Simple deployments <ul style="list-style-type: none"> <li>Minimum 3 ESX hosts with vSAN Storage or minimum 2 ESX hosts with external storage</li> </ul> </li> <li>High availability deployments <ul style="list-style-type: none"> <li>minimum 4 ESX hosts</li> </ul> </li> </ul> </li> <li>Any supported vSphere storage type.<br/>If the default cluster includes multiple datastore types, VCF determines the primary datastore using the following priority: <ol style="list-style-type: none"> <li>vSAN</li> <li>NFS v3</li> <li>VMFS</li> <li>NFS 4.1</li> <li>iSCSI</li> <li>vVols*</li> </ol> <p><b>Note:</b></p> <p>If you converge a cluster that runs vCenter in an environment with multiple mounted datastores of the same type, the datastore on which the vCenter runs is designated as the principal storage for the cluster. Starting with VCF 9.0.1, the datastore with the largest amount of free space is designated as the principal storage for the cluster.</p> <p>If you are using a JSON specification file to converge, you can use the <code>existingDatastoreName</code> property to override the priority order and use a specific datastore.</p> </li> <li>vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 3 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones</li> <li>Two-node vSAN clusters for remote offices/branch offices (ROBO) with two ESX hosts and one witness host for VCF simple deployments</li> </ul> <p><b>Note:</b></p> <p>For two-node vSAN clusters, the admission control policy should be configured to <math>\geq 50\%</math> for both memory and CPU.</p> <ul style="list-style-type: none"> <li>vSAN OSA clusters, where deduplication and compression are either both activated or both deactivated</li> </ul> |               |

| Category | Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Not Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <ul style="list-style-type: none"> <li>vSAN OSA clusters with activated compression only</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Network  | <ul style="list-style-type: none"> <li>Virtual Distributed Switch (VDS) 8.0 or later</li> </ul> <p><b>Note:</b></p> <p>You can move to a vSphere Distributed Switch. See <a href="#">Migrating from Standard to Distributed vSwitch</a>.</p> <ul style="list-style-type: none"> <li>Ports aligned with <a href="#">VMware Ports and Protocols</a></li> <li>vCenter instance with temporary IP address</li> <li>Statically assigned VMkernel IP addresses</li> <li>Dynamically allocated NSX Host TEP interfaces</li> <li>Dedicated network for VMware vMotion</li> <li>vSphere Distributed Switches (VDS) with enabled link aggregation control protocol (LACP)</li> <li>ESX hosts with a combination of standard switches and VDS when the VMkernel interfaces are on the VDS</li> <li>ESX hosts with multiple physical uplinks of minimum of 10Gbps, that are assigned to a vSphere Distributed Switch</li> <li>ESX hosts with a single pNIC with a minimum of 10 Gbps speed, that is connected to a standard switch</li> <li>Clusters using shared vSphere Distributed Switches</li> <li>ESX hosts with a single Physical Network Adapter (pNIC)</li> <li>Starting with VCF 9.0.1:             <ul style="list-style-type: none"> <li>vCenter instances with existing NSX registrations without ELM</li> <li>vCenter instances without ELM and with existing NSX Local Managers that are part of an NSX Federation configuration</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Cisco virtual switches</li> <li>vCenter instances without a vSphere Distributed Switch</li> <li>Custom vCenter ports for client connections</li> <li>Environments using ports not aligned with <a href="#">VMware Ports and Protocols</a></li> <li>Dynamically allocated VMkernel IP addresses</li> </ul> <p><b>Note:</b></p> <p>You can move to statically assigned IP addresses. See <a href="#">Edit a VMkernel Adapter Configuration</a>.</p> <ul style="list-style-type: none"> <li>Shared networks for VMware vMotion</li> <li>Multiple NSX Manager instances managing a single vCenter instance</li> <li>NSX Bare Metal Edge nodes</li> </ul>                                                                                                                                                                      |
| Compute  | <ul style="list-style-type: none"> <li>Virtual machine of vCenter instance hosted on one of the clusters that is managed by the same instance</li> </ul> <p><b>Note:</b></p> <p>You can move the virtual machine to a supported location by using VMware vMotion. See <a href="#">Migrating Virtual Machines Between vCenter Systems</a>.</p> <ul style="list-style-type: none"> <li>Clusters using vSphere Configuration Profiles</li> <li>Clusters using vSphere Lifecycle Manager images</li> </ul> <p><b>Note:</b></p> <p>You can convert clusters that use baselines to use images before upgrading to ESX 9.0. See <a href="#">Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images</a>.</p> <ul style="list-style-type: none"> <li>Clusters using fully automated Dynamic Resource Scheduler (DRS)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Clusters managed by Dell VxRail</li> <li>vCenter instances with ELM</li> </ul> <p><b>Note:</b></p> <p>If your vCenter instance does not have existing NSX registrations, you can manually upgrade your vCenter instances to version 9.0.x and then break ELM for all vCenter instances. See <a href="#">Deactivate Enhanced Link Mode from vCenter Using the cmsso-util break-elm Utility</a>.</p> <ul style="list-style-type: none"> <li>vCenter instance VM hosted on a cluster that is managed by a different vCenter instance</li> <li>Clusters using baselines for lifecycle management</li> <li>Clusters using manual or partially automated Dynamic Resource Scheduler (DRS)</li> </ul> <p><b>Note:</b></p> <p>You can set the automation level to fully automated. See <a href="#">Edit Cluster Settings</a>.</p> |

| Category | Supported                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Not Supported                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <ul style="list-style-type: none"> <li>Standalone ESX hosts and single ESX host clusters, if your vCenter instance has an additional cluster that meets the cluster requirements</li> </ul> <p><b>Note:</b></p> <p>You can manually perform the lifecycle management of the standalone ESX hosts and single ESX host clusters by using the vSphere Client.</p> <ul style="list-style-type: none"> <li>Clusters using fully automated Dynamic Resource Scheduler (DRS)</li> </ul> | <ul style="list-style-type: none"> <li>Partial import of clusters             <ul style="list-style-type: none"> <li>All clusters managed by a vCenter instance are imported together</li> </ul> </li> <li>Clusters that use vCenter High Availability (VCHA)</li> </ul> |

\* vSphere Virtual Volumes (vVols) framework is deprecated in version 9 and using it is not recommended.

Some functionalities are not available for converged clusters:

- Adding hosts to imported clusters without the use of the vSphere Client
- Password Management of imported ESX hosts via the Password Management Console of VCF Operations

**Note:**

You can use the workaround in the Broadcom Knowledge Base Article [388859](#) to create the missing ESXi root credential entries in SDDC Manager to enable password management for these hosts.

## Supported Scenarios to Converge to VCF

You can converge the following non-VCF deployments to a VCF 9.0.x platform.

During the converge process, you use VCF Installer to specify deployment information for your environment such as your existing vSphere infrastructure, VMware Aria infrastructure, and other information in the deployment wizard. Any missing VMware Cloud Foundation software components are automatically deployed and configured using the information provided. Existing vSphere infrastructure is used to create the management domain of the VCF Instance.

You can download and complete the *Planning and Preparation Workbook* to help plan and gather all the information required for deployment.

- [Converging a vCenter Instance and ESX Hosts](#)
- [Converging a vCenter Instance, ESX Hosts, and VMware Aria Operations](#)
- [Converging a vCenter Instance, ESX Hosts, VMware Aria Suite Lifecycle, and VMware Aria Operations](#)
- [Converging a vCenter Instance, ESX Hosts, VMware Aria Lifecycle, and VMware Aria Automation](#)
- [Converging a vCenter Instance, ESX Hosts, VMware Aria Lifecycle, VMware Aria Operations, and VMware Aria Automation](#)

## Converging a vCenter Instance and ESX Hosts

To successfully converge your existing vCenter instance and ESX hosts to a management domain, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).

| Item                                                                   | VCF 9.0.0 Requirements                                                                       | VCF 9.0.1 and 9.0.2 Requirements                                                                               |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware vCenter 8.0 Update 3</li> <li>VMware ESX 8.0 Update 1</li> </ul> |

| Item | VCF 9.0.0 Requirements | VCF 9.0.1 and 9.0.2 Requirements                                   |
|------|------------------------|--------------------------------------------------------------------|
|      |                        | <ul style="list-style-type: none"> <li>VMware NSX 4.2.1</li> </ul> |

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

- Ensure your vCenter instance is running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
- Ensure your ESX hosts are running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.
- Deploy the VCF Installer appliance.
 

See [Deploy the VMware Cloud Foundation Installer Appliance](#).
- Download binaries to the VCF Installer appliance.
 

See [Downloading Binaries to the VCF Installer Appliance](#).
- Deploy VCF 9.0.x by using your existing components as starting building blocks.
 

See [Deploy Components by Using VCF Installer to Complete the Converging to VCF Process](#).
- Deploy the remaining components.
 

See [Completing the Deployment of Your Platform](#).

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your VCF environment. See [Licensing Overview](#).
- If you have VMware Aria components that are not subject to upgrade to version 9.0, import configurations from the old appliances to the new ones. See [Deploying the VCF Operations for Logs Appliance for VCF](#).
- Import your existing infrastructure as a workload domain. See [Import an Existing vCenter to Create a Workload Domain](#).
- Upgrade your workload domains. See [Upgrading Workload Domains to VMware Cloud Foundation 9.0](#).
- Integrate your workload domains in VCF Automation. See [Add Workload Domains to Your Cloud Organization Structure](#).

## Converging a vCenter Instance, ESX Hosts, and VMware Aria Operations

To successfully converge your existing vCenter instance, ESX hosts, and VMware Aria Operations, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

- Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).

- Verify that your vCenter instance is registered in VMware Aria Operations. See [Configure a vCenter Server Cloud Account in VMware Aria Operations](#).
- Verify that you activated SSH. See [Activate or deactivate SSH on VMware Aria Suite Lifecycle](#).

| Item                                                                   | VCF 9.0.0 Requirements                                                                                                            | VCF 9.0.1 and 9.0.2 Requirements                                                                                                                                              |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware Aria Operations 8.18</li> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware Aria Operations 8.18</li> <li>VMware vCenter 8.0 Update 3</li> <li>VMware ESX 8.0 Update 1</li> <li>VMware NSX 4.2.1</li> </ul> |

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

### Note:

You can have multiple VMware Aria Operations instances in your VCF environment. By following this procedure, you choose a main instance to create the VCF fleet and connect the instance to the VCF Operations fleet management appliance. This instance will have full VCF Operations functionality. For all subsequent VMware Aria Operations instances, you must perform the upgrade manually. These will not be registered with the VCF Operations fleet management appliance and will have limited functionality.

- Upgrade your VMware Aria Operations to VCF Operations version 9.0.x.
 

See [Upgrading VMware Aria Operations to VCF Operations 9 without VMware Aria Suite Lifecycle](#).
- Ensure your vCenter instance is running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
- Ensure your ESX hosts are running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.
- Deploy the VCF Installer appliance.
 

See [Deploy the VMware Cloud Foundation Installer Appliance](#).
- Download binaries to the VCF Installer appliance.
 

See [Downloading Binaries to the VCF Installer Appliance](#).
- Deploy VCF 9.0.x by using your existing components as starting building blocks.
 

See [Deploy Components by Using VCF Installer to Complete the Converging to VCF Process](#).
- Deploy the remaining components.
 

See [Completing the Deployment of Your Platform](#).

- Manually upgrade your subsequent Aria Operations instances to VCF Operations.

See [Upgrading VMware Aria Operations 8.18 to 9.0 without VMware Aria Suite Lifecycle](#).

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your VCF environment. See [Licensing Overview](#).
- Import your existing infrastructure as a workload domain. See [Import an Existing vCenter to Create a Workload Domain](#).
- Upgrade your workload domains. See [Upgrading Workload Domains to VMware Cloud Foundation 9.0](#).
- Integrate your workload domains in VCF Automation. See [Add Workload Domains to Your Cloud Organization Structure](#).

## Converging a vCenter Instance, ESX Hosts, VMware Aria Suite Lifecycle, and VMware Aria Operations

To successfully converge your existing vCenter instance, ESX hosts, VMware Aria Suite Lifecycle, and VMware Aria Operations, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

- Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).
- Verify that your vCenter instance is registered in VMware Aria Operations. See [Configure a vCenter Server Cloud Account in VMware Aria Operations](#).
- Verify that you activated SSH. See [Activate or deactivate SSH on VMware Aria Suite Lifecycle](#).

| Item                                                                   | VCF 9.0.0 Requirements                                                                                                                                                | VCF 9.0.1 and 9.0.2 Requirements                                                                                                                                                                                |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware Aria Lifecycle 8.18</li> <li>VMware Aria Operations 8.18</li> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware Aria Lifecycle 8.18</li> <li>VMware Aria Operations 8.18</li> <li>VMware vCenter 8.0 Update</li> <li>VMware ESX 8.0 Update 1</li> <li>VMware NSX 4.2.1</li> </ul> |

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

### Note:

VMware Aria Suite Lifecycle is not upgraded to version 9.0.x. After the VCF Operations fleet management appliance is deployed, the existing connection with VMware Aria Operations is automatically imported to the new appliance.

- Apply VMware Aria Suite Lifecycle 8.18 Patch 6 or later.

See [Patch VMware Aria Suite Lifecycle](#).

- Upgrade your VMware Aria Operations to VCF Operations version 9.0.x.

See [Upgrade to VCF Operations 9.0](#).

- Ensure your vCenter instance is running the supported version for your convergence scenario.

- If you are converging to VCF 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
- If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.0 or later. See [Upgrading the vCenter Appliance](#).
- If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.

- Ensure your ESX hosts are running the supported version for your convergence scenario.

- If you are converging to VCF 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
- If you are converging to VCF 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.

- Deploy the VCF Installer appliance.

See [Deploy the VMware Cloud Foundation Installer Appliance](#).

- Download binaries to the VCF Installer appliance.

See [Downloading Binaries to the VCF Installer Appliance](#).

- Deploy VCF 9.0.x by using your existing components as starting building blocks.

See [Deploy Components by Using VCF Installer to Complete the Converging to VCF Process](#).

- Deploy the remaining components.

See [Completing the Deployment of Your Platform](#).

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your VCF environment. See [Licensing Overview](#).
- If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
- Import your existing infrastructure as a workload domain. See [Import an Existing vCenter to Create a Workload Domain](#).
- Upgrade your workload domains. See [Upgrading Workload Domains to VMware Cloud Foundation 9.0](#).
- Integrate your workload domains in VCF Automation. See [Add Workload Domains to Your Cloud Organization Structure](#).
- If your environment has additional VMware Cloud Foundation Operations Collector appliances deployed in other data centers, you can add all appliances as deployment targets and import VCF Operations to VCF Operations fleet management. See [Add a Deployment Target and Import VCF Operations to VCF Operations fleet management](#).

## Converging a vCenter Instance, ESX Hosts, VMware Aria Lifecycle, and VMware Aria Automation

To successfully converge your existing vCenter instance, ESX hosts, VMware Aria Lifecycle, and VMware Aria Automation, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

- Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).
- Verify that you activated SSH. See [Activate or deactivate SSH on VMware Aria Suite Lifecycle](#).

| Item                                                                   | VCF 9.0.0 Requirements                                                                                                                      | VCF 9.0.1 and 9.0.2 Requirements                                                                                                                                                        |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware Aria Automation 8.18.1 Patch 3</li> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware Aria Automation 8.18.1 Patch 3</li> <li>VMware vCenter 8.0 Update 3</li> <li>VMware ESX 8.0 Update 1</li> <li>VMware NSX 4.2.1</li> </ul> |

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

**Note:**

VMware Aria Lifecycle is not upgraded to version 9.0.x. After the VCF Operations fleet management appliance is deployed, you must manually import the existing connection with VMware Aria Automation to the new appliance.

**Note:**

If your environment has multiple VMware Aria Automation instances, only the first instance can be managed by your VCF Operations fleet management appliance. Any subsequent VMware Aria Automation instances must be managed by other dedicated VCF Operations fleet management appliances.

- Ensure your vCenter instance is running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
- Ensure your ESX hosts are running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.
- Deploy the VCF Installer appliance.
 

See [Deploy the VMware Cloud Foundation Installer Appliance](#).
- Download binaries to the VCF Installer appliance.
 

See [Downloading Binaries to the VCF Installer Appliance](#).
- Deploy VCF 9.0.x by using your existing components as starting building blocks.
 

See [Deploy Components by Using VCF Installer to Complete the Converging to VCF Process](#).
- Manually deploy the remaining components.
 

See [Manual Deployment of Components to Complete Your VCF Platform](#).
- Import your VMware Aria Automation in VCF Operations and upgrade to VCF Automation version 9.0.x.
 

See [Import and Upgrade VMware Aria Automation in VCF Operations 9.0](#) and [Upgrade to VCF Automation](#).

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your VCF environment. See [Licensing Overview](#).

- If you have VMware Aria components that are not subject to upgrade to version 9.0, import configurations from the old appliances to the new ones. See [Deploying the VCF Operations for Logs Appliance for VCF](#) and [Import and Upgrade from 6.13 to VCF Operations for Networks](#).
- If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
- Import your existing infrastructure as a workload domain. See [Import an Existing vCenter to Create a Workload Domain](#).
- Upgrade your workload domains. See [Upgrading Workload Domains to VMware Cloud Foundation 9.0](#).
- Integrate your workload domains in VCF Automation. See [Add Workload Domains to Your Cloud Organization Structure](#).

## Converging a vCenter Instance, ESX Hosts, VMware Aria Lifecycle, VMware Aria Operations, and VMware Aria Automation

To successfully converge your existing vCenter instance, ESX hosts, VMware Aria Lifecycle, VMware Aria Operations, and VMware Aria Automation, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

- Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).
- Verify that your vCenter instance is registered in VMware Aria Operations. See [Configure a vCenter Server Cloud Account in VMware Aria Operations](#).
- Verify that you activated SSH. See [Activate or deactivate SSH on VMware Aria Suite Lifecycle](#).

| Item                                                                   | VCF 9.0.0 Requirements                                                                                                                                                                                               | VCF 9.0.1 and 9.0.2 Requirements                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware Aria Lifecycle 8.18</li> <li>VMware Aria Operations 8.18</li> <li>VMware Aria Automation 8.18.1 Patch 3</li> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware Aria Lifecycle 8.18</li> <li>VMware Aria Operations 8.18</li> <li>VMware Aria Automation 8.18.1 Patch 3</li> <li>VMware vCenter 8.0 Update 3</li> <li>VMware ESX 8.0 Update 1</li> <li>VMware NSX 4.2.1</li> </ul> |

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

**Note:**

VMware Aria Suite Lifecycle is not upgraded to version 9.0.x. After the VCF Operations fleet management appliance is deployed, the existing connections with VMware Aria Operations and VMware Aria Automation are automatically imported to the new appliance.

- Apply VMware Aria Suite Lifecycle 8.18 Patch 6 or later.
 

See [Patch VMware Aria Suite Lifecycle](#).
- Upgrade your VMware Aria Operations to VCF Operations version 9.0.x.
 

See [Upgrade to VCF Operations 9.0](#).
- Import your VMware Aria Automation in VCF Operations and upgrade to VCF Automation version 9.0.x.
 

See [Import and Upgrade VMware Aria Automation in VCF Operations 9.0](#) and [Upgrade to VCF Automation](#).



- Ensure your vCenter instance is running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
- Ensure your ESX hosts are running the supported version for your convergence scenario.
  - If you are converging to VCF 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
  - If you are converging to VCF 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.
- Deploy the VCF Installer appliance.
 

See [Deploy the VMware Cloud Foundation Installer Appliance](#).
- Download binaries to the VCF Installer appliance.
 

See [Downloading Binaries to the VCF Installer Appliance](#).
- Deploy VCF 9.0.x by using your existing components as starting building blocks.
 

See [Deploy Components by Using VCF Installer to Complete the Converging to VCF Process](#).
- Deploy the remaining components.
 

See [Completing the Deployment of Your Platform](#).

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your VCF environment. See [Licensing Overview](#).
- If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
- Import your existing infrastructure as a workload domain. See [Import an Existing vCenter to Create a Workload Domain](#).
- Upgrade your workload domains. See [Upgrading Workload Domains to VMware Cloud Foundation 9.0](#).
- Integrate your workload domains in VCF Automation. See [Add Workload Domains to Your Cloud Organization Structure](#).
- If your environment has additional VMware Cloud Foundation Operations Collector appliances deployed in other data centers, you can add all appliances as deployment targets and import VCF Operations to VCF Operations fleet management. See [Add a Deployment Target and Import VCF Operations to VCF Operations fleet management](#).

## Deploy Components by Using VCF Installer to Complete the Converging to VCF Process

By using VMware Cloud Foundation Installer, you use your existing vSphere and Aria infrastructure to converge it to a new VCF fleet or a new VCF Instance in an existing VCF fleet. During the process, VCF Installer deploys all required components that you do not have in your existing environment and uses your vCenter instance and ESX hosts to create the management domain for the new instance.

Verify that you meet the requirements and have completed the pre-requisite tasks based on your converge scenario. See [Supported Scenarios to Converge to VCF](#).

A VCF fleet refers to an environment consisting of one or more VCF Instances and standalone vCenter instances that are managed by the same VCF Operations instance. If you have a VCF fleet, you can converge your infrastructure to a new VCF Instance within the fleet by connecting to your existing VCF Operations instance or you can deploy a new VCF

Operations instance to create a new VCF fleet. If you have an existing VCF Operations instance, you can also use it to create a new VCF fleet.

You can use the VCF Installer deployment wizard or a custom JSON specification file. For more information, see [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#).

VCF Installer allows you to skip VCF Automation during the initial deployment and deploy it as a Day-N operation. See [Deploying VCF Automation as a Day-N Operation](#).

- In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.  
Enter the `admin@local` user and password you provided when you deployed the appliance and click **Log In**.
- Under **Deploy**, select **Deployment Wizard > VMware Cloud Foundation**.
- Select **Deploy a new VCF fleet** or **Deploy a VCF Instance in an existing VCF fleet**.
- On the **Existing Components** page, select your existing infrastructure components.  
  
When you connect to your existing components, VCF Installer validates them to ensure compatibility.  
  
When you select an existing VMware vCenter infrastructure, the **Storage**, **Hosts**, **Networks**, and **Distributed Switch** pages of the wizard are not available.
- On the **General information** page, enter the configuration details for your deployment.

| Option                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCF Instance Name                                                     | The VCF Instance Name is used to provide a unique name when creating the VMware Cloud Foundation Adapter within VCF Operations. A descriptive name provides for easy identification when you have multiple VCF Instances.                                                                                                                                                                                                                                                                                                                                                                  |
| Management domain name                                                | This name is used as prefix for the default names of objects created or deployed by the VCF Installer. You can change these default names as you progress through the deployment wizard.                                                                                                                                                                                                                                                                                                                                                                                                   |
| Custom networking configuration for VCF Operations and VCF Automation | By default, the VCF Installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVPG) as vCenter.<br>To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF Installer deploys the other components. See <a href="#">Deploying VCF Management Components with Custom Networking</a> .<br><b>Note:</b><br>This option is not available when deploying a VCF Instance in an existing VCF fleet or when using an existing VCF Operations instance. |
| Deployment model                                                      | You can choose the deployment model for any new appliances that get deployed. <b>High Availability (Three-node)</b> is recommended for production environments.<br><br><b>Note:</b> The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance.<br><br>See the VMware Cloud Foundation <a href="#">Design</a> documentation for more information.                                                                                                                                                                       |



| Option                                         | Description                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <b>Note:</b> New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).                                                                                                                                                                          |
| Password creation                              | You can auto-generate passwords for newly installed appliances if you don't want to manually enter passwords. After a successful deployment, you can view the generated passwords.                                                                                                                                  |
| Customer Experience Improvement Program (CEIP) | Select an option to activate or deactivate CEIP across all deployed components.<br>VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See <a href="#">Customer Experience Improvement Program</a> for more information. |

6. On the **VCF Operations** page, enter the required details.

| Scenario                                                    | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A new VCF fleet with a new VCF Operations instance          | Enter all the required information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| A new VCF fleet with an existing VCF Operations instance    | <ol style="list-style-type: none"> <li>Connect to your existing VCF Operations instance, provide the required details, and confirm the certificate thumbprint. <ul style="list-style-type: none"> <li>If a VCF Operations fleet management appliance is connected to the VCF Operations instance, enter the administrator password.</li> <li>If no VCF Operations fleet management appliance is connected to the VCF Operations instance, enter the required information to deploy a new one.</li> </ul> </li> <li>Enter details for the new VCF Operations collector appliance.</li> </ol> <p><b>Note:</b><br/>If you use a Load Balancer, verify that you use the Primary node IP.</p> |
| A new VCF Instance with an existing VCF Operations instance | <ol style="list-style-type: none"> <li>Connect to the VCF Operations instance that manages the VCF fleet in which you want to deploy this VCF Instance.</li> <li>Enter details for the new VCF Operations collector appliance.</li> </ol> <p><b>Note:</b><br/>Your existing VCF Operations instance must be connected to VCF Automation and VCF Operations fleet management.</p>                                                                                                                                                                                                                                                                                                         |

**Note:**

The **VCF Operations** page does not appear if you selected the **Advanced: Custom networking configuration for VCF Operations and VCF Automation** check box.

7. If you deploy a new VCF fleet without existing VCF Operations, on the **VCF Automation** page, enter the required details.

If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and no additional information is required.

Select the **I want to connect a VCF Automation instance later** checkbox if you want to use an existing Aria Automation 8.x instance that needs to be upgraded to VCF Automation 9.0. When deployment is complete, upgrade

Aria Automation to VCF Automation and then use VCF Operations to import the VCF Automation instance. See [Import an Existing VMware Aria Automation Instance Into VCF Operations and Upgrade to VCF Automation 9.0](#).

You can also select this check box to deploy VCF Automation as Day-N operation, instead of using the VCF Installer deployment workflows. See [Deploying VCF Automation as a Day-N Operation](#).

**Note:**

The **VCF Automation** page does not appear if you selected **Advanced: Custom networking configuration for VCF Operations and VCF Automation** check box on the **General information** page or if VCF Automation is not discovered within your VCF Operation instance.

When creating a new VCF Automation instance in a High Availability deployment, you must specify 4 node IP addresses. Three of the IP addresses are used for active nodes and the fourth IP address is used when recreating a node for upgrade purposes. The node IPs do not require DNS records.

8. On the **vCenter** page, enter the required details of your existing vCenter instance.

To set up the single sign-on with identity broker, select a vCenter instance where identity broker or Identity Manager is used as the identity provider.

9. On the **NSX Manager** page, enter the required details for NSX Manager appliance.

If you selected an existing VMware NSX Manager on the **Existing Components** page, the **NSX Manager** page is not available.

10. On the **SDDC Manager** page, enter the required details.

11. On the **Review** page, review the deployment information.

To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

12. On the **Validate & Deploy** page, review the validation information.

The VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warnings | Review any warning messages. To proceed with your deployment, address the warnings or click <b>Acknowledge</b> .                                                                                                                                                                                                                                                                                                            |
| Errors   | Use the Review all failed validations and resolve the issues by using the provided remediation information.<br>You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations.<br>Alternatively, you can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage. |

13. Click **Deploy**.

- If there are errors during the deployment progress, review the failed tasks in the **Tasks** panel, address the issues, and retry the deployment.
- You can download the JSON specification file for the deployment.

**Note:**

The JSON specification file does not include passwords.

- Click **Review Passwords** to review the passwords. You can also retrieve the passwords using the [SDDC Manager API](#) and [VCF Operations API](#).

## Converge a vCenter Instance and ESX Hosts to vSphere Foundation Platform

To successfully converge your existing vCenter instance and ESX hosts, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

Verify that your existing configuration is supported and you satisfy the minimum component versions. See *Supported Configurations* in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#).

| Item                                                                   | vSphere Foundation 9.0.0 Requirements                                                        | vSphere Foundation 9.0.1 and 9.0.2 Requirements                                                                |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Minimum component versions prior to running the VCF Installer workflow | <ul style="list-style-type: none"> <li>VMware vCenter 9.0</li> <li>VMware ESX 9.0</li> </ul> | <ul style="list-style-type: none"> <li>VMware vCenter 8.0 Update 3</li> <li>VMware ESX 8.0 Update 1</li> </ul> |

During the procedure, you use VCF Installer to specify deployment information for your existing vCenter instance and ESX hosts, and to deploy VCF Operations to complete the converging to VMware vSphere Foundation platform. You can also manually deploy VCF Operations, see [Deploy VCF Operations Nodes](#).

You can download and complete the *Planning and Preparation Workbook* to help plan and gather all the information required for a VMware vSphere Foundation deployment.

You can use the VCF Installer deployment wizard or a custom `JSON` specification file. For more information, see [Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation](#).

Instead of using the VCF Installer workflows, you can manually deploy VCF Operations, and add your vSphere Foundation license in VCF Operations.

### Important:

If you have existing VMware vCenter, VMware ESX, and VMware Aria Operations instance, you manually upgrade the components to version 9.0.x. See [Upgrading the vCenter Appliance](#), [Overview of the ESX Host Upgrade Process](#), and [Upgrade to VCF Operations 9.0](#). Then you use VCF Operations to license and monitor your VMware vSphere Foundation platform.

- Ensure your vCenter instance is running the supported version for your convergence scenario.
  - If you are converging to vSphere Foundation 9.0.0, your vCenter instance must be version 9.0 or later. See [Upgrading the vCenter Appliance](#).
  - If you are converging to vSphere Foundation 9.0.1 or 9.0.2 and your vCenter instance is version 8.0 Update 3 or later, proceed without upgrading.
- Ensure your ESX hosts are running the supported version for your convergence scenario.
  - If you are converging to vSphere Foundation 9.0.0, your ESX hosts must be version 9.0 or later. See [Overview of the ESX Host Upgrade Process](#).
  - If you are converging to vSphere Foundation 9.0.1 or 9.0.2 and your ESX hosts are version 8.0 Update 1 or later, proceed without upgrading.
- Deploy the VCF Installer appliance.
 

See [Deploy the VMware Cloud Foundation Installer Appliance](#).
- Download bundles for the VCF Installer appliance.
 

See [Downloading Binaries to the VCF Installer Appliance](#).

- Use VCF Installer to deploy your vSphere Foundation platform.

- In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.

Enter the `admin@local` user and password you provided when you deployed the appliance and click **Log In**.

- Under **Deploy**, select **Deployment Wizard > VMware vSphere Foundation**.
- On the **Existing Components** page, select **VMware vCenter**.

When you select an existing VMware vCenter infrastructure, the **Storage**, **Hosts**, **Networks**, and **Distributed Switch** pages of the wizard are not available.

- On the **General information** page, enter the configuration details for your deployment.

Additional information:

| Option                                         | Description                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Password creation                              | You can auto-generate passwords for newly installed appliances if you don't want to manually enter passwords. After a successful deployment, you can view the generated passwords.                                                                                                                                  |
| Customer Experience Improvement Program (CEIP) | Select an option to activate or deactivate CEIP across all deployed components.<br>VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See <a href="#">Customer Experience Improvement Program</a> for more information. |

- On the **VCF Operations** page, enter the required details.
- On the **vCenter** page, enter the required details of your existing vCenter instance.

VCF Installer validates your existing vCenter instance to ensure compatibility.

### Note:

An NSX Manager cannot be connected to the vCenter.

To set up the single sign-on with identity broker, select a vCenter instance where identity broker or Identity Manager is used as the identity provider.

- On the **Review** page, review the deployment information.

To make changes, navigate back to the relevant page in the deployment wizard. You can also download the `JSON` specification file, make the necessary changes, and then upload the modified `JSON` specification file from the VCF Installer homepage.

- On the **Validate & Deploy** page, review the validation information.

The VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

|          |                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------|
| Warnings | Review any warning messages. To proceed with your deployment, address the warnings or click <b>Acknowledge</b> . |
| Errors   | Use the Review all failed validations and resolve the issues by using the provided remediation information.      |

You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. Alternatively, you can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

i) Click **Deploy**.

- If there are errors during the deployment progress, review the failed tasks in the **Tasks** panel, address the issues, and retry the deployment.
- You can download the JSON specification file for the deployment.

**Note:**

The JSON specification file does not include passwords.

- You can click **Review Passwords** to review the passwords.
- Deploy the remaining components to complete your vSphere Foundation platform. See [Deploying the VCF Operations for logs Appliance for VMware vSphere Foundation](#) and [Deploying VCF Operations Orchestrator](#).
- License your vSphere Foundation environment. See [Licensing Overview](#).
- Register your vCenter instance in VCF Operations. See [Configuring a vCenter Account in VCF Operations](#).

## Upgrading Your VCF Management Domain to VMware Cloud Foundation 9.0

As a mandatory part of the upgrade to version 9, you must upgrade your VMware Cloud Foundation management domain to version 9.0. Before you upgrade, it is important to understand how VCF 9.0 is different from earlier versions. Upgrading your workload domains is not a mandatory part of the upgrade and can be performed later as a Day-N procedure.

VMware Cloud Foundation 9.0 includes more components that were optional in VCF 5.x.

Before version 9, VCF always included the following core components:

- SDDC Manager
- NSX Manager
- vCenter
- ESX

Optionally, you could deploy the management components, part of the VMware Aria suite.

Starting with VCF 9, the management components VCF Operations (Aria Operations), VCF Operations fleet management (Aria Suite Lifecycle), and VCF Automation (Aria Automation) are now part of the VCF fleet. A comprehensive list of components for VCF 9.0 is:

- Core components (upgraded using SDDC Manager):
  - SDDC Manager
  - NSX Manager
  - vCenter
  - ESX
- Management components (upgraded manually or deployed in version 9 prior to upgrading the core components):
  - VCF Operations
  - VCF Operations fleet management
  - VCF Operations collector (for new deployments deploy after the SDDC Manager upgrade)
  - VCF Automation (can be deployed as a Day-N operation after the upgrade)

**Note:**

The following upgrade procedures use the SDDC Manager UI to upgrade VCF. The SDDC Manager UI is being deprecated and will be removed in a future release. After your upgrade to VCF 9.0 is complete, use VCF Operations to perform lifecycle management activities.

For information on how to upgrade your workload domains, see [Upgrading VMware Cloud Foundation 5.x Workload Domains to 9.0](#).

### VMware Cloud Foundation 9.0 Management Domain Upgrade Overview

You can perform a sequential or skip-level upgrade to VCF 9.0 from VCF 5.0 or later. If your platform is at a version earlier than 5.0, you must upgrade the management domain and all workload domains to VCF 5.0 or later before you can upgrade to VCF 9.0.

The process depends on which, if any, VMware Aria management components you have deployed as part of VCF 5.x.

- [Preparing Your Management Components for VCF 9 Core Components Upgrade](#)
- [Upgrade the Management Domain Core Components to VMware Cloud Foundation 9.0](#)
- (Optional) [Completing the Deployment of Your Platform](#)

### Updating VCF to Maintenance Releases

When you update your VCF environment to a maintenance release in the same minor or major line, for example, from 9.0.0.0 to 9.0.1.0 or 9.0.2.0, update components per the order documented here: [Update Order for Components for Maintenance Releases](#).

## Preparing Your Management Components for VCF 9 Core Components Upgrade

Before you can proceed with the upgrade of core components, there are mandatory management components required on version 9 and optional upgrade tasks depending on your current infrastructure.

You must have the following components (either upgraded or deployed) on version 9:

- VCF Operations
- VCF Operations fleet management appliance
- VCF Operations collector (cloud proxy) appliance (for new deployments, deploy after the SDDC Manager upgrade)

Optional component upgrades:

- VMware Aria Automation / VCF Automation
- VMware Aria Automation Orchestrator / VCF Operations orchestrator
- VMware Aria Operations for Networks / VCF Operations for networks

Components that do not support upgrade to version 9 and must be re-deployed later:

- VMware Aria Operations for Logs / VCF Operations for logs

## Upgrading VMware Aria Operations to VCF Operations 9.0 without VMware Aria Suite Lifecycle

You manually update your existing Aria Operations deployments if they are not connected to VMware Aria Suite Lifecycle.

When you perform a software update, make sure you use the correct PAK file for your cluster. Take a snapshot of the cluster before you update, but you must remember to delete the snapshot after the update is complete.

**Note:** During the upgrade process, any user modifications made to the default content, including alert definitions, symptoms, recommendations, policy definitions, views, dashboards, widgets, and reports, will be overwritten. To preserve such customizations and facilitate easy restoration after the upgrade, you must clone, export or create a backup of your content.

Internal certificates are renewed when you upgrade a cluster. Automatic root-CA certificate renewal will be available when cloud proxy is upgraded to higher versions. After each product upgrade, the cluster will have a new root-CA certificate with a 5-year validity period.

**Note:** Automatic certificate renewal does not affect custom certificates.

Starting with VCF Operations 9.0, the VCF Operations Pre-upgrade Readiness Assessment Tool (Assessment Tool) is no longer available when upgrading VMware Aria Operations to VCF Operations 9.0 without VMware Aria Suite Lifecycle. The VCF Operations Pre-upgrade Readiness Assessment Tool (Assessment Tool) is now available with VMware Aria Suite Lifecycle. For more information, see [Upgrade VMware Aria Operations to VCF Operations 9.0](#).

## Obtain the Software Update PAK File

You upgrade your VCF Operations environment and cloud proxy using a PAK file.

### Download the PAK Files

You download the upgrade PAK file to upgrade your VCF Operations cluster. In case modifications are required, you can manually update the hosts file after completing the software update.

To download the PAK file for VCF Operations, go to [Broadcom Support Portal](#) and select the correct version from the drop-down list.

Download the *VMware Cloud Foundation Operations* PAK file from the Broadcom Support Portal, to update the VCF Operations environment and your cloud proxy together.

## Create a Snapshot Before You Upgrade VCF Operations

Create a snapshot of each node in a cluster before you update a VCF Operations cluster. After the update completes, you must delete the snapshot to avoid performance degradation.

For more information about snapshots, see the *vSphere Virtual Machine Administration* documentation.

1. Log in to the VCF Operations Administrator interface at `https://<primary-node-FQDN-or-IP-address>/admin`.
2. Click **Take Offline** under the cluster status.
3. When all nodes are offline, open the vSphere client.
4. Right-click a VCF Operations virtual machine.
5. Click **Snapshot** and then click **Take Snapshot**.
  - a) Name the snapshot. Use a meaningful name such as "Pre-Update."
  - b) Deselect the **Snapshot the Virtual Machine Memory** check box.
  - c) Deselect the **Ensure Quiesce Guest File System (Needs VMware Tools installed)** check box.
  - d) Click **OK**.
6. Repeat these steps for each node in the cluster.

Start the update process as described in [Updating your VCF Operations Software](#).

### Preserving Customized Content

When you upgrade VCF Operations, it is important that you upgrade the current versions of content types that allow you to alert and monitor the objects in your environment. With upgraded alert definitions, symptom definitions, and recommendations, you can alert on the various states of objects in your environment and identify a wider range of

problem types. With upgraded views, you can create dashboards and reports to easily identify and report on problems in your environment.

It is recommended that you do not edit any **out-of-the-box-content**, for example the alert definitions, symptom definitions, recommendations, dashboards, or views in VCF Operations. If you did not customize any of the **out-of-the-box-content** that were provided with previous versions of VCF Operations, you do not need to back them up and can start the upgrade process. After the upgrade, the **out-of-the-box-content** is updated to the latest version.

If you need to customize any of the existing **out-of-the-box-content**, that was provided with VCF Operations, clone the content and then edit the cloned content. To clone, customize, and preserve the **out-of-the-box-content**, perform the following steps.

1. Clone **out-of-the-box-content** and then customize the cloned content. Create a backup of the customized content before the upgrade. For more information, see [Creating a Backup](#) and [Image-Based Backup for VCF Operations](#).
2. Upgrade VCF Operations to the latest version. For more information, see [Updating your VCF Operations Software](#).
 

The customized versions of alert definitions, symptom definitions, recommendations, dasboards, and views are preserved.

After the upgrade, review the upgraded **out-of-the-box-content** in VCF Operations and compare them with the preserved customized content to determine whether to keep your previously modified versions, or to use the upgraded versions.

## Updating your VCF Operations Software

If you have already installed VCF Operations, you can update your software when a newer version becomes available.

- Create a snapshot of each node in your cluster. See [Create a Snapshot Before You Upgrade VCF Operations](#) for details.
- Obtain the PAK file for your cluster. See [Obtain the Software Update PAK File](#) for details.
- Before you install the PAK file, or upgrade your VCF Operations instance, clone any customized content to preserve it. Customized content can include alert definitions, symptom definitions, recommendations, and views. Then, during the software update, you select the option named **Install the PAK file even if it is already installed**. For more information, see [Preserving Customized Content](#)
- VCF Operations update operation has a validation process that identifies issues before you start to update your software. Although it is good practice to run the pre-update check and resolve any issues found, users who have environmental constraints can deactivate this validation check.
 

To deactivate the pre-update validation check, perform the following steps:

  - Edit the update file to `/storage/db/pakRepoLocal/bypass_prechecks_VMware AriaOperationsManagerEnterprise-buildnumberofupdate.json`.
  - Change the value to TRUE and run the update.

**Note:** If you deactivate the validation, you might encounter blocking failures during the update itself.

**Note:** Installation might take several minutes or even a couple hours depending on the size and type of your clusters and nodes.

1. Log in to the primary node VCF Operations administrator interface of your cluster at `https://primary-node-FQDN-or-IP-address/admin`.
2. Click **Software Update** in the left pane.
3. Click **Install a Software Update** in the main pane.
4. Follow the steps in the wizard to locate and install your PAK file.
 

This updates the OS on the virtual appliance and restarts each virtual machine.

5. Read the **End User License Agreement** and **Update Information**, and click **Next**.

6. Click **Install** to complete the installation of software update.

**Note:** After you click **Install**, the installer will restart the VCF Operations administrator interface, and you will be logged out. Log in once again to the VCF Operations administrator interface when it is ready, and follow the update status in the software update page.

7. Log back into the primary node administrator interface.

The main Cluster Status page appears and cluster goes online automatically. The status page also displays the Bring Online button, but do not click it.

8. Clear the browser caches and if the browser page does not refresh automatically, refresh the page.

The cluster status changes to Going Online. When the cluster status changes to Online, the upgrade is complete.

**Note:** If a cluster fails and the status changes to offline during the installation process of a PAK file update, then some nodes become unavailable. To fix this, you can access the administrator interface and manually take the cluster offline and click **Finish Installation** to continue the installation process.

9. Click **Software Update** to check that the update is done.

A message indicating that the update completed successfully appears in the main pane.

**Note:**

When you update VCF Operations to a latest version, all nodes get upgraded by default.

If you are using cloud proxies, the cloud proxy upgrades start after the VCF Operations upgrade is complete successfully. For more information, see the Monitoring the Health of Cloud Proxies from the Admin UI topic in the *VMware Aria Operations Configuration Guide*.

Delete the snapshots you made before the software update.

**Note:** Multiple snapshots can degrade performance, so delete your pre-update snapshots after the software update completes.

## Upgrading Management Components with VMware Aria Suite Lifecycle to Version 9

You can upgrade your version 8 VMware Aria components, deployed as part of a VCF 5.x or vSphere 8.x environments, to version 9. The upgrade procedure uses both the VMware Aria Suite Lifecycle 8.18 appliance and the VCF Operations fleet management appliance that is deployed as part of the VMware Aria Operations upgrade to VCF Operations version 9.0.

**Table 816: Version 8 Aria Component Upgrade Scenarios**

The following Aria component versions are supported for upgrade using the VMware Aria Suite Lifecycle 8.18 appliance.

An outline of the upgrade path is provided for each deployment scenario.

| Aria component deployed     | Supported versions for upgrade        | Upgrade path                                                                                                                                                                                                                              |
|-----------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VMware Aria Operations only | VMware Aria Operations 8.14 or later. | <ol style="list-style-type: none"> <li>Download and apply VMware Aria Suite Lifecycle 8.18 Patch 6 or later.</li> <li>Use VMware Aria Suite Lifecycle to upgrade to VCF Operations and install the fleet management appliance.</li> </ol> |
| VMware Aria Automation only | VMware Aria Automation 8.18 or later  | <ol style="list-style-type: none"> <li>Install VMware Aria Operations 8.18</li> <li>Download and apply VMware Aria Suite Lifecycle 8.18 Patch 6 or later.</li> </ol>                                                                      |

| Aria component deployed                                                                    | Supported versions for upgrade                                                                                                     | Upgrade path                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                            |                                                                                                                                    | <ol style="list-style-type: none"> <li>Use VMware Aria Suite Lifecycle to upgrade to VCF Operations and install the fleet management appliance.</li> <li>Use the fleet management appliance to upgrade to VCF Automation.</li> </ol>                                                                                                                |
| VMware Aria Operations for Networks with VMware Aria Operations and VMware Aria Automation | VMware Aria Operations 8.14 or later.<br>VMware Aria Automation 8.18 or later<br>VMware Aria Operations for Networks 6.13 or later | <ol style="list-style-type: none"> <li>Download and apply VMware Aria Suite Lifecycle 8.18 Patch 6 or later.</li> <li>Use VMware Aria Suite Lifecycle to upgrade to VCF Operations and install the fleet management appliance.</li> <li>Use the fleet management appliance to upgrade to VCF Automation and VCF Operations for networks.</li> </ol> |

### Important:

Because you will use the VCF Operations fleet management appliance to upgrade management components, you must have VMware Aria Operations installed in your environment so that you can upgrade it to VCF Operations. If VMware Aria Operations is not installed, follow these instructions to add it to your *VMware Aria Suite Lifecycle* environment: [Add a product to an existing private cloud environment](#).

After upgrading your existing VMware Aria components, you must deploy any missing management components before upgrading your management domain. See [Deploy VCF Operations for VMware Cloud Foundation 9 Core Components Upgrade](#).

VMware Aria Operations for Logs and Workspace One Access (formerly VMware Identity Manager) cannot be upgraded to version 9 and must be manually deployed after VCF Operations is at version 9. See [Completing the Deployment of Your Platform](#). VMware Aria Operations for Logs users who want to migrate data from version 8 to version 9 can perform the migration after installing VCF Operations for logs.

## Download Patch and Upgrade Binaries and Patch VMware Aria Suite Lifecycle

Before starting the upgrade process, download VMware Aria Suite Lifecycle 8.18 Patch 6 or later. You also need the binaries that VMware Aria Suite Lifecycle uses to upgrade VMware Aria Operations to VCF Operations version 9, and to deploy the VCF Operations fleet management appliance.

To upgrade to VCF Operations using VMware Aria Suite Lifecycle, you must apply the latest patch to your existing VMware Aria Suite Lifecycle 8.18 installation.

Perform the following steps to download the binaries that you need and apply the latest VMware Aria Suite Lifecycle 8.18 patch.

### Important:

Before starting the procedure, review the [VMware Aria Suite Lifecycle 8.18 Release Notes](#) for current information about the latest patch.

Download binaries.

1. Log in to the [Broadcom Support Portal](#).
2. Download VMware Aria Suite Lifecycle 8.18 Patch 6 or later.
  - a) On **My Downloads**, search for the `VMware Aria Suite` product name and click **Show Results**.
  - b) Select **VMware Aria Suite**.
  - c) Click the **Solutions** tab, select **VMware Aria Suite - Enterprise**, and choose the 2019 release.
  - d) Select the `vrslcm-8.18.0-PATCH` solution with the latest patch number and download the `.patch` file.
3. Download the binaries for the upgrade to VCF Operations.
  - a) On **My Downloads** search for the `VMware Cloud Foundation` product name and click **Show Results**.
  - b) Select **VMware Cloud Foundation**.
  - c) Click the **Product** tab and select **VMware Cloud Foundation 9**. Choose the 9.0.0.0 release.
  - d) On the **Primary Downloads** tab, next to VMware Cloud Foundation Operations, click **View Group**.
  - e) Download the `Operations-Upgrade-9.0.0.X.XXXXXXXX.pak` PAK file for the VCF Cloud Foundation Operations upgrade.
4. Download the binaries for the deployment of the VCF Operations fleet management appliance.
  - a) On **My Downloads** search for the `VMware Cloud Foundation` product name and click **Show Results**.
  - b) Select **VMware Cloud Foundation**.
  - c) Click the **Product** tab and select **VMware Cloud Foundation 9**. Choose the 9.0.0.0 release.
  - d) On the **Primary Downloads** tab, next to VMware Cloud Foundation Operations fleet management, click **View Group**.
  - e) Download the `VCF-OPS-Lifecycle-Manager-Appliance-9.0.0.X.XXXXXXXX.ova` OVA file for the VCF Cloud Foundation Operations fleet management deployment.
5. Use WinSCP or a similar file transfer utility to log in to your VMware Aria Suite Lifecycle virtual appliance.
6. Copy the downloaded binaries to the `/data` partition.
 

Apply the VMware Aria Suite Lifecycle 8.18 patch.
7. Log in to VMware Aria Suite Lifecycle.
8. From the **Lifecycle Operations** dashboard, navigate to **Settings > Binary Mapping** and click **Patch Binaries**. If any binaries from a previous installation are listed, delete them.
9. To map the binary, click **Add Binaries**.
  - In **Source Location**, enter the path to the patch binary or `/data` and click **Discover**.
  - When the patch binary appears, click **Add**.

The binary mapping request begins.
10. After binary mapping completes, navigate to **Settings > System Patches** and click **Create Snapshot**.
 

Creating a snapshot is mandatory before patching.
11. To install the patch, click **New Patch**.
  - a) Select the VMware Aria Suite Lifecycle 8.18 2 or later and click **Next**.
  - b) Review the installation summary and click **Install**.

The installation request moves through patch installation and services restart.

The VMware Aria Suite Lifecycle 8.18 patch has been applied. To verify that the latest patch has been applied:

- Click **Settings > System Details** and verify the Patch Version shows Patch 2 or later.
- Click **Settings > System Patches** and verify the Product Patch Level shows Patch 2 or later.

## Upgrade VMware Aria Operations to VCF Operations 9.0

The upgrade to VCF Operations 9.0 upgrades your VMware Aria Operations instance to VCF Operations 9, installs the VCF Operations fleet management appliance, and transfers the existing VMware Aria Operations and VMware Aria Automation integrations from the VMware Aria Suite Lifecycle appliance to the newly deployed fleet management appliance. You later use that VCF Operations instance to upgrade other management components.

- Verify that your environment includes VMware Aria Operations version 8.14 or later.
- Verify that you have downloaded the following binary files to the `/data` partition on your VMware Aria Suite Lifecycle appliance.
  - OVA file: `VCF-OPS-Lifecycle-Manager-Appliance-9.0.0.X.XXXXXXXX.ova`
  - PAK file: `Operations-Upgrade-9.0.0.X.XXXXXXXX.pak`
- Verify that you have an FQDN for the fleet management appliance. The FQDN must resolve both ways.
- Verify that you have a 15 character or longer password to use for both the `root` and `admin@local` users of the VCF Operations fleet management appliance.
- To deploy the VCF Operations fleet management appliance, verify that the following hardware requirements are satisfied:
  - 4 vCPU
  - 12 GB RAM
  - 200 GB Disk

Perform the following steps to upgrade to VCF Operations using the patched version of VMware Aria Suite Lifecycle.

1. Log in to VMware Aria Suite Lifecycle.
2. From the **Lifecycle Operations** dashboard, navigate to **Settings > Binary Mapping** and click **Product Binaries**.
3. To map the binaries, click **Add Binaries**.
  - In **Base Location**, enter the directory to the binaries or `/data` and click **Discover**.
  - Select the `VCF-OPS-Lifecycle-Manager-Appliance-9.0.0.X.XXXXXXXX.ova` and `Operations-Upgrade-9.0.0.X.XXXXXXXX.pak` files and click **Add**.

The binary mapping request begins and takes a few seconds to complete.

4. From the **Lifecycle Operations** dashboard, click **Environments**. On the **Aria Operations** environment, click **View Details** and select **Upgrade**.

The upgrade starts with an inventory sync process that ensures the VMware Aria Suite Lifecycle and the Aria Operations environments are in sync.

When the **Proceed to Upgrade** advisory appears, read the recommendation.

- The recommended action is to click **Trigger Inventory Sync** and click **Submit**. After the inventory sync completes, click **Upgrade** again and click **Proceed**.
- If your product inventory is already synced, click **Proceed**.

The upgrade proceeds through several stages.

5. The Product Version and Repository URL fields are populated for version 9.0. Click **Next**.
6. Choose the VCF license type then click **Next**.
  - VMware Cloud Foundation: Provides the license to install the VCF Operations fleet management appliance with the VMware Aria Operations upgrade to VCF Operations.
  - VMware vSphere Foundation: Provides the license for the VMware Aria Operations upgrade to VCF Operations.
7. Click **Run Assessment**.
 

This starts the upgrade assessment tool for VMware Aria Operations. After the assessment completes, click **Next**.



8. Click **Take product snapshot**. Retaining the product snapshot is optional.
9. Enter the Infrastructure properties needed to deploy the VCF Operations fleet management appliance.
  - Select vCenter server: Enter a value to place the appliance.
  - Select Cluster: Enter a value to place the appliance.
  - Folder: Enter a value to place the appliance.
  - Resource pool: Enter a value to place the appliance.
  - Select Network
  - Select Datastore
  - Select Disk Mode
  - VM Name: Enter a name for the new appliance that is being deployed.
  - FQDN: Enter the FQDN of the appliance.
  - IP Address: Enter the IP address of the appliance.
  - Admin Password: Enter the password that `admin@local` uses to log in to the VCF Operations fleet management native UI.
  - Root Password: Must be a 15-character password, or Precheck will fail
10. Enter the Network details.
  - Domain Name: Top-level domain, such as `broadcom.com`.
  - Domain Search Path: Top-level domain, such as `broadcom.com`.
  - DNS Servers: To select the DNS server that you want to configure VCF Operations with, click **Edit Server Selection**. If the DNS server that you want is not available, click **Add New Server** to add and select it.
  - NTP: To select the NTP server that you want to VCF Operations with, click **Edit Server Selection**. If the NTP server that you want is not available, click **Add New Server** to add and select it.
  - IPv4 Details:
    - IPv4 gateway: Default gateway of the network for VCF Operations to join.
    - IPv4 netmask: Netmask of the network for VCF Operations to join.
11. Click **Run Precheck**.
12. After reviewing the summary, click **Submit**.

The request moves through several task steps during which the Aria Operations instance and cloud proxies are upgraded to version 9.

**Note:**

The cloud proxy type remains the same as it was before the upgrade. For example, a cloud proxy type that was classic remains the same classic cloud proxy type in version 9. To access features of the unified cloud proxy that is available in version 9, deploy a unified cloud proxy. See [Configuring Cloud Proxies in VCF Operations](#).

Upgrading to VCF Operations removes the Aria Operations environment from VMware Aria Suite Lifecycle and installs the fleet management appliance node. VCF Operations is listed as a new deployment on the **Overview** tab under **Fleet Management > Lifecycle**.

## Import an Existing VMware Aria Automation Instance Into VCF Operations and Upgrade to VCF Automation 9.0

To upgrade to VCF Automation 9.0, you first import VMware Aria Automation into VCF Operations and then use VCF Operations to manage the upgrade to VCF Automation.

Before starting the import and upgrade:

- If your VMware Identity Manager certificate is due to expire within one year of date of import, replace the certificate. This allows VMware Aria Suite Lifecycle to retrust the VMware Identity Manager certificate within VMware Aria Automation before VMware Aria Automation is removed from VMware Aria Suite Lifecycle as part of the import process. See [Replace your Workspace ONE Access certificate by using VMware Aria Suite Lifecycle](#). NOTE: Workspace ONE Access is another name for VMware Identity Manager.
- Because the upgrade introduces architectural changes that could affect existing cloud account and integration configurations, verify that that your environment does not include any unsupported endpoints by reviewing the precheck requirements and remediation steps outlined in [KB 389563](#).
- Verify that the upgrade binaries for VCF Automation have been downloaded successfully. See [Downloading VCF Management Components Binaries](#).

### Import the Aria Automation 8.x Instance In VCF Operations 9.0

As part of the import process, you choose an Aria Automation instance to bring into VCF Operations. That is the instance that you will upgrade to VCF Automation 9.0.

On the **Overview** tab under **Fleet Management > Lifecycle**, click **Add** on the tile for VCF Automation. The import moves through the following stages.

1. Deployment Settings: Select **Import from legacy Fleet Management**.
2. VMware Aria Suite Lifecycle Configuration: Enter values for your existing legacy environment.
  - FQDN: The VMware Aria Suite Lifecycle FQDN that is managing the Aria Automation instance that you would like to upgrade.
  - Username: `admin@local`
  - Admin Password: Password for `admin@local`
  - Root Password: Root password for VMware Aria Suite Lifecycle.
3. Select Aria Automation Instance:
 

Select the Aria Automation instance to be deployed. Only the first instance can be managed by this VCF Operations fleet management appliance. Any subsequent Aria Automation instances must be imported into other VCF Operations fleets or other VCF deployments.

If multiple Aria Automation deployments exist, select the most critical deployment and click **Next**. For any additional Aria Automation deployments, repeat the import and upgrade process with a separate VCF Operations deployment.
4. Review
  - From VMware Aria Suite Lifecycle : FQDN of the Aria Suite Lifecycle legacy environment that the component is being imported from.
  - To VCF Operations: FQDN of the VCF Operations fleet that the component is being imported into.
  - Component VMware Aria Automation: FQDN of the Aria Automation instance that is to be imported and integrated.

After reviewing the details of the components and environments that are being integrated, check the box to consent to the integration and trigger the import of VMware Aria Automation into VCF Operations.

As the import process progresses, control of VMware Aria Automation is transferred from VMware Aria Suite Lifecycle to VCF Operations, and after the import completes, the Automation tile no longer appears in VMware Aria Suite Lifecycle .

### Upgrade VMware Aria Automation to VCF Automation 9.0

After importing and integrating the VMware Aria Automation instance into VCF Operations, proceed to upgrade the instance.

**Important:**

If after completing the import process, you realize that your upgrade cannot continue due to specific endpoints, delete Automation from VCF Operations and import VMware Aria Automation back into VMware Aria Suite Lifecycle .



To upgrade to VCF Automation 9.0, your VCF Operations account must have the **VMware Aria Suite Lifecycle** role assigned in vSphere with the following permissions:

| Privilege                  | Value                                                     |                             |
|----------------------------|-----------------------------------------------------------|-----------------------------|
| CNS                        | Searchable                                                |                             |
| Datastore                  | All privileges                                            |                             |
| Global                     | Manage Custom Attribute                                   |                             |
|                            | Set Custom Attribute                                      |                             |
|                            | Settings                                                  |                             |
| Host                       | Local Operations                                          | Add Host to vCenter         |
|                            |                                                           | Create Virtual Machine      |
|                            |                                                           | Delete Virtual Machine      |
|                            |                                                           | Reconfigure Virtual Machine |
|                            | Inventory                                                 | Modify cluster              |
|                            | Configuration                                             | Change settings             |
| Network                    | Assign network                                            |                             |
| Permissions                | Modify permission                                         |                             |
|                            | Modify role                                               |                             |
| Resource                   | Assign vApp to Resource Pool                              |                             |
|                            | Assign Virtual Machine to Resource Poo                    |                             |
| VM Storage Policies        | View VM storage policies                                  |                             |
| vApp                       | All privileges                                            |                             |
| Virtual Machines           | All privileges                                            |                             |
| Content Library            | All privileges                                            |                             |
| Service Account Management | Administer (Create / Delete accounts and Reset passwords) |                             |
|                            | Create / Delete service accounts                          |                             |
|                            | Reset password / Get status of password (valid / invalid) |                             |

Upgrading to VCF Automation starts with creating an upgrade plan. This ensures that the correct binary is mapped to the upgrade. After creating the plan, you start the upgrade process.

**Note:**

After you upgrade VMware Aria Automation 8.x to VCF Automation 9.0, the VCF Operations fleet management appliance starts managing VCF Automation 9.0, but VMware Aria Suite Lifecycle 8.x continues to manage VMware Identity Manager. Do not delete VMware Identity Manager as it is still an authentication source for VCF Automation 9.0. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x.

1. On the **Components** tab under **Fleet Management > Lifecycle**, **automation** appears as an integrated component. Click **Plan Upgrade**.

2. In the **Plan Upgrade** window that appears:

- a) Use the drop down menu to select the Target Version, such as 9.0.0.0 for the automation component.

**Note:**

If the drop down menu is empty, the mapped binaries are incorrect. For example, install binaries might have been mapped instead of upgrade binaries. To display the correct target versions, you must download the correct binary type. See [Downloading VCF management components](#).

- b) Click **Create Plan**.

On the **Components** tab, the action for the automation component changes from Manage to Upgrade.

3. Click the **Upgrade** action button for automation.

4. On the **Proceed to Upgrade** advisory that appears:

- a) Click the checkbox to acknowledge the warnings about upgrade failures or broken integrations.  
 b) Click **Trigger Inventory Sync**.  
 c) After inventory sync complete successfully, click **Proceed**.

The upgrade process begins. Most fields are automatically completed with values discovered and imported from the integrated Aria Automation instance, so you can accept default values at each stage or input new values, if needed.

5. Infrastructure includes the following prepopulated settings for your review.

- Deployment Target: vCenter where VMware Aria Automation is deployed.
- Select Cluster
- Select Folder (optional)
- Select Resource Pool (optional)
- Select Network
- Select Datastore

6. Network includes the following pre-populated settings for your review.

- Domain Name
- Domain Search Path
- DNS Servers: If no DNS servers are listed, click to add a new server.
- Time Sync Mode
- NTP Servers: If no NTP servers are listed, click to add a new server.

7. Components include the following settings:

- FQDN: Use the prepopulated VCF Automation FQDN.
  - Certificate: Use the prepopulated certificate.
  - Password: Select the password to be used for the VCF Automation vmware-system-user and admin accounts. If the desired password is not a selection, click **ADD PASSWORD** at the upper right and enter a 15 character password.
- Cluster FQDN (for clustered deployments): Use the pre-populated VCF Automation FQDN.
- Controller type:
- Node Prefix: Enter a user-defined prefix for naming newly deployed VCF Automation virtual machines. To avoid conflicts and maintain proper node identification within the cluster, ensure that the node prefix is unique.
- Primary VIP: The IP address of the first node in the VMware Aria Automation instance.
- Internal Cluster CIDR: Choose a CIDR block that does not conflict with your existing infrastructure.
- Additional VIPs: The IP addresses of the second and third nodes in the VMware Aria Automation instance.
- Cluster Node IP Pool: Enter the IP addresses to use for deploying the new virtual machines that will host the VCF Automation components. The number of IP addresses depends on the size of your deployment:
  - For a small deployment, enter two IP addresses.
  - For a medium or large deployment, enter four IP addresses.

**Tip:**

When entering values for the Cluster Node IP Pool, note how an Aria Automation 8.x deployment will be transitioned into VCF Automation 9.0.

| Automation deployment  | 8.x cluster size | 9.0 cluster size |
|------------------------|------------------|------------------|
| Small, single node     | Medium           | Small (1-node)   |
|                        | Extra large      | Large (3-nodes)  |
| Clustered, three nodes | Medium           | Medium (3-node)  |
|                        | Extra large      | Large (3-nodes)  |

VMware Aria Suite Lifecycle 8.

8. Click **RUN PRECHECK**.

**Note:**

If the data validation is not successful, perform the recommended actions and run the precheck again.

9. After a successful precheck run, review the upgrade summary and click **Submit**.

An upgrade takes approximately one hour to complete. If the upgrade does not succeed, perform the following steps to remove the new VCF Automation instance:

1. Power on the existing Aria Automation 8.x.
2. To start the services on all nodes, log in to any appliance using SSH and run the following command.  

```
/opt/scripts/deploy.sh
```
3. Use VMware Aria Suite Lifecycle 8.18 Patch 6 or later to import Aria Automation 8.x and re-register with VMware Identity Manager.
4. [Import and Upgrade VMware Aria Automation in VCF Operations 9.0.](#)

## Import an Existing VMware Aria Operations for Networks Instance Into VCF Operations and Upgrade to VCF Operations for Networks

Use VCF Operations to import and upgrade VMware Aria Operations for Networks 6.13.

**Note:**

You can only import and upgrade one instance of VMware Aria Operations for Networks to VCF Operations for networks.

### Add upgrade binaries in VCF Operations

Before upgrading to VCF Operations for networks, you log in to your VCF Operations instance to ensure that your offline depot is active with a path that points to the location where the VCF Operations for networks binaries are stored. See [Lifecycle Management of VCF Management Components](#).

In VCF Operations, on the Binary Management tab, verify that the VMware Cloud Foundation Operations for networks version 9.0.0.0 is listed and click to download the binary. See [Downloading VCF Management Components Binaries](#).

### Import VMware Aria Operations for Networks

On the **Overview** tab under **Fleet Management > Lifecycle**, click **Add** on the tile for VCF Operations for networks. The import moves through the following stages.

**Note:**

You can only deploy or import one VMware Aria Operations for Networks instance into the VCF Operations fleet management appliance

1. Deployment Settings: Select **Import from legacy Fleet Management**
2. Lifecycle Configuration
  - VMware Aria Suite Lifecycle FQDN
  - Username
  - Admin Password
  - Root Password
3. Select Aria Operations for Networks Platform instance  
 Select the instance that you want to deploy. You can only deploy or import one VMware Aria Operations for Networks instance into the VCF Operations fleet management appliance.
4. Review
  - From VMware Aria Suite Lifecycle: FQDN of the Aria Suite Lifecycle legacy environment that the component is being imported from.
  - Component VMware Aria Operations for Networks platform: FQDN of the VMware Aria Operations for Networks instance that is to be imported and integrated.

After reviewing the details of the components and environments that are being integrated, check the box to acknowledge that this integration is irreversible and click **Submit**.

After the import task completes, the **Overview** tab reappears with a tile that displays a new deployment of VCF Operations for networks.

- On the tile, click **Manage**.
- On the Components page that appears, click **Plan Upgrade**.

The **Overview** tab reappears with a tile that displays a pending upgrade of VMware Aria Operations for Networks Version 6.13.0 to VCF 9.0.0.0.

## Upgrade to VCF Operations for Networks

The upgrade process automatically completes fields with values discovered and imported from VMware Aria Operations for Networks, so you can accept default values to perform the upgrade.

From the **Overview** tab, click **Upgrade** on the VCF Operations for networks tile.

To ensure that VCF Operations fleet management appliance and the imported Aria Operations for Networks environment are in sync, the upgrade triggers an inventory sync.

- Click **Trigger Inventory Sync**, then click **Submit**.
- When all stages of the inventory sync are complete, it is safe to start the upgrade.

Return to the Overview tab, click **Upgrade**, then click **Proceed**. The upgrade moves through the following stages.

- Product Version and Repository URL are populated by default.
- Snapshot: Take product snapshot (default)
- Precheck: To validate various properties of VCF Operations for networks, click **Run Precheck**.
- Upgrade Summary: Shows the details of the upgrade. To start the upgrade, click **Submit**.

The **Tasks** tab shows the status of the upgrade.

## Upgrading a VCF Operations Orchestrator Deployment

You can upgrade your VMware Aria Automation Orchestrator deployment to VCF Operations orchestrator by using a mounted ISO image.

- Verify that you are upgrading a VMware Aria Automation Orchestrator 8.18.1 deployment to VCF Operations orchestrator 9. Upgrades from earlier product versions are not supported.
- If you are upgrading from VMware Aria Automation Orchestrator 8.18.1 U3, U4, or U5, see Broadcom KB Article [429226](#).

- Download and mount the ISO image.
  - Download the ISO image from the official Broadcom download site.
  - Connect the CD-ROM drive of the Automation Orchestrator Appliance virtual machine in vSphere. See the *vSphere Virtual Machine Administration* documentation.

### Note:

After connecting the CD-ROM drive, navigate to your Automation Appliance VM settings page and verify that **Connect At Power On** is enabled.

- Mount the ISO image to the CD-ROM drive of the Automation Orchestrator Appliance virtual machine in vSphere. See the *vSphere Virtual Machine Administration* documentation.
- Log in to the Automation Orchestrator Appliance command line as **root**.
  - Run the `blkid` command, and note the device name for the Automation Orchestrator Appliance CD-ROM drive.
  - Mount the CD-ROM drive.

```
mount /dev/xxx /mnt/cdrom
```

### Important:

For clustered Automation Orchestrator deployments, you must perform steps 2 and 3 on all nodes in the cluster.

- Back up your Automation Orchestrator deployment by taking a virtual machine snapshot. See [Take a Snapshot of a Virtual Machine](#).

### CAUTION:

Automation Orchestrator 8.x does not currently support memory snapshots. Before taking the snapshot of your deployment, verify that the **Snapshot the virtual machine's memory** option is deactivated.

- To initiate the upgrade, run the `vracli upgrade exec -y --repo cdrom://` command on one of the nodes in your deployment.

### Note:

For Automation Orchestrator deployments authenticated with vSphere, you must provide the credentials of the user who registered your deployment with the vCenter Single Sign-On (SSO) service. You can verify the identity of this user either in the `/data/vco/usr/lib/vco/app-server/conf/sso.properties` path in the virtual appliance, or in the output of the `vracli vro authentication` command. You can either enter the password manually or export the your password as a environmental variable. This can be useful for scenarios where you are using an automated script to upgrade multiple Automation Orchestrator deployments. To export the SSO password, run the `export VRO_SSO_PASSWORD=your_sso_password` command.

During the upgrade, you are automatically logged out of your terminal, because the Automation Orchestrator Appliance reboots.

- Log in to the Automation Orchestrator Appliance command line as **root** and follow the upgrade progress by running the `vracli upgrade status --follow` command.

**CAUTION:** The `vracli upgrade status --follow` command can occasionally display a false error message that indicates that the upgrade has failed. To troubleshoot this problem, see step 8.

- Optional: If you receive an error message while running the `vracli upgrade status --follow` command, follow these steps:

- Verify that you receive the following error message:

```
Running health check after upgrade for nodes and pods.
Health check after upgrade for nodes and pods failed.
... Upgrade terminated due to critical error. Follow the upgrade guide to recover
the system. ...
```

- Navigate to the `/var/log/vmware/prelude/upgrade-report-latest` and confirm that you receive the following error:

```
Pod: vco-app-xxxx is not in Ready or Completed state. All pods must be in either of these states
```

- Run the `kubect1 get pods -n prelude -w | grep -E 'vco|orchestration-ui'` command and verify that the status of all 3 `vco-app` pods and the `orchestration-up-app` pod is `RUNNING`.

**Note:** It can take up to 5-10 minutes after receiving the error message for all pods to enter the `RUNNING` state.

```
orchestration-ui-app-xxxx 1/1 Running 0 5h42m
vco-app-xxxx 3/3 Running 0 5h47m
```

- Run the `curl -k https://<your_orchestrator_FQDN>/vco/api/healthstatus` command and verify that the health check is returning a `RUNNING` status.

```
{"state":"RUNNING","health-status":{"state":"OK","time":1615296823325},"instance-id":"your_orchestra-tor_FQDN"}
```

### Important:

The preceding command must run in an environment different from the Automation Orchestrator command line. You can run the command from the command line of a different virtual machine. You can also view the health status information in a browser by navigating directly to [https://<your\\_orchestrator\\_FQDN>/vco/api/healthstatus](https://<your_orchestrator_FQDN>/vco/api/healthstatus).

You have upgraded your Automation Orchestrator 8.18.1 deployment to VCF Operations orchestrator 9. If you encounter a notification stating that the upgrade has failed, see [False Upgrade Failure Notification](#).

Validate that the Automation Orchestrator upgrade was successful by running the `vracli version` command in the command line of the appliance. By running this command, you can validate the product version and build number of VCF Operations orchestrator.

### Upgrade Failure Notification During VMware Aria Automation Orchestrator Upgrade

The upgrade log indicates that the upgrade process has failed, but the individual nodes of the deployment are upgraded.

After the upgrade script finishes running, you receive the following message in your Automation Orchestrator Appliance indicating that the upgrade has failed:

```
Upgrade failed and left the system in non-working state. Check the error report below
to correct the problem. Once addressed, you can continue the upgrade by running 'vracli
upgrade exec --resume'
```

However, the upgrade log lists that the nodes of your Automation Orchestrator deployment are upgraded.

```
Hostname: <your_orchestrator_node_FQDN>
Status: Upgraded
Cluster Member: Yes
Version Before: <build_before_upgrade>
Version After: <build_after_upgrade>
Description: The node is upgraded successfully.
```

To resolve this problem, verify that the nodes are running, and resume the upgrade.

1. Verify that your Automation Orchestrator nodes are running.

```
kubectl get all pods
```

2. If your Automation Orchestrator nodes are running, resume the upgrade process.

```
vracli upgrade exec --resume
```

### Deploy VCF Operations for VMware Cloud Foundation 9 Core Components Upgrade

If you are planning to upgrade a VMware Cloud Foundation 5.x instance that has no VMware Aria Suite components installed, you must prepare for the upgrade by deploying the VCF Operations nodes and the VCF Operations fleet management appliance. You do not deploy the VCF Operations Collector (cloud proxy) before your SDDC Manager is upgraded to version 9.

#### Note:

If your VCF instance has some VMware Aria Suite components installed, first upgrade those components. See [Upgrading Management Components with VMware Aria Suite Lifecycle](#).

You can deploy VCF Automation after upgrading the management domain.

### Deploy VCF Operations Nodes

Use the vSphere Client to deploy the VCF Operations primary, replica, and data nodes.

Perform the following steps to deploy VCF Operations.

1. Download the VCF Operations 9.0 Install OVA files from the [Broadcom Support Portal](#).
  - a) Log in to the [Broadcom Support Portal](#).
  - b) Browse to My Downloads.
  - c) Search for VMware Cloud Foundation and select **VMware Cloud Foundation 9.0**.
  - d) Accept the terms and conditions.
  - e) Select **VCF Operations** and download the VCF Operations Install OVA files.
2. Log in to vCenter using the vSphere Client.
3. Start the **Deploy OVF Template** wizard.

On the “Select an OVF template”, specify the local location of the source VCF Operations Install OVA template and follow the prompts until you are asked to enter a name for the node.

Deploy nodes for VCF Operations.

4. Provide input to deploy each node.

For Node Name, do not include nonstandard characters such as underscores (\_) in node names. If creating a multi-node cluster, use a unique name for each node.

- a) Select the size configuration. Your selection does not affect the disk size.
- b) Select the disk format:

- Thin Provisioned
- Thick Provisioned Lazy Zeroed
- Thick Provisioned Eager Zeroed

- c) From the drop-down menu, select a Destination Network.
- d) Specify networking properties.

For static IPv4 or static IPv6, specify the associated Domain Name, Domain Search Path, and Domain Name Servers values.

- e) Specify a time zone.

You must configure all nodes with the same time zone.

- f) Optional: (optional) Specify a FIPS setting and click **Next**.

To deploy a FIPS activated VCF Operations setup, check **Activate FIPS Mode**

- g) Review the settings and click **Finish**.

After deployment, note the node FQDN or IP address.

To create a multi-node cluster, repeat the steps to deploy each node.

5. Power on the VCF Operations appliance and wait until it bootstraps completely.

Create and add a primary node in VCF Operations.

6. Navigate to the name or IP address of the primary node.

- a) Click **New Installation > Next**.
- b) Enter and confirm a password for the admin user and click **Next**.
  - The *admin* user account name cannot be changed.
  - Passwords require a minimum of fifteen characters, one uppercase letter, one lowercase letter, one digit, and one special character.
- c) Use the certificate included with VCF Operations or to install one of your own.
  - To use the included certificate, click **Use the default certificates** and review the certificate information. If it meets the requirements for VCF Operations, click **Next**.
  - To install your own certificate, click **Install a certificate**.
    - a. Click **Browse** to locate the certificate file.
    - b. Click **Open** to load the file in the Certificate text box.
    - c. Click **Next**
- d) Enter a name for the primary node such as `ops-primary`.
- e) For NTP server cluster synchronization:
  - To have VCF Operations manage its own synchronization, leave the shared IP address blank. All nodes will synchronize with the primary and replica node.
  - Or enter a URL or IP address such as `nist.time.gov`, and click **Add**.
- f) Configure VCF Operations availability.

To install VCF Operations with availability, activate the Availability Mode and select **High Availability or Continuous Availability**. To continue your installation on full capacity, click **Next**.

g) To add a node, click the plus sign, specify the following information:

1. Enter the Node Name and Node Address.
2. Select the Current Cluster Role.
3. Click **Next**

h) Click **Finish**.

The administration interface appears. It takes a moment for VCF Operations to finish adding the primary node.

After creating the primary node, you can create and add data nodes before starting the cluster or you can start a single-node cluster.

7. To start the cluster, click **Start VCF Operations**.

#### Note:

Depending on the size of the cluster and nodes, deployment might take 10 to 30 minutes to start. Do not make any changes or perform any actions on cluster nodes while the cluster is starting.

VCF Operations 9.0 is deployed in evaluation mode. Before adding other components, assign the VCF entitlement type. See [Registering VCF Operations with VCF Business Services Console](#).

### Deploy the VCF Operations fleet management Appliance

Use vCenter to deploy the VCF Operations fleet management appliance.

Perform the following steps to deploy the VCF Operations fleet management appliance.

1. Download the VCF Operations fleet management OVA from the [Broadcom Support Portal](#).

- a) Log in to the [Broadcom Support Portal](#).
- b) Browse to My Downloads.
- c) Search for `VMware Cloud Foundation` and select **VMware Cloud Foundation 9.0**.
- d) Accept the terms and conditions.
- e) Select **VCF Operations** and download the VCF Operations fleet management appliance OVA.

2. Log in to vCenter using the vSphere Client.

3. Start the **Deploy OVF Template** wizard.

On the "Select an OVF template", specify the local location of the source VCF Operations fleet management OVA template.

Configure the fleet management appliance.

4. Follow prompts until you are asked to enter a name for the node.

a) For Compute Resource

Provide the storage or datastore where the appliance will be deployed.

b) Select and configure a network for the appliance.

- Hostname. Do not include nonstandard characters such as underscores (`_`) in node names. Always use the FQDN.
- Initial Root Password. The password must be 15 characters or more.
- Initial Admin Password. The password must be 15 characters or more.
- Select **Enable SSH service on the appliance**.
- Select **Enable firstboot for vrlcm**.
- Select **Join CEIP**.
- Host Network Default Gateway. Specify the gateway to which the appliance will connect.
- Domain Name. Specify the Domain Name of the appliance. This is not the hostname. For example, if the hostname is `ops1cm.ops.com`, `ops.com` is the domain name.
- Domain Search Path. Typically the same as the domain name.
- Network 1 IP address. IP address of the virtual appliance. Should resolve to the FQDN of the appliance.
- Network 1 IP NetMask. Netmask of the virtual appliance.

c) Review the Settings and click **Finish**.

5. Power on the fleet management appliance and wait until it bootstraps completely.

### Register the VCF Operations fleet management Appliance with VCF Operations

Use the VCF Operations admin UI to register the fleet management appliance.

Perform the following steps to register the fleet management appliance. The process takes about five minutes to complete.

1. Launch the VCF Operations admin UI, for example: `https://vcfops-master-node-fqdn/admin/`
2. Log in as admin user.
3. On **System Status > Fleet Management**, click **Connect**.
4. To connect the fleet management node, provide:
  - Node address: Enter the FQDN of the VCF Operations fleet management appliance.
  - Admin Password: Enter the VCF Operations fleet management admin password.

- Click **Test Connection**.
- Review and accept the certificate. Click **Next**.
- Enter the VCF Operations admin password.
- To start the registration process, click **Finish**.

### Add a VCF Endpoint into VCF Operations

To integrate VMware Cloud Foundation with VCF Operations, add a VCF endpoint into VCF Operations.

Perform the following steps to add a VCF endpoint and enable collection for that endpoint.

- Log in to VCF Operations.
- On **Administration > Integrations**, click **Add**.
- Under Account Types, select **VMware Cloud Foundation** and enter the following information:
  - Cloud Account Information
    - Name: Alias of SDDC-Manager
    - Description
    - Physical Datacenter (Optional)
  - VCF Credentials
    - SDDC Manager: FQDN of the SDDC Manager
    - Credential: If not present, click to add one.
    - Collector/Group: Use the Default Collector Group, or select a cloud proxy.
 Click **Validate Connection**.

Click **Add** to save the adapter instance

- On **Administration > Integrations**, click to open the VMware Cloud Foundation account. The newly added endpoint appears. Click the ellipses for the endpoint and select **Enable Collection**.

You configure and deploy the remaining VCF Operations components after the SDDC Manager upgrade. You can now proceed to [Upgrade the Management Domain Core Components to VMware Cloud Foundation 9.0](#).

## Upgrade the Management Domain Core Components to VMware Cloud Foundation 9.0

To upgrade to VMware Cloud Foundation 9.0, the management domain must be at VMware Cloud Foundation 5.0 or higher. If your platform is at a version lower than 5.0, you must upgrade the management domain and all workload domains to 5.0 or later before you can upgrade to 9.0.

You can upgrade workload domains before or after upgrading the management domain, as long as all components in the workload domain are compatible.

Upgrade the core components in the management domain in the following order:

- SDDC Manager and VMware Cloud Foundation services.
- NSX Manager and NSX Global Manager instances (if applicable).
- vCenter instance.
- ESX hosts.

After all upgrades have completed successfully:

- Remove the VM snapshots you took before starting the update.
- Back up the newly installed components.

## VMware Cloud Foundation Core Components Upgrade Prerequisites

Before you upgrade the VMware Cloud Foundation management domain, verify that the following prerequisites are satisfied.

**Note:** You cannot upgrade VMware Cloud Foundation on Dell VxRail to VMware Cloud Foundation 9.0.

**Table 817: Upgrade Prerequisites**

| Prerequisite                                                                                                                      | Additional Information                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Allocate a temporary IP address for each vCenter appliance upgrade.                                                               | Required for each vCenter upgrade. Must be allocated from the management subnet. The IP address can be reused.                                                                             |
| Verify that no vCenter appliance is joined to an Active Directory domain.                                                         | See <a href="#">KB 373004</a> .                                                                                                                                                            |
| Verify there are no expired or expiring passwords.                                                                                | Review the password management dashboard in SDDC Manager.                                                                                                                                  |
| Verify there are no expired or expiring certificates.                                                                             | Review the <b>Certificates</b> tab in SDDC Manager for each workload domain.                                                                                                               |
| Verify ESX host TPM module status.                                                                                                | <b>[Conditional]</b> If ESX hosts have TPM modules in use, verify they are running the latest 2.0 firmware. If not in use they must be disabled in the BIOS. See <a href="#">KB 312159</a> |
| Verify ESX hardware is compatible with target version.                                                                            | See <a href="#">ESX Requirements</a> and <a href="#">Broadcom Compatibility Guide</a> .                                                                                                    |
| Manually update the vSAN HCL database to ensure that it is up-to-date.                                                            | See <a href="#">KB 2145116</a>                                                                                                                                                             |
| Back up SDDC Manager and all vCenter and NSX Manager instances.                                                                   | Take file-based backups or image-based backups of SDDC Manager and all vCenter and NSX Manager instances. Take a cold snapshot of SDDC Manager.                                            |
| Verify that there are no failed workflows in your system and none of the SDDC Manager resources are in activating or error state. | <b>CAUTION:</b><br>If any of these conditions are true, contact Broadcom Support before starting the upgrade.                                                                              |
| Download the upgrade bundles.                                                                                                     | See <a href="#">Downloading VMware Cloud Foundation Upgrade Bundles for Core Components</a> .                                                                                              |
| Review the <i>Release Notes</i> for known issues related to upgrades.                                                             |                                                                                                                                                                                            |
| In the vSphere Client, verify there are no active alarms on hosts or vSphere clusters.                                            |                                                                                                                                                                                            |

## Downloading VMware Cloud Foundation Upgrade Bundles for Core Components

Before you can upgrade to VMware Cloud Foundation 9.0, you must download the upgrade bundles for each VMware Cloud Foundation core component that requires an upgrade.

Downloading upgrade bundles requires an internet connection. The method you use to download bundles depends on how you access the internet in your environment.

- Online depot: SDDC Manager can connect to the internet, either directly or through a proxy.
- Offline depot: SDDC Manager cannot connect to the internet, but can connect to a computer with internet access.
- No depot: SDDC Manager cannot connect to the internet or to a computer with internet access.

### Online Depot

When you connect to an online depot, you can download upgrade bundles directly to SDDC Manager.



Downloading upgrade bundles using an online depot happens in two stages. Initially, only the SDDC Manager upgrade bundle is available. After you upgrade SDDC Manager to 9.0, you can download the remaining 9.0 upgrade bundles.

See [Connect SDDC Manager to a Depot for Accessing Bundles](#).

### Offline Depot

When you connect to an offline depot, you must use the VCF Download Tool to download upgrade bundles to the computer that is hosting the offline depot. Once the bundles are downloaded to the offline depot, SDDC Manager can use them for upgrades.

See [Connect SDDC Manager to a Depot for Accessing Bundles](#) and [Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot](#).

### No Depot

If SDDC Manager cannot connect to an online or offline depot, you must install the VCF Download Tool on a computer with internet access, download the upgrade bundles, and upload the bundles to SDDC Manager.

See [Download VMware Cloud Foundation 9.0 Upgrade Bundles Without an Online or Offline Depot](#).

## Connect SDDC Manager to a Depot for Accessing Bundles

Connect to an online or offline depot to access software bundles, compatibility data, and more.

To connect to an online depot, SDDC Manager must be able to access the internet, either directly or through a proxy server. See [Configure a Proxy Server for Accessing VMware Cloud Foundation Bundles](#). If you have not already done so, you must configure SDDC Manager to use a download token. See [KB 390098](#).

To connect to an offline depot, you must first configure it. See [KB 312168](#) for information about the requirements and process for creating an offline depot. To download bundles to an offline depot using the VCF Download Tool, see [Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot](#).

See [Public URL list for SDDC Manager](#) for information about the URLs that must be accessible to download bundles.

SDDC Manager supports two types of software depots:

- Online depot
- Offline depot

You can only connect SDDC Manager to one type of depot. If SDDC Manager is connected to an online depot and you configure a connection to an offline depot, the online depot connection is disabled and deleted.

1. In the navigation pane, click **Administration > Depot Settings**.
2. Connect SDDC Manager to an online depot or an offline depot.

| Depot Type | Configuration Steps                                                                                                                                                                                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Online     | <ol style="list-style-type: none"> <li>1. Click <b>Authenticate</b> for the VMware Depot.</li> <li>2. Type your Broadcom Support Portal user name and password.</li> <li>3. Click <b>Authenticate</b></li> </ol>                                                                                |
| Offline    | <ol style="list-style-type: none"> <li>1. Click <b>Set Up</b> for the Offline Depot.</li> <li>2. Enter the following information for the offline depot: <ul style="list-style-type: none"> <li>— Hostname</li> <li>— Port number</li> <li>— Username</li> <li>— Password</li> </ul> </li> </ol> |

| Depot Type | Configuration Steps    |
|------------|------------------------|
|            | 3. Click <b>Save</b> . |

SDDC Manager attempts to connect to the depot. If the connection is successful, SDDC Manager starts looking for available bundles. To view available bundles, click **Lifecycle Management > Bundle Management** and then click the **Bundles** tab. It may take some time for all available bundles to appear.

### Configure a Proxy Server for Accessing VMware Cloud Foundation Bundles

If SDDC Manager does not have direct internet access, you can configure a proxy server to access bundles. SDDC Manager supports proxy servers with authentication.

1. In the navigation pane, click **Administration > Proxy Settings**.
2. Click **Set Up Proxy**.
3. Toggle the **Enable Proxy** setting to the on position.
4. Select **HTTP** or **HTTPS**.
5. Enter the proxy server IP address and port number.
6. If your proxy server requires authentication, toggle the **Authentication** setting to the on position and enter the user name and password.

**Note:** VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.
7. Click **Save**.

You can now download bundles as described in [Download VMware Cloud Foundation 9.0 Upgrade Bundles Using an Online or Offline Depot](#).

### Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot

VMware Cloud Foundation 9.0 supports an offline depot that you can connect to from multiple instances of SDDC Manager. Use the VCF Download Tool to download bundles and ESX components to the offline depot which any SDDC Manager connected to the offline depot can access.

- [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).
  - Generate a download token on the Broadcom Support Portal.
  - The offline depot must have:
    - The latest version of the VCF Download Tool. You can download it from the Broadcom Support portal.
    - Internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
    - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
  - Connect SDDC Manager to the offline depot. See [Connect SDDC Manager to a Depot for Accessing Bundles](#).
- Note:** You can also connect SDDC Manager to the offline depot after you download bundles to the offline depot.

You can use the VCF Download Tool to download upgrade bundles and ESX components to the offline depot. If the offline depot can only access the internet using a proxy server, use the proxy server options as described in [VCF Download Tool Update Manager Download Service \(UMDS\) Commands](#).

### Note:

VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

In the steps below, where applicable, replace the following values.

| Value                           | Description                                     |
|---------------------------------|-------------------------------------------------|
| <code>vcf-source-version</code> | Your current VCF version. For example, 5.2.1.0. |

| Value                                  | Description                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>offline_depot_root_dir</code>    | The download location for the SDDC Manager upgrade bundle. For example, <code>/var/www</code> .                                                        |
| <code>url:port</code>                  | The URL and port of the offline depot. For example, <code>https://10.111.222.10:8282</code> .                                                          |
| <code>download-token</code>            | Your download token from the Broadcom Support Portal.                                                                                                  |
| <code>token_file_path</code>           | Path to a text file that contains a single line with your download token from the Broadcom Support Portal. For example, <code>/root/token.txt</code> . |
| <code>absolute_path_to_zip_file</code> | The path to the third-party offline zip bundle file.                                                                                                   |

1. On the computer hosting offline depot, run the following command to download the SDDC Manager upgrade bundle.

```
./lcm-bundle-transfer-util --setUpOfflineDepot -sv vcf-source-version --offlineDepotRootDir offline_depot_root_dir --offlineDepotUrl url:port --depotDownloadToken download-token
```

2. Run the following command to download the remaining upgrade bundles.

```
./vcf-download-tool binaries download --depot-store offline_depot_root_dir --depot-download-token-file token_file_path --vcf-version 9.0.0.0 --lifecycle-managed-by SDDC_MANAGER_VCF --type UPGRADE
```

**Note:** Use the same `offline_depot_root_dir` that you used to download the SDDC Manager upgrade bundle.

3. Use Update Manager Download Service (UMDS) to download the ESX components required to create an upgrade image in vSphere Lifecycle Manager.

- a) Install UMDS.

```
./vcf-download-tool umds install
```

- b) Configure the patch store for the offline depot.

```
./vcf-download-tool umds run -S --patch-store offline_depot_root_dir/umds-patch-store/
```

**Note:**

Use the same `offline_depot_root_dir` that you used to download the SDDC Manager upgrade bundle.

- c) Enable downloading of ESX host updates.

```
./vcf-download-tool umds run -S --enable-host
```

- d) (Optional) Configure UMDS to connect to the websites of third-party vendors to download vendor-specific ESX host patches and notifications.

```
./vcf-download-tool umds run -S --add-url https://host_url/index.xml --url-type HOST
```

- e) Download updates based on the current configuration.

```
./vcf-download-tool umds run vmware-umds -D
```

- f) (Optional) Import third-party offline zip files.

```
./vcf-download-tool umds run --import --offline-bundle absolute_path_to_zip_file
```

After the bundles and ESX components are available in the offline depot, you can use the SDDC Manager UI to apply the bundles to workload domains and use the vSphere Client to create a vSphere Lifecycle Manager image for upgrading ESX. Multiple instances of SDDC Manager UI can connect to the same offline depot.

### Download VMware Cloud Foundation 9.0 Upgrade Bundles Using an Online or Offline Depot

After you connect SDDC Manager to an online or offline depot, you can start to download upgrade bundles.

Connect SDDC Manager to an online or offline depot. See [Connect SDDC Manager to a Depot for Accessing Bundles](#).

If you are using an offline depot, [Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot](#).

Downloading upgrade bundles happens in two stages. Initially, only the SDDC Manager upgrade bundle is available. After you upgrade SDDC Manager to 9.0, you can download the remaining 9.0 upgrade bundles.

When you download bundles, SDDC Manager verifies that the file size and checksum of the downloaded bundles match the expected values.

1. In the navigation pane, click **Lifecycle Management > Bundle Management**.

**Note:** After you upgrade SDDC Manager to version 9.0, **Bundle Management** is renamed to **Binary Management**.

2. Click the **Bundles** tab to view available bundles.

**Note:**

If you just connected SDDC Manager to a depot, it can take some time for bundles to appear.

To view more information about the bundle, click **View Details**.

3. For the bundle you want to download, do one of the following:

- Click **Download Now** for an immediate download.  
The bundle download begins right away.
- Click **Schedule Download** to schedule a download.  
Select the date and time for the bundle download and click **Schedule**.

4. Click the **Download History** tab to see the downloaded bundles.

### Download VMware Cloud Foundation 9.0 Upgrade Bundles Without an Online or Offline Depot

If the SDDC Manager does not have access to an online or offline depot, you can use the VCF Download Tool to download bundles and ESX component data to a different computer and then upload the bundles and data to the SDDC Manager appliance.

- A Windows or Linux computer with internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts for long-running operations.

**Note:** The VCF Download Tool is the only supported method for downloading bundles. Do not use third-party tools or other methods to download bundles.

Using the VCF Download Tool to download VMware Cloud Foundation 9.0 upgrade bundles involves the following steps:

- Download the latest version of the tool.
- Generate a download token on the Broadcom Support Portal.
- On a computer with access to the internet, download the SDDC Manager upgrade bundle, the other upgrade bundles, and the ESX component data required to create an upgrade image in vSphere Lifecycle Manager.
- Copy the bundles and other required files to the SDDC Manager appliance.
- On the SDDC Manager appliance, upload the bundles and other required files to the internal LCM repository.

This procedure is for upgrading VCF 5.x to version 9.0 without an online or offline depot. If you plan to upgrade VCF 9.0 to version 9.0.x, see [Download Upgrade Binaries for a Subset of Core Components Without an Online or Offline Depot](#).

If the computer with internet access can only access the Internet using a proxy server, use the following options when downloading:

| Option                           | Description                             |
|----------------------------------|-----------------------------------------|
| <code>--proxyServer, --ps</code> | Provide the proxy server FQDN and port. |

| Option                           | Description                                                                                                                                                                                                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | For example: <code>--proxyServer proxy.example.com:3128</code> .                                                                                                                                                                                                                                               |
| <code>--proxyHttps</code>        | Add this option if the proxy server uses HTTPS.<br>To use this option, the proxy certificate must be added to Bundle Transfer Utility JRE default trust store. For example:<br><pre>/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore /opt/obtu/jre/lin64/lib/se- curity/cacerts</pre> |
| <code>--proxyUser</code>         | For a proxy server that requires authentication, enter the user name.                                                                                                                                                                                                                                          |
| <code>--proxyPasswordFile</code> | For a proxy server that requires authentication, enter the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password.<br>For example, <code>--proxyPasswordFile ../../password.txt</code> .                                                         |

**Note:** VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

Example that combines the options:

```
./lcm-bundle-transfer-util --download --manifestDownload --depotDownloadToken download-token --proxyServer
proxy.example.com:3128 --proxyUser vmwuser --proxyPasswordFile ../../password.txt --proxyHttps
```

1. Download the most recent version of the VCF Download Tool on a computer with internet access.
  - a) Log in to the Broadcom Support Portal and browse to **My Downloads > VMware Cloud Foundation**.
  - b) Click the version of VMware Cloud Foundation to which you are upgrading.
  - c) Click **Drivers & Tools**.
  - d) Click the download icon for the VCF Download Tool.
  - e) Extract `vcf-download-tool-<version.build>.tar.gz`.
  - f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.  
Note that the directory includes both the Bundle Transfer Utility (`lcm-bundle-transfer-util`) and the VCF Download Tool (`vcf-download-tool`).
2. Use the Bundle Transfer Utility to download the SDDC Manager upgrade bundle and other artifacts to the computer with internet access.

In the commands below, replace the following values where applicable.

| Value                              | Description                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>download-token</code>        | Your download token from the Broadcom Support Portal.                                                        |
| <code>current-vcf-version</code>   | Current version of SDDC Manager. For example, <code>5.2.1.0</code> .                                         |
| <code>target-vcf-version</code>    | Target version of SDDC Manager. For example, <code>9.0.0.0</code> .                                          |
| <code>/path/to/download_dir</code> | Path to the directory where the bundle files should be downloaded. For example, <code>/var/opt/obtu</code> . |

| Value | Description                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | This directory must have read and write permissions. If you do not specify the download directory, binaries are downloaded to the default directory with read and write permissions. |

- a) Download the manifest file.

This is a structured metadata file that contains information about the VMware product versions included in the release Bill of Materials.

```
./lcm-bundle-transfer-util --download --manifestDownload --depotDownloadToken download-token --output-
Directory /path/to/download_dir
```

Note the location to which the Bundle Transfer Utility downloads the manifest. For example:

```
Validating the depot user credentials...
Downloading LCM Manifest to: /root/PROD2/evc/vmw
Successfully completed downloading file
```

Later when you upload the manifest, you replace *Manifest-Directory* in the `--sourceManifestDirectory` *Manifest-Directory* attribute with the location.

- b) Download the bundle manifests.

```
./lcm-bundle-transfer-util --download --bundleManifests --depotDownloadToken download-token --bundle-
ManifestsDir /path/to/download_dir/bundleManifests
```

- c) Download the compatibility data.

```
./lcm-bundle-transfer-util --download --compatibilityMatrix --depotDownloadToken download-token --out-
putDirectory /path/to/download_dir
```

- d) Download the Product Version Catalog.

```
./lcm-bundle-transfer-util --download productVersionCatalog --depotDownloadToken download-token --out-
putDirectory /path/to/download_dir
```

- e) Download the vSAN HCL file.

```
./lcm-bundle-transfer-util --vsanHclDownload --outputDirectory /path/to/download_dir
```

- f) Download the SDDC Manager upgrade bundle.

```
./lcm-bundle-transfer-util --download --depotDownloadToken download-token --sv current-vcf-version --
p target-vcf-version --outputDirectory /path/to/download_dir
```

- g) Specify the bundles to download.

Enter one of the following options:

- `all`
- `install`
- `patch`

You can also enter a comma-separated list of bundle names to download specific bundles. For example: `bundle-38371, bundle-38378`.

Download progress for each bundle is displayed. Wait until all bundles are downloaded successfully.

3. Use the VCF Download Tool to download the remaining upgrade bundles and other artifacts to the computer with internet access.

In the command below, replace the following values.

| Value                     | Description                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>binaries_dir</code> | Path to the directory where the bundle files should be downloaded. For example, <code>/var/opt/obtu</code> . |

| Value                        | Description                                                                                                                                                                                                                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | This directory must have read and write permissions. If you do not specify the download directory, binaries are downloaded to the default directory with read and write permissions.<br><b>Note:</b><br>Download the upgrade bundles to the same directory to which you downloaded the SDDC Manager upgrade bundle. |
| <code>token_file_path</code> | Path to a text file that contains a single line with your download token from the Broadcom Support Portal.                                                                                                                                                                                                          |

```
./vcf-download-tool binaries download --depot-download-token-file token_file_path --vcf-version 9.0.0.0 --
lifecycle-managed-by SDDC_MANAGER_VCF --type UPGRADE --depot-store /path/to/download_dir
```

4. Use Update Manager Download Service (UMDS) to download the ESX components required to create an upgrade image in vSphere Lifecycle Manager.

a) Install UMDS.

```
./vcf-download-tool umds install
```

b) Enable downloading of ESX host updates.

```
./vcf-download-tool umds run -S --enable-host
```

c) (Optional) Configure UMDS to connect to the websites of third-party vendors to download vendor-specific ESX host patches and notifications

```
./vcf-download-tool umds run -S --add-url https://host_url/index.xml --url-type HOST
```

d) Download updates based on the current configuration.

```
./vcf-download-tool umds run vmware-umds -D
```

e) Export the downloaded data.

```
./vcf-download-tool umds run -E --export-store repository_path
```

Replace `repository_path` with the location of your repository, which can be a path to a portable media drive. Export the downloads to the portable media drive that you can use to physically transfer the patches to the SDDC Manager appliance after it upgraded to 9.0.

See [Use UMDS on SDDC Manager to Get ESX Component Data](#) for information about importing ESX component data after you upgrade to SDDC Manager 9.0.

5. Copy the following files/directories to the SDDC Manager appliance.

- Manifest file
- Bundles manifests
- Compatibility data file (`VmwareCompatibilityData.json`)
- vSAN HCL
- Entire bundle output directory
- Product version catalog

If you have direct access to the SDDC Manager appliance, use `rsync`. Alternatively, you must copy the files to a portable media and connect the media to the appliance.

Replace `binaries_dir` with the directory to which you downloaded the SDDC Manager upgrade bundle. For example, `/var/opt/obtu`.

Replace `sddc_binaries_dir` with any location on the SDDC Manager appliance that has enough free space available. For example, use `/nfs/vmware/vcf/nfs-mount/upload`.

```
rsync -aP /path/to/upload_dir vcf@sddc-manager:/path/to/upload_dir
```

6. Copy the VCF Download Tool to the SDDC Manager appliance.

a) SSH in to the SDDC Manager appliance using the `vcf` user account.

b) Enter `su` to switch to the root user.

c) Create the `lcm-tools` directory.

```
mkdir /home/vcf/lcm-tools
```

**Note:** If the `/home/vcf/lcm-tools` directory already exists with an older version of the VCF Download Tool, you need to delete contents of the existing directory before proceeding.

d) Copy the VCF Download Tool file (`vcf-download-tool-<version.build>.tar.gz`) that you downloaded in step 1 to the `/home/vcf/lcm-tools` directory.

e) Extract the contents of `vcf-download-tool-<version.build>.tar.gz`.

```
tar -xvf vcf-download-tool-<version.build>.tar.gz
```

f) Set the permissions for the `lcm-tools` directory.

```
cd /home/vcf/
chown vcf_lcm:vcf -R lcm-tools
chmod 750 -R lcm-tools
```

7. From the SDDC Manager appliance, use the Bundle Transfer Utility to upload the bundles and artifacts.

In the commands below, replace the following values, where applicable.

| Value                                     | Description                                                             |
|-------------------------------------------|-------------------------------------------------------------------------|
| <code>Manifest-Directory</code>           | The location where the Bundle Transfer Utility downloaded the manifest. |
| <code>FQDN</code>                         | The FQDN address of your SDDC Manager.                                  |
| <code>Username</code>                     | Use your vSphere SSO credentials.                                       |
| <code>compatibility-file-directory</code> | The path to directory of the compatibility file.                        |
| <code>hcl-file-path</code>                | The path to the HCL file.                                               |

| Value                            | Description                                              |
|----------------------------------|----------------------------------------------------------|
| <code>/path/to/upload_dir</code> | Path to the directory where the bundle files are copied. |

a) Upload the manifest file.

```
./lcm-bundle-transfer-util --update --sourceManifestDirectory /path/to/upload_dir --sddcMgrFqdn FQDN --sddcMgrUser Username
```

b) Upload the bundle manifests.

```
./lcm-bundle-transfer-util --upload --bundleManifests --bundleManifestsDir /path/to/upload_dir/bundle-Manifests --sddcMgrFqdn FQDN --sddcMgrUser Username
```

c) Upload the compatibility file.

```
./lcm-bundle-transfer-util --update --compatibilityMatrix --inputDirectory /path/to/upload_dir --sddcMgrFqdn FQDN --sddcMgrUser Username
```

d) Upload the Product Version Catalog.

```
./lcm-bundle-transfer-util --upload productVersionCatalog --inputDirectory /path/to/upload_dir --sddcMgrFqdn FQDN --sddcMgrUser Username
```

e) Upload the HCL file.

```
./lcm-bundle-transfer-util --vsanHclUpload --inputDirectory /path/to/upload_dir --sddcMgrFqdn FQDN --sddcMgrUser Username
```

f) Upload the bundle directory.

```
./lcm-bundle-transfer-util --upload --bundleDirectory /path/to/upload_dir
```

Perform Update Precheck in SDDC Manager

You must perform a precheck in SDDC Manager before applying an update to ensure that your environment is ready for the update.

Bundle-level pre-checks for vCenter are available in VMware Cloud Foundation.

Note:

Because ESX bundle-level pre-checks only work in minor-version upgrades (for example: from ESX 8.x through 8.y), these prechecks do not run in VMware Cloud Foundation.

If you silence a vSAN Skyline Health alert in the vSphere Client, SDDC Manager skips the related precheck and indicates which precheck it skipped. Click **RESTORE PRECHECK** to include the silenced precheck. For

RESTORE PRECHECK

| Resource Name | Description                                                                                                                                                                                                                                                    | Health Status                                                                                | Error Description                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| SDDC-Cluster1 | Checks the age of the VMware Hardware Compatibility Guide database used for the HCL checks. Shows warning or error when it is older than 90 or 180 days, respectively. VMware updates the VCG frequently, so it is important to keep the local copy up-to-date |  SILENCED | Check skipped because 'com.vmware.vsan.health' is silenced in vSAN |

example:

You can also silence failed vSAN prechecks in the SDDC Manager UI by clicking **Silence Precheck**. Silenced prechecks do not trigger warnings or block upgrades.

Important:

Only silence alerts if you know that they are incorrect. Do not silence alerts for real issues that require remediation.

- 1. In the navigation pane, click **Inventory > Workload Domains**.
- 2. On the Workload Domains page, click the workload domain where you want to run the precheck.
- 3. On the domain summary page, click the **Updates** tab.

Note:

It is recommended that you precheck your workload domain prior to performing an upgrade.

4. Click **RUN PRECHECK** to select the components in the workload domain you want to precheck.

- a) You can select to run a Precheck only on vCenter or the vSphere cluster. All components in the workload domain are selected by default. To perform a precheck on certain components, choose **Custom selection**.

**Precheck**

Run a general precheck or precheck for a specific release version. You can precheck across the entire workload domain or select specific components.

**Target Version** ⓘ General Upgrade Readiness ▾

**Run Precheck on**

☒ Entire Workload Domain ☐ Custom selection

Datagrid lists the components that are going to be prechecked.

| <input type="checkbox"/>            | Component                        |
|-------------------------------------|----------------------------------|
| <input checked="" type="checkbox"/> | sddc-manager.vrack.vsphere.local |
| <input checked="" type="checkbox"/> | vip-nsx-mgmt.vrack.vsphere.local |
| <input checked="" type="checkbox"/> | SDDC-Cluster1                    |
| <input checked="" type="checkbox"/> | vcenter-1.vrack.vsphere.local    |
| <input checked="" type="checkbox"/> | 4                                |

**RUN PRECHECK** **CANCEL**

- b) If there are pending upgrade bundles available, then the "Target Version" dropdown contains "General Upgrade Readiness" and the available VMware Cloud Foundation versions to upgrade to. If there is an available VMware Cloud Foundation upgrade version, there will be extra checks - bundle-level prechecks for hosts, vCenter, and so

forth. The version specific prechecks will only run prechecks on components that have available upgrade bundles

**sddcId-1001** | **ACTIONS** ▾

MANAGEMENT ✓ ACTIVE Version : 4.5.0.0

Starting VCF 4.4 release, SSH Service on ESXi hosts is disabled on new and upgraded VCF deployment

Summary Services **Updates 1** Update History Hosts Clusters Edges

[< BACK TO UPDATES](#)

**Precheck**

Run a general precheck or precheck for a specific release version. You can precheck across the entire workload domain or select specific components.

**Target Version** ⓘ 5.0.0.0 ▾

**Precheck Scope** ⓘ

All components in version - 5.0.0.0

All components in version - 5.0.0.0

SDDC\_MANAGER\_VCF - 5.0.0.0-20646706

SDDC\_MANAGER\_VCF - 5.0.0.0-20646706

NSX - 4.0.2.0.0-20606912

**VCENTER - 8.0.0.20000-20646706**

ESX\_HOST - 8.0.0-20646706

Custom selection

|                                     |                                  |
|-------------------------------------|----------------------------------|
| <input checked="" type="checkbox"/> | vip-nsx-mgmt.vrack.vsphere.local |
| <input checked="" type="checkbox"/> | vcenter-1.vrack.vsphere.local    |
| <input checked="" type="checkbox"/> | SDDC-Cluster1                    |
| <input checked="" type="checkbox"/> | 4                                |

**RUN PRECHECK** **CANCEL**

downloaded.

5. When the precheck begins, a progress message appears indicating the precheck progress and the time when the precheck began.



## Precheck

In Progress

 72% Completed

Started on Dec 21, 2022, 10:02:29 AM

Precheck report will be shown once completed.

**Note:** Parallel precheck workflows are supported. If you want to precheck multiple domains, you can repeat steps 1-5 for each of them without waiting for step 5 to finish.

6. Once the Precheck is complete, the report appears. Click through **ALL**, **ERRORS**, **WARNINGS**, and **SILENCED** to filter and browse through the results.

## Precheck

### Results

 174 Passed

 7 Errors

 3 Warnings

 0 Silenced

[RETRY ALL FAILED RESOURCES](#)

### Report

**ALL** ERRORS WARNINGS SILENCED

Version: General Upgrade Readiness (4 component)

- >  SDDC Manager sddc-manager.vrack.vsphere.local
- >  NSX vip-nsx-mgmt.vrack.vsphere.local
- >  ESXi Host Cluster SDDC-Cluster1
- >  vCenter vcenter-1.vrack.vsphere.local

| Resource Name   | Description | Health Status |
|-----------------|-------------|---------------|
| Precheck Report |             |               |

7. To see details for a task, click the expander arrow.

If a precheck task failed, fix the issue, and click **Retry Precheck** to run the task again. You can also click **RETRY ALL FAILED RESOURCES** to retry all failed tasks.

8. If the workload domain contains a host that includes pinned VMs, the precheck fails at the Enter Maintenance Mode step. If the host can enter maintenance mode through the vCenter UI, you can suppress this check for NSX and ESX in VMware Cloud Foundation by following the steps below.
- Log in to SDDC Manager by using a Secure Shell (SSH) client with the user name vcf and password.
  - Open the `/opt/vmware/vcf/lcm/lcm-app/conf/application-prod.properties` file.
  - Add the following line to the end of the file:
 

```
lcm.nsx.suppress.dry.run.emm.check=true
lcm.esx.suppress.dry.run.emm.check.failures=true
```
  - Restart the lcm service by typing the following command in the console window.
 

```
systemctl restart lcm
```
  - After the lcm service is restarted, run the precheck again.

The precheck result is displayed at the top of the Upgrade Precheck Details window. If you click **Exit Details**, the precheck result is displayed at the top of the Precheck section in the Updates tab.

Ensure that the precheck results are green before proceeding. Although a failed precheck will not prevent the upgrade from proceeding, it may cause the update to fail.

## Upgrade SDDC Manager to 9.0

The VMware Cloud Foundation upgrade bundle upgrades the SDDC Manager appliance and VMware Cloud Foundation services.

- Download the VMware Cloud Foundation upgrade bundle for your target release. See [Downloading VMware Cloud Foundation Upgrade Bundles for Core Components](#).
- Ensure you have a recent successful backup of SDDC Manager using an external SFTP server.
- Ensure you have taken a snapshot of the SDDC Manager appliance.
- Ensure you have recent successful backups of the components managed by SDDC Manager.
- [Perform Update Precheck in SDDC Manager](#) and resolve any issues.

- In the navigation pane, click **Inventory > Workload Domains**.
- On the Workload Domains page, click the management domain and then click the **Updates** tab.
- In the **Available Updates** section, click **Plan Upgrade**, select a target version, and click **Confirm**.
- Click **Update Now** or **Schedule Update** next to the VMware Cloud Foundation Upgrade bundle.
- If you selected **Schedule Update**, select the date and time for the bundle to be applied and click **Schedule**.

If you clicked **Update Now**, the Update Status window displays the components that will be upgraded and the upgrade status. Click **View Update Activity** to view the detailed tasks. After the upgrade is completed, a green bar with a check mark is displayed.

- Click **Finish**.

When the update completes successfully, you are logged out of the SDDC Manager UI and must log in again.

If you are using an online depot, download the remaining upgrade bundles. See [Download VMware Cloud Foundation 9.0 Upgrade Bundles Using an Online or Offline Depot](#).

**Note:** After you upgrade to SDDC Manager 9.0, bundles are referred to as binaries.

## SDDC Manager Functionality During an Upgrade to VMware Cloud Foundation 9.0

During the upgrade to VMware Cloud Foundation 9.0, some SDDC Manager functionality may be limited during each phase of the upgrade. Prior to initiating the upgrade determine if you must perform any of these tasks.

### Upgrade States and Terminology

- Source BoM** - Prior to initiating the upgrade, all components are at VMware Cloud Foundation version 5.0, 5.1, 5.2.

- SDDC Manager only** - You have upgraded VCF Operations and SDDC Manager to 9.0, but none of the other BoM components.
- Split BOM** - Management domain or workload domains are only partially upgraded to VMware Cloud Foundation 9.0.
- Mixed 5.2.x / 9.0 BoM** - Some domains (management or workload) are completely upgraded to VMware Cloud Foundation 9.0 and at least one workload domain is at the Source 5.x BoM version.
- Target BoM** - All components are at VMware Cloud Foundation 9.0.

When a VMware Cloud Foundation instance is in Source BoM or Target BoM, the features available within SDDC Manager are as expected for that given release. However when in a Mixed BoM the operations available vary per workload domain depending on which state the domain itself is in.

The following table indicates the available SDDC Manager functionalities during an upgrade.

**Table 818: Available SDDC Manager Functionalities During Upgrade**

| Category                | Feature                                                                                                                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Backup / Restore        | Configure and perform Backup / Restore                                                                                                                                                                                                                  |
| CEIP                    | Activate / Deactivate CEIP                                                                                                                                                                                                                              |
| Certificate Management  | View / Generate / Upload / Install                                                                                                                                                                                                                      |
| NSX Edge Cluster        | Expand edge cluster                                                                                                                                                                                                                                     |
| DNS / NTP configuration | Validate / Configure DNS<br>Validate / Configure NTP                                                                                                                                                                                                    |
| Hosts                   | Commission / Decommission Host                                                                                                                                                                                                                          |
| Licensing               | Update License Key Information<br>Add License Key<br>Relicensing<br>License check                                                                                                                                                                       |
| LCM                     | Connect to VMware or Dell Depot / Download Bundles<br>LCM Pre checks<br>Schedule Bundle Download<br>Install vCenter Patch<br>Install ESXi Patch<br>Install NSX Patch                                                                                    |
| Networking              | Create / Edit / Delete Network Pool                                                                                                                                                                                                                     |
| Password Management     | Rotate / Update                                                                                                                                                                                                                                         |
| User Operations         | Add / Remove User / Group                                                                                                                                                                                                                               |
| Workload Domain         | Add / Remove ESXi Host<br>Add / Remove vSphere Cluster<br>Add 5.x workload domain<br>Remove 5.x workload domain<br>Stretch a vSphere Cluster<br>Unstretch a vSphere Cluster<br>Expand a Stretched vSphere Cluster<br>Shrink a Stretched vSphere Cluster |

Apply VMware Cloud Foundation Configuration Updates

VMware Cloud Foundation Configuration Updates identify and resolve any discrepancies between the intended/prescribed configuration and the actual configuration, ensuring that the deployment aligns with the recommended configuration. This process includes reconciling the configuration for 2nd party software components listed in the VMware Cloud Foundation Bill of Materials (BOM).

Configuration updates may be required after you apply software updates. Once a configuration update becomes available, you can apply it immediately or wait until after you have applied all software updates. Configuration Updates must be performed during a maintenance window. Configuration Updates can be applied to multiple domains in parallel. However, if a Configuration Update is in progress, another configuration update on the same domain should not be attempted.

- 1. In the navigation pane, click **Inventory > Workload Domains**.
- 2. On the Workload Domains page, click the workload domain name and then click the **Updates** tab.
- 3. Click **Run Precheck** to run the upgrade precheck.  
Resolve any issues before proceeding with the upgrade.
- 4. Expand **Available Configuration Updates**, click **Apply All**.

▼ Available Configuration Updates 4

Configuration updates may be required after you apply software updates. Once a configuration update becomes available, you can apply it immediately or wait until after you have applied all software updates.

| Description                                                                                                | Type    | Resolution Type |
|------------------------------------------------------------------------------------------------------------|---------|-----------------|
| Remove das.includeFTcomplianceChecks option from all clusters HA configuration                             | FEATURE | DC              |
| ESXi advanced options for ESXi Hosts Addition Drift                                                        | FEATURE | DC              |
| Register SDDC Manager as an extension in a domain vCenter                                                  | FEATURE | DC              |
| Creates a Distributed Virtual Port Group to enable traffic isolation between management VMs and ESXi hosts | FEATURE | CL              |

- **FEATURE:** Configuration change required for a new feature.
- **FIX:** Configuration change associated with a fix for a defect.

- 5. Check the progress of a configuration update by clicking the task in the Tasks

Tasks

Task

Configuration Updates

- 6. After the configuration updates are successfully applied, they will no longer appear in the

▼ Available Configuration Updates

No available updates

Pending Configuration Updates do not block future BOM upgrades.

Upgrading NSX to 9.0

After, you have upgraded SDDC Manager to version 9, you must upgrade your NSX Manager cluster. It is recommended that you upgrade by using the streamlined, orchestrated process by using SDDC Manager. In rare cases the manual option for upgrade by using Upgrade Coordinator is also available.

Determine Your Upgrade Type

You can select between the following upgrade types for your NSX deployment. The upgrade type you choose determines the method used to perform the upgrade.

| Upgrade Type      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Upgrade Method               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Automated Upgrade | <ul style="list-style-type: none"> <li>Recommended for most upgrade scenarios.</li> <li>Upgrades edge nodes in serial order within a cluster.</li> <li>Upgrades hosts in serial order.</li> <li>Upgrades NSX VIBs automatically as part of ESX host upgrade.</li> </ul>                                                                                                                                                                                                         | SDDC Manager                 |
| Manual Upgrade    | <ul style="list-style-type: none"> <li>Used for customized upgrade scenarios.</li> <li>Can upgrade unit groups of edge nodes or groups of hosts in serial or parallel order. Within a cluster or unit group, edge nodes are always upgraded in serial order.</li> <li>Can alternate between upgrading groups of edge nodes and groups of hosts.</li> <li>Can upgrade NSX VIBs on hosts separately from ESX upgrade. This option is available only for ESX 8.x hosts.</li> </ul> | NSX Manager and SDDC Manager |

### Points to Consider when Upgrading

- Do not perform upgrade tasks in SDDC Manager and at the same time. When an upgrade runs in SDDC Manager, the upgrade coordinator in the UI reflects the progress of the upgrade and vice versa.
- If you are performing an out-of-band upgrade, that is, instead of using VCF operations to perform the upgrade you choose to upgrade through any of the component's console (such as NSX Manager or vCenter) you still must upgrade in the following sequence:
  - VCF Operations
  - NSX Upgrade Coordinator
  - NSX Manager
  - Edge nodes
  - vCenter
  - NSX Finalize Upgrade
  - ESX

### Checklist for Automated NSX Upgrade

Use the following checklist to track your work through the automated upgrade process.

| Automated Upgrade Task                                                                                                                    | Instructions                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaluate the operational impact of the upgrade.                                                                                           | See <a href="#">Operational Impact of the Upgrade</a> .                                                                                                                                                                                                                     |
| Run the pre-checks to verify that system components are ready for the upgrade.                                                            | See <a href="#">Run the NSX Upgrade Pre-check Bundle</a> .                                                                                                                                                                                                                  |
| Complete the pre-upgrade tasks.                                                                                                           | See <a href="#">Pre-Upgrade Tasks for Upgrading NSX</a> .                                                                                                                                                                                                                   |
| Verify that the NSX environment is in a healthy state.                                                                                    | See <a href="#">Verify the Current State of NSX</a> .                                                                                                                                                                                                                       |
| Make the required changes in your system configuration if your deployment contains features that are not supported in the target release. | See <a href="#">Manage Unsupported Features</a> .                                                                                                                                                                                                                           |
| In SDDC Manager, upgrade NSX Manager, edge nodes, vCenter, and the hosts.                                                                 | See: <ul style="list-style-type: none"> <li><a href="#">Automated Upgrade to NSX 9 by Using SDDC Manager</a></li> <li><a href="#">Upgrade to vCenter 9.0 for VMware Cloud Foundation</a></li> <li><a href="#">Upgrade to ESX 9.0 for VMware Cloud Foundation</a></li> </ul> |
| Verify the upgrade.                                                                                                                       | See <a href="#">Verify the Upgrade</a> .                                                                                                                                                                                                                                    |

### Upgrading NSX in a Federated Environment

If NSX Federation is configured between two VCF Instances, VMware Cloud Foundation does not manage the lifecycle of the NSX Global Manager nodes. You must manually upgrade the NSX Global Manager nodes for each instance.

#### Download the NSX Global Manager Upgrade Bundle

VMware Cloud Foundation does not manage the lifecycle of NSX Global Manager instances. You must download the NSX upgrade bundle manually to upgrade the NSX Global Managers.

- Log in to the Broadcom Support Portal and browse to **My Downloads > VMware NSX**.
- Click the version of NSX to which you are upgrading.
- Locate the **NSX *version* Upgrade Bundle** and verify that the upgrade bundle filename extension ends with `.mub`. The upgrade bundle filename has the following format `vmware-nsx-upgrade-bundle-versionnumber.buildnumber.mub`.
- Click the download icon to download the upgrade bundle to the system where you access the NSX Global Manager UI.

#### Upgrade the Upgrade Coordinator for NSX Federation

The upgrade coordinator runs in the NSX Manager. It is a self-contained web application that orchestrates the upgrade process of hosts, NSX Edge cluster, NSX Controller cluster, and the management plane.

The upgrade coordinator guides you through the upgrade sequence. You can track the upgrade process and, if necessary, you can pause and resume the upgrade process from the UI.

- In a web browser, log in to the NSX Global Manager for the domain at `https://nsx_gm_vip_fqdn/`.
- Select **System > Upgrade** from the navigation panel.
- Click **Proceed to Upgrade**.
- Navigate to the upgrade bundle `.mub` file you downloaded or paste the download URL link.
  - Click **Browse** to navigate to the location you downloaded the upgrade bundle file.
  - Paste the VMware download portal URL where the upgrade bundle `.mub` file is located.
- Click **Upload**.
 

When the file is uploaded, the **Begin Upgrade** button appears.
- Click **Begin Upgrade** to upgrade the upgrade coordinator.
 

**Note:**

Upgrade one upgrade coordinator at a time.
- Read and accept the EULA terms and accept the notification to upgrade the upgrade coordinator..
- Click **Run Pre-Checks** to verify that all NSX components are ready for upgrade.
 

The pre-check checks for component connectivity, version compatibility, and component status.
- Resolve any warning notifications to avoid problems during the upgrade.

#### Upgrade NSX Global Managers for VMware Cloud Foundation

Manually upgrade the NSX Global Managers when NSX Federation is configured between two SDDC Manager instances.

You must upgrade NSX Global Managers before you can upgrade any VMware Cloud Foundation instances in the NSX Federation, including NSX Local Managers.

1. In a web browser, log in to the NSX Global Manager for the domain at [https://nsx\\_gm\\_vip\\_fqdn/](https://nsx_gm_vip_fqdn/).
2. Select **System > Upgrade** from the navigation panel.
3. Click **Start** to upgrade the management plane and then click **Accept**.
4. On the Select Upgrade Plan page, select **Plan Your Upgrade** and click **Next**.  
The NSX Manager UI, API, and CLI are not accessible until the upgrade finishes and the management plane is restarted.

### Automated Upgrade to NSX 9 by Using SDDC Manager

You upgrade NSX in the management domain and workload domains by using the recommended orchestration and automation that SDDC Manager provides.

You can upgrade NSX in workload domains before or after upgrading NSX in the management domain.

Prior to VMware Cloud Foundation 9.0, host clusters were upgraded as part of the NSX upgrade. In VMware Cloud Foundation 9.0, host clusters are upgraded as part of the ESX upgrade. As a result of this change, ESX hosts no longer go into maintenance mode during an NSX upgrade.

Upgrading NSX involves the following components:

- Upgrade Coordinator
- NSX Manager cluster
- NSX Edges/Clusters (if deployed)

1. In the navigation pane, click **Inventory > Workload Domains**.
2. On the Workload Domains page, click the domain you are upgrading and then click the **Updates/Patches** tab.

When you upgrade NSX components for a selected workload domain, those components are upgraded for all workload domains that share the NSX Manager cluster.

3. Click **Precheck** to run the upgrade precheck.

Resolve any issues before proceeding with the upgrade.

#### Note:

The NSX precheck runs on all workload domains in your environment that share the NSX Manager cluster.

4. Configure the update.
  - a) In the Available Updates section, click the **Configure Update** button.
  - b) Review the Introduction and click **Next**.
  - c) On the NSX Edge Clusters page, select the NSX Edge clusters you want to upgrade and click **Next**.

By default, all NSX Edge clusters are upgraded. To select specific NSX Edge clusters, select the **Upgrade only NSX Edge clusters** check box and select the **Enable edge selection** option. Then select the NSX Edges you want to upgrade.

- d) On the Review page, review your settings and click **Run Precheck**.  
The precheck begins. Resolve any issues until the precheck succeeds.
  - e) After the precheck succeeds, click **Schedule Update** and select an option.  
If the precheck report includes any errors or warnings, you must select the checkbox to proceed.
5. Monitor the upgrade progress. See [Monitoring VMware Cloud Foundation Updates](#).  
If a component upgrade fails, the failure is displayed across all associated workload domains. Resolve the issue and retry the failed task.

When all NSX workload components are upgraded successfully, a message with a green background and check mark is displayed.

#### Note:

NSX 9.0 functionality is not available until all ESX hosts in all vSphere clusters that share an NSX Manager are upgraded to 9.0 and the NSX finalize procedure of the ESX upgrade completes.

After the upgrade is complete, allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade might be canceled or auto-reconciliation might fail.

### Manual Upgrade to NSX 9 by Using Upgrade Coordinator

The automated upgrade method using SDDC Manager is recommended for most NSX upgrade scenarios. However, a manual upgrade method using NSX Manager is also available for scenarios that require a customized NSX upgrade.

#### Note:

Starting with VMware Cloud Foundation 9.0, the NSX VIBs are packaged as part of the ESX image by default. Therefore, the automated upgrade of these VIBs occurs as part of the ESX host upgrade process performed through SDDC Manager.

If you use NSX Manager instead to perform a manual upgrade of ESX 8.x hosts, only the NSX VIBs are upgraded. You must then use SDDC Manager separately to upgrade ESX on the hosts.

### Understanding the Types of Manual NSX Upgrade

There are two types of manual upgrade using NSX Manager.

#### Out-of-band Upgrade

In an out-of-band NSX upgrade, you first converge your existing infrastructure to a management domain as described in [Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform](#). Then you manually upgrade NSX using NSX Manager. Since the NSX instance has been converged to a VCF management domain, it receives a valid VCF license entitlement.

Although this upgrade method is available for scenarios that require a customized upgrade, the automated upgrade method using SDDC Manager is recommended for most NSX upgrade scenarios.

#### Standalone Upgrade

A standalone NSX upgrade refers to either of the following scenarios:

- You attempt to upgrade NSX manually without converging your existing infrastructure to a management domain.
- You attempt to upgrade NSX using an upgrade sequence that deviates from the prescribed sequence outlined in the [Checklist for Manual NSX Upgrade](#).

In a standalone upgrade, the NSX instance is not properly converged to a VCF management domain. As a result, NSX cannot fetch a valid VCF license entitlement and enters into an evaluation mode of limited duration. Once this evaluation mode expires, license enforcement will go into effect, restricting your access to NSX features.

### Checklist for Manual NSX Upgrade

Use the following checklist to track your work through the manual upgrade process.

| Manual Upgrade Task                             | Instructions                                            |
|-------------------------------------------------|---------------------------------------------------------|
| Evaluate the operational impact of the upgrade. | See <a href="#">Operational Impact of the Upgrade</a> . |

| Manual Upgrade Task                                                                                                                                                                                                                         | Instructions                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Run the pre-checks to verify that system components are ready for the upgrade.                                                                                                                                                              | See <a href="#">Run the NSX Upgrade Pre-check Bundle</a> .                                                                                                                                           |
| Complete the pre-upgrade tasks.                                                                                                                                                                                                             | See <a href="#">Pre-Upgrade Tasks for Upgrading NSX</a> .                                                                                                                                            |
| Verify that the NSX environment is in a healthy state.                                                                                                                                                                                      | See <a href="#">Verify the Current State of NSX</a> .                                                                                                                                                |
| Make the required changes in your system configuration if your deployment contains features that are not supported in the target release.                                                                                                   | See <a href="#">Manage Unsupported Features</a> .                                                                                                                                                    |
| Download the latest NSX upgrade bundle (.mub file).                                                                                                                                                                                         | See <a href="#">Download the NSX Upgrade Bundle</a> .                                                                                                                                                |
| In NSX Manager, upgrade the upgrade coordinator.                                                                                                                                                                                            | See <a href="#">Update the Upgrade Coordinator in NSX Manager</a> .                                                                                                                                  |
| In NSX Manager, upgrade the Management Plane.                                                                                                                                                                                               | See <a href="#">Upgrade the Management Plane Through NSX Manager</a> .                                                                                                                               |
| In NSX Manager, create upgrade unit groups of edge nodes, define the order of upgrade for each unit group, and complete the edge node upgrades.                                                                                             | See <a href="#">Upgrade the NSX Edge Cluster Through NSX Manager</a> .                                                                                                                               |
| If you want to manually upgrade the NSX VIBs on host transport nodes, create upgrade unit groups of hosts in NSX Manager. Define the order of upgrade for each unit group and complete the upgrade of NSX VIBs on the host transport nodes. | See <a href="#">Configure Hosts for a Manual Upgrade</a> .                                                                                                                                           |
| (Optional) In NSX Manager, download the Consistency Check Report.                                                                                                                                                                           | See <a href="#">Download the Consistency Check Report</a> .                                                                                                                                          |
| In SDDC Manager, upgrade vCenter and the ESX hosts.                                                                                                                                                                                         | See:<br><ul style="list-style-type: none"> <li><a href="#">Upgrade to vCenter 9.0 for VMware Cloud Foundation</a></li> <li><a href="#">Upgrade to ESX 9.0 for VMware Cloud Foundation</a></li> </ul> |
| In NSX Manager, finalize the upgrade.                                                                                                                                                                                                       | See <a href="#">Finalize Upgrade in NSX Manager</a> .                                                                                                                                                |
| Verify the upgrade.                                                                                                                                                                                                                         | See <a href="#">Verify the Upgrade</a> .                                                                                                                                                             |

### Operational Impact of the NSX Upgrade by Using Upgrade Coordinator

The duration of the NSX upgrade process depends on the number of components you have to upgrade in your infrastructure. It is important to understand the operational state of NSX components during an upgrade.

The upgrade of components occurs in the following order:

1. Management Plane
2. Edge nodes
3. vCenter
4. Hosts

### Management Plane Upgrade

| During Upgrade                                                                                                                                                                                                                                                                                                                 | After Upgrade                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Do not make any configuration changes during the Management plane upgrade.</li> <li>User interface is unavailable for a short period.</li> <li>When upgrading from NSX 4.x, API service is momentarily unavailable. When upgrading from VCF 9.x, API downtime is negligible.</li> </ul> | <ul style="list-style-type: none"> <li>Configuration changes are allowed.</li> <li>New features introduced in the upgrade are not configurable until all edge and host nodes are upgraded and the Finalize Upgrade step is complete.</li> <li>From the Upgrade Coordinator, verify that the upgrade process has completed. Perform configuration tasks only after the upgrade process is complete.</li> </ul> |

| During Upgrade                                                                                                                                                                                                                                                              | After Upgrade |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <ul style="list-style-type: none"> <li>When upgrading from NSX 4.x, you may experience longer than usual workload traffic downtime for VMs undergoing migration through vSphere vMotion. When upgrading from VCF 9.x, this possible downtime is further reduced.</li> </ul> |               |

### NSX Edge Cluster Upgrade

| During Upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | After Upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>During the edge upgrade, you might experience the following traffic interruptions: <ul style="list-style-type: none"> <li>North-south datapath is affected if the edge is part of the datapath.</li> <li>East-west traffic between tier-1 routers using edge firewall, NAT, or load balancing.</li> <li>Temporary Layer 2 and Layer 3 interruption.</li> </ul> </li> <li>Configuration changes are not blocked on NSX Manager but might be delayed.</li> <li>The hardware versions of edge node VMs change or remain the same as follows: <ul style="list-style-type: none"> <li>For edge nodes auto-deployed through the NSX Manager UI or the API, the VM hardware is automatically updated to the maximum version supported by the underlying ESX host.</li> <li>For edge nodes manually deployed through vSphere Client, the VM hardware remains unchanged at the same version.</li> </ul> </li> </ul> <p>For more details, see Broadcom KB <a href="#">319091</a>.</p> | <ul style="list-style-type: none"> <li>Configuration changes are allowed.</li> <li>Upgraded edge cluster is compatible with the older versions of the hosts.</li> <li>New features introduced in the upgrade are not configurable until all the edge and host nodes are upgraded and the Finalize Upgrade step is complete.</li> <li>Run post checks to make sure that the upgraded edge cluster and NSX do not have any problems.</li> </ul> |

### Hosts Upgrade

The following considerations apply only when using NSX Manager to upgrade the NSX VIBs on ESX 8.x hosts.

To upgrade ESX 7.x hosts, you must first upgrade to ESX 8.x.

| During Upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | After Upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>For standalone ESX hosts or ESX hosts that are part of a disabled DRS cluster, place hosts in maintenance mode.</li> <li>For ESX hosts that are part of a fully automated DRS cluster, if the host is not in maintenance mode, the upgrade coordinator requests the host to be put in maintenance mode. The vSphere DRS tool migrates the VMs to another host in the same cluster during the upgrade and places the host in maintenance mode.</li> <li>For ESX host, for an in-place upgrade you do not need to power off the tenant VMs.</li> </ul> | <ul style="list-style-type: none"> <li>Power on or return the tenant VMs of standalone ESX hosts or ESX hosts that are part of a disabled DRS cluster that were powered off before the upgrade.</li> <li>New features introduced in the upgrade are not configurable until all the edge and host transport nodes are upgraded and the Finalize Upgrade step is complete.</li> <li>Run post checks to make sure that the upgraded hosts and NSX do not have any problems.</li> </ul> |



| During Upgrade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | After Upgrade |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <ul style="list-style-type: none"> <li>During an in-place upgrade: <ul style="list-style-type: none"> <li>For VMs attached to an NSX logical switch or VMs connected to the distributed portgroup of a VDS prepared for NSX for vSphere, vmotion of VMs is not supported from and to the host on which an upgrade is in progress.</li> <li>Creating new VMs is also not supported on hosts on which an upgrade is in progress.</li> </ul> </li> </ul> <p><b>Note:</b></p> <p>NSX In-place Upgrade for vSphere Lifecycle Manager baseline clusters is supported with VMware Cloud Foundation 5.2.1 and later. For more information, refer to "Upgrade NSX for VMware Cloud Foundation 5.2.x" in the <i>VMware Cloud Foundation Lifecycle Management</i> guide.</p> <ul style="list-style-type: none"> <li>Configuration changes are allowed on NSX Manager.</li> <li>You may experience brief disruption in traffic during in-place upgrade of the hosts. For critical applications that cannot handle packet loss, maintenance mode upgrade is recommended.</li> </ul> |               |

### Limitations on In-Place Upgrade

The following considerations apply:

- For automated upgrades through SDDC Manager:
  - In-place upgrade is integrated with the live patch process in VMware Cloud Foundation lifecycle management.
  - When upgrading to an NSX hot patch or express patch, VMware Cloud Foundation will initiate a live patch when possible and trigger the in-place upgrade.
- For manual upgrades through NSX Manager:
  - In-place upgrade is supported only for ESX 8.x hosts in non-vLCM clusters.
  - When using in-place upgrade, serial upgrade within the cluster is required.
  - For vSAN clusters, NSX in-place upgrade does not check vSAN health. The absence of a check means that in the event of a failure, you may experience a breach of vSAN FTT and/or some data loss.

In-place upgrade is not supported in the following scenarios:

- More than 1000 vNICs are configured on the ESX host and the VM's vNICs connect to a single VDS. If the host has multiple VDS for NSX, this vNIC limit is per VDS.
- Layer 7 firewall rules or Identity Firewall rules are enabled.
- Service Insertion has been configured to redirect north-south traffic or east-west traffic. See *Security* in the *NSX Administration Guide* for information on uninstalling service insertion.
- A VProbe-based packet capture is in progress.
- The `nsx-cfgagent` service is not running on the host.
- IDS/IPS or distributed malware prevention is enabled for your NSX environment.

### Pre-Upgrade Tasks for Upgrading NSX

Before you upgrade NSX, perform the pre-upgrade tasks to ensure that the upgrade is successful.

- Ensure that your transport node profiles have the appropriate transport zones added to them. NSX Manager may not display the list of transport node profiles if any transport node profiles do not have transport zones added to them.
- Run the Certificate Analyzer, Results and Recovery (CARR) script. The goal of running the CARR script is to ensure that NSX certificates, such as transport node (TN) certificates, do not expire within 90 days. For more information, see Broadcom KB article [369034](#).
- Ensure that you back up the NSX Manager before you start the upgrade process. See [Backing Up and Restoring NSX Manager or Global Manager](#).
- Ensure that your host OS is supported for NSX Manager. For more information, see the [Broadcom Compatibility Guide](#).
- Disable automatic backups before you start the upgrade process. For more information, see [Configure Backups](#).
- Terminate any active SSH sessions or local shell scripts that you may have invoked on the NSX Manager or the NSX edge nodes, before you begin the upgrade process.
- Ensure that the appropriate communication ports are open from the host transport nodes and edge nodes to the NSX Manager nodes. For more information on ports, see [VMware Ports and Protocols](#).
- Delete all expired user accounts before you begin the upgrade. Upgrade for NSX on vSphere fails if your exception list for vSphere lockdown mode includes expired user accounts. If your host is part of the vLCM-enabled cluster, several users such as lldp-vim-user, nsx-user, mux-user, and da-user, are created automatically and added to the exception users list on an ESX host when NSX VIBS are installed. For more information on accounts with access privileges in lockdown mode, see *Specifying Accounts with Access Privileges in Lockdown Mode* in the *vSphere Security* Guide. For more details on these NSX user accounts on the ESX host, see [Knowledge Base article 324186](#).
- VCF 9.x does not support the multi NSX feature. Ensure that you switch off the multi NSX features for all the compute managers registered with NSX Manager. Alternatively, you can map the NSX Manager instances to individual vCenter instances before upgrading.
- Starting with VCF 9.0, a service account is created on vCenter for NSX to communicate with a registered compute manager. If you are upgrading to VCF 9.0 and any of the registered compute managers do not have a service account, a pre-check error is raised. To proceed with the upgrade, you must enable the **Create Service Account** setting on each registered compute manager.
- Ensure that you have the supported hardware version for your NSX edge VMs. For more information, see Knowledge Base article 319091: [NSX Edge VM Hardware Version upgrade](#).

### Verify the Current State of NSX

Before you begin the upgrade process, it is important to test the NSX working state. Otherwise, you cannot determine if the upgrade caused post-upgrade problems or if the problem existed before the upgrade.

### Note:

Do not assume that everything is working before you start to upgrade the NSX infrastructure.

1. Identify and record the administrative user IDs and passwords.
2. Verify that you can log in to the NSX Manager web user interface.
3. Check the **Dashboard**, system overview, host transport nodes, edge transport nodes, NSX edge cluster, HA status of the edge, and all logical entities to make sure that all the status indicators are green, deployed, and do not show any warnings.
4. Validate North-South connectivity by pinging out from a VM.
5. Validate that there is an East-West connectivity between any two VMs in your environment.
6. Record BGP states on the edge devices.
  - `get logical-routers`
  - `vrf <vrf>`
  - `get bgp`
  - `get bgp neighbor`

### Manage Unsupported Features

Certain features in the source version of NSX may not be supported in the target version. If your deployment contains an unsupported feature, make the required changes to your system configuration before starting the upgrade. The upgrade precheck step of the upgrade process returns an error if it detects an unsupported feature.

### Multi-NSX

Starting with VCF 9.0, the multi-NSX feature is not supported. Ensure that before upgrading to version 9.0 or later, you have switched off the multi-NSX feature for compute managers added to your NSX Manager, else the NSX Manager pre-check step returns an error.

Alternatively, you can map the NSX Manager instances to individual vCenter instances before upgrading.

### Compute Managers Without Service Accounts

Starting with VCF 9.0, a service account is created on vCenter for NSX to communicate with a registered compute manager. If you are upgrading to VCF 9.0 and any of the registered compute managers do not have a service account enabled, a pre-check error is thrown.

To proceed with the upgrade, you must enable the service account for those compute managers. Note that for compute managers that already have a service account feature enabled, the same service account is used post upgrade for those compute managers. The old privileges are deleted for such a service account and the predefined privileges are added.

### Manager API

Starting with VCF 9.0, the Manager APIs for logical networking and security are removed. Before proceeding with the upgrade, use the Policy promotion tool in the source NSX deployment to migrate any Manager objects in your environment to the equivalent Policy objects.

In addition, perform the following steps for specific use cases.

### Retrieve Manager IDs of logical objects

After the upgrade, Manager IDs will no longer be available. To ensure the operation of any API that requires the Manager ID of a logical object as an input, perform these steps before starting the upgrade:

1. Create the object using the equivalent Policy API.
2. Retrieve the Unique ID value of the Policy object. The Unique ID is equivalent to the Manager ID of the logical object.
3. Use the retrieved Unique ID as input to the API requiring the Manager ID.

For example, the API for transport node profile creation takes the Manager IDs of IP pool and logical switch objects as inputs. After creating the IP pool and segment objects using Policy APIs, you can input the Unique ID values of these objects to the API for transport node profile creation.

### Perform intermediate upgrade for mixed-mode configurations

A configuration that contains a combination of policy and manager objects is known as a mixed-mode configuration. Some examples of mixed-mode configurations are:

- NAT rules created through Manager APIs attached to routers created through Policy APIs
- Groups created through Policy APIs used in DFW rules created through Manager APIs

If your source deployment uses mixed-mode configurations, you must use the following workflow to upgrade to VCF 9.0 or later.

1. Upgrade the deployment to NSX 4.2.1 or later.
2. In NSX 4.2.1 or later, use the [Promotion feature](#) to promote the mixed-mode objects to policy objects.
3. Upgrade to VCF 9.0 or later.

### Convert mixed-mode IP allocations

The upgrade precheck process in the NSX Manager upgrade coordinator does not detect mixed-mode IP allocations or warn you of the presence of this unsupported condition.

Before performing the upgrade, verify whether mixed-mode IP allocations exist in your environment. If present, use the [Promotion feature](#) in the source NSX deployment to migrate the mixed-mode IP allocations to Policy equivalents.

### Replace member types related to grouping

Certain member types are deprecated or no longer supported. If your source deployment uses any of these member types, replace them with their supported equivalents which are functionally the same.

| Deprecated or Unsupported Member Type | Supported Equivalent |
|---------------------------------------|----------------------|
| IPSET (deprecated)                    | Group                |
| Logical Switch (not supported)        | Segment              |
| Logical Port (not supported)          | Segment Port         |

### Remove any remaining Manager objects after the upgrade

After the upgrade, NSX Manager will raise an alarm if it detects any pure Manager logical objects that have not been migrated to Policy equivalents.

### Run the NSX Upgrade Pre-check Bundle

The NSX upgrade pre-check bundle `.pub` allows you to run the latest upgrade pre-check operations on your deployment during the planning phase of the upgrade. These operations check for component connectivity, version compatibility, and component status among other environment readiness indicators, for your current upgrade plan.

Run the upgrade pre-checks at least one week in advance of the planned upgrade maintenance window. Running the pre-checks in advance allows you sufficient time to address and resolve issues identified by the pre-checks.

To run the NSX upgrade pre-checks, do the following:

1. Download the `.pub` upgrade pre-check bundle file.
2. Upload the `.pub` bundle file to NSX Manager.
3. Run the upgrade pre-checks in NSX Manager.

### Download the Pre-check Bundle

Ensure that you download the pre-check bundle version that corresponds to the target VCF version.

#### Note:

You can also navigate to the pre-check bundle and save the URL. When you upload to NSX Manager, paste the URL so that the pre-check bundle is uploaded from the Broadcom Support portal.

1. Locate the NSX build on the [Broadcom Support portal](#).
2. Navigate to the pre-check bundle file and click **Read More**. Verify that the pre-check bundle filename extension ends with `.pub`.

The filename will be similar to `VMware-NSX-upgrade-bundle-ReleaseNumberNSXBuildNumber.pub`.

3. Download the NSX pre-check bundle to the same system you use to access the NSX Manager user interface.

### Upload the Pre-check Bundle to NSX Manager

Uploading the pre-check bundle file upgrades the upgrade coordinator in NSX Manager to the target version so that you can verify that all system components are ready for upgrade.

1. From your browser, log in as a local admin user to the NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`. You can log in as a local admin user to any one of the NSX Manager nodes.
2. Select **System > Upgrade** from the navigation panel.
3. Click **Upgrade**.
4. Click **Browse** to navigate to the location where you downloaded the pre-check bundle `.pub` file.
5. Click **Upload**.  
When the upload process finishes, the **Prepare for Upgrade** button appears.
6. Click **Prepare for Upgrade** to upgrade the upgrade coordinator.  
The EULA appears.
7. Read and accept the EULA terms.
8. Accept the notification to upgrade the upgrade coordinator.  
Upgrading the upgrade coordinator might take 10–20 minutes, depending on your network speed. If the network times out, reload the pre-check bundle.

### Run the Upgrade Pre-checks

Run the pre-checks when you start, change, or reset your upgrade plan.

1. In the upgrade coordinator in NSX Manager, click **Run Pre-Checks**.
2. View the list of pre-checks that are performed with the API call:  
`GET https://<nsx-manager>/api/v1/upgrade/upgrade-checks-info`
3. Acknowledge the issues as required and refresh your browser if needed.
4. Resolve issues detected by the pre-checks.
  - a. In the **NSX Manager** section, click the **Pre Check Status** issues to see the issue details.
  - b. In the **Edges** section, click the **Pre Check Status** issues to see the issue details.

#### Important:

If the allocated CPU cores or memory of the edge node is less than the standard edge node form factor, an alarm is generated to warn you. In this scenario, you must acknowledge this type of alarm to proceed with the edge upgrade which will result in the upgrade coordinator changing the edge CPU cores and memory to match the standard edge node form factor. After the upgrade, these type of alarms are cleared on the next edge transport node refresh action. For more details, see [Edge Node Upgrade Process Through the Upgrade Coordinator](#).

- c. In the **Hosts** section, click the **Pre Check Status** issues to see the issue details.  
You might have to place some of the hosts in maintenance mode.

5. (Optional) Click **Download Pre-Check Results** to download a CSV file with details about pre-check errors for each component and their status.
6. (Optional) Click **View Upgrade History** to view information about previous upgrades.

### Download the NSX Upgrade Bundle

The upgrade bundle `.mub` contains all the files to upgrade the NSX infrastructure. Before you begin the upgrade process, you must download the correct upgrade bundle version.

You can also navigate to the upgrade bundle and save the URL. When you upgrade the upgrade coordinator, paste the URL so that the upgrade bundle is uploaded from the Broadcom Support portal.

1. Locate the NSX build on the [Broadcom Support portal](#).
2. Navigate to the upgrade bundle file and click **Read More**.
3. Verify that the upgrade bundle filename extension ends with `.mub`.  
The upgrade bundle filename has a format similar to `VMware-NSX-upgrade-bundle-ReleaseNumberNSXBuildNumber.mub`.
4. Download the NSX upgrade bundle to the same system you are using to access the NSX Manager user interface.

### Configure NSX Edge Failover Pre-checks

Edge failover pre-checks ensure that routing services can continue without downtime across NSX edge clusters during an upgrade. As each edge node prepares to shut down for the upgrade, pre-checks validate the availability of peer nodes in that edge cluster and alert the user if a peer is not ready for failover.

By default, edge failover pre-checks are deactivated during upgrades and you must activate them using the NSX API. You can also selectively deactivate failover pre-check warnings for specific edges, edge clusters, and hosts due to known issues in your environment.

In addition, you can use the API to run failover pre-checks prior to an upgrade, as part of your pre-upgrade planning process.

### Running Failover Pre-checks Before an Upgrade

To plan for an upgrade and identify areas of failover readiness that require attention, you can run the failover pre-checks on demand. Use the following API call and request payload.

```
POST https://<nsx-manager>/api/v1/upgrade/pre-upgrade-checks/host/planned-maintenance

{
 "vcenter-uuid": "<compute manager UUID>",
 "entity-id": "<host moref ID>"
}
```

The pre-checks will run and report any failover readiness issues in the response body.

### Failover Pre-check Workflows During an Upgrade

The failover pre-checks run during both the host upgrade and edge upgrade steps of the upgrade process.

- When the host upgrade step begins, the pre-checks consider all the peers of edges on the candidate host to be upgraded. If a peer is found not ready for failover, the upgrade pauses on the candidate host until the user makes the peer edge available for failover. After this host completes its upgrade, the failover pre-checks run sequentially on the next candidate host.
- When the edge upgrade step begins, the pre-checks validate the peers of each candidate node to be upgraded. If a peer itself has recently been upgraded and is still starting up, the upgrade process implicitly waits for the peer to become available before retrying the pre-check.

### Activate Edge Failover Pre-checks

To activate the edge failover pre-checks, use the following API call and request payload:

```
PATCH https://<NSX Manager>/policy/api/v1/system-config

{
 "keyValuePairs" : [{
 "key" : "EdgeFailoverReadiness_DisablePreChecks",
 "value" : false
 }]
}
```

By default, the `EdgeFailoverReadiness_DisablePreChecks` key is set to `true`, which deactivates all edge failover pre-checks.

### Deactivate Edge Failover Pre-checks

To deactivate some or all of the edge failover pre-checks, use the API call `PATCH https://<NSX Manager>/policy/api/v1/system-config` with a request payload consisting of key/value pairs as described in the following table.

| Key                                                       | Value                                                                                                                                                                             | Description                                                                                                  |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| EdgeFailoverReadiness_DisablePreChecks                    | true (default)                                                                                                                                                                    | Deactivates all edge failover pre-checks.                                                                    |
|                                                           | false                                                                                                                                                                             | Activates edge failover pre-checks, with the option to selectively turn off certain pre-checks.              |
| EdgeFailoverReadiness_DisablePreChecksForVmotionableEdges | false (default)                                                                                                                                                                   | Activates failover pre-checks for all edges on a given host, including edges eligible for vMotion migration. |
|                                                           | true                                                                                                                                                                              | Deactivates failover pre-checks for edges eligible for vMotion migration.                                    |
| EdgeFailoverReadiness_DisablePreCheckForEdgeClusterID     | Comma-separated list of edge cluster UUIDs                                                                                                                                        | Deactivates failover pre-checks for the specified edge clusters.                                             |
| EdgeFailoverReadiness_DisablePreCheckForEdgeNodeID        | Comma-separated list of edge node UUIDs                                                                                                                                           | Deactivates failover pre-checks for the specified edge nodes.                                                |
| EdgeFailoverReadiness_DisablePreCheckForHostNodeID        | Comma-separated list of hosts, represented in either of these formats: <ul style="list-style-type: none"> <li>Host UUID</li> <li>Compute manager UUID : Host MO-REF ID</li> </ul> | Deactivates failover pre-checks for the specified hosts.                                                     |

For example, the following request activates failover pre-checks in general, but turns off pre-checks for these entities:

- Edge node with UUID 5b21679b-87bd-49c9-80fb-a9964bab34e9
- Edge node with UUID 5b21679b-87bd-49c9-80fb-a9964bab3413
- Host with MO-REF `host-11` on vCenter with UUID b07c9a01-16eb-43fb-968a-44c57ca5e270
- Host with UUID 4ac05527-94fb-4f12-9ed6-b11fbf48a1dd

```
PATCH https://<NSX Manager>/policy/api/v1/system-config

{
 "keyValuePairs" : [{
 "key" : "EdgeFailoverReadiness_DisablePreChecks",
```

```
 "value" : false
 },
 {
 "key" : "EdgeFailoverReadiness_DisablePreCheckForEdgeNodeID",
 "value" : "5b21679b-87bd-49c9-80fb-a9964bab34e9, 5b21679b-87bd-49c9-80fb-a9964bab3413"
 },
 {
 "key" : "EdgeFailoverReadiness_DisablePreCheckForHostNodeID",
 "value" : "b07c9a01-16eb-43fb-968a-44c57ca5e270:host-11, 4ac05527-94fb-4f12-9ed6-b11fbf48a1dd"
 }
}
```

### Update the Upgrade Coordinator in NSX Manager

After you finish the prerequisites for a manual upgrade, your first step is to update the NSX Manager upgrade coordinator to initiate the upgrade process.

- Complete all the checklist tasks related to preparing for the upgrade, as described in [Upgrading NSX to 9.0](#).
- Download the upgrade pre-check bundle (.pub) and [run the upgrade pre-checks](#).
- [Download the upgrade bundle](#).
- Make necessary adjustments to your system configuration to [manage unsupported features](#). The pre-check step of the upgrade coordinator will raise an error if it detects an unsupported feature in the source deployment.

The upgrade coordinator runs in NSX Manager. It is a self-contained web application that orchestrates the upgrade process of the Management Plane, NSX edge clusters, and NSX VIBs on the host transport nodes.

The upgrade coordinator guides you through the proper upgrade sequence. You can track the upgrade process and if necessary you can pause and resume the upgrade process from the user interface.

- From your browser, log in as a local admin user to the NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`. You can log in as a local admin user to any one of the NSX Manager nodes.
- Select **System > Upgrade** from the navigation panel.
- Click **Upgrade**.
- Click **Browse** to navigate to the location where you [downloaded the upgrade bundle](#).pub file.
- Click **Upload**.

Upgrading the upgrade coordinator might take 10–20 minutes, depending on your network speed. If the network times out, reload the upgrade bundle.

When the upload process finishes, the **Prepare for Upgrade** button appears.

- Click **Prepare for Upgrade** to upgrade the upgrade coordinator.

**Note:** Do not initiate multiple simultaneous upgrade processes for the upgrade coordinator.

The EULA appears.

7. Read and accept the EULA terms.
8. Accept the notification to upgrade the upgrade coordinator.
9. Optional: If a patch release becomes available after the upgrade coordinator is updated, upload the latest upgrade bundle and upgrade the upgrade coordinator.
10. Click **Run Pre-Checks** to verify that all the NSX components are ready for upgrade.
 

**Note:** You must run the pre-checks when you change or reset your upgrade plan, or upload a new upgrade bundle.

This action checks for component connectivity, version compatibility, and component status among other environment readiness checks, for your current upgrade plan.
11. Optional: View the list of pre-checks that are performed with the API call `GET https://<nsx-manager>/api/v1/upgrade/upgrade-checks-info`.
12. Acknowledge the issues as required and refresh your browser if needed.
13. Resolve issues detected from the pre-check.
  - a) In the **NSX Manager** section, click the **Pre Check Status** issues to see the issue details.
  - b) In the **Edges** section, click the **Pre Check Status** issues to see the issue details.

**Important:**

If the allocated CPU cores or memory of the edge node is less than the standard edge node form factor, an alarm is generated to warn you. In this scenario, you must acknowledge this type of alarm to proceed with the edge upgrade which will result in the upgrade coordinator changing the edge CPU cores and memory to match the standard edge node form factor. After the upgrade, these type of alarms are cleared on the next edge transport node refresh action. For more details, see [Edge Node Upgrade Process Through the Upgrade Coordinator](#).

- c) In the **Hosts** section, click the **Pre Check Status** issues to see the issue details.

You might have to place some of the hosts in maintenance mode.

You can click **Download Pre-Check Results** to download a CSV file with details about pre-check errors for each component and their status.

14. Optional: Click **View Upgrade History** and view information about previous NSX Manager upgrades.

**Note:**

If new NSX Manager nodes are added after the upgrade coordinator is upgraded, upload the upgrade bundle to the newly added nodes and upgrade the upgrade coordinator again.

Proceed to [Upgrade the Management Plane](#).

### Upgrade the Management Plane Through NSX Manager

The upgrade sequence upgrades the Management Plane at the beginning. If required, you can continue to make configuration changes while the Management Plane upgrade is in progress.

#### Update the upgrade coordinator.

NSX supports only an odd number of nodes in a cluster. A three-node cluster is the recommended configuration for fault tolerance. NSX Manager nodes are upgraded sequentially due to the rolling upgrade feature so that two nodes are up all the time.

You can push new configurations while the NSX Manager upgrade is in progress. Do not add or delete NSX Manager nodes during an upgrade.

The upgrade process also includes taking a local backup of NSX Manager nodes that can be used to restore or roll back the system. Once all pre-checks are complete, the system saves the configuration backup followed by local backup of all the nodes in the cluster. To check if the system saved the local backup, you can go to the root admin and check the following folder:

- `/config_bak/backup/<unified_app_version>/cluster-node-backups` - if the upgrade is from NSX 4.1.0 and later versions.
- `/image/backup/<unified_app_version>/cluster-node-backups` - if the upgrade is from NSX 4.0.x versions.

The rollback backup is also saved at `/config_bak`. If the local backup fails for any reason, the upgrade is stopped.

**Important:**

If a rollback is required, roll back the NSX Manager nodes using the backup before starting the upgrade of edge nodes and host transport nodes.

If you need to roll back the system after the upgrade of edge nodes or host transport nodes has started, you must contact Broadcom Support to first roll back the upgraded edge and host transport nodes before you can roll back the NSX Manager nodes.

**Note:**

Starting with VCF 9.0, the multi-NSX feature is not supported. Ensure that before upgrading to version 9.0 or later, you have switched off the multi NSX feature for compute managers added to your NSX Manager. Alternatively, you can map the NSX Manager instances to individual vCenter instances before upgrading.

When the Management Plane upgrade begins, the upgrade coordinator pins the NSX API and NSX Manager UI to the source version. This operation ensures that your access to the NSX API and NSX Manager UI remains unchanged during the upgrade process.

**Note:**

Some API methods removed from the target version cannot be accessed during the upgrade.

Also, when performing a rolling upgrade of NSX Manager nodes from version 4.x to version 9.0 or later, you cannot create the following types of objects during the rolling upgrade period:

- Edge transport node
- Edge cluster high-availability profile
- Failure domain

The create functionality will become available after the upgrade coordinator completes the migration of logical to policy object paths in the management plane.

1. Back up the NSX Manager.

See the *NSX Administration Guide*.

2. Click **Start** to upgrade the Management plane.

3. Accept the upgrade notification.

You can safely ignore any upgrade related errors such as, HTTP service disruption that appears at this time. These errors appear because the NSX Manager node may be rebooting during the upgrade. You can continue to monitor the progress of the upgrade from the UI of any of the other NSX Manager nodes.

If you are using a Virtual IP address, the UI remains accessible but you need to re-authenticate yourself after all the nodes have been upgraded.

4. In case of upgrade errors, NSX may prompt you to roll back the upgrade. The rollback is performed on all the NSX Manager nodes:

1. Run the following command from a root shell on all the NSX Manager nodes :  
`/etc/init.d/corfu-server stop`
2. Run the following command as an admin user on all the NSX Manager nodes:  
`node-rollback run-step step1_start_rollback`
3. Run the following command as an admin user on any one of the NSX Manager nodes:  
`node-rollback run-step step2_restore_data`  
If you encounter any errors, perform the troubleshooting steps described later on this page.
4. Run the following command as an admin user on all the NSX Manager nodes:  
`node-rollback run-step step3_exit_rollback`

The rollback applies only to the Management Plane upgrade. Your NSX Manager nodes return to the version prior to starting the Management Plane upgrade.

Proceed to [upgrade the edge clusters](#).

After you upgrade the Management Plane, you can join the Customer Experience Improvement Program (CEIP) for NSX. See *Customer Experience Improvement Program* in the *NSX Administration Guide* for more information, including how to join or leave the program.

#### Troubleshooting the Management Plane rollback

If you encounter errors during a rollback, perform these troubleshooting steps:

1. Execute `corfu_tool_runner` to delete the record from the registry table:  
`/opt/vmware/bin/corfu_tool_runner.py -t RegistryTable -n CorfuSystem --port 9000 -o deleteRecord --key-ToDelete="{\"namespace\": \"CorfuSystem\", \"tableName\": \"CompactionControlsTable\"}"`
2. Run compaction verification:  
`/opt/vmware/bin/corfu_compactor_upgrade_runner.py --runs 3 --lock false`
3. Stop `corfu-server` on all NSX Manager nodes.
4. Start `corfu-server` on all NSX Manager nodes.
5. Resume with the restore command as an admin user on any one of the NSX Manager nodes:  
`node-rollback run-step step2_restore_data`

#### Upgrade the NSX Edge Cluster Through NSX Manager

After the Management Plane upgrade is complete, the NSX Manager upgrade coordinator upgrades the NSX edge cluster based on the ordering of upgrade unit groups that you have defined. Upgrade unit groups consist of edge nodes from the same cluster. You can reorder upgrade unit groups and add or remove an upgrade unit group from the upgrade sequence.

- Verify that the edge nodes are members of an NSX edge cluster.  
Any edge nodes that are not members of an NSX edge cluster are flagged during the upgrade coordinator pre-check and prevent the upgrade from proceeding.
- Familiarize yourself with the upgrade impact during and after the edge cluster upgrade. See [Operational Impact of the Upgrade by Using Upgrade Coordinator](#).
- [Upgrade the Management Plane](#).

#### Note:

You cannot move an edge node from one upgrade unit group to another because the upgrade unit group membership adheres to the edge cluster membership before the upgrade.

The edge nodes are upgraded in serial mode so that when the upgrading node is down, the other nodes in the edge cluster remain active to continuously forward traffic.

The maximum limit of simultaneous upgrade of Edge upgrade unit groups is twenty.

#### Note:

NSX edge virtual machine hardware will be upgraded to version vmx-20 during the edge upgrade.

1. Enter the edge cluster upgrade plan details.

| Option                                       | Description                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Serial</b>                                | Upgrade all the Edge upgrade unit groups consecutively.<br>This menu item is selected by default. This selection is applied to the overall upgrade sequence.                                                                                                                                                                   |
| <b>Parallel</b>                              | Upgrade all the Edge upgrade unit groups simultaneously.<br>For example, if the overall upgrade is set to the parallel order, the Edge upgrade unit groups are upgraded together and the edge nodes are upgraded one at a time.<br><b>Note:</b> The maximum limit of simultaneous upgrade of Edge upgrade unit groups is five. |
| <b>When an upgrade unit fails to upgrade</b> | Selected by default so that you can fix an error on the edge node and continue the upgrade.<br>You cannot deselect this setting.                                                                                                                                                                                               |
| <b>After each group completes</b>            | Select to pause the upgrade process after each Edge upgrade unit group finishes upgrading.                                                                                                                                                                                                                                     |

2. Optional: Reorder the upgrade sequence of an Edge upgrade unit group.

For example, if you configure the overall group upgrade as serial, you can reorder the Edge upgrade unit groups serving internal networks or Edge upgrade unit groups interfacing with external networks to be upgraded first.

**Note:** It is recommended to follow an order so that critical standby nodes are upgraded before active nodes. This order allows only one failover to occur during upgrade for critical nodes.

- a) Select the Edge upgrade unit group and click the **Actions** tab.
- b) Select **Reorder** from the drop-down menu.
- c) Select **Before** or **After** from the drop-down menu.
- d) Click **Save**.

You can also do this with the following REST API. Reorder an upgrade unit within the upgrade unit group by placing it before/after the specified upgrade unit.

| METHOD  | POST                                                                                                                                        |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI     | <code>https://&lt;nsx-mgr&gt;/api/v1/upgrade/upgrade-unit-groups/&lt;groupId&gt;/upgrade-unit/&lt;upgrade-unit-id&gt;?action=reorder</code> |
| Payload | <code>{ \"id\": \"&lt;upgrad-uniti-id&gt;\", \"is_before\": \"false\" }</code>                                                              |

3. Optional: Click **Reset** to revert to the default state.

**CAUTION:** After reset, you cannot restore your previous configuration. Also, a reset causes the Edge upgrade unit group order to allow Edge nodes with standby logical routers to be upgraded first and followed by Edge nodes with active logical routers. This order reduces failovers.



4. Click **Start** to upgrade the edge cluster.

5. Monitor the upgrade process.

You can view the overall upgrade status and progress details of each Edge upgrade unit group. The upgrade duration depends on the number of Edge upgrade unit groups you have in your environment.

You can pause the upgrade to configure the Edge upgrade unit group that is not upgraded and restart the upgrade.

6. Click **Run Post Checks** to verify whether the Edge upgrade unit groups were successfully upgraded.

If some Edge upgrade unit groups failed to upgrade, resolve the errors.

7. Optional: In the NSX Manager, select **System > Overview** and verify that the product version is updated on each edge node.

If the process is successful, you can proceed to [upgrade vCenter and ESX hosts through SDDC Manager](#). Alternatively, you can use NSX Manager to [upgrade only the NSX VIBs on the hosts](#).

If there are upgrade errors, you must resolve the errors. See [Troubleshooting Upgrade Failures](#).

### NSX Edge Node Upgrade Process Through the Upgrade Coordinator

The upgrade process for edge cluster nodes depends on how and when the edge nodes were deployed.

#### Upgrade Process for Auto Deployed Edge Nodes

For auto deployed edge nodes managed by NSX Manager, the upgrade coordinator checks the edge node configuration settings and updates them if the settings do not match the standard values from the current NSX version.

When an NSX version introduces an improvement to the edge node configuration, that improvement only applies to edge nodes deployed after the upgrade. Edge nodes deployed prior to the NSX upgrade are not automatically updated. You can use the upgrade coordinator to update auto deployed edge nodes to the latest configuration settings.

The upgrade workflow for auto deployed edge nodes is as follows:

1. The upgrade bundle file is uploaded and prepared by the upgrade coordinator.
2. The edge node is placed into maintenance mode.
3. The OS is downloaded to the edge node.
4. The OS is installed on the edge node.
5. The OS switch is performed on the edge node.
6. Using vSphere APIs, the edge node virtual machine is powered off.

#### Note:

VMware vCenter should be running and reachable during this workflow.

7. Upgrade coordinator updates the edge node virtual machine configuration parameters and the allocation of CPU cores and memory.

The following edge virtual machine configuration attributes are added or updated before powering on an edge virtual machine in the operating system:

— VMX file properties:

```
"ethernet0.ctxPerDev": "3",
"ethernet1.ctxPerDev": "3",
"ethernet2.ctxPerDev": "3",
"ethernet3.ctxPerDev": "3",
"ethernet4.ctxPerDev": "3",
"ethernet0.udpRSS": "1",
"ethernet1.udpRSS": "1",
"ethernet2.udpRSS": "1",
"ethernet3.udpRSS": "1",
```

```
"ethernet4.udpRSS": "1",
"ethernet0.pnicFeatures": "4",
"ethernet1.pnicFeatures": "4",
"ethernet2.pnicFeatures": "4",
"ethernet3.pnicFeatures": "4",
"ethernet4.pnicFeatures": "4",
"featMask.vm.cpuid.PDPE1GB": "Val:1",
"snapshot.maxSnapshots": "0"
```

— OVF file property:

```
"memoryReservationLockedToMax": "true"
```

— CPU

— Memory

For the allocation of CPU cores, memory, and storage, the upgrade coordinator changes the edge node values to match the standard form factor values.

**Note:** The edge node CPU core and memory values are only changed to match the form factor values if the existing edge node values are less than the form factor values. If the edge node values are greater than the form factor values, then no changes are made. Ensure that there are sufficient resources in the vSphere resource pool and cluster.

Table 819: Supported Edge Form Factors for Upgrade

| Edge Form Factor   | Memory | CPU Cores | Disk Space |
|--------------------|--------|-----------|------------|
| Small Form Factor  | 4 GB   | 2         | 200 GB     |
| Medium Form Factor | 8 GB   | 4         | 200 GB     |
| Large Form Factor  | 32 GB  | 8         | 200 GB     |
| Extra Large Form   | 64 GB  | 16        | 200 GB     |

8. The edge node virtual machine is powered on.
9. Check for required reboot if there is a GRUB update, and if required, the OS is rebooted into the updated GRUB.
10. Upgrade coordinator verifies edge node version.
11. Users from the old OS are migrated to the edge node.
12. The edge node exits maintenance mode.
13. The upgrade process is completed and the edge node upgrade is marked as successful.

#### Upgrade Process for Manually Deployed Edge Nodes

For manually deployed edge nodes, the upgrade coordinator only updates the OS and does not change the edge parameters or CPU core and memory values.

The upgrade workflow for manually deployed edge nodes is as follows:

1. The upgrade bundle file is uploaded and prepared by the upgrade coordinator.
2. The edge node is placed into maintenance mode.
3. The OS is downloaded to the edge node.
4. The OS is installed on the edge node.
5. The OS switch is performed on the edge node.
6. The edge node reboots into the new OS.
7. Upgrade coordinator verifies edge node version.
8. The edge node exits maintenance mode.

**Important:** Before upgrading manually deployed edge nodes, it is important to set the VMX file property "featMask.vm.cpuid.PDPE1GB": "Val:1" of the edge node virtual machine in order to prevent upgrade failure.

## Upgrade NSX VIBs on Hosts Through the Upgrade Coordinator

You can upgrade the NSX VIBs on hosts in your environment using the upgrade coordinator.

- Verify that you have configured the overall hosts upgrade plan. See [Configure Hosts for a Manual Upgrade](#).
- If the ESX hosts are part of a disabled DRS cluster or are standalone hosts, verify that they are placed in maintenance mode.
- For ESX hosts that are part of a fully automated DRS cluster, if the host is not in maintenance mode, the upgrade coordinator requests the host to be put in maintenance mode. vSphere DRS migrates the VMs to another host in the same cluster during the upgrade and places the host in maintenance mode.
- For ESX host, for an in-place upgrade you do not need to power off the tenant VMs.
- For a stateless ESX host, log in to vCenter and update the ESX image with the NSX kernel modules.

### Note:

The upgrade coordinator only upgrades the NSX VIBs on hosts. To [upgrade ESX on hosts](#), you must use SDDC Manager instead.

1. Click **Start** to upgrade the hosts.
2. Click **Refresh** and monitor the upgrade process.

You can view the overall upgrade status and specific progress of each host upgrade unit group. The upgrade duration depends on the number of host upgrade unit groups you have in your environment.

Wait until the in progress upgrade units are successfully upgraded. You can then pause the upgrade to configure the host upgrade unit group that is not upgraded and resume the upgrade.

3. Click **Run Post Checks** to make sure that the upgraded hosts and NSX do not have any problems.

### Note:

If a host upgrade unit failed to upgrade and you removed the host from NSX, refresh the upgrade coordinator to view all the successfully upgraded host upgrade units.

If a host fails during the upgrade, reboot the host and try the upgrade again.

4. After the upgrade is successful, verify that the latest version of NSX packages is installed on the vSphere hosts.  
For vSphere hosts, enter `esxcli software vib list | grep nsx`
5. Power on the tenant VMs of standalone ESX hosts that were powered off before the upgrade.
6. Migrate the tenant VMs on hosts managed by vCenter that are part of the enabled DRS cluster to the appropriate host.
7. Power on or return the tenant VMs of ESX hosts that are part of a disabled DRS cluster that were powered off before the upgrade.

Proceed to [Finalize Upgrade](#).

You can finalize the upgrade only after the upgrade process finishes successfully. If some of the hosts are disabled, you must enable and upgrade them before you proceed.

If there are upgrade errors, you must resolve the errors. See [Troubleshooting Upgrade Failures](#).

### Configure Hosts for a Manual Upgrade

When performing a manual upgrade through the upgrade coordinator in NSX Manager, you can customize the upgrade sequence of the hosts, exclude certain hosts from the upgrade, or pause the upgrade at various stages of the upgrade process.

- If the ESX hosts are part of a disabled DRS cluster or are standalone hosts, verify that they are placed in maintenance mode.

- For ESX hosts that are part of a fully automated DRS cluster, if the host is not in maintenance mode, the upgrade coordinator requests the host to be put in maintenance mode. vSphere DRS migrates the VMs to another host in the same cluster during the upgrade and places the host in maintenance mode.
- For ESX host, for an in-place upgrade you do not need to power off the tenant VMs.
- Verify that the transport zone or transport node N-VDS name does not contain spaces.  
If there are spaces, create a transport zone with no spaces in the N-VDS name. You must reconfigure all the components that are associated with the old transport zone to use the new transport zone and delete the old transport zone.
- Verify that your vSAN environment is in good health before you use the in-place upgrade mode.

See the *Place a Host in Maintenance Mode* section of the *vSphere Resource Management* guide.

All the existing standalone ESX hosts, vCenter, and managed ESX hosts are grouped in separate host upgrade unit groups by default.

Before you upgrade the hosts, you can select to update the hosts in parallel or serial mode. The maximum limit for a simultaneous upgrade is twenty host upgrade unit groups and five hosts per group.

### Note:

Host upgrade unit group with hosts that belong to the same vCenter cluster can be upgraded serially.

You can customize the host upgrade sequence before the upgrade. You can edit a host upgrade unit group to move a host to a different host upgrade unit group that upgrades immediately and another host to a host upgrade unit group that upgrades later. If you have a frequently used host, you can reorder the host upgrade sequence within a host upgrade unit group so it is upgraded first and move the least used host to upgrade last.

1. Enter the host upgrade plan details.

You can configure the overall group upgrade order to set the host upgrade unit groups to be upgraded first.

| Option                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Serial</b>                                | Upgrade all the host upgrade unit groups consecutively.<br><br>This menu item is selected by default and applied to the overall upgrade sequence. This selection is useful to maintain the step-by-step upgrade of the host components.<br><br>For example, if the overall upgrade is set to serial and the host upgrade unit group upgrade is set to parallel, the host upgrade unit group is upgraded one after the other. The hosts within the group are updated in parallel. |
| <b>Parallel</b>                              | Upgrade all the host upgrade unit groups simultaneously.<br><br>You can upgrade up to five hosts simultaneously.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>When an upgrade unit fails to upgrade</b> | Select to pause the upgrade process if any host upgrade fails.<br><br>This selection allows you to fix the error on the host upgrade unit group and resume the upgrade.                                                                                                                                                                                                                                                                                                          |
| <b>After each group completes</b>            | Select to pause the upgrade process after each host upgrade unit group finishes upgrading.                                                                                                                                                                                                                                                                                                                                                                                       |

2. Optional: Change the host upgrade unit group upgrade order.

If you configure the overall group upgrade in the serial order, the upgrade waits for a host upgrade unit group upgrade to finish before proceeding to upgrade the second host upgrade unit group. You can reorder the host upgrade unit group upgrade sequence to set a host upgrade unit group to upgrade first.

- a) Select the host upgrade unit group and click the **Actions** tab.
- b) Select **Reorder** from the drop-down menu.
- c) Select **Before** or **After** from the drop-down menu.

3. Optional: Remove a host upgrade unit group from the upgrade sequence.

- a) Select the host upgrade unit group and click the **Actions** tab.
- b) Select **Change State** from the drop-down menu.
- c) Select **Disabled** to remove the host upgrade unit group.

4. Optional: Change the individual host upgrade unit group upgrade sequence.

By default, the upgrade sequence is set to the parallel order.

- a) Select the host upgrade unit group and click the **Actions** tab.
- b) Select **Change Upgrade Order** from the drop-down menu.
- c) Select **Serial** to change the upgrade sequence.

5. Optional: Change the host upgrade unit group upgrade mode.

- Select **Maintenance** mode.

For standalone ESX hosts and ESX hosts that are part of a disabled DRS cluster, place the hosts into maintenance mode.

For ESX hosts that are part of a fully automated DRS cluster, if the host is not in maintenance mode, the upgrade coordinator requests the host to be put in maintenance mode. vSphere DRS migrates the VMs to another host in the same cluster during the upgrade and places the host in maintenance mode.

- Select **In-place** mode to avoid powering off and placing a host into maintenance mode before the upgrade.

For standalone ESX hosts and ESX hosts that are part of a disabled DRS cluster, you do not need to place the hosts into maintenance mode.

For ESX hosts that are part of a fully automated DRS cluster, you do not need to place the host into maintenance mode.

**Note:** During upgrade the host might experience a packet drop in the workload traffic.

- Use the API call PUT `https://<nsx-manager>/api/v1/upgrade/upgrade-unit-groups/<group-id>` and enable the upgrade coordinator to restart the ESX host.

The `rebootless_upgrade:true` parameter states that after the ESX host upgrade, the host is not rebooted.

By default, the upgrade coordinator does not restart the ESX host. This mode is used for troubleshooting purposes.

- Use the API call PUT `https://<nsx-manager>/api/v1/upgrade/upgrade-unit-groups/<group-id>` and upgrade vCenter-managed ESX hosts that are part of a DRS cluster with vSAN configured.

The `ensure_object_accessibility` parameter requires vSAN to assume control of data accessibility while a vCenter-managed ESX host that is part of a DRS cluster is placed in maintenance mode for the upgrade.

The `evacuate_all_data` parameter requires vSAN to take all the data from a vCenter-managed ESX host that is part of a DRS cluster to another managed ESX host that is part of a DRS cluster while placed in maintenance mode for the upgrade.

The `no_action` parameter requires vSAN to take no action while the vCenter-managed ESX host that is part of a DRS cluster is placed in maintenance mode for the upgrade.

For more information about the parameters, see the *Update the upgrade unit group* section of the *NSX API Guide*.

6. For vSphere Lifecycle Manager-enabled clusters, select one of the following options:

- **NSX only Upgrade:** Use this option if you want to upgrade only NSX. The Upgrade Coordinator runs the entire upgrade process including the remediation of hosts.
- **Stage in vSphere Lifecycle Manager:** Use this option if you want to upgrade NSX along with ESX hosts and other solutions. You need to remediate the hosts using vSphere Lifecycle Manager. After remediation, you can monitor the upgrade from the Upgrade Coordinator.

7. Click **Reset** to discard your custom upgrade plan and revert to the default state.

**CAUTION:** You cannot restore your previous upgrade configuration.

If you register a new host transport node during the upgrade, you must click **Reset** to view the status of the recently added host and to continue the upgrade process.

Determine whether to add, edit, or delete host upgrade unit groups or to upgrade host upgrade unit groups. See [Manage Host Upgrade Unit Groups](#) or [Upgrade NSX VIBs on Hosts Through the Upgrade Coordinator](#).

### Manage Host Upgrade Unit Groups

You can edit and delete an existing host upgrade unit group before you start the upgrade or after you pause the upgrade.

- Verify that you have configured the overall hosts upgrade. See [Configure Hosts for a Manual Upgrade](#).
- If the ESX hosts are part of a disabled DRS cluster or are standalone hosts, verify that they are placed in maintenance mode.
- For ESX hosts that are part of a fully automated DRS cluster, if the host is not in maintenance mode, the upgrade coordinator requests the host to be put in maintenance mode. vSphere DRS migrates the VMs to another host in the same cluster during the upgrade and places the host in maintenance mode.
- For ESX host, for an in-place upgrade you do not need to power off the tenant VMs.

Hosts in a ESX cluster appear in one host upgrade unit group in the upgrade coordinator. You can move these hosts from one host upgrade unit group to another host upgrade unit group.

**Note:** If any of the hosts are part of a vSAN enabled cluster, retain the default upgrade unit groups without recreating any groups.

## 1. Create a host upgrade unit group.

- Click **Add** to include existing hosts into a host upgrade unit group.
- Toggle the **State** button to enable or disable the host upgrade unit group from the upgrade.
- Select an existing host and click the arrow icon to move that host to the newly created host upgrade unit group.  
If you select an existing host that was part of a host upgrade unit group, the host is moved to the new host upgrade unit group.
- Select whether to upgrade the host upgrade unit group in parallel or serial mode.
- Select the upgrade mode.  
See step 5 of [Configure Hosts for a Manual Upgrade](#).
- Optional: Select **Reorder** from the drop-down menu to reposition the host upgrade unit groups.
- Optional: Select **Before** or **After** from the drop-down menu.

## 2. Move an existing host to another host upgrade unit group.

If an enabled DRS ESX cluster is part of the upgrade, then a host upgrade unit group is created for the hosts managed by this cluster.

- Select a host upgrade unit group.
- Select a host.
- Click the **Actions** tab.
- Select **Change Group** from the drop-down menu to move the host to another host upgrade unit group.
- Select the host upgrade unit group name from the drop-down menu to move the host to.
- Optional: Select **Reorder** from the drop-down menu to reposition the host within the host upgrade unit group.
- Optional: Select **Before** or **After** from the drop-down menu.

## 3. Delete a host upgrade unit group.

You cannot delete a host upgrade unit group that has hosts. You must first move the hosts to another group.

- Select the host upgrade unit group.
- Select a host.
- Click the **Actions** tab.
- Select **Change Group** from the drop-down menu to move the host to another host upgrade unit group.
- Select the host upgrade unit group name from the drop-down menu to move the host to.
- Select the host upgrade unit group you want to remove and click **Delete**.
- Accept the notification.

Upgrade the newly configured hosts. See [Upgrade NSX VIBs on Hosts Through the Upgrade Coordinator](#).

**Upgrade vCenter and ESX Hosts Through SDDC Manager**

After upgrading all the edge clusters, you can use SDDC Manager to upgrade vCenter and the ESX hosts.

[Upgrade all the edge transport nodes](#). You cannot proceed with ESX host upgrades in SDDC Manager until all the edge nodes have finished upgrading.

The upgrade tool in SDDC Manager guides you through the process of upgrading vCenter and the ESX hosts. For detailed instructions, see the following:

- [Upgrade to vCenter 9.0 for VMware Cloud Foundation](#)
- [Upgrade to ESX 9.0 for VMware Cloud Foundation](#)

Starting with VMware Cloud Foundation 9.0, the NSX VIBs are packaged as part of the ESX image by default. Therefore, the upgrade of these VIBs occurs as part of the ESX host upgrade process performed through SDDC Manager.

The upgrade coordinator in NSX Manager reflects the status of the host upgrade performed through SDDC Manager.

**Download the Consistency Check Report**

The optional Consistency Check Report allows you to view a list of errors and warnings that occurred during the upgrade so that you can perform remediation actions as needed.

**Note:**

The optional Consistency Check Report is different from the mandatory pre-check operations that run at the start of the upgrade workflow.

The upgrade coordinator in NSX Manager continuously monitors the system during the upgrade process for indicators of possible failure, such as:

- Policy objects that have not upgraded successfully
- Unusual increase in traffic drops in a cluster, indicating a possible failure in the cluster
- Unusual decrease in forwarding throughput in a cluster, indicating a possible failure in the cluster

You can retrieve the Consistency Check Report during or after the upgrade to view the list of possible failures. To retrieve the report, do one of the following:

- In the upgrade coordinator in NSX Manager, click the **Consistency Check Report** button to download the `.csv` report file.
- Call the consistency check API:

```
GET https://{nsx-mgr}/api/v1/upgrade/consistency-report?component_type={MP, EDGE, HOST, ALL}
```

For more information, see the [NSX API Guide](#).

**Finalize Upgrade Through NSX Manager**

As the last step in the manual upgrade process, you must finalize the upgrade.

Verify that you have upgraded all of the following components:

- [Management Plane](#)
- All [edge nodes](#)
- All host transport nodes, either by upgrading [both the NSX VIBs and ESX](#) through SDDC Manager or by upgrading [only the NSX VIBs](#) through NSX Manager.

After the upgrades of the Management Plane, edge transport nodes, and hosts are complete, the upgrade coordinator in NSX Manager finalizes the upgrade by unpinning the NSX API and NSX Manager UI from the source version. This operation enables your access to the target version of the NSX API and NSX Manager UI.

The upgrade coordinator also validates that any vSphere Supervisor cluster running on the NSX instance is compatible with the target VCF version. If the validation fails, the Finalize Upgrade step pauses until you have upgraded vSphere Supervisor to a compatible version (for example, vSphere Supervisor 9.0.0.0 is compatible with VCF 9.0.0.0). This validation applies only to vSphere Supervisor, not to any other type of Kubernetes cluster running on the NSX instance.

In the upgrade coordinator in NSX Manager, click **Finalize Upgrade**.

**Note:**

The upgraded NSX functionality does not become available until after the Finalize Upgrade step succeeds.

- Check the cluster status and verify that the services have started from the NSX Manager user interface.
- [Verify the upgrade](#) or [troubleshoot upgrade failures](#), depending on the upgrade status.

**Verify the NSX Upgrade**

After you upgrade NSX, you can verify whether the versions of the upgraded components have been updated. For more information on the NSX Manager, see "Overview of the NSX Manager" in the *NSX Administration Guide*.

Perform a successful upgrade.

- From your browser, log in as a local admin user to an NSX Manager at <https://nsx-manager-ip-address/login.jsp?local=true>.
- Select **System > Upgrade**.
- Verify that the overall upgrade version, component version, and initial and target product version are accurate.
  - Optional: Verify that the Dashboard, fabric hosts, edge cluster, transport nodes, and all logical entities status indicators are green, normal, deployed, and do not show any warnings.
  - Optional: Verify the status of several components.
    - Fabric nodes installation
    - Transport node Local Control Plane (LCP) and Management plane agent connectivity
    - Routers connectivity
    - NAT rules
    - DFW rules
    - DHCP lease
    - BGP details
    - Flows in the IPFIX collector
    - TOR connectivity to enable the network traffic

The status of the upgrade appears as Successful.

**Note:**

If you upgraded from NSX 4.1.x or earlier, you will see new [system-generated transport zones](#) in addition to the transport zones originally configured in your source deployment.

- For clusters prepared for networking and security, you will see new security transport zones. The system generates these security transport zones by default.
- For clusters prepared for NSX on DVPs, you will see new VLAN transport zones. The system generates these VLAN transport zones by default.

- Modify the default admin password expiration.

If the password expires, you will be unable to log in and manage components. Additionally, any task or API call that requires the administrative password to execute will fail. By default, passwords expire after 90 days. For more information, see Knowledge Base article 338942: [NSX admin password expired](#).

- Reset the expiration period.

You can set the expiration period for between 1 and 9999 days.

```
nsxcli set user admin password-expiration <1 - 9999>
```

- Optional: You can disable password expiry so the password never expires.

```
nsxcli clear user audit password-expiration
```

- If you have VIDM enabled, access your the local account at <https://nsx-manager-ip-address/login.jsp?local=true>.
- Verify CPU and Memory values for edge VMs.

After upgrading, log in to the vSphere Client to verify if your existing edge VMs are configured with the following CPU and Memory values. If they are not, edit the VM settings to match these values.

| NSX Edge Form Factor | Memory | vCPU |
|----------------------|--------|------|
| NSX Edge Small VM    | 4 GB   | 2    |
| NSX Edge Medium VM   | 8 GB   | 4    |
| NSX Edge Large VM    | 32 GB  | 8    |

- (For upgrades from NSX 4.1.x or earlier) Create a new certificate for the NSX cluster VIP, either self-signed or signed by a third-party Certificate Authority, that includes the Subject Alternative Name and Subject Alternative IP for all NSX Manager nodes in the cluster.

This step ensures a successful repository sync for any NSX Manager node that is redeployed in the future. For more information, see KB article 395087: [After redeploying NSX Manager, repo sync is in failed status](#).

- [Download the Consistency Check Report](#) and remediate any failures detected during the upgrade.

### Troubleshooting Upgrade Failures

You can review the support bundle log messages to identify the upgrade problem.

You can also perform any of the following debugging tasks.

- Log in to the NSX Manager CLI as root user and navigate to the upgrade coordinator log files `/var/log/upgrade-coordinator/upgrade-coordinator.log`.
- Navigate to the system log files `/var/log/syslog` or API log files `/var/log/proton/nsxapi.log`.
- Configure a remote logging server and send log messages for troubleshooting. See the *NSX Administration Guide*.

**Note:**

If you are unable to troubleshoot the failure and want to revert to the previous working version of NSX, contact Broadcom support.

### Collect Support Bundles

You can collect support bundles on registered cluster and fabric nodes and download the bundles to your machine or upload them to a file server.

If you choose to download the bundles to your machine, you get a single archive file consisting of a manifest file and support bundles for each node. If you choose to upload the bundles to a file server, the manifest file and the individual bundles are uploaded to the file server separately.

- From your browser, log in as a local admin user to an NSX Manager at <https://nsx-manager-ip-address/login.jsp?local=true>.
- Select **System > Support Bundle**.
- Select the target nodes.
 

The available types of nodes are **Management Nodes**, **Edges**, and **Hosts**.
- Optional: Specify log age in days to exclude logs that are older than the specified number of days.
- Optional: Toggle the switch that indicates whether to include or exclude core files and audit logs.
 

**Note:** Core files and audit logs might contain sensitive information such as passwords or encryption keys.
- Optional: Select the check box to upload the bundles to a remote file server.
- Click **Start Bundle Collection** to start collecting support bundles.
 

Depending on how many log files exist, each node might take several minutes.
- Monitor the status of the collection process.
 

The status tab shows the progress of collecting support bundles.
- Click **Download** to download the bundle if the option to send the bundle to a file remote server was not set.

The bundle collection may fail for a manager node if there is not enough disk space. If you encounter an error, check whether older support bundles are present on the failed node. Log in to the NSX Manager UI of the failed manager node using its IP address and initiate the bundle collection from that node. When prompted by the NSX Manager, either download the older bundle or delete it.

### Upgrade Fails Due to a Timeout

An event during the upgrade process fails and the message from the Upgrade Coordinator indicates a timeout error.

During the upgrade process, the following events might fail because they do not complete within a specific time. The Upgrade Coordinator reports a timeout error for the event and the upgrade fails.

| Event                                               | Timeout Value |
|-----------------------------------------------------|---------------|
| Putting a host into maintenance mode                | 4 hours       |
| Waiting for a host to reboot                        | 32 minutes    |
| Waiting for the NSX service to be running on a host | 13 minutes    |

- For the maintenance mode issue, log in to vCenter and verify the status of tasks related to the host. Resolve any problems.
- For the host reboot issue, check the host to see why it failed to reboot.
- For the NSX service issue, log in to the NSX Manager UI, select **System > Appliances** and see if the host has an installation error. If so, you can resolve it from the NSX Manager UI. If the error cannot be resolved, you can refer to the upgrade logs to determine the cause of the failure.

### Upgrade Fails Due to Insufficient Space in Bootbank on ESX Host

NSX upgrade might fail if there is insufficient space in the bootbank or in the alt-bootbank on an ESX host.

Unused VIBs on the ESX host might be relatively large in size and therefore use up significant disk space. The unused VIBs can result in insufficient space in the bootbank or in the alt-bootbank during upgrade.

Uninstall the VIBs that are no longer required and free up additional disk space.

For more information on locating and deleting VIBs, see [Knowledge Base article 324231](#).

### Failure to Upload the Upgrade Bundle

The upgrade bundle fails to upload because of insufficient disk space.

- In the NSX Manager CLI, delete the unused files located at `/image/vmware/nsx/file-store/*` and `/image/core/*`.  
**Note:** Ensure that you do not delete the `/image/upgrade-coordinator-tomcat` folder or other folders located at `/image`.
- From your browser, log in as a local admin user to an NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`.
- Select **System > Support Bundle** and delete any unused support bundles.
- Delete TAR files that contain PCAP files.
- Reupload the upgrade bundle and continue with the upgrade process.

### Backup and Restore During Upgrade

The Management Plane stops responding during the upgrade process and you need to restore a backup that was taken while the upgrade was in progress.

The Upgrade Coordinator has been upgraded and the Management Plane stops responding. You have a backup that was created while the Management Plane upgrade was in progress.

You can restore the system from the local backup taken by the upgrade process just before upgrading the first NSX Manager. The local backup is available at `/storage_location/backup/unified_app_version/cluster-node-backups` on all manager nodes. Note that the `storage_location` is `/image` if the upgrade is from NSX 4.0.x versions. For upgrades from NSX 4.1.0 and later versions, the `storage_location` is `/config_bak`.

If you have deployed a new NSX Manager node in the source version, upload the upgrade bundle (.mub file) and update the upgrade coordinator before restoring the backup.

If you want to use the local backup for restore, call BCM GSS for assistance. Copy the backup file from NSX Manager to an SFTP location and then perform the following steps to restore the system.

- Log in to NSX Manager as a root user.
- Change the directory to the storage location.  
`cd /image` - if upgrading from NSX 4.0.x versions.  
`cd /config_bak` - if upgrading from NSX 4.1.0 and later versions.
- Run the following command to copy the backup file to an SFTP server.  
`scp -rp backup/<unified_app_version>/* user@<SFTP server IP address>:<backup_path>`
- Run the following command to view the generated passphrase.  
`cat .backup_keystore/.keyfile`
- Select the passphrase to copy and save it at a secure location.

While copying the file, you must maintain the same directory structure on the SFTP location as on NSX Manager.

Now, perform the following steps to restore the system.

- Deploy your Management Plane node with the same IP address that the backup was created from.
- Upload the upgrade bundle that you used at the beginning of the upgrade process.
- Upgrade the Upgrade Coordinator.
- Restore the backup taken during the upgrade process.
- Upload a new upgrade bundle if necessary.
- Continue with the upgrade process.

### Loss of Controller Connectivity after Host Upgrade

Controller connectivity is lost after you upgrade your hosts.

After upgrading your host, when running post checks, your **Note Status** shows loss of connectivity to the controller.

- Open an SSH session to the ESX host experiencing the issue and confirm that none of the three NSX controllers are in a connected state. Run the `nsxcli -c get controllers` command.

Example response:

| Controller IP | Port | SSL     | Status       | Is Physical Master | Session State | Controller FQDN |
|---------------|------|---------|--------------|--------------------|---------------|-----------------|
| 192.168.60.5  | 1235 | enabled | disconnected | true               | down          | nsxmgr.corp.com |

In a working configuration, two controllers display the not used status and one controller has the connected status. If the NSX Controller shows connected, refresh the UI and confirm that the status is green. If the controller shows not connected, continue to the next step.

- Open an SSH session to one of the NSX Manager nodes as admin and run the `get certificate api thumbprint` command.

The command output is a string of alphanumeric numbers that is unique to this NSX Manager.

- On the ESX host, push the host certificate to the Management Plane:

```
ESX1> nsxcli -c push host-certificate <NSX Manager IP or FQDN> username admin thumbprint <thumbprint obtained in step #1>
```

When prompted, enter the admin user password for the NSX Manager. See the *NSX Command-Line Interface Reference* for more information.



4. Confirm the controller status is connected.

```
ESX1> nsxcli -c get controllers
```

Confirm the controller connection state is green on the UI for this Transport Node.

If this issue continues, restart the following NSX services on the ESX host:

```
ESX1> /etc/init.d/nsx-opsagent restart
ESX1> /etc/init.d/nsx-proxy restart
```

Upgrade Coordinator User Interface is Inaccessible

The Upgrade Coordinator user interface may not be accessible.

You cannot access the Upgrade Coordinator user interface or APIs.

Internal service dependencies may cause the Upgrade Coordinator user interface to become inaccessible.

Run the following command to restart the Upgrade Coordinator service:

```
restart service install-upgrade
```

Upgrade to vCenter 9.0 for VMware Cloud Foundation

The upgrade bundle for VMware vCenter is used to upgrade the vCenter instances managed by SDDC Manager. Upgrade vCenter in the management domain before upgrading vCenter in workload domains.

- Download the VMware vCenter upgrade bundle. See [Downloading VMware Cloud Foundation Upgrade Bundles for Core Components](#).
  - Take a file-based backup of the vCenter appliance before starting the upgrade. See [On-Demand Back Up of vCenter](#).
- Note:**

- After taking a backup, do not make any changes to the vCenter inventory or settings until the upgrade completes successfully.
- If your workload domain contains Workload Management (vSphere with Tanzu) enabled clusters, the supported target release depends on the version of Kubernetes (K8s) currently running in the cluster. Older versions of K8s might require a specific upgrade sequence. See [KB 92227](#) for more information.

- In the navigation pane, click **Inventory > Workload Domains**.
- On the Workload Domains page, click the domain you are upgrading and then click the **Updates** tab.
- Click **Precheck** to run the upgrade precheck.

Resolve any issues before proceeding with the upgrade.

- In the Available Updates section, click **Configure Update**.
- Review the upgrade mechanism information and click **Next**.

A vCenter Reduced Downtime Upgrade (RDU) uses a migration-based approach. In this approach, a new vCenter appliance is deployed and the current vCenter data and configuration is copied to it.

During the preparation phase of a reduced downtime upgrade, the source vCenter Appliance and all resources remain online. The only downtime occurs when the source vCenter Appliance is stopped, the configuration is switched over to the target vCenter, and the services are started. The downtime is expected to take approximately 5 minutes under ideal network, CPU, memory, and storage provisioning.

- Select a backup option and click **Next**.
- Provide a temporary network to be used only during the upgrade and click **Next**.

|           |                                           |
|-----------|-------------------------------------------|
| Automatic | Automatically assign network information. |
|-----------|-------------------------------------------|

|        |                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------|
| Static | Enter an IP address, subnet mask, and gateway. The IP address must be in the management subnet. |
|--------|-------------------------------------------------------------------------------------------------|

- Schedule the update and click **Next**.  
Select scheduling options for the preparation and switchover phases of the upgrade.

**Note:** If you are scheduling the switchover phase, you must allow a minimum of 4 hours between the start of preparation and the start of switchover.

- Review the upgrade settings and click **Finish**.
- Monitor the upgrade progress. See [Monitoring VMware Cloud Foundation Updates](#).  
A Cancel button is available once a vCenter RDU upgrade has started. If you cancel the upgrade, the configuration reverts to the source vCenter instance.
- After the upgrade is complete, remove the old vCenter appliance.  
If the upgrade fails, resolve the issue and retry the failed task. If you cannot resolve the issue, restore vCenter using the file-based backup. See [Restoring a vCenter Instance](#). vCenter RDU upgrades perform automatic rollback if the upgrade fails.

After the upgrade is complete, allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade may be cancelled or auto-reconciliation may fail.

Once the upgrade successfully completes, use the vSphere Client to change the vSphere DRS Automation Level setting back to the original value (before you took a file-based backup) for each vSphere cluster that is managed by the vCenter. See [KB 87631](#) for information about using VMware PowerCLI to change the vSphere DRS Automation Level.

After you successfully upgrade vCenter, transition any clusters that use vSphere Lifecycle Manager baselines to use vSphere Lifecycle Manager images before upgrading ESX. See [Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images](#).

Finish Configuring a Newly Deployed VCF Operations Instance as Part of Upgrade

If you have deployed a new instance of VCF Operations as part of the upgrade process, you must finish the configuration and deployment of the remaining components. If you have upgraded your VCF Operations instance to VCF Operations 9.0, these configurations are already present and you can skip that section and can proceed to [Apply VMware Cloud Foundation Configuration Updates](#).

Add a Deployment Target and Import VCF Operations to VCF Operations fleet management

- Use VCF Operations to add vCenter deployment targets. Then import the VCF Operations component into fleet management.
- After integrating your VCF instance in VCF Operations, wait at least 15 minutes for the VCF endpoint that you added to VCF Operations to complete a collection cycle.
  - Verify that SSH is enabled on the nodes for import.
  - Verify that you have upgraded to SDDC Manager 9.0.

The deployment target is the vCenter instance where the VCF Operations instance is deployed. Perform the following steps to add that vCenter to the fleet management appliance inventory and import VCF Operations into VCF Operations fleet management.

1. Log in to VCF Operations.
2. Add the vCenter instance as a deployment target.
  - a) Select **Fleet Management** > **Lifecycle**, and click the **Settings** tab.
  - b) Under **Preferences**, select **Deployment Targets** and click **Add Deployment Target**.
  - c) Add the information for the deployment target.
    - Select the vCenter where VCF Operations is deployed or go to **Administration** > **Integrations** to add it.

**Note:**

To add the deployment target, the SDDC and vCenter must be version 9.0 or later.
    - Click **Validate**.
    - Click **Add**.

The vCenter is added to the fleet management appliance inventory.

3. Import VCF Operations in the fleet management appliance.
  - a) Select **Fleet Management** > **Lifecycle** and click the **Overview** tab.
  - b) On the VCF Operations component tile, **Pending Import** appears. Click **Import**.
  - c) For configuration, add component properties. Most fields are prepopulated and require no action.
    - VCF Operations FQDN.
    - Admin password of VCF Operations.
    - Root password of VCF Operations. Select the root password that is being used. If the root password is the same as the admin password and the admin password is not listed as an option, click **ADD PASSWORD** to enter the Operations Admin password.
    - vCenter Server. Check the box next to the name of the vCenter that was added as a deployment target.
    - Accept the certificate.
  - d) Review the summary and click **Submit**.

If the workflow fails while looking for a new VCF collector, select a new Management vCenter and click **Submit**.

After VCF Operations is imported, it becomes a component that you can manage in VCF Operations fleet management. As examples of the types of Day-N operations that you can manage, see [Scaling Up and Scaling Out Management Nodes in VCF](#).

### Deploy the VCF Operations collector

Use VCF Operations fleet management to deploy the VCF Operations collector. After the deployment completes, reconfigure the VCF Instance to use the collector for new and existing domains.

Verify that the VCF Operations binary is downloaded and available. When the VCF Operations binary is downloaded, it will also map the binary for the VCF Operations collector. See [Downloading VCF management components into Binary Management](#).

#### Note:

The VCF Operations collector is the same as the VCF Operations Cloud Proxy that appears in the VCF Operations UI.

The following procedure shows how to deploy the VCF Operations collector using VCF Operations fleet management.

Alternatively, you can also deploy the VCF Operations collector using VCF Operations and vCenter. See [Configuring Cloud Proxies in VCF Operations](#).

1. Log in to VCF Operations.
2. Select **Fleet Management** > **Lifecycle** and click the **Components** tab.
3. Under **Integrated Components**, click **operations**.
4. On the **operations** details page, click **Add Nodes**.  
The **Proceed to Add Nodes** dialog box appears.
5. Click **Trigger Inventory Sync** to synchronize the VCF Operations fleet management appliance with the latest data from the cluster.
  - a) Click **Submit** to start the inventory sync.
  - b) When the inventory sync completes, click **Proceed**.
6. On the **Infrastructure** page, review the existing infrastructure and click **Next**.
7. On the **Network** page, review the existing network information, add DNS and NTP servers, and click **Next**.
8. On the **Components** page, click the plus sign to add a cloud proxy (collector).
  - a) VM Name: Enter a display name of the cloud proxy.
  - b) Enter the FQDN of the cloud proxy.
  - c) Enter the IP Address of the cloud proxy.
  - d) Select the deployment type:
    - Small Unified
    - Standard Unified

For information about cloud proxy types, see [Collecting Data with Cloud Proxy in VCF Operations](#).
9. To run prechecks, click **Next**.
10. On the **Summary** page, click **Submit** to deploy the cloud proxy (collector).
11. After the deployments completes, update the VCF instance integration to use the new collector for new workload domains.
  - a) Select **Administration** > **Integrations** and click **Accounts**.
  - b) Expand **VMware Cloud Foundation** and expand the integration that you want to configure.
  - c) At the VCF instance level, click the **vertical ellipsis** and click **Edit**.
  - d) On the **Cloud Account Information** page, from the **Collector / Group** drop-down menu, select the newly deployed collector, click **Validate Connection**, and click **Save**.
12. For the existing domains of the same VCF Instance, update the integrations for all vCenter instances to use the new collector/collector group.
  - a) Select **Administration** > **Integrations** and click **Accounts**.
  - b) Expand **VMware Cloud Foundation** and expand the integration that you want to configure.
  - c) At the domain level, click the **vertical ellipsis** and click **Edit**.
  - d) On the **Connect vCenter** page, from the **Collector / Group** drop-down menu, select the newly deployed collector, click **Validate Connection**, and click **Save**.
  - e) Repeat the step for all remaining workload domains in that VCF instance.

### Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images

vSphere Lifecycle Manager baselines are not supported with VMware Cloud Foundation 9.0 or later. Before you can upgrade to ESX 9.0 or later, you must transition any clusters that use vSphere Lifecycle Manager baselines to use vSphere Lifecycle Manager images.

You can transition to vSphere Lifecycle Manager images by using on of the following methods.

- A PowerShell script
  - Menu-driven interface
  - Command-line interface (CLI)
- The SDDC Manager API

After you upgrade SDDC Manager to version 9.0 or later, you must transition all clusters in the management domain and in all workload domains. If the transition is not possible or you have vSphere Supervisor (Workload Management) enabled, transition any remaining vSphere Lifecycle Manager baseline clusters after upgrading vCenter and before upgrading ESX.

Learn about the planning considerations, prerequisites, and tasks for transitioning ESX clusters and standalone hosts from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images. This process is for transitioning ESX clusters and standalone hosts that are already managed by SDDC Manager. For ESX clusters and standalone hosts that are not managed by SDDC Manager, you can transition to vSphere Lifecycle Manager images using the vSphere Client. See [Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images](#).

**Warning:**

For clusters and standalone hosts that are managed by SDDC Manager, the transition must be performed by SDDC Manager supported mechanisms only.

### Prerequisites for Transitioning to vSphere Lifecycle Manager Images

Before you can transition a vSphere cluster or standalone host from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images, you must meet certain prerequisites.

#### vSphere Lifecycle Manager Method Transition Prerequisites

| Category                                    | Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDDC Manager version                        | Clusters: 5.2.2 or later<br>Standalone hosts: 9.1 or later                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Network access for SDDC Manager and vCenter | HTTPS (TCP 443)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| SDDC Manager access privileges              | Administrator (ADMIN)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ESX hosts                                   | Ensure all the ESX hosts that you are attempting to transition are healthy before beginning the workflow.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| vSphere Lifecycle Manager image             | To use the PowerShell script or SDDC Manager API, you must have at least one vLCM image available. While these images are originally authored in vCenter, the required format has changed: <ul style="list-style-type: none"> <li>• Version 8.0.3 and earlier: Images must be created a zero-host, vLCM-managed cluster.</li> <li>• Version 9.0 and later: Images should be authored using the vCenter Image Library.</li> </ul> For information about creating a temporary cluster with a vLCM image, see <a href="#">Creating and Managing vSphere Lifecycle Manager Clusters</a> . Choose the documentation section corresponding to the version in the following table.<br><b>Note:</b> |

| Category                      | Prerequisite                                                                                                                                                                                                                                                                                                                                             |                                                                             |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|                               | vCenter Version                                                                                                                                                                                                                                                                                                                                          | Documentation Section                                                       |
|                               | 8.0 Update 3                                                                                                                                                                                                                                                                                                                                             | Create a Cluster That Uses a Single Image by Importing an Image from a Host |
|                               | 8.0 through 8.0 Update 2                                                                                                                                                                                                                                                                                                                                 | Create a Cluster That Uses a Single Image by Composing a New Image          |
|                               | For information about creating an Image Library vLCM image in vCenter 9, see <a href="#">Create an Image in Image Library Manually</a> .<br><b>Note:</b><br>The ESX version in the image must match the current version of the ESX hosts in the cluster you are transitioning.<br><b>Note:</b><br>Use a descriptive name for the images that you create. |                                                                             |
| PowerShell version            | 7.2 or later<br>See <a href="#">Installing Powershell</a> .                                                                                                                                                                                                                                                                                              |                                                                             |
| PowerShell script             | Download the script from <a href="#">KB 385617</a> .                                                                                                                                                                                                                                                                                                     |                                                                             |
| VCF PowerCLI version          | 9.1 or later.<br><b>Important:</b><br>VMware.PowerCLI conflicts with VCF.PowerCLI. VMware.PowerCLI must be removed before installing VCF.PowerCLI.<br><a href="#">Uninstall VMware.PowerCLI</a> .<br><a href="#">Installing VCF.PowerCLI</a> .                                                                                                           |                                                                             |
| Accurate client time settings | <a href="#">Windows Time Settings</a><br><a href="#">MacOS Time Settings</a><br>Linux time settings are different depending on your distribution                                                                                                                                                                                                         |                                                                             |

The processes for transitioning to vSphere Lifecycle Manager images support the following topologies.

#### VMware Cloud Foundation Architecture Support

| Topology                                                 | vSAN Ready Nodes Minimum Workload Domain Version |
|----------------------------------------------------------|--------------------------------------------------|
| Standard architecture Cluster                            | 5.0                                              |
| Standalone host                                          | 5.0 *                                            |
| Consolidated architecture Cluster                        | 5.1                                              |
| vSAN Stretched Cluster                                   | 5.1                                              |
| DR Protected Cluster                                     | 5.1                                              |
| vSphere Supervisor (Workload Management) enabled Cluster | Split BOM 9.0 **                                 |
| Heterogeneous Hardware Cluster                           | 9.0 ***                                          |

\* Standalone host transition support is only possible in combination with SDDC Manager 9.1 or later.

\*\* The workload domain vCenter must first be upgraded to vCenter 9.0 or later, while ESX must remain on version 8.0 Update 1a or later.

\*\*\* If a vSphere cluster contains heterogeneous hardware, it must first be upgraded to ESX 9.0 or later before it can be transitioned to vSphere Lifecycle Manager images. See Broadcom knowledge base article [424294](#).

### Important Considerations

- You cannot combine the process of transitioning to vSphere Lifecycle manager images with the process of upgrading ESX.
- vSphere clusters that use vSphere Lifecycle Manager baselines must be transitioned to use images before they can be upgraded to ESX 9.0 or later.
- During the transition process, you cannot perform any other operations on the cluster or their underlying hosts until the transition is complete. This restriction includes, but is not limited to ESX password rotation.

### Maximum Compliance Checks and Transitions for Clusters

- A maximum of 25 simultaneous cluster compliance checks might be run per SDDC Manager. Extra compliance checks are discarded.
- A maximum of 25 simultaneous cluster transitions might be run per SDDC Manager. Extra transition requests are discarded.

### Maximum Compliance Checks and Transitions for Standalone Hosts

- A maximum of 25 simultaneous standalone host compliance checks might be run per Workload Domain. Extra compliance checks are discarded.
- A maximum of 25 simultaneous standalone host transitions might be run per Workload Domain. Extra transition requests are discarded.

### Image Seeding Support for vLCM Image Auto-Generation

- A minimum workload domain version of 5.2.1 is required to enable the option of image seeding (to complement the option of constructing and applying your own vLCM image).

### Using Image Seeding

Baseline-managed standalone hosts or clusters with compatible ESX and vCenter sBOM support allow for image seeding. This process auto-generates a vLCM image-based on a single host (standalone or cluster member). In a cluster environment, the created image must be uniform across all hosts, with matching VIBs. Upon completion, the image is stored in the vCenter depot.

### Transitioning vSphere Lifecycle Manager Baseline Clusters to vSphere Lifecycle Manager Image Clusters Using PowerShell

You can use the PowerShell script through a menu-driven interface or a command-line interface (CLI). The menu-driven interface provides a guided transition process and is best suited for smaller environments. The command-line interface is useful for larger environments and can be used to operate against a single cluster at a time or, at scale, using JSON files.

#### Attention:

After you start a task with PowerShell, entering `Ctrl-C` or exiting PowerShell will not halt many tasks (for example, image import, compliance checks, or transition tasks). Those tasks continue to run in the background in SDDC Manager or vCenter.

#### Notice:

If the script execution system does not trust the certificate of the SDDC Manager, execute the following lines before running the script:

```
Set-PowerCLIConfiguration -InvalidCertificateAction Ignore -Scope User
$Env:SkipCertificateCheck="enabled"
```

### Logging

The script records all actions to time-stamped files in a `logs` sub-directory that is created under the directory containing the script. This functionality facilitates troubleshooting.

If you run the script and this directory does not exist, it is automatically created.

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -Connect
LogFolder not found, creating C:\Users\Admin\logs
```

### Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Menu-Driven Interface

The menu-driven interface provides a guided experience for the vLCM baseline to vLCM image management transition process.

The PowerShell script imports vSphere Lifecycle Manager images from the vCenter Image Catalog in vCenter 9.0 or later and from vSphere clusters and standalone-hosts in vCenter 8.0.3 and earlier.

#### Notice:

If the vLCM image is vSphere cluster-based and contains no hosts, the script will delete the vSphere cluster after completing the import and safety checks.

#### Attention:

SDDC Manager requires images be unique in two regards: name and manifest.

Avoid duplicate image names across vCenters. If you import an image that is named `esx-803b` from vCenter `m01-vc01.example.com`, followed by an image that is named `esx-803b` from vCenter `w01-vc01.example.com`, SDDC Manager rejects the second image.

SDDC Manager ensures the images in its catalog are unique not just based on their names, but also on the image metadata. For example, vCenter `m01-vc01.example.com` has vLCM image that is called `esx-803-m01` comprised of base image ESX 8.0.3a and vendor-addon Vendor A Version 1.2.3. vCenter `w01-vc01.example.com` has a vLCM image that is called named `esx-803-w01` also comprised of base image ESX 8.0.3a and vendor-addon Vendor A Version 1.2.3. If you import `esx-803-m01` followed by `esx-803-w01`, SDDC manager rejects `esx-803-w01`. This validation step occurs during the final stage—after the full image has been uploaded to SDDC Manager.

1. Launch the script.

```
.\VcfBaselineClusterTransition.ps1
```

2. Connect to SDDC Manager and select a vCenter that manages the vSphere clusters you want to transition.

- a) From the main menu of the script, select `Connect to SDDC Manager` and select `vCenter`.

If a credentials file is not found on disk, you are prompted to provide the SDDC Manager FQDN and administrative credentials. If an SDDC credential file is present, the script uses those credentials to connect to the SDDC Manager referenced in the file.

- b) Select the vCenter(s) to which you want to connect.

You can connect to all vCenters, a single vCenter, or a comma-delimited list of multiple vCenters.

The following vCenter(s) were detected in SDDC Manager "vcf0

| Id | vCenter              | Workload Domain |
|----|----------------------|-----------------|
| -- | -----                | -----           |
| 1  | <ALL VCENTERS>       | N/A             |
| 2  | m01-vc01.example.com | m01             |
| 3  | w01-vc01.example.com | w01             |

This script can only transition vLCM baseline-managed cluste

Enter an id, a comma-delimited list of ids of vCenters to co

[INFO] Preparing to connect to vCenter(s)...

[INFO] Successfully connected to vCenter "m01-vc01.example.c

[INFO] Successfully connected to vCenter "w01-vc01.example.c

3. Import a vSphere Lifecycle Manager image into SDDC Manager from any connected vCenter.

Manually importing a vLCM image into SDDC Manager is only required if your clusters or standalone hosts do not support image seeding.

- a) From the main menu of the script, `Import vLCM images from vCenter(s) into SDDC Manager`.
- b) Select the ID of the image to import and wait for the task to complete.

[INFO] Scanning vCenter(s) for vLCM images...

| Id | vCenter           | Image Name           | vCenter Name |
|----|-------------------|----------------------|--------------|
| -- | -----             | -----                | -----        |
| 1  | esx-803b-vendor-a | m01-vc01.example.com |              |
| 2  | esx-803b-vendor-b | m01-vc01.example.com |              |

Enter an id or comma-delimited list of ids of image(s) to im

[INFO] Successfully created SDDC Manager image "esx-803b-ven

[INFO] Safety check passed – No ESX hosts detected in cluste

If the import task fails with the error message `An image named <image-name> already exists in the SDDC Manager`, rename the source image in vCenter or delete the conflicting image from SDDC Manager before

retrying. To delete the conflicting image in SDDC Manager, return to the main menu and select (Optional) Delete SDDC Manager Image.

c) Repeat these steps for all the images you want to import.

4. Check vSphere Lifecycle Manager baseline clusters or standalone hosts for compliance with a vSphere Lifecycle Manager image.

Be mindful of the executing logic:

The menu-driven interface always processes clusters sequentially. Each compliance check of a cluster must finish before the next one begins. Parallel checks are only possible using the command-line interface with the `-Parallel` option.

Standalone hosts, however, are processed in parallel, so long as the following conditions are met:

- Each batch does not exceed 25 standalone hosts
- Each batch contains identical images or uses image seeding
- Each batch uses the same workload domain vCenter.

#### Attention:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation. The transition instead can be performed live.

| Solution                       | Type      | VIB/Component Name |
|--------------------------------|-----------|--------------------|
| HA (High Availability)         | VIB       | vmware-fdm         |
| NSX                            | Component | nsx-lcp-bundle     |
| Spherelet (vSphere Supervisor) | Component | VMware-Spherelet   |

| Solution            | Type | VIB/Component Name |
|---------------------|------|--------------------|
| vSphere Replication | VIB  | vmware-hbr-agent   |

For more information, see Broadcom knowledge base article [311882](#).

- From the main menu of the script, select `Check existing cluster(s)/standalone host(s)' vLCM image compliance`.
- Enter a comma-delimited list of the IDs of the vSphere clusters and/or standalone hosts that you want to scan for compliance.
- When image seeding is supported across all selected clusters, you have the following two options.
  - Yes: Automatically generate an image based on the current cluster state
  - No: Manually select a vLCM image from the SDDC Manager inventory.

The script returns a high-level status and summary and saves a JSON file with the full findings for each vSphere cluster or standalone host.

- Review the summary report to determine if a vSphere cluster or standalone host is ready to be transitioned.

Resolve any issues for vSphere clusters with a status of `INCOMPATIBLE` or `UNKNOWN`.

| Status        | Description                                                                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NON_COMPLIANT | The vSphere cluster or standalone host is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.                                                                                                                                                         |
| INCOMPATIBLE  | The vSphere cluster or standalone host cannot be transitioned to the selected vSphere Lifecycle Manager image without taking extra steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve one or more issues. |
| UNKNOWN       | Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster, or the standalone host, or the is/are healthy and connected to vCenter. Contact Broadcom Support if you are unable to resolve the issue.                                                                          |
| COMPLIANT     | All hosts are confirmed to be compliant the vLCM image.                                                                                                                                                                                                                                                       |

#### Important:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation. The transition instead can be performed live.

| Solution                       | Type      | VIB/Component Name |
|--------------------------------|-----------|--------------------|
| HA (High Availability)         | VIB       | vmware-fdm         |
| NSX                            | Component | nsx-lcp-bundle     |
| Spherelet (vSphere Supervisor) | Component | VMware-Spherelet   |



| Solution            | Type | VIB/Component Name |
|---------------------|------|--------------------|
| vSphere Replication | VIB  | vmware-hbr-agent   |

For more information, see Broadcom knowledge base article [311882](#).  
Example of a compliance check of a vSphere cluster using a specified image.

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...

Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected Mode

 Id Resource Name Resource Type vCenter Name
 --
 1 m01-cl01 Cluster m01-vc01.example.com
 2 w01-cl01 Cluster w01-vc01.example.com
 3 esxi-1.example.com Standalone Host w01-vc01.example.com
 4 esxi-2.example.com Standalone Host w01-vc01.example.com

[INFO] Select the vLCM baseline cluster(s)/standalone-host(s) to check for compliance.
Enter an id, a comma-delimited list of ids, or 'c' to cancel: 1

[INFO] Checking for image seeding support...

All selected clusters and standalone-hosts are eligible for image seeding. Would you like to seed them using image seeding?
[Y] Yes [N] No [?] Help (default is "Y"): N

[INFO] vLCM images available in SDDC Manager "sddc-manager.vrack.vsphere.local"

 Id SDDC Manager Image Name Base Image Components
 --
 1 esx-803-vendor-a 8.0.3-24280767 N/A
 2 esx-803-vendor-b 8.0.3-24280767 N/A

Please enter the id of an image or press 'c' to cancel: 1

[INFO] Beginning compliance check for Cluster "m01-cl01" in workload domain "m01"
[INFO] Check image compliance of cluster m01-cl01 completed successfully.
[INFO] Cluster "m01-cl01" in Workload Domain "m01" has status "NON_COMPLIANT"
[WARNING] Remediation involving maintenance mode or a reboot, or both, is required.
[INFO] Summary of compatibility check cluster "m01-cl01" in Workload Domain "m01"

 Component Name Component Vendor Version Change Impact

 N/A N/A No MAINTENANCE_MODE_REQUIRED WARNING
```

Example of a compliance check for a vSphere cluster using image seeding.

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...

Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected Mode

 Id Resource Name Resource Type vCenter Name
 --
1 m01-cl01 Cluster m01-vc01.example.com
2 w01-cl01 Cluster w01-vc01.example.com
3 esxi-1.example.com Standalone Host w01-vc01.example.com
4 esxi-2.example.com Standalone Host w01-vc01.example.com

[INFO] Select the vLCM baseline cluster(s)/standalone-host(s) to check for compliance.

Enter an id, a comma-delimited list of ids, or 'c' to cancel: 2

[INFO] Checking for image seeding support...

All selected clusters and standalone-hosts are eligible for image seeding. Would you like to proceed with image seeding for them using image seeding?
[Y] Yes [N] No [?] Help (default is "Y"): Y

[INFO] Image seeding workflow selected.

[INFO] Auto-generating vLCM image for Cluster "w01-cl01" in workload domain "w01".

[INFO] Auto-generating vLCM image for "w01-cl01" in workload domain "w01" and standalone hosts.

[INFO] Check image compliance of cluster w01-cl01 completed successfully.

[INFO] Cluster "w01-cl01" in Workload Domain "w01" has status "NON_COMPLIANT".

[WARNING] Remediation involving maintenance mode or a reboot, or both, is required.

[INFO] Summary of compatibility check cluster "w01-cl01" in Workload Domain "w01":

Component Name Component Vendor Version Change Impact

N/A N/A No MAINTENANCE_MODE_REQUIRED WAITING FOR REMEDIATION

N/A N/A No N/A WAITING FOR REMEDIATION

N/A N/A No MAINTENANCE_MODE_REQUIRED IN PROGRESS

MSX ICD Bundle VMware Yes MAINTENANCE_MODE_REQUIRED IN PROGRESS
```

5. Transition `NON_COMPLIANT` (ready to transition) vSphere clusters or standalone hosts to vSphere Lifecycle Manager images.

**CAUTION:**

Once you start a compliance or transition task, it cannot be paused or stopped. The task continues to run on the SDDC manager even if the Powershell window is closed.

- From the main menu, Transition vLCM baseline (VUM) cluster/standalone host to vLCM image management.
- Enter a comma-delimited list of the IDs of the vSphere clusters or standalone hosts that you want to transition to vSphere Lifecycle Manager images.
- Review the compliance findings from those vSphere clusters and/or standalone hosts, and if you wish to proceed, enter `y` to begin the workflow.
- Once you select your clusters and/or standalone hosts, you might choose to temporarily override remediation policies for the transition operation. See [Configuring Remediation Settings](#) for further details.

Example of a cluster transition with an override remediation policy

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone hosts...
```

```
Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected
```

|   | Id                 | Resource Name | Resource Type   | vCenter Name         |
|---|--------------------|---------------|-----------------|----------------------|
| 1 | m01-cl01           |               | Cluster         | m01-vc01.example.com |
| 2 | w01-cl01           |               | Cluster         | w01-vc01.example.com |
| 3 | esxi-1.example.com |               | Standalone Host | w01-vc01.example.com |
| 4 | esxi-2.example.com |               | Standalone Host | w01-vc01.example.com |

```
[INFO] Select the vLCM baseline clusters/standalone hosts to transition to vLCM image management:
```

```
Enter an id, a comma-delimited list of ids in order, or 'c' to cancel: 1
```

```
Confirm you have reviewed the image compliance findings stored in "C:\Users\A\Documents\image-compliance-findings.html":
```

```
Would you like to continue?
```

```
[Y] Yes [N] No [?] Help (default is "N"): Y
```

```
[INFO] Clustert "m01-cl01" transition to vLCM image management accepted.
```

```
Would you like to use the default host remediation options?
```

```
[Y] Yes [N] No [?] Help (default is "Y"): N
```

```
[INFO] For information more information on host remediation, please visit: https://www.vmware.com/resources/compatibility/sig/figuring-remediation-settings.html
```

```
Enter your preferred remediation action (FAIL or RETRY) or press 'c' to cancel: RETRY
```

```
Enter the number of times to retry the failed operation (between 3 and 100) or press 'c' to cancel: 3
```

```
Enter the seconds to wait between retry operations (between 300 and 3600) or press 'c' to cancel: 300
```

```
Choose one preRemediationPowerAction from the following list: POWER_OFF_VMS, POWER_OFF_HOSTS, POWER_OFF_BOTH: POWER_OFF_VMS
```

```
Do you want to enable Quickboot? Enter yes, no, or 'c' to cancel: yes
```

```
[INFO] Beginning vLCM image transition for Standalone Host "m01-cl01" in Workload Domain "m01-vc01"
```

```
...
```

- Monitor the status of the transition task.

If the task fails, try the following steps:

- Return to the main menu .
- Select Option 6, (Optional) Retry incomplete transition tasks..
- Wait some time then select Option 9,(Optional) Show cluster transition status..
- If the process succeeds, no further action is required, if it fails, contact Broadcom Support.

#### 6. Disconnect from SDDC Manager and vCenter.

Close the PowerShell window or enter `Q` to quit.

### Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Command-Line Interface

The PowerShell script command-line interface (CLI) is useful for larger environments. The CLI can be used to transition a single resource command-line arguments or multiple resources using JSON input files.

See [PowerShell Script CLI Options for Transitioning to vSphere Lifecycle Manager Images](#) for information about all the CLI options.

#### Attention:

For bulk operations using a JSON file, use a JSON validator to ensure that your file is properly formed before using it as an input.

#### Attention:

Appending `-Silence` to any command suppresses screen output and any progress indicators. Pertinent data is still sent to the log file and task operation status might still be queried through relevant check commands. This process can be useful in scripting operations.

1. Run the script with `-Version` and compare it to the version listed on [KB 385617](#) to determine if you are running the latest copy of the script.

```
PS C:\Users\Admin>.\VcfBaselineClusterTransition.ps1 -Version
```

2. Connect to SDDC Manager and its vCenter instances.

You can provide your SDDC Manager credentials interactively or by including the path to a JSON with the credentials.

| Command                                                                                 | Description                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -Connect</code>                                | Enter the following information for SDDC Manager: <ul style="list-style-type: none"> <li>• FQDN</li> <li>• ADMIN user name</li> <li>• ADMIN password</li> </ul> Optionally, you can choose to save these credentials in a JSON file. |
| <code>.\VcfBaselineClusterTransition.ps1 -Connect -JsonInput <i>path_to_file</i></code> | For the <i>path_to_file</i> .json, enter the location of the file containing the SDDC Manager credentials. For example:<br>C:\Users\Admin\SddcManagerCredentials.json                                                                |

The script connects to SDDC Manager and all vCenter instances that it manages.

**Note:** If you see the message Failed to connect to vCenter, verify that you can access the vCenter from PowerShell and that it returns a 200 status code.

```
Invoke-WebRequest -Method HEAD -SkipCertificateCheck -Uri https://vCenter_FQDN | Select-Object StatusCode
```

If it does not, verify you can login to that vCenter from a browser. If you cannot, contact Broadcom Support.

3. Identify vSphere clusters and standalone hosts across all workload domains that are managed using vSphere Lifecycle Manager baselines.

| Command                                                                                                | Description                                                                                                        |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ShowBaselineResources</code>                                 | Displays a list of baseline resources (clusters and standalone hosts).                                             |
| <code>.\VcfBaselineClusterTransition.ps1 -ShowBaselineResources -JsonOutput <i>path_to_file</i></code> | Displays a list of baseline-managed resources (clusters and standalone hosts) and then saves the output to a file. |

```
PS C:\Users\Administrator> .\VcfBaselineClusterTransition.ps1 -ShowBaselineResources
```

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...
```

```
[INFO] vLCM Baseline (VUM) Managed Clusters/Standalone Hosts:
```

| Resource Name      | Resource Type   | vCenter Name         | Component |
|--------------------|-----------------|----------------------|-----------|
| m01-cl01           | Cluster         | m01-vc01.example.com | NOT_      |
| w01-cl01           | Cluster         | w01-vc01.example.com | NOT_      |
| esxi-1.example.com | Standalone Host | w01-vc01.example.com | NOT_      |
| esxi-2.example.com | Standalone Host | w01-vc01.example.com | NOT_      |

4. Display and/or save all the vSphere Lifecycle Manager images in connected vCenter(s).

| Command                                                                                              | Description                                                       |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter</code>                                 | Displays vCenter images on screen.                                |
| <code>.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter -JsonOutput <i>path_to_file</i></code> | Displays vCenter images on screen and saves the output to a file. |

| Command                                                                                                       | Description                                                                           |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter -JsonOutput <i>path_to_file</i> -Silence</code> | Only saves vCenter image output to a file, and does not display output to the screen. |

PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -ShowIma

[INFO] Scanning vCenter(s) for vLCM images...

[INFO] vLCM images in attached vCenter(s):

```

vCenter Image Name vCenter Name

esx-803b-vendor-a m01-vc01.example.com
esx-803b-vendor-b m01-vc01.example.com

```

5. Import vSphere Lifecycle Manager images into SDDC Manager from connected vCenter instances.

| Command                                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ImportImages-FromVcenter -VcenterImageName <i>image_name</i> -VcenterName <i>vCenter_FQDN</i></code>          | Import a single image from vCenter and display progress on screen.                                                                                                                                                                                                                                                                                               |
| <code>.\VcfBaselineClusterTransition.ps1 -ImportImages-FromVcenter -VcenterImageName <i>image_name</i> -VcenterName <i>vCenter_FQDN</i> -Silence</code> | Import a single image from vCenter and do not display progress on screen.                                                                                                                                                                                                                                                                                        |
| <code>.\VcfBaselineClusterTransition.ps1 -ImportImages-FromVcenter -JsonInput <i>Path_To_File</i> -VcenterName <i>vCenter_FQDN</i></code>               | Import multiple images from vCenter from a JSON input file serially and display progress on screen.<br>Example JSON file for bulk import:<br>[<br>{<br>"VcenterImageName": " <i>image_name</i> ",<br>"VcenterName": " <i>vCenter_FQDN</i> "<br>},<br>{<br>"VcenterImageName": " <i>another_image_name</i> ",<br>"VcenterName": " <i>vCenter_FQDN</i> "<br>}<br>] |

| Command                                                                                                                                             | Description                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ImportImages-FromVcenter -JsonInput <i>path_to_file</i> -VcenterName <i>vCenter_FQDN</i> -Parallel</code> | Import multiple images from vCenter from a JSON input file in parallel. |

**Note:**

If the script is called without the flag `-Parallel`, it attempts to delete empty vSphere clusters after a successful import. The script does not delete vSphere clusters with hosts or images from the vCenter image catalog.

To check the task status, run the following command:

```
.\VcfBaselineClusterTransition.ps1 -CheckTaskStatus -TaskType SddcManagerImageUpload
```

**Note:** If the import task fails with the error message `An image named <image-name> already exists in the SDDC Manager`, you can rename the source image in vCenter or delete the conflicting image from SDDC Manager before retrying. To delete the conflicting image in SDDC Manager, run the following command:

```
.\VcfBaselineClusterTransition.ps1 -DeleteImageFromSddcManager -SddcManagerImageName image_name
```

6. Review the images available in SDDC Manager.

| Command                                                                  | Description                                                              |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>./VcfBaselineClusterTransition.ps1 -ShowImagesInSddcManager</code> | Displays a list of all vSphere Lifecycle Manager images in SDDC Manager. |



| Command                          | Description                                                                     |
|----------------------------------|---------------------------------------------------------------------------------|
| Get-Content logs/Image_Name.json | Displays detailed information about a specific vSphere Lifecycle Manager image. |

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -ShowImages
[INFO] vLCM images available in SDDC Manager "sfo-vcf01.sfo.ra
```

| Id | SDDC Manager Image Name | Base Image     |
|----|-------------------------|----------------|
| 1  | esx-803b-vendor-a       | 8.0.3-24280767 |
| 2  | esx-803b-vendor-b       | 8.0.3-24280767 |

```
PS C:\Users\Admin> Get-Content .\logs\esx-803b-vendor-a.json
{
 "add_on": {
 "name": "VEND-ESXi",
 "version": "803.123456-A02"
 },
 "alternative_images": null,
 "base_image": {
 "version": "8.0.3-0.35.24280767"
 },
 "components": null,
 "hardware_support": null,
 "removed_components": null,
 "solutions": null
}
```

## 7. Attention:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation. The transition instead can be performed live.

| Solution                       | Type      | VIB/Component Name |
|--------------------------------|-----------|--------------------|
| HA (High Availability)         | VIB       | vmware-fdm         |
| NSX                            | Component | nsx-lcp-bundle     |
| Spherelet (vSphere Supervisor) | Component | VMware-Spherelet   |
| vSphere Replication            | VIB       | vmware-hbr-agent   |

For more information, see Broadcom knowledge base article [311882](#).

Check vSphere Lifecycle Manager baseline-managed vSphere clusters for compliance with a vSphere Lifecycle Manager image.

| Command                                                                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .\VcfBaselineClusterTransition.ps1 -ComplianceCheck -ResourceType [Standalone Host Cluster] -ResourceName Resource_Name -WorkloadDomainName workload_domain -SddcManagerImageName image_name | Executes a compliance check on a baseline-managed resource (cluster or host) against the vSphere Lifecycle Manager image, displaying the results once the process is complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| .\VcfBaselineClusterTransition.ps1 -ComplianceCheck -JsonInput path_to_file                                                                                                                  | <p>Performs sequential compliance checks on multiple vSphere clusters and/or standalone hosts against vSphere Lifecycle Manager images using a JSON input file.</p> <p><b>Important:</b></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>If your "ResourceName" is a cluster, the "ResourceType" must be defined as a "Cluster".</li> <li>If your "ResourceName" is a standalone host, the "ResourceType" must be defined as a "Standalone Host".</li> <li>If "SddcManagerImageName" is not specified, host seeding is assumed.</li> </ul> <p>Example JSON input file:</p> <pre>[   {     "ResourceName": "standalone_host_name",     "ResourceType": "Standalone Host",     "WorkloadDomainName": "workload_domain",     "SddcManagerImageName": "image_name",   },   {     "ResourceName": "vsphere_cluster_name",     "ResourceType": "Cluster",     "WorkloadDomainName": "workload_domain",     "SddcManagerImageName": "image_name"   } ]</pre> |



| Command                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                    | ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>.\VcfBaselineClusterTransition.ps1 -ComplianceCheck -JsonInput path_to_file -Parallel</code> | Performs parallel compliance checks for multiple baseline-managed clusters and/or standalone hosts against vSphere Lifecycle Manager images using a JSON input file, without displaying results upon completion.<br><br>When performing compliance checks using the parallel or silent flags, use the following command to verify the status of the operation.<br><br><code>.\VcfBaselineClusterTransition.ps1 -CheckTaskStatus -TaskType ComplianceCheck</code> |

```
PS C:\Users\Administrator> ./VcfBaselineClusterTra
```

```
[INFO] No SDDC Manager image named specified for "
[INFO] No SDDC Manager image named specified for "
[INFO] Auto-generating vLCM image for Standalone H
[INFO] Auto-generating vLCM image for Standalone H
```

8. For compliance checks performed in parallel or with the silence flag, review the compliance results after the compliance check is complete.

| Command                                                                                                                                                                                               | Description                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ShowAllResources</code>                                                                                                            | Displays a summary of the compliance findings for all vSphere clusters and standalone hosts.   |
| <code>.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ShowAllResources -ShowExtendedResults</code>                                                                                       | Displays a summary and detailed compliance findings for all vSphere and standalone hosts.      |
| <code>.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ResourceName resource_name -ResourceType [Cluster Standalone Host] -WorkloadDomainName workload_domain</code>                      | Displays a summary of the compliance findings for a specific cluster or standalone host.       |
| <code>.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ResourceName resource_name -ResourceType [Cluster Standalone Host] -WorkloadDomainName workload_domain -ShowExtendedResults</code> | Displays a summary and detailed compliance findings for a specific cluster or standalone host. |

9. Review the compliance findings to determine if a vSphere cluster is ready to be transitioned.

Resolve any issues for vSphere clusters or standalone hosts with a status of `INCOMPATIBLE` or `UNKNOWN`.

|                            |                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>NON_COMPLIANT</code> | The vSphere cluster or standalone host is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned. |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>INCOMPATIBLE</code> | The vSphere cluster or standalone host cannot be transitioned to the selected vSphere Lifecycle Manager image without taking extra steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve one or more issues. |
| <code>UNKNOWN</code>      | Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster, or the standalone host, or the is/are healthy and connected to vCenter. Contact Broadcom Support if you are unable to resolve the issue.                                                                          |
| <code>COMPLIANT</code>    | All hosts are confirmed to be compliant the vLCM image.                                                                                                                                                                                                                                                       |

10. Transition `NON_COMPLIANT` (ready to transition) clusters to vSphere Lifecycle Manager images.

| Command                                                                                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.\VcfBaselineClusterTransition.ps1 - CreateHostRemediationOptionsFile -JsonOutput path_to_file</code>                                                                                                          | (Optional) Create an override host remediation option setting for the transition process.                                                                                                                                                                                                                                                                                                                                |
| <code>./VcfBaselineClusterTransition.ps1 -TransitionResource -ResourceName resource_name -ResourceType [Cluster Standalone Host] -WorkloadDomainName workload_name</code>                                            | Transitions a single cluster or standalone host to use a Lifecycle Manager image and displays progress.                                                                                                                                                                                                                                                                                                                  |
| <code>./VcfBaselineClusterTransition.ps1 -TransitionResource -ResourceName resource_name -ResourceType [Cluster Standalone Host] -WorkloadDomainName workload_domain -HostRemediationOptionsFile path_to_file</code> | Transition a cluster or standalone host to a Lifecycle Manager image using a custom remediation file. Progress is displayed during the process.                                                                                                                                                                                                                                                                          |
| <code>/VcfBaselineClusterTransition.ps1 -TransitionResources -JsonInput path_to_file</code>                                                                                                                          | Transitions multiple vSphere clusters sequentially to use a vSphere Lifecycle Manager image and displays progress.<br><br>Example JSON input file:<br><pre>[ {   "ResourceName": "vSphere_cluster_name",   "ResourceType": "Cluster",   "WorkloadDomainName": "workload_domain", }, {   "ResourceName": "standalone host name",   "ResourceType": "Standalone Host",   "WorkloadDomainName": "workload_domain" } ]</pre> |

| Command                                                                                                | Description                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>./VcfBaselineClusterTransition.ps1 -TransitionResources -JsonInput path_to_file -Parallel</code> | Transitions multiple vSphere clusters and/or standalone hosts in parallel to use a vSphere Lifecycle Manager image and does not display progress. |

You can run compliance checks in parallel, but there is a concurrent task limit of 25 compliance and transition tasks. If you exceed this limit, the task is rejected, rather than queued.

#### Important:

The system transitions the cluster or standalone hosts using the last image that you checked against. If you run the compliance check multiple times against multiple images, run `./VcfBaselineClusterTransition.ps1 -ShowBaselineResources` to review the final association before proceeding.

#### CAUTION:

Once you start a compliance or transition task, it cannot be paused or stopped. The task continues to run on the SDDC manager even if the Powershell window is closed.

- a) Run one of the transition commands listed in the preceding table.

Example of a standalone host transition with a host remediation file.

```
PS C:\Users\Administrator> .\VcfBaselineClusterTr
-ResourceType "Standalone Host" -WorkloadDomainNa

[INFO] No errors detected in HostRemediationOptio

[INFO] Preparing for cluster(s)/standalone-host(s)

[INFO] Beginning vLCM image transition for Standa
[INFO] Standalone Host "esx-1.example.com" in Wor
management.

[INFO] Full details of the transition written to
26.log.
```

- b) For parallel and silenced operations, check the status of the transition.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions
```

11. If the transition task fails, try the following steps.

- a) Identify which vSphere cluster and/or standalone host has failed to transition to vLCM image management.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions
```

- b) Retrieve instructions on how to retry the transition.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions -ResourceName resource_name -ResourceType [Clus-
ter|Standalone Host] -WorkloadDomain workload_domain
```

- c) Resume the failed task using the instructions.

```
.\VcfBaselineClusterTransition.ps1 -RetryTransition -TaskId task_id
```

- d) Run the check transitions command to see if the vSphere cluster and/or standalone host transition is In Progress.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions
```

- e) Periodically re-run `-CheckTransitions` to see if the transition completes successfully or it fails. If it fails again, contact Broadcom Support.

12. Disconnect from the SDDC Manager and all vCenter instances.

```
./VcfBaselineClusterTransition.ps1 -Disconnect
```

#### PowerShell Script CLI Options for Transitioning to vSphere Lifecycle Manager Images

This list covers the CLI options for the PowerShell script that transitions clusters or standalone hosts from baselines to vSphere Lifecycle Manager images.

You can run the script with the `-Help` argument to see these options.

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -Help
```

For examples of how to use these options, see [Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Command-Line Interface](#).

| Parameter                   | Argument                                                                                                                                                               | Importance                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| -CheckTaskStatus            | -TaskType SddcManagerImageUpload<br>or<br>-TaskType ComplianceCheck                                                                                                    | Required                                                                                |
|                             | -Silence                                                                                                                                                               | Optional                                                                                |
| -CheckTransitions           | -ResourceName <i>resource_name</i><br>-ResourceType [Cluster Standalone Host]<br>-WorkloadDomainName <i>workload_domain</i>                                            | Optional                                                                                |
|                             | -Silence                                                                                                                                                               | Optional                                                                                |
| -Connect                    | -JsonInput <i>path_to_file</i>                                                                                                                                         | Optional                                                                                |
| -ComplianceCheck            | -ResourceName <i>resource_name</i><br>-ResourceType [Cluster Standalone Host]<br>-SddcManagerImageName <i>image_name</i><br>-WorkloadDomainName <i>workload_domain</i> | You must provide either all of the individual arguments or use the -JsonInput argument. |
|                             | -JsonInput <i>path_to_file</i>                                                                                                                                         |                                                                                         |
|                             | -Parallel                                                                                                                                                              | Optional                                                                                |
|                             | -Silence                                                                                                                                                               | Optional                                                                                |
| -DeleteImageFromSddcManager | -SddcManagerImageName <i>image_name</i>                                                                                                                                | Required                                                                                |
|                             | -Silence                                                                                                                                                               | Optional                                                                                |
| -DeleteTemporaryCluster     | -ResourceName <i>resource_name</i>                                                                                                                                     | Required                                                                                |

| Parameter                | Argument                                                                                                                                                      | Importance                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|                          | -ResourceType [Cluster Standalone Host]<br>-VcenterName <i>vCenter_FQDN</i>                                                                                   |                                                                                         |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -Disconnect              | -Silence                                                                                                                                                      | Optional                                                                                |
| -Help                    |                                                                                                                                                               | N/A                                                                                     |
| -ImportImagesFromVcenter | -VcenterImageName <i>image_name</i><br>-VcenterName <i>vCenter_FQDN</i><br>-JsonInput <i>path_to_file</i>                                                     | You must provide either all of the individual arguments or use the -JsonInput argument. |
|                          | -Parallel                                                                                                                                                     | Optional                                                                                |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -ReviewComplianceResults | -ResourceName <i>resource_name</i><br>-ResourceType [Cluster Standalone Host]<br>-WorkloadDomainName <i>workload_domain</i>                                   | Optional                                                                                |
|                          | -ShowExtendedResults                                                                                                                                          | Optional                                                                                |
|                          | -ShowAllResources                                                                                                                                             | Optional                                                                                |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -RetryTransition         | -TaskId <Task Id>                                                                                                                                             | Required                                                                                |
| -ShowBaselineResources   | -JsonOutput <i>path_to_file</i>                                                                                                                               | Optional                                                                                |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -ShowImagesInSddcManager | -Silence                                                                                                                                                      | Optional                                                                                |
| -ShowImagesInVcenter     | -JsonOutput <i>path_to_file</i>                                                                                                                               | Optional                                                                                |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -TransitionResource      | -ResourceName <i>resource_name</i><br>-ResourceType [Cluster Standalone Host]<br>-WorkloadDomainName <i>workload_domain</i><br>-JsonInput <i>path_to_file</i> | You must provide either all of the individual arguments or use the -JsonInput argument. |
|                          | -Parallel                                                                                                                                                     | Optional                                                                                |
|                          | -Silence                                                                                                                                                      | Optional                                                                                |
| -Version                 |                                                                                                                                                               | N/A                                                                                     |

### Transition to vSphere Lifecycle Manager Images Using the SDDC Manager API

If you do not want to use the PowerShell script to transition vSphere clusters to vSphere Lifecycle manager images, you can use the SDDC Manager API instead.

The following procedure describes how to use the SDDC Manager Developer Center to transition a single vSphere cluster or a set of standalone hosts to vSphere Lifecycle Manager Images.

#### CAUTION:

The API workflow does not contain the same guardrails and error handling as the PowerShell script. Using the SDDC Manager API is only recommended if the PowerShell script does not meet your business requirements.

### Cluster API Workflow

## 1. Retrieve the names and IDs of vSphere clusters that are managed using vSphere Lifecycle Manager baselines.

- Log in to the SDDC Manager UI.
- Click **Developer Center > API Explorer**.
- Expand **Clusters** and click **GET /v1/clusters**.
- Enter the value `false` for the `isImageBased` parameter and click **Execute**.
- Click **Download** to save the results.

You can rename the saved file using a more meaningful name.

- Review the file to identify the vSphere cluster ID associated with each vSphere cluster name.

Example:

```
{
 "elements": [
 {
 "id": "7b0fadd9-c6dd-46d6-ab12-4a2cce93bb59",
 "domain": {
 "id": "108355fd-e758-4b19-ac93-1c5efc6fd7df"
 },
 "name": "m01-cl01",
 "status": "ACTIVE",
 "primaryDatastoreName": "m01-m01-cl01-vsan01",
 "primaryDatastoreType": "VSAN",
 "hosts": [
 {
 "id": "a392202a-1c3a-491e-96ea-b3175a4d10"
 },
 {
 "id": "f3f180cc-69cd-43a9-8c27-ed9018d410"
 },
 {
 "id": "fff2e3aa-2322-4a25-9b85-2bfd9bcb42"
 }
],
 "isStretched": false,
 "isDefault": true,
 "failuresToTolerate": 1,
 "isImageBased": false,
 "capacity": {
 ...
 }
 }
]
}
```

2. Import vSphere Lifecycle Manager images from vCenter into SDDC Manager using the SDDC Manager UI.
  - a) In the navigation pane, click **Lifecycle Management > Image Management**.
  - b) Click **Import Image**.
  - c) Select **Import from a vCenter**.
  - d) From the **vCenter** drop-down menu, select the vCenter containing your vSphere Lifecycle Manager image.

Click **Refresh** to see the current list of images or vSphere clusters.

- e) Select the image (vCenter 9.0 and later) or vSphere cluster (vCenter 8.0.3 and below).

For vCenter 8.0.3 and below, enter a name for your image.

- f) Click **Import**.

You can monitor the status in the **Tasks** panel.

- g) Repeat the process for each unique image you want to import.

3. Check a vSphere Lifecycle Manager baseline-managed vSphere cluster for compliance with a vSphere Lifecycle Manager image.

You can run a vSphere cluster compliance check multiple times, but only the latest check (and associated image) is used for transitioning the vSphere cluster.

#### Attention:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation. The transition instead can be performed live.

| Solution                       | Type      | VIB/Component Name |
|--------------------------------|-----------|--------------------|
| HA (High Availability)         | VIB       | vmware-fdm         |
| NSX                            | Component | nsx-icp-bundle     |
| Spherelet (vSphere Supervisor) | Component | VMware-Spherelet   |
| vSphere Replication            | VIB       | vmware-hbr-agent   |

For more information, see Broadcom knowledge base article [311882](#).

- a) Click **Lifecycle Management > Image Management**.
- b) Identify the name of the vSphere Lifecycle Manager image that you want to use for the compliance check.
- c) Click **Developer Center > API Explorer**.
- d) Expand **Personalities** and click **GET /v1/personalities**.
- e) Enter the name of the image for the `personalityName` parameter and click **Execute**.
- f) Retrieve the `personalityId` from the response.
- g) Expand **Clusters** and click **PATCH /v1/clusters/{id}**.
- h) Enter the vSphere cluster ID for the `id` parameter.
- i) Enter the following information for the `body` parameter.

Use the personality ID that you retrieved for the `clusterImageId` value.

```
{
 "clusterImageComplianceCheckSpec": { "clusterImageId": "personalityId" }
}
```

- j) Click **Execute**.

4. Review the compliance findings to determine if a cluster is ready to be transitioned.

Resolve any issues for clusters with a status of `INCOMPATIBLE` or `UNKNOWN`.

| Status        | Description                                                                                                                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NON_COMPLIANT | The vSphere cluster is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.                                                                                                                                                        |
| INCOMPATIBLE  | The vSphere cluster cannot be transitioned to the selected vSphere Lifecycle Manager image without taking additional steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve the issue(s). |
| UNKNOWN       | Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster are available. Contact Broadcom Support if you are unable to resolve the issue.                                                                                                                |
| COMPLIANT     | The vSphere cluster has already successfully been transitioned to vSphere Lifecycle Manager image management and no further action is required.                                                                                                                                           |

- a) Expand **Clusters** and click **GET /v1/clusters/{id}/image-compliance**
- b) Enter the vSphere cluster ID for the `id` parameter.
- c) Click **Execute**.
- d) Click **Download** to save the results.
- e) Review the results using the tool of your choice.

5. Transition `NON_COMPLIANT` (ready to transition) clusters to vSphere Lifecycle Manager images.

- a) Expand **Clusters** and click **PATCH /v1/clusters/{id}**.
- b) Enter the vSphere cluster ID for the `id` parameter.
- c) Enter the following information for the `body` parameter.

```
{ "clusterTransitionSpec": { } }
```

Optionally, you can specify more options:

```
{
 "clusterTransitionSpec":
 {
 "remediationOptionsSpec":
 {
 "disableDpm": false,
 "disableHac": false,
 "enableQuickboot": true,
 "evacuateOfflineVms": true,
 "preRemediationPowerAction": "DO_NOT_CHANGE_VMS_POWER_STATE", "remediationFailureAction":
 {
 "action": "RETRY",
 "retryCount": 3,
 "retryDelay": 1200
 }
 }
 }
}
```

6. Monitor the task in the **Tasks** panel.

If the task completes successfully, you can repeat these steps for any additional vSphere clusters you want to transition.

If the task fails, resolve the issues and try again.

**Troubleshooting API Error Messages**

Review the following common error messages that you might see when transitioning vSphere clusters from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images. Refer to the remediation information for the next steps.

The underlying cause of a failed step might be within an asynchronous task or the original API call.

| Error Message                                                                                                             | Possible cause                                                                                                                                                          | Remediation                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Personality <Image Name> with same specification already exists                                                           | A vSphere Lifecycle Manager image with the exact same parameters already exists in SDDC Manager. You cannot import duplicate images, even if they have different names. | No action required. Use the existing image in SDDC Manager to transition the vSphere cluster(s).                                                 |
| Failed to import the image. Personality with the specified name/id already exists.                                        | A vSphere Lifecycle Manager image with the same name already exists in SDDC Manager.                                                                                    | For vCenter 8.0.3 and below, repeat the workflow with a new name.<br>For vCenter 9.0 and above, rename the source image and repeat the workflow. |
| Highest ESX version <Version> in cluster: <Cluster Name> does not match version: <Version> in cluster image: <Image Name> | The vSphere Lifecycle Manager base image does not match the ESX version running on the hosts in the vSphere cluster.                                                    | Update the vSphere Lifecycle Manager image to match the ESX version on the hosts.                                                                |

**Use UMDS on SDDC Manager to Get ESX Component Data**

SDDC Manager 9.0 acts as an Update Manager Download Service (UMDS) repository in vSphere Lifecycle Manager and it is automatically added as a download source for ESX component data in vSphere. Depending on how you access binaries, you may need to perform some steps using UMDS on the SDDC Manager appliance in order to have access to all ESX component information.

- SDDC Manager is upgraded to 9.0.
  - For an online depot: [Connect SDDC Manager to a Depot for Accessing Bundles](#).
  - For an offline depot: [Connect SDDC Manager to a Depot for Accessing Bundles](#) and [Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot](#).
  - For no depot:
    - [Download VMware Cloud Foundation 9.0 Upgrade Bundles Without an Online or Offline Depot](#).
    - Confirm that the compatibility data bundle used by vSphere Lifecycle Manager (`vvs-consolidated-bundle.zip`) is present in the `/nfs/vmware/vcf/nfs-mount/umds/patch-store/vvs` directory on the SDDC Manager appliance.
- If it is not:

| Task                                                                                                   | Details                                                |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Download the compatibility data to a computer with an internet connection using the VCF Download Tool. | <code>./vcf-download-tool umds run -D -m</code>        |
| Copy <code>vvs-consolidated-bundle.zip</code> to the SDDC Manager appliance.                           | You can copy it to any location that has enough space. |

| Task                                                                         | Details                                                                                                                                                        |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Import the compatibility data file using UMDS on the SDDC Manager appliance. | Navigate to <code>/opt/vmware/vmware-umds/bin</code> and import the file.<br><br><code>vmware-umds -I --offline-bundle <i>absolute_path_to_zip_file</i></code> |

vSphere Lifecycle Manager must have all the required ESX component data before you can create an image to use to upgrade ESX.

The steps you need to perform depend on your depot configuration.

| Depot Type    | Next Steps                                                                                                                                                                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Online depot  | <ul style="list-style-type: none"> <li>Configure UMDS to connect to the websites of third-party vendors to download ESX component data (optional).</li> <li>Import third-party offline zip files (optional)</li> </ul>                                  |
| Offline depot | No additional steps required.                                                                                                                                                                                                                           |
| No depot      | <ul style="list-style-type: none"> <li>Import the ESX component data that you exported in <a href="#">Download VMware Cloud Foundation 9.0 Upgrade Bundles to an Offline Depot</a>.</li> <li>Import third-party offline zip files (optional)</li> </ul> |

- SSH in to the SDDC Manager appliance using the `vcf` user account.
- Navigate to `/opt/vmware/vmware-umds/bin`.
- (Optional) Configure UMDS to connect to the websites of third-party vendors to download ESX component data.

**Note:** For online depot only.

- Add the URL.

```
vmware-umds -S --add-url https://host_url/index.xml --url-type HOST
```

- Download ESX component data.

```
./vmware-umds -D
```

- (Optional) Import a single third-party add-on or component as an offline zip file.

**Note:** For online depot and no depot.

```
vmware-umds -I --offline-bundle absolute_path_to_zip_file
```

Replace `absolute_path_to_zip_file` with the path to the zip file. For example: `/tmp/VMware-ESX-9.0.0-24197573-vib-test-certs.zip`.

- Import the ESX component data that you previously exported.

**Note:** For no depot only.

- Copy the exported data to a location on the SDDC Manager appliance.
- Run the following command:

```
vmware-umds --import --import-store repository_path
```

Replace `repository_path` with the location to which you copied the data.



## Upgrade to ESX 9.0 for VMware Cloud Foundation

Prior to VMware Cloud Foundation 9.0, the product supported both vSphere Lifecycle Manager baseline clusters and vSphere Lifecycle Manager image clusters. Starting with VMware Cloud Foundation 9.0, only vSphere Lifecycle Manager image clusters are supported.

- All vSphere clusters in a workload domain must be managed by vSphere Lifecycle Manager images before they can be upgraded to ESX 9.0. If any clusters are still using legacy vSphere Lifecycle Manager baselines, they must be transitioned to use vSphere Lifecycle Manager images before proceeding. See [Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images](#).
- Confirm that the ESX passwords are valid.
- Ensure that the domain for which you want to perform cluster-level upgrade does not have any ESX hosts or vSphere clusters in an error state. Resolve the error state or remove the ESX hosts and vSphere clusters with errors before proceeding.
- You must upgrade NSX Manager and vCenter before you can upgrade ESX hosts with a vSphere Lifecycle Manager image.
- Create a vSphere Lifecycle Manager image using the vSphere Client. See [Working with Image Library](#).
  - Use the ESX version specified in the VMware Cloud Foundation BOM.
  - You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but may include different vendor or firmware and driver add-ons. When you create alternate images, you also specify the rules that are used to filter the hosts to which the alternative image is assigned.
  - If you plan to use the **Import from a file** option to import a vSphere Lifecycle Manager image, you must first export the files from the vSphere Client.

The process of upgrading to ESX 9.0 involves the following steps:

- Importing one or more images into SDDC Manager from vSphere.
- Using that image to upgrade to ESX 9.0.

**Note:** For the 9.0 release, VMware ESXi has been rebranded to VMware ESX.

Starting with VMware Cloud Foundation 9.0, NSX kernel modules (VIB files) are installed as part of ESX host cluster upgrades.

If you are using external (non-vSAN) storage, the following procedure updates the ESX hosts attached to the external storage. However, updating and patching the storage software and drivers is a manual task and falls outside of SDDC Manager lifecycle management. To ensure supportability after an ESX upgrade, consult the vSphere HCL and your storage vendor.

You can configure multiple cluster update tasks with different schedules without having to wait for the first cluster update to complete. For example, you could schedule an update for `cluster-1` and `cluster-2` to run immediately and then configure another update task for `cluster-3` and schedule it to run overnight.

- Import a vSphere Lifecycle Manager image into SDDC Manager.
  - In the SDDC Manager UI, click **Lifecycle Management** > **Image Management**.
  - Click **Import Image**.
  - Select the import method, provide the required information, and click **Import**.

| Option                | Description                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------|
| Import from a vCenter | Select the vCenter where you created the image from the drop-down menu.<br>Select the upgrade image. |

| Option             | Description                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Import from a file | Enter a name for the image and then click <b>Select File</b> to import each of the required files.<br>Select the JSON, ZIP, and ISO files from your local computer. The ISO file is optional. |

You can view status in the **Tasks** panel.

- In the navigation pane, click **Inventory** > **Workload Domains**.
- On the Workload Domains page, click the domain you are upgrading and then click the **Updates** tab.
- Click **Precheck** to run the upgrade precheck.  
Resolve any issues before proceeding with the upgrade.
- In the Available Updates section, click **Configure Update** for ESX.
- Read the introductory information and click **Next**.
- Select the clusters to update and click **Next**.  
The default setting is to upgrade all clusters. To upgrade specific clusters, select **Custom selection** and select the clusters to upgrade.  
For VCF 9.0 and 9.0.2, if you are upgrading a stretched cluster, a virtual vSAN Witness Host is also upgraded. A physical vSAN Witness Host must be upgraded manually, before upgrading the ESX hosts in the stretched cluster.  
For VCF 9.0.1, you must manually upgrade the vSAN Witness Host, before upgrading the ESX hosts in the stretched cluster.
- Click **Assign Image**, select an imported image to assign to the selected clusters, and click **Next**.
- Select the upgrade options and click **Next**.  
By default, the selected clusters are upgraded in parallel. If you selected more than ten clusters to be upgraded, the first ten are upgraded in parallel and the remaining clusters are upgraded sequentially. To upgrade all selected clusters sequentially, select **Enable sequential cluster upgrade**.  
Select **Enable Quick Boot** to reduce the upgrade time by skipping the physical reboot of the host.  
Select **Migrate Powered Off and Suspended VMs** to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster.  
Select **Enforce Live Patch** when the cluster image includes a Live Patch. With the **Enforce Live Patch** option, vSphere Lifecycle Manager does not place the hosts in the cluster into maintenance mode, hosts are not rebooted, and there is no need to migrate the virtual machines running on the hosts in the cluster.
- Review the settings, and click **Finish** or **Run Precheck**.  
VMware Cloud Foundation runs a cluster image hardware compatibility and compliance precheck. Resolve any reported issues before proceeding.
- After the precheck succeeds, click **Schedule Update**, select a scheduling option, and click **Finish**.  
If the precheck report includes any errors or warnings, you must select the checkbox to proceed.

12. Monitor the upgrade progress. See [Monitoring VMware Cloud Foundation Updates](#).

If you are upgrading multiple clusters in parallel and one or more of the cluster upgrades fails, the upgrade continues on the remaining clusters. In addition, you can configure a new upgrade task for the failed cluster(s) without having to wait for the original upgrade task to finish.

**Note:**

NSX 9.0 functionality is not available until all ESX hosts in all vSphere clusters that share an NSX Manager are upgraded to 9.0 and the NSX Finalize subtask of the ESX upgrade step succeeds. If the NSX Finalize subtask fails, the ESX upgrade can still reach completion but a subtask error is displayed in the task panel.

After the upgrade is complete, allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade may be cancelled or auto-reconciliation may fail.

Upgrade the vSAN Disk Format for vSAN clusters. Your vSAN cluster continues to run smoothly if you use a previous disk format version. For best results, upgrade the objects to use the latest on-disk format. The latest on-disk format provides the complete feature set of vSAN. See [Update vSAN On-Disk Format Versions](#).

## Upgrade vSphere Distributed Switch Versions

[Optional] Upgrade the vSphere Distributed Switch in the vSphere Client to take advantage of features that are available only in the later versions.

ESX and vCenter upgrades are completed.

1. On the vSphere Client Home page, click **Networking** and navigate to the vSphere Distributed Switch.
2. Right-click the vSphere Distributed Switch and select **Upgrade > Upgrade Distributed Switch**.
3. Select the vSphere Distributed Switch version that you want to upgrade the switch to and click **Next**.

The vSphere Distributed Switch is successfully upgraded.

## Update vSAN On-Disk Format Versions

Upgrade the vSAN on-disk format version using the vSphere Client to take advantage of features that are available only in the later versions.

- ESX and vCenter upgrades are completed.
- Verify that the disks are in a healthy state. Navigate to the Disk Management page to verify the object status.
- Verify that your ESX hosts are not in maintenance mode. When upgrading the disk format, do not place the ESX hosts in maintenance mode.
- Verify that there are no component rebuilding tasks currently in progress in the vSAN cluster. For information about vSAN resynchronization, see vSphere Monitoring and Performance.
- The upgrade may cause temporary resynchronization traffic and use additional space by moving data or rebuilding object components to a new data structure.

1. In the vSphere Client, navigate to the vSAN cluster.
2. Click the **Configure** tab.
3. Under **vSAN**, select **Disk Management**.
4. Click **Pre-check Upgrade**. The upgrade pre-check analyzes the cluster to uncover any issues that might prevent a successful upgrade. Some of the items checked are host status, disk status, network status, and object status. Upgrade issues are displayed in the **Disk pre-check status text** box.
5. Click **Upgrade**.
6. Click **Yes** on the Upgrade dialog box to perform the upgrade of the on-disk format.

vSAN successfully upgrades the on-disk format. The On-disk Format Version column displays the disk format version of storage devices in the cluster.

## Upgrade vSAN File Service

When you upgrade the file service, the upgrade is performed on a rolling basis.

Ensure that the following are upgraded:

- ESX Hosts
- vCenter
- vSAN disk format

During the upgrade, the file server containers running on the VMs which are undergoing upgrade fails over to other virtual machines. The file shares remain accessible during the upgrade. During the upgrade, you might experience some interruptions while accessing the file shares.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. Under **File Service**, click **Edit**.
5. Click **Check upgrade** from the list.
6. In the Upgrade File Service dialog box, select one of the following deployment options and then click **Upgrade**.

| Option                    | Action                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Automatic approach</b> | This is the default option. This option lets the system search and download the OVF. After the upgrade begins, you cannot cancel the task.<br><br><b>Note:</b><br>vSAN requires internet connectivity for this option.                                                                                                                                           |
| <b>Manual approach</b>    | This option allows you to browse and select an OVF that is already available on your local system. After the upgrade begins, you cannot cancel the task.<br><br><b>Note:</b> If you select this option, you should upload all the following files:<br><ul style="list-style-type: none"> <li>• VMware-vSAN-File-Services-Appliance-x.x.x.x-x_OVF10.mf</li> </ul> |

| Option | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <ul style="list-style-type: none"> <li>VMware-vSAN-File-Services-Appliance-x.x.x.x-x-<code>_OVF10.cert</code></li> <li>VMware-vSAN-File-Services-Appliance-x.x.x.x-x-<code>system.vmdk</code></li> <li>VMware-vSAN-File-Services-Appliance-x.x.x.x-x-<code>cloud-components.vmdk</code></li> <li>VMware-vSAN-File-Services-Appliance-x.x.x.x-x-<code>log.vmdk</code></li> <li>VMware-vSAN-File-Services-Appliance-x.x.x.x-x-<code>_OVF10.ovf</code></li> </ul> |

## Monitoring VMware Cloud Foundation Updates

You can monitor in-progress updates for VMware Cloud Foundation components.

- In the In-Progress Updates section, click **View Status** to view the high-level update progress and the number of components to be updated.  
Details of the component being updated is displayed.
- Click the arrow to see a list of tasks being performed to update the component.  
After each task is completed successfully, it shows a green check mark.  
When all tasks to update a component have been completed, the update status for the component is displayed as **Updated**.  
If a component update fails, the status is displayed as **Failed**. The reason for the failure as well as remediation steps are displayed. Resolve the issues and retry the update.

## Accessing the VMware Cloud Foundation Update History

The Update History page displays all updates applied to a workload domain.

- In the navigation pane, click **Inventory > Workload Domains**.
- Click the name of a workload domain and then click the **Update History** tab.  
All updates applied to this workload domain are displayed. If an update bundle was applied more than once, click **View Past Attempts** to see more information.

## Accessing the VMware Cloud Foundation Upgrade Log Files

You can check the log files for failed upgrades to help troubleshoot and resolve issues.

- SSH in to the SDDC Manager appliance with the `vcf` user name and enter the password.
- To access upgrade logs, navigate to the `/var/log/vmware/vcf/lcm` directory.
  - `lcm-debug` log file contains debug level logging information.
  - `lcm.log` contains information level logging.

**Note:** Once SDDC Manager is upgraded to 9.0, all logging information will be sent to the `lcm-debug` log file and `lcm.log` will no longer be updated.